CHAPTER
FOURTEEN

PROCESS RESPONSES AND TRANSFER FUNCTIONS

Mathematics may be defined as the subject in which we never know what we are talking about, nor whether what we are saying is true.

—Bertrand Russell



CHAPTER HIGHLIGHTS

As we have already discussed, PLCs control machines and processes via discrete, analog, and special I/O interfaces that communicate with the real world. With the aid of control software, a user can program a PLC to control any process through these I/O interfaces. In our discussion, however, we have not yet explained process control in its true form, as it applies to the behavior and control of a manufacturing activity. Therefore, we will dedicate this chapter to the explanation of basic process control concepts. In the next chapter, we will explain how these concepts apply to a process control operation.

14-1 PROCESS CONTROL BASICS

Process control is the regulation of designated process parameters to within a specified target range or to a set target value called the **set point**. Process control is most often used in product manufacturing, because many factors, such as color, composition, and density, must be accurate for a product to be well made. Therefore, to implement a quality product, process control is used to monitor and correct process parameters by analyzing the state of dynamic variables. Dynamic variables are process characteristics, such as temperature, flow, and pressure, that vary with time. Through its I/O interfaces, a PLC can regulate these dynamic variables to a desired set point, thus implementing process control.

Figure 14-1 illustrates the basic concept of process control using a reactor tank in which steam controls the temperature in the tank. In this case, the temperature must be maintained at a target value, or set point, of 125°C. Because it varies with time, the temperature is the dynamic variable, which is also called the **process variable** (*PV*). The steam level, which regulates the process variable (i.e., raises and lowers the temperature), is called the **control variable** (*CV*). The valve that controls the amount of steam entering the reactor tank's jacket is called the **control element**, or *final output field device*, because the more the controller opens the valve, the more the steam increases the temperature.

Figure 14-2 shows the block diagram of the process control system illustrated in Figure 14-1. The PLC reads the process variable from the system (i.e., obtains feedback) and compares it with the set point to determine how well the temperature is being regulated. This configuration is known as a closed-loop system, because the controller uses feedback to monitor the system. An open-loop system does not use feedback, so the controller does not receive process variable data. Figure 14-3 illustrates the configuration of an open-loop system.

If the temperature reading in the process in Figure 14-2 is low, the controller will adjust the control variable by opening the valve to allow steam to enter the tank, thereby raising the temperature. The controller will then recheck

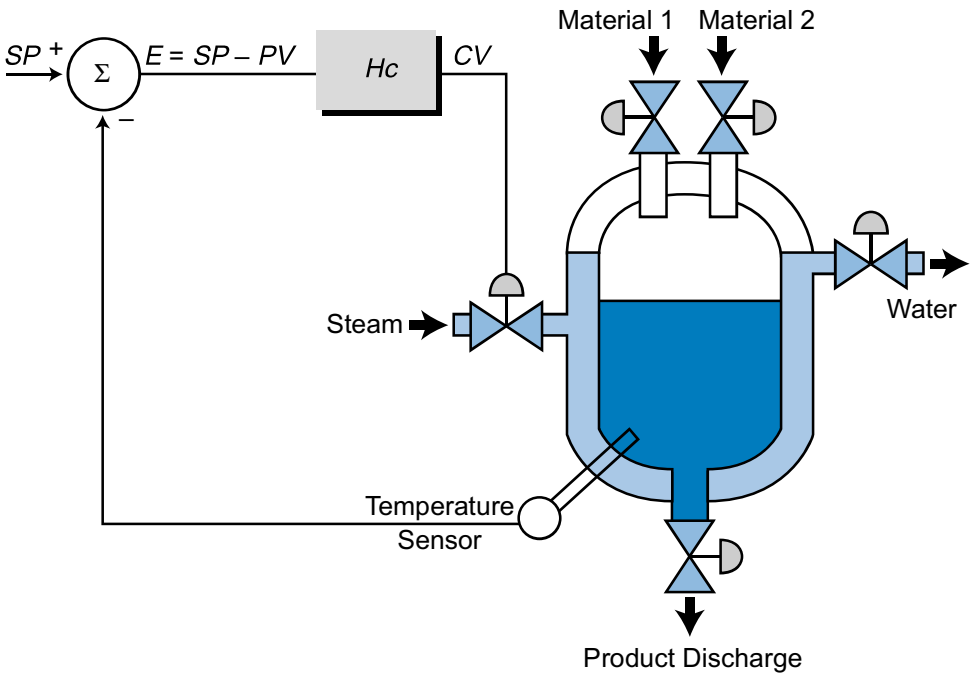


Figure 14-1. Reactor tank control system.

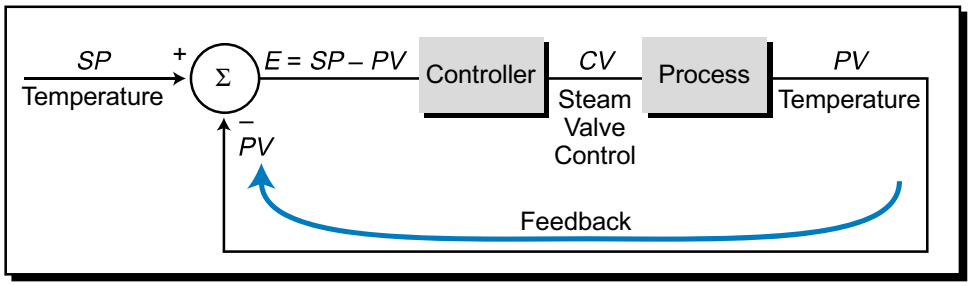


Figure 14-2. Block diagram of the closed-loop reactor tank system.

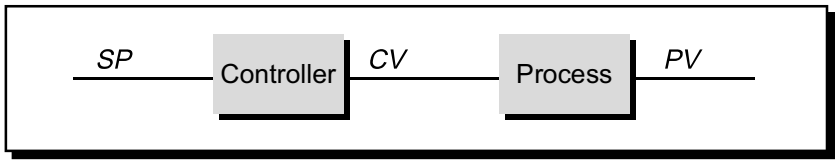


Figure 14-3. Open-loop process control system.

the process variable. If the temperature is still low, it will again open the steam valve to increase the temperature. The controller will repeat this process until the actual temperature (process variable) is as close as possible to the target temperature (set point value). The difference between the process variable and the set point is called the **error**. The error can be either positive or negative, depending on whether the process variable is too high or too low. However, regardless of the sign of the error, the controller still performs the same basic function—adjusting the process variable until it equals the set point (i.e., making the error equal to zero). Once equality is achieved, the

process is said to be *regulated*. As we will discuss later, many factors can disturb the system, thus altering the process variable. Therefore, the controller must adjust the control variable to correct for errors created by these factors, as well as to correct for errors due to a change in the set point.

In a PLC-based system, a control system block diagram like the one shown in Figure 14-2 can be expanded to include interfaces that control the field output devices, as well as those that read process variable input data (see Figure 14-4). Figure 14-5 shows a control system that uses a PID interface (discussed in Chapter 8) to implement process control independent of the PLC. The next chapter will further explain PID control.

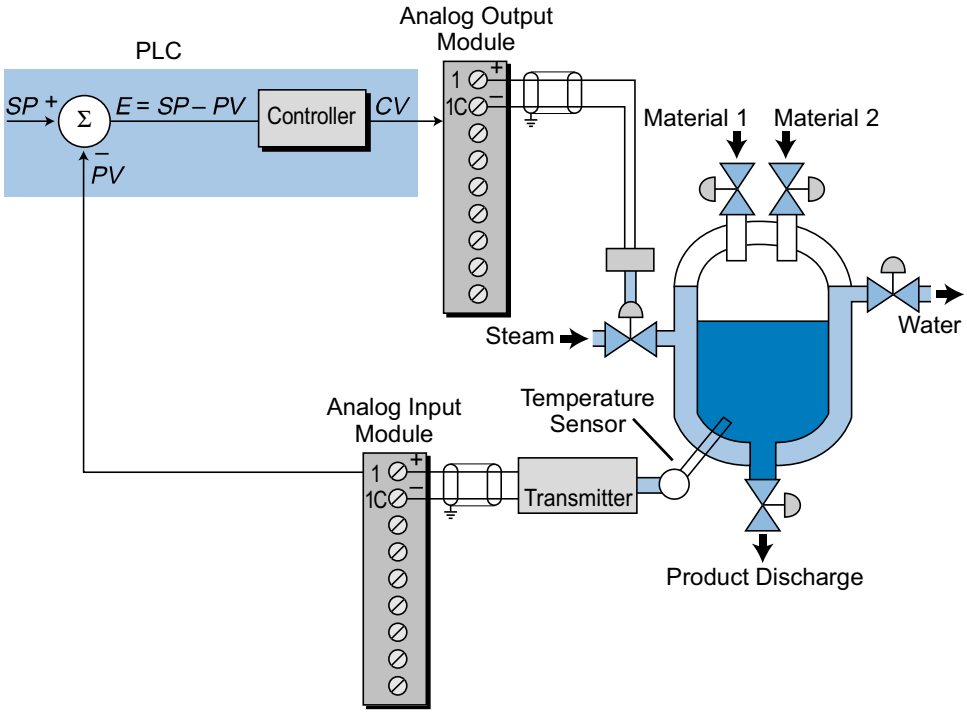


Figure 14-4. Control system block diagram including I/O interfaces.

The adjustment of the control variable according to data obtained by reading the process variable and analyzing the error between it and the set point is referred to as the **control loop**. Most control loops are affected by *disturbances*, which influence the process and alter the process variable (see Figure 14-6). To understand disturbances, let's examine a simple control loop example—a car's cruise control mechanism. As shown in Figure 14-7, once the cruise control has been set at a target speed (set point), the system will maintain that speed by keeping the accelerator (control variable) at a constant level. However, if the system experiences a disturbance, such as pavement with higher friction or an uphill climb, the system will increase the control variable (i.e., increase acceleration) to maintain the set point speed. This is demonstrated by the fact that the accelerator pedal of a car

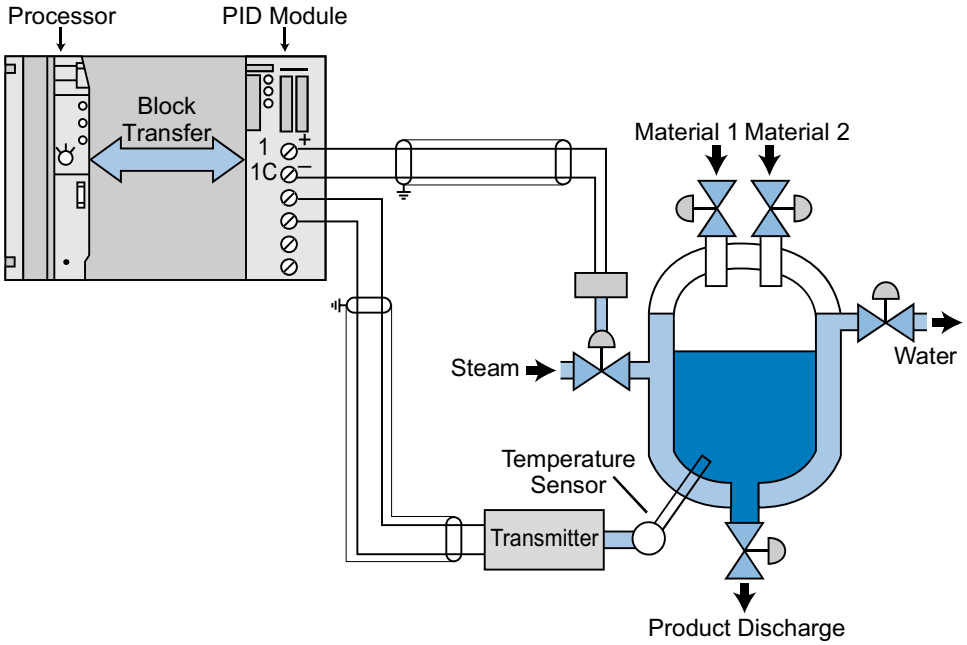


Figure 14-5. Control system using a PID interface.

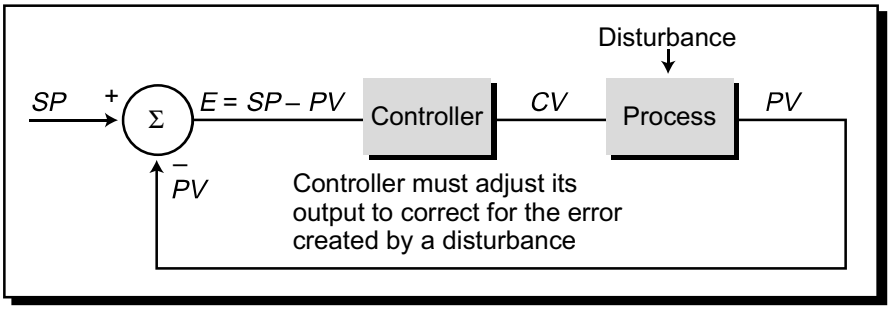


Figure 14-6. A control loop with a disturbance.

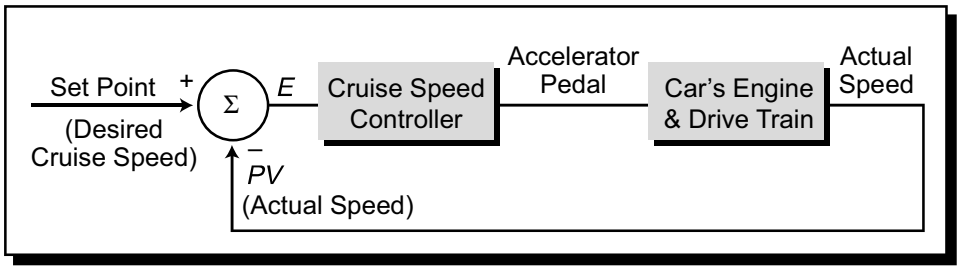


Figure 14-7. Cruise control process loop.

under cruise control is depressed further when the car is going uphill than when it is going downhill. Figure 14-8 illustrates how a cruise control system compensates for disturbances. The components that form this simple system respond to keep the process variable at the set point by adjusting the control variable to maintain the error at zero during the disturbance. This, in fact, is the main function of process control—monitoring the error signal generated by the system and adjusting the outputs accordingly.

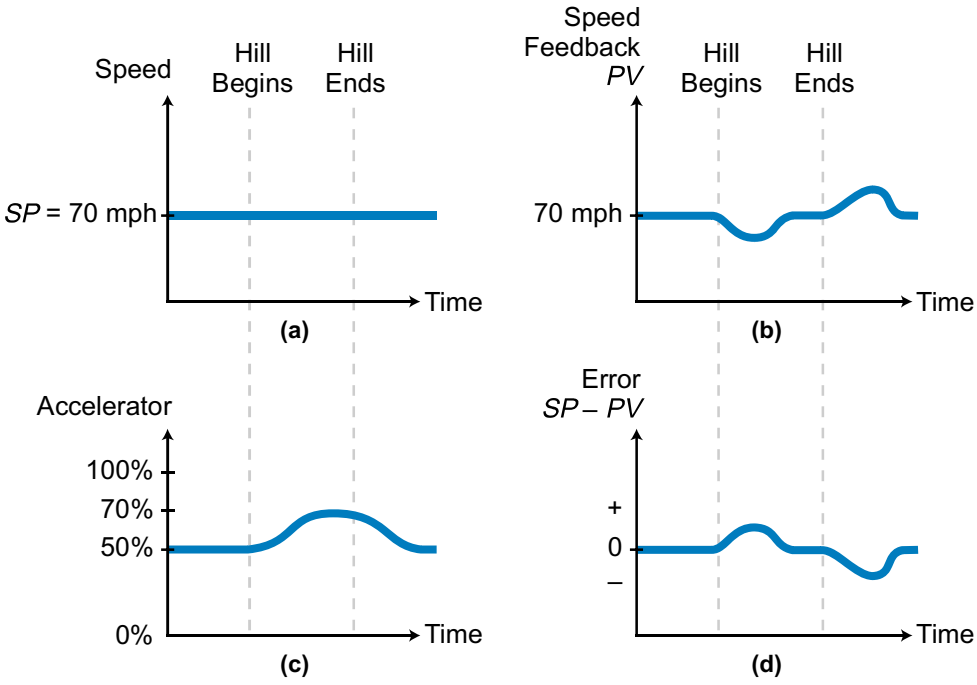


Figure 14-8. Cruise control compensation graphs showing **(a)** the set point speed, **(b)** the reaction of the process variable to the disturbance, **(c)** the reaction of the control variable to the disturbance, and **(d)** the error.

14-2 CONTROL SYSTEM PARAMETERS

As we just discussed, a controller calculates the process error (*E*) as the difference between the set point (*SP*) and the process variable (*PV*). It then uses this error data as the input for its control computations. It uses this input data to apply control to the process by manipulating the control variable (*CV*) so as to eliminate the error. The way it does this depends on the controller's mode (covered in the next chapter) and the degree of error. Therefore, a thorough understanding of the relationship between error and the control variable is beneficial when designing and applying a process control application.

ERROR

The control deviation, or error, between the set point and the process variable is given by the equation:

$$= \quad -$$

where:

$$=$$

$$=$$

$$=$$

This equation can also be represented as:

$$E = PV - SP$$

where the set point is subtracted from the process variable. Both equations give the same magnitude (value) of error, but with different signs. The first error equation ($E = SP - PV$) is used in *negative feedback* control loops, where the process variable is fed back into the system and subtracted from the set point for error correction control. Negative feedback is used in closed-loop control systems instead of positive feedback, because negative feedback reduces the error in the system while positive feedback magnifies it. As shown in Figure 14-9a, positive feedback results in a feedback signal that is in phase with the error deviation, thus enhancing the system error. Negative feedback, on the other hand, produces a signal that is directly out of phase with the error deviation (see Figure 14-9b). This reversal of phase causes the system error to decrease as the control variable regulates the process. All feedback controllers produce a 180° phase shift in the feedback signal to provide negative feedback.

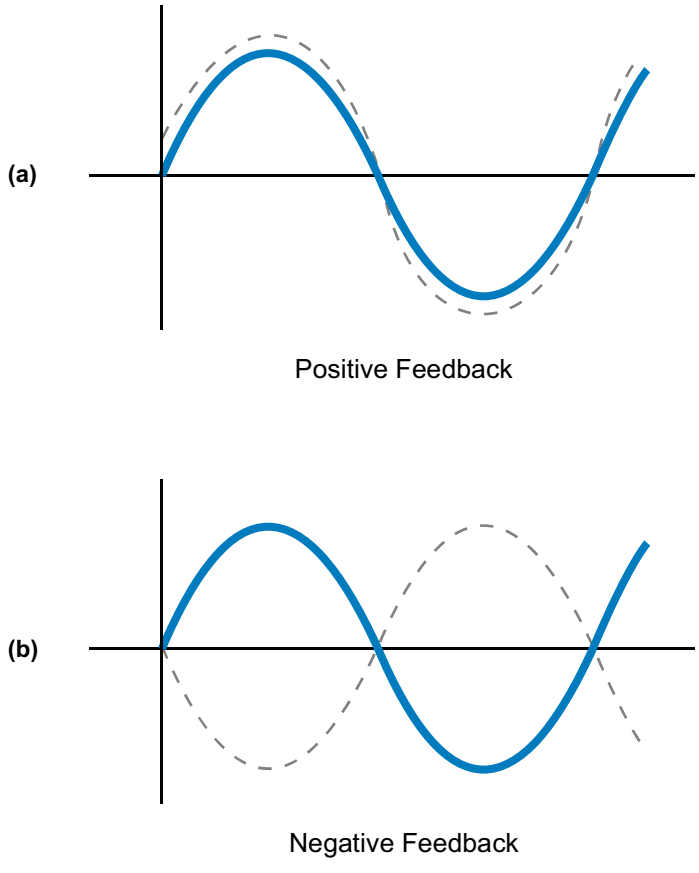


Figure 14-9. (a) Positive and **(b)** negative feedback.

INTERPRETATION OF ERROR

The representation of error as the difference between the set point and the process variable provides a "natural" value; that is, a value expressed in the units being measured. For example, a PLC that receives process variable and set point data in the form of analog counts will express the error as a function of analog counts. Likewise, a system with a set point of 125°C and a process variable of 120°C will have an error of 5°C. However, the system cannot determine if 5°C is an acceptable error because it does not know how close the error is to zero relative to the variable range. Therefore, another way for the controller to calculate error is as a percentage of the target set point. This is expressed as:

$$E = \frac{SP - PV}{SP}$$

Using this equation, the error for the previous temperature example would be:

$$E = \frac{125°C - 120°C}{125°C}$$
$$= 4\%$$

This indicates that the 5°C system error is within 4% of the set point target. This percentage value provides more information than the 5°C error value; however, the controller requires even more information to adjust the process correctly.

The expression of error as a percentage of the process variable range provides an even more indicative value of error. The range of *PV* indicates the maximum and minimum values that the process value can have. Figure 14-10 illustrates the error as a percentage of the process variable range. Mathematically, this can be expressed as:

$$E\% = \frac{SP - PV}{PV_{min} - PV_{max}}$$

where:

$E\%$ = the error as a percentage of PV range
SP = the set point
PV = the process variable
PV_{max} = the maximum value of PV
PV_{min} = the minimum value of PV

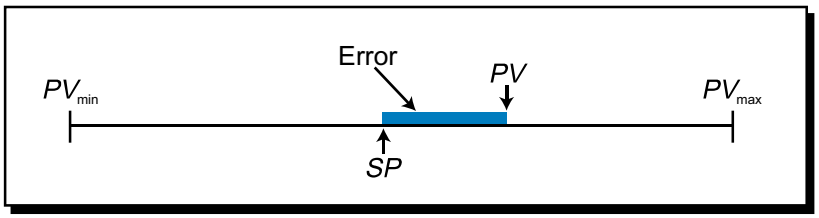


Figure 14-10. Error as a percentage of the process variable range.

Note that, in this equation, the sign of the terms PV_{min} and PV_{max} (positive and negative, respectively) are such that the error will be positive if the process variable is above the set point and negative if it is below the set point. This error representation provides additional information about the magnitude of the error.

EXAMPLE 14-1

A process with a temperature set point of 180°C has a process variable input of 168°C (see Figure 14-11). Express the error as a percentage of range given that the process variable has a range of **(a)** 100°C to 200°C and **(b)** 50°C to 350°C.

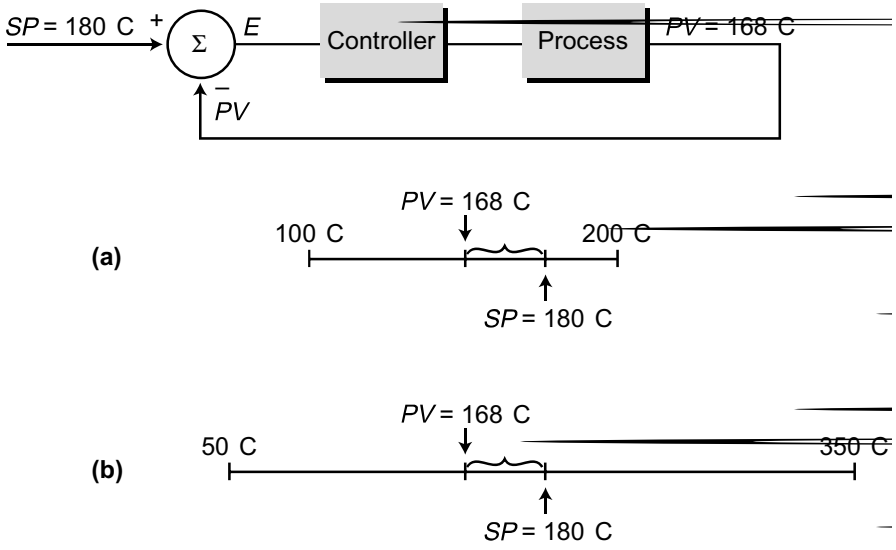


Figure 14-11. Process control loop for Example 14-1 given **(a)** a 100–200°C process variable range and **(b)** a 50–350°C range.

SOLUTION

(a) The value of the error as a percentage of range (E%) is expressed as:

$$E = \frac{SP - PV}{PV_{min} - PV_{max}}$$

For a process variable range of 100 to 200°C, the error is:

$$E\% = \frac{180°C - 168°C}{100°C - 200°C} = \frac{12°C}{-100°C} = -0.12 = -12\%$$

(b) For a process variable range of 50 to 350°C, the error is:

$$E\% = \frac{180°C - 168°C}{50°C - 350°C} = \frac{12°C}{-300°C} = -0.04 = -4\%$$

Although the actual natural value of the error is the same in both parts (a) and (b)—i.e., 12°C—the magnitude of the error in the first case (12%) is three times greater than the magnitude of the error in the second case (4%). The negative sign of the error calculations indicates that the process variable is lower than the set point.

THE CONTROL VARIABLE

During the control of a process, the controller calculates the error value and adjusts the control variable accordingly to bring the error to zero. Like the error, the value of the control variable can also be expressed as a percentage of range; however, the control variable is expressed in terms of the full range of the controller's output (i.e., the control field device). This range of the controller output is defined as:

$$CV\% = \frac{CV_{actual} - CV_{min}}{CV_{max} - CV_{min}}$$

where:

$CV\%$ = the control variable value as a percentage of its range
CV_{actual} = the actual value of the controller output
CV_{max} = the maximum value of the controllable signal
CV_{min} = the minimum value of the controllable signal

The order and the sign of the nominator and denominator terms in this equation result in a control variable percentage value that is always positive, since the value of CV_{actual} cannot be less than its minimum possible value.

EXAMPLE 14-2

The PLC system shown in Figure 14-12 has an analog output module that sends a 0–10 VDC signal to an electric-to-pneumatic (I/P) converter. The I/P converter controls a steam valve that regulates the process to a set point of 140°C. The range of the controller output is from 0 to 4095 counts, which provides a range of 20 to 220°C in steam temperature control. The process variable has a value of 130°C. Find the percentage of controller output as a function of voltage.

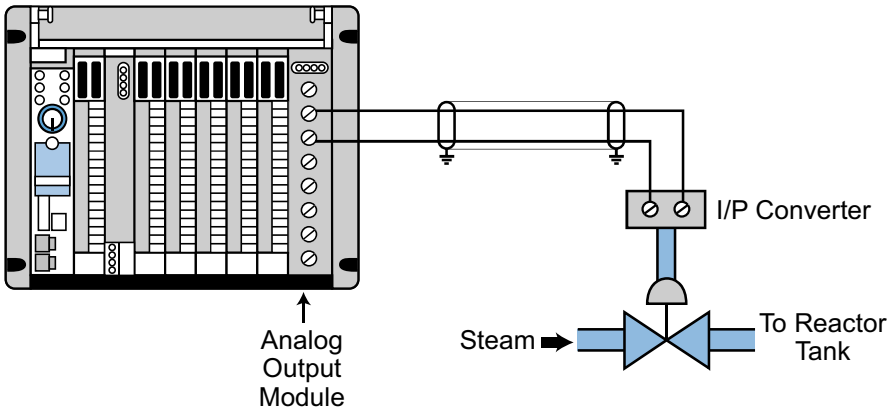


Figure 14-12. Process regulated by an I/P converter.

SOLUTION

Figure 14-13 illustrates the relationship between the control variable output and the controllable range of temperature. Since the relationship between the controller output and the temperature is linear, the equation of the control variable as a function of voltage is represented by:

$$CV_{volt} = \left(\frac{V_{max} - V_{min}}{Temp_{max} - Temp_{min}}\right)T - CV_{(T=0)}$$

$$CV_{volt} = \left(\frac{10\ V - 0\ V}{220°C - 20°C}\right)T - 1$$

$$= \left(\frac{10}{200}\right)T - 1$$

$$= \left(\frac{T}{20}\right) - 1$$

where T is the given value of the temperature and CV_{volt} is the output of the controller in voltage. Note that this equation takes the form of the equation of a line, $Y = mX + b$ (see Appendix E). At a temperature of 140°C, then, the controller output in voltage would be:

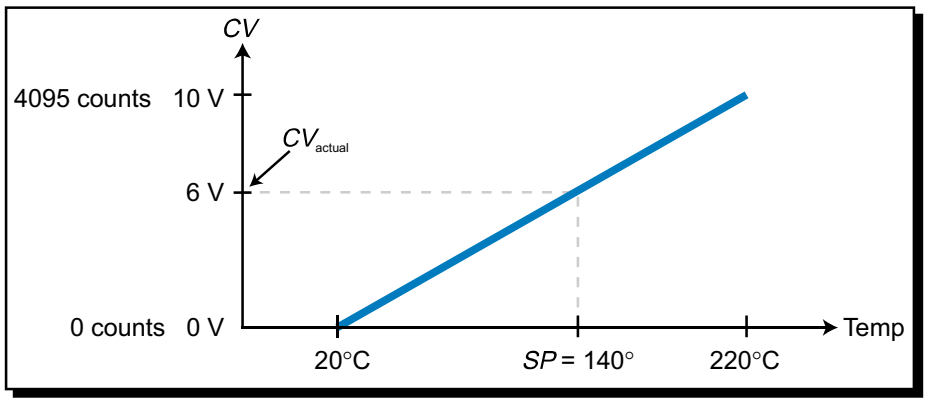


Figure 14-13. Relationship between control variable output and temperature range.

$$CV_{volt} = \left(\frac{140}{20}\right) - 1$$
$$= 6 \text{ volts}$$

So, the control variable in voltage as a percentage of total output would be:

$$CV\% = \frac{CV_{actual} - CV_{min}}{CV_{max} - CV_{min}}$$
$$= \frac{6\text{ V} - 0\text{ V}}{10\text{ V} - 0\text{ V}}$$
$$= 0.60 = 60\%$$

If the minimum temperature output of 20°C corresponded to a controller output of 1 volt instead of 0 volts, the percentage output would be different because the value of the control variable (CV_{actual}) would change, thus changing the percentage result (see Figure 14-14). A temperature of 140°C would require a 6.4 volt output, which as a percentage of the range would become:

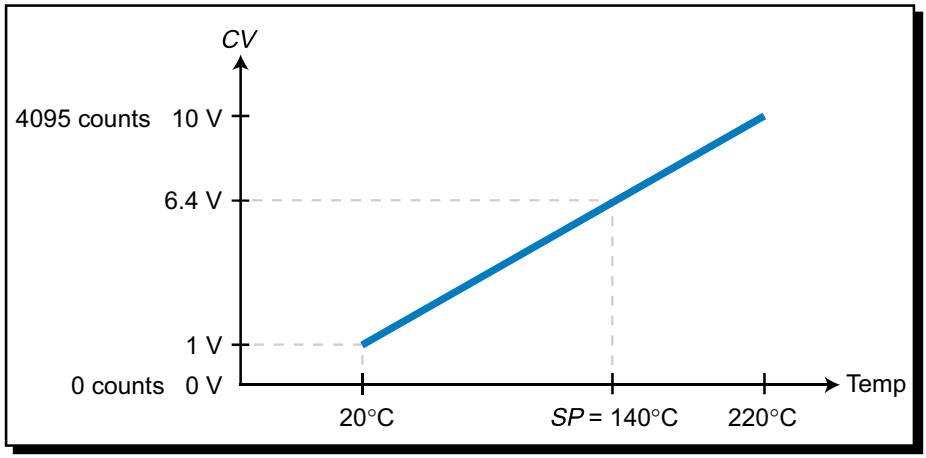


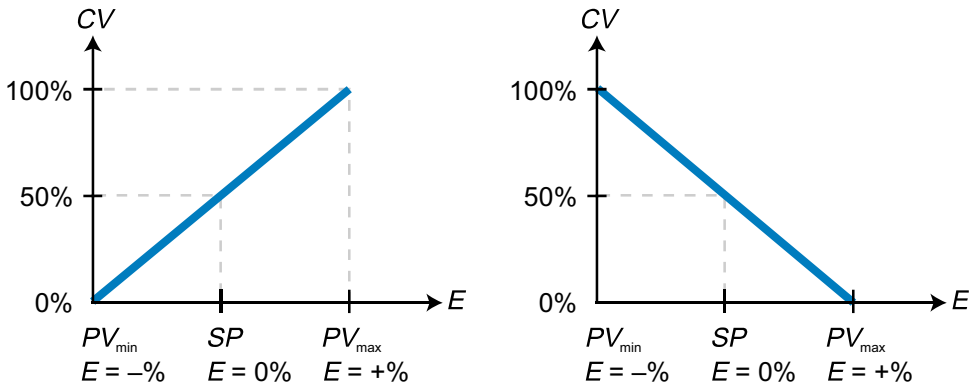
Figure 14-14. Minimum temperature output of 20°C corresponding to a 1 V control variable output.

$$CV\% = \frac{6.4\text{ V} - 0\text{ V}}{10\text{ V} - 0\text{ V}} = \frac{6.4}{10} = 0.64 = 64\%$$

Having a minimum control output value that is greater than zero is common in many process control systems, because most systems require that the control element be constantly on to regulate the process variable. Note, however, that the range of the control variable is still from 0 to 10 V.

ERROR AND THE CONTROL VARIABLE

The control variable can affect the error in two ways, as illustrated in Figure 14-15. If the error becomes more positive as the control variable *increases*, the action of the controller is called *direct acting*. On the other hand, if the error becomes more positive as the control variable *decreases*, the action of the controller is called *reverse acting.* Table 14-1 illustrates the relationship between error and direct- and reverse-acting controllers. The error values refer to the error as a percentage of the process variable range, meaning that a positive error corresponds to a process variable that is greater than the set point and a negative error corresponds to one that is less than the set point. Note that the equation $E = SP - PV$ yields error values that will have a sign that is opposite of those just described. Therefore, the correct way to identify the type of controller action (direct or reverse) is through the error computed as a percentage of the full range. Chapter 15 discusses reverse- and direct-acting controller modes in more detail.



Direct Acting
As percentage of error *increases*, the control variable *increases*

Reverse Acting
As percentage of error *increases*, the control variable *decreases*

Figure 14-15. Direct and reverse action in process control.

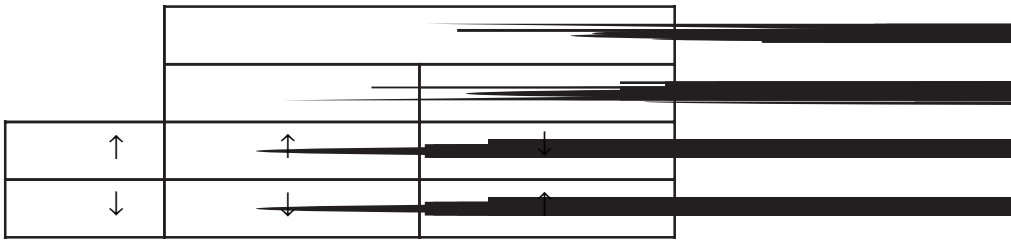


Table 14-1. Relationship between error and the control variable in direct- and reverse-acting controllers.

ERROR DEADBAND

The purpose of a process control system is to keep the error value as close to zero as possible. However, all systems have an allowable fluctuation in error, meaning that the error can vary from zero by a certain amount without hampering the final product. Figure 14-16 illustrates this fluctuation allowance, which is called the **error deadband**. Within the deadband, the controller treats the error as if it were zero, meaning that it will not make any corrective actions to the control variable. If the value of the error deviates from zero by more than the deadband, the controller will initiate a correction by changing its output (*CV*). The error deadband is a user-specified value, which the PLC stores in one of its registers.

Most PLCs that offer process control loop manipulation software have upper and lower limit alarms, which engage when the process variable deviates from the error deadband. An error will trigger these alarms, as shown

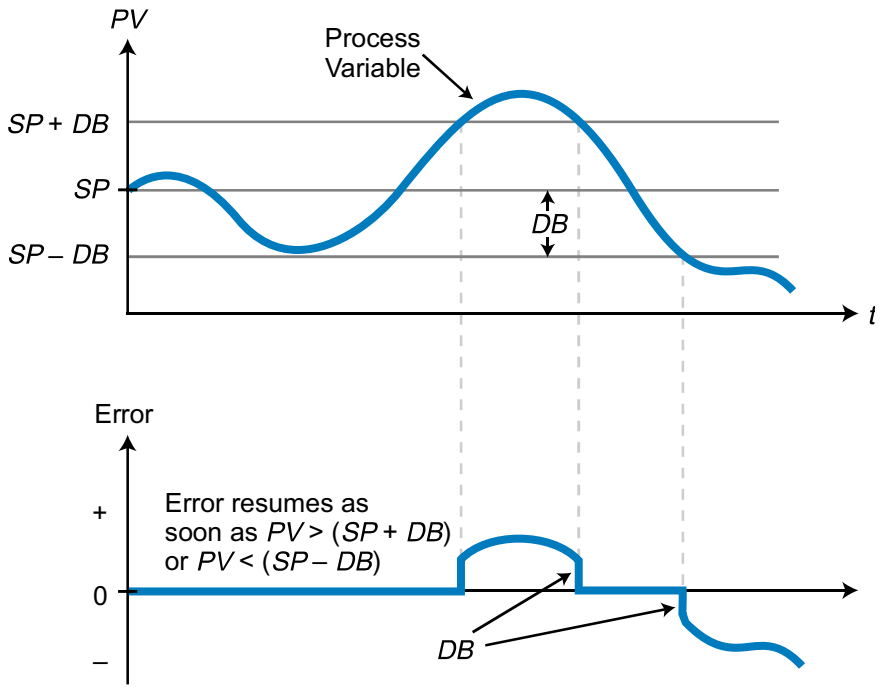


Figure 14-16. Error deadband.

in Figure 14-17, when the error condition exists for more than a specified amount of time. Once the error exceeds the upper or lower limit, a timer starts timing and triggers the alarm after it times out. The alarm indicator will remain ON until the process variable returns to within the specified range.

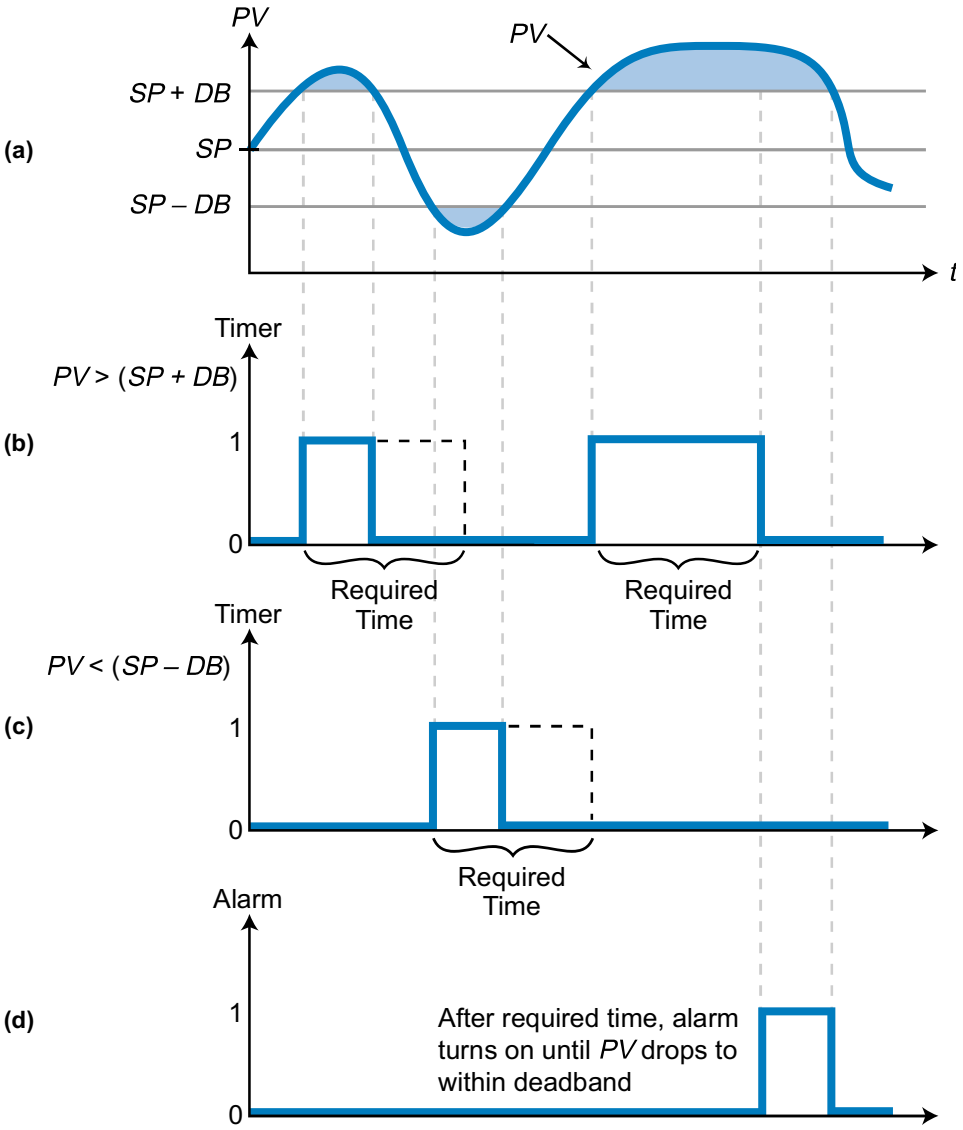


Figure 14-17. (a) Control loop process variable, **(b)** timer for upper limit alarm, **(c)** timer for lower limit alarm, and **(d)** alarm activation.

14-3 PROCESS DYNAMICS

The term *dynamics*, as used in process control, refers to the changes that occur in a process. These changes involve the response of the process system to changes in its input (*CV*), which occur when disturbances are present, or to changes in its set point (see Figure 14-18). Process dynamics does not refer

to the behavior of the process when the error is zero (or within the error deadband). At this point, the process is said to be at **steady state**. Instead, process dynamics deals with the system's (process variable's) reaction to corrective actions taken by the controller to bring the error to zero after it senses that the error is too large. Therefore, the analysis of process dynamics explores the relationship between the control variable and the process variable. This relationship is important during the "tuning," or adjustment, of system parameters, which we will discuss in the next chapter.

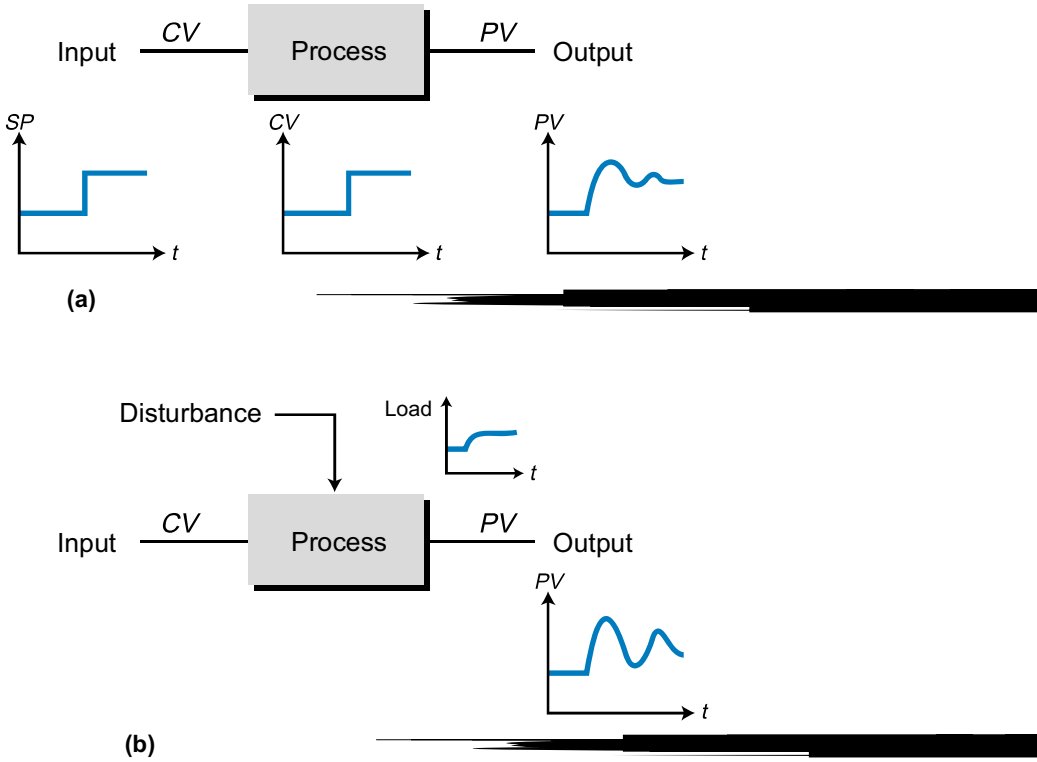
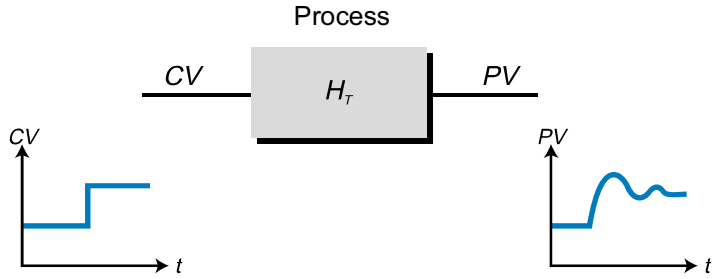


Figure 14-18. Process change due to **(a)** a change in set point and **(b)** a disturbance.

TRANSFER FUNCTIONS AND TRANSIENT RESPONSES

A process responds via the process variable (*PV*) to a change in input (*CV*) in a dynamic manner according to the characteristics of the process. These process characteristics, which include factors such as delay time and inherent physical responses of the process, are defined by a **transfer function**, represented by the term H_T (see Figure 14-19). A transfer function is an equation that describes a process in terms of response over time, as well as calculates the outcome of the process variable. Therefore, the value of the term H_T equals the value of the process variable at a particular control variable value and time, given the characteristics of the process. Every process has its own unique transfer function based on its particular characteristics, and for most processes, the transfer function equation is not known. Thus, certain

assumptions must be made about the process to estimate H_T. Experimentation can also be used to approximate the outcome of H_T (i.e., the process variable response) to a forced change in the process input. This experimental change in process input is called a **step test** and the response is called a **step response**.



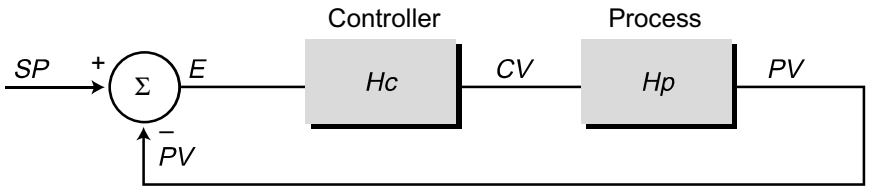
Change in *CV*...through H_T...affects change in *PV*

$$H_T = \frac{PV}{CV}$$

Figure 14-19. Transfer function.

The most important aspect of a transfer function is not so much its composition or form, but its response to sudden process input changes created by disturbances. This behavioral response of a process is called a **transient response**, and it includes the time required for the output to reach a steady-state final value given a sudden change in input. Transient responses provide much information about the dynamics of a process and, therefore, about the transfer function.

As shown in Figure 14-20, a closed-loop control system includes two transfer functions—one that defines the controller (*Hc*) and another that defines the process (*Hp*). The input to the controller's transfer function is the error signal (*E*), and its output is the control variable (*CV*). This control variable becomes the input to the process's transfer function, whose output is the process variable (*PV*). In this chapter, we will discuss the process's transfer function and its behavior. In the next chapter, we will discuss the controller's transfer function and the different forms that it can take.



$$Hc = \frac{CV}{E}$$

$$Hp = \frac{PV}{CV}$$

Figure 14-20. A closed-loop control system with two transfer functions.

To better understand transient responses and the information we can obtain from them, let's explore a basic example of an open-loop hot-water heater system. Figure 14-21 shows this process, while Figure 14-22 shows the corresponding process block diagram. The transfer function of the process depends on many factors, such as the rate of flow of the steam, the temperature of the steam, the temperature of the incoming water at the inlet, the ambient temperature, and the inflow and outflow rates of the water. Regardless of all these process factors, the controller must maintain the temperature in the tank (the process variable) as close as possible to the user-defined set point by manipulating the control variable. For this example, let's assume that the temperature in the tank (at steady state) is at a set point of 65°C, the temperature range spans from 15°C to 93°C, and the steam control valve is at 55% of its open position.

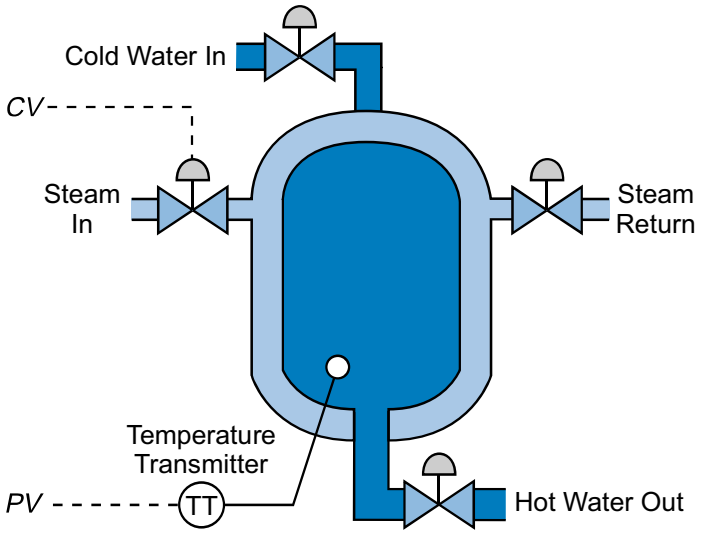


Figure 14-21. Hot-water heater system.

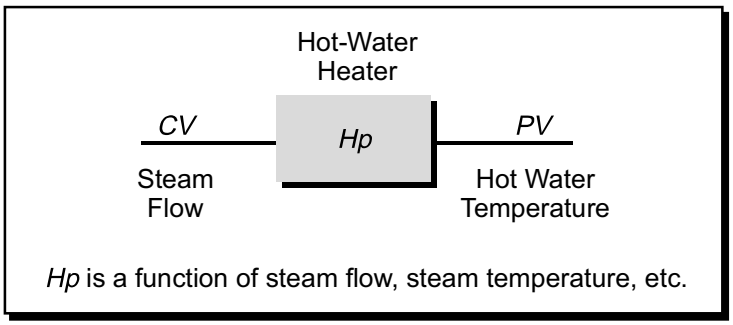


Figure 14-22. Water heater process diagram.

With a step change in valve position from 55% to 75% open, the temperature (*PV*) in the tank will begin to heat up (see Figure 14-23). After 15 minutes, the process variable will increase to 81°C. The process variable's behavior during

this 15-minute period is the transient response of the transfer function *Hp*. Note that the transient response is very smooth, heating the water slowly over the 15 minutes until steady state is once again achieved.

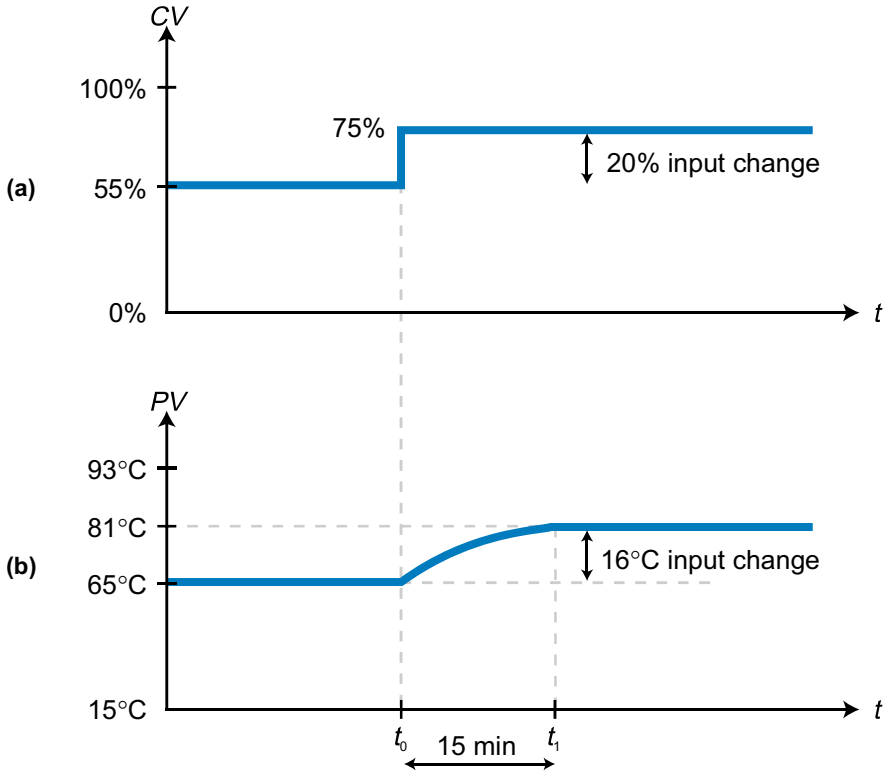


Figure 14-23. (a) Control variable step change and **(b)** its corresponding process variable change.

PROCESS GAIN

The **process gain**, represented by the term *K*, defines the ratio between process output and process input. This gain is another dynamic element that is observed in a transient response. It is calculated by dividing the change in process output over a period of time by the corresponding change in process input. Thus, the process gain is equal to the change in the process variable divided by the change in the control variable:

$$K = \frac{PV_{\text{final}} - PV_{\text{initial}}}{CV_{\text{final}} - CV_{\text{initial}}}$$

Hence, for the previous hot-water tank example, *K* would be the change in the tank temperature divided by the corresponding change in the valve output over the 15-minute period:

$$K = \frac{81°C - 65°C}{75\% - 55\%}$$
$$= \frac{16°C}{20\%}$$
$$= 0.8°C/\%$$

So, the process gain is 0.8°C/%. This means that the process variable (temperature) changes 0.8°C for every one percent of change in the control variable (steam valve). The process gain calculation of 0.8°C/% (see Figure 14-24) is only valid for the process conditions under which it was calculated (i.e., a constant inflow and outflow of water—Q_{in} and Q_{out}, respectively—at a constant water temperature). Under these conditions, the mass of water in the heater tank remains constant and the steam heating system's gain (0.8°C/%) operates linearly over the span of the temperature range. If either the input water flow Q_{in} or any other parameter changes, the gain of the system will also change.

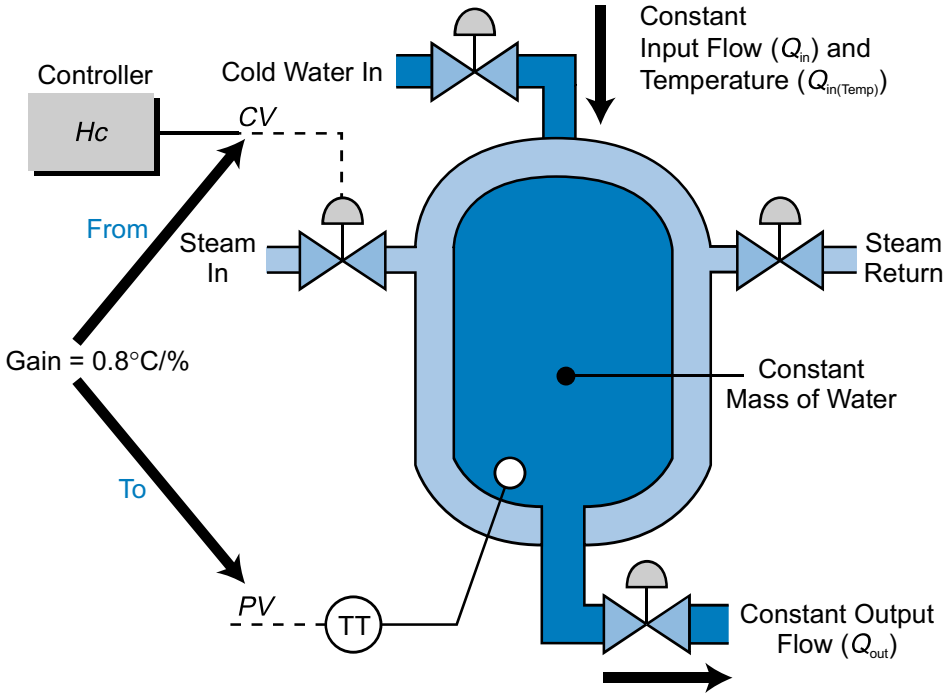


Figure 14-24. Process gain in the hot-water tank example.

DEAD TIME

The perfect response of a process variable to a step change in the control variable is instantaneous, as shown in Figure 14-25. In this type of perfect system, the process's transfer function is equal to 1, meaning that a control variable input immediately results in an equal process variable output. In

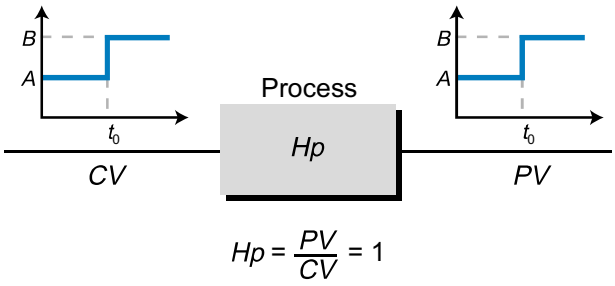


Figure 14-25. Instantaneous response of the process output to a step change in input.

reality, however, characteristics distinctive to the process influence the relationship between the control variable and process variable, resulting in a transfer function that is not equal to 1. One of these process characteristics is the inherent delay associated with an output's response to an input (see Figure 14-26). This delay is called the **dead time**. A typical example of dead time occurs when you first turn on the hot-water knob in the shower. The water will not become hot until all the cool water runs out of the pipe and the water from the hot-water heater reaches the shower head. The time that elapses between turning the knob and receiving the hot water is the dead time.

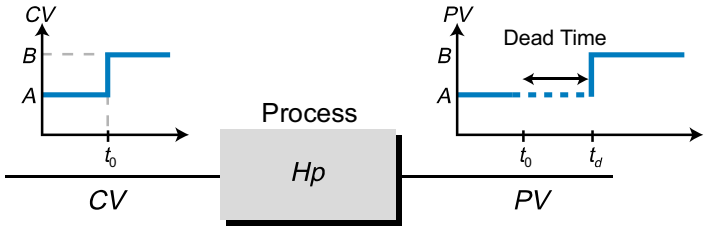


Figure 14-26. Dead time delay.

A dead time delay occurs not only when a control variable input is introduced into the system, but also when an existing input is increased or decreased. Figure 14-27 illustrates a trailing-end dead time delay for a stepped-down input. In this system, a dead time delay occurs between the decrease of the input and the response of the output to the change in input. In addition to process dead time delay, sensor measuring devices also introduce a dead time delay into a system, because a time lapse occurs between the moment the analog process reading is taken and the moment the voltage or current value is available at the transmitter's output (see Figure 14-28). Although the sensor delay is, for the most part, small in comparison to the process variable dead time, it is still an important part of the transfer function because it affects the process. This is especially important in fast-reacting, closed-loop systems, such as servo motor and other positioning applications.

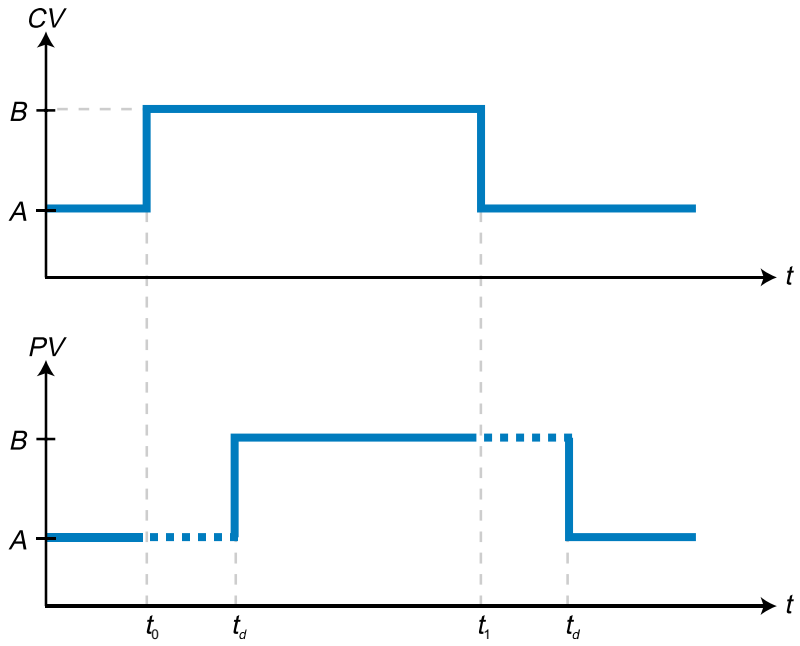


Figure 14-27. Trailing-end dead time delay.

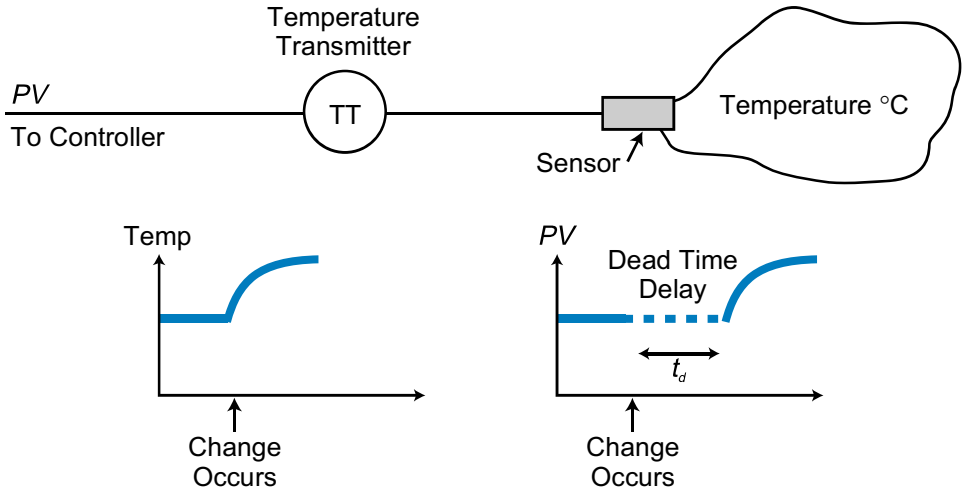


Figure 14-28. Sensor dead time delay.

LAG TIME

Dead time is not the only delay associated with a process and its transfer function. Another delay is **lag time**. Unlike a dead time delay, which is the delay between a change in input and the initial response of the process variable to the input change, a lag time delay occurs when a process variable exhibits a time lapse between its initial response to the input variable and its optimal response to it. Lag time occurs due to process characteristics that are contained in the transfer function. An example of lag time delay can be

observed in a car's cruise control mechanism (see Figure 14-29). If you are driving at 70 mph under cruise control and disengage the control because you see a police car, the car's speed will decrease. Once you pass the police car and press the cruise control's resume button, the car will start to accelerate immediately; however, a delay will occur before the car reaches the set point speed (70 mph). This delay is the lag time. Note that, in this example, the dead time is minimal, because once the resume button is pressed, the control variable (accelerator) increases and the speed (process variable) begins to react almost immediately.

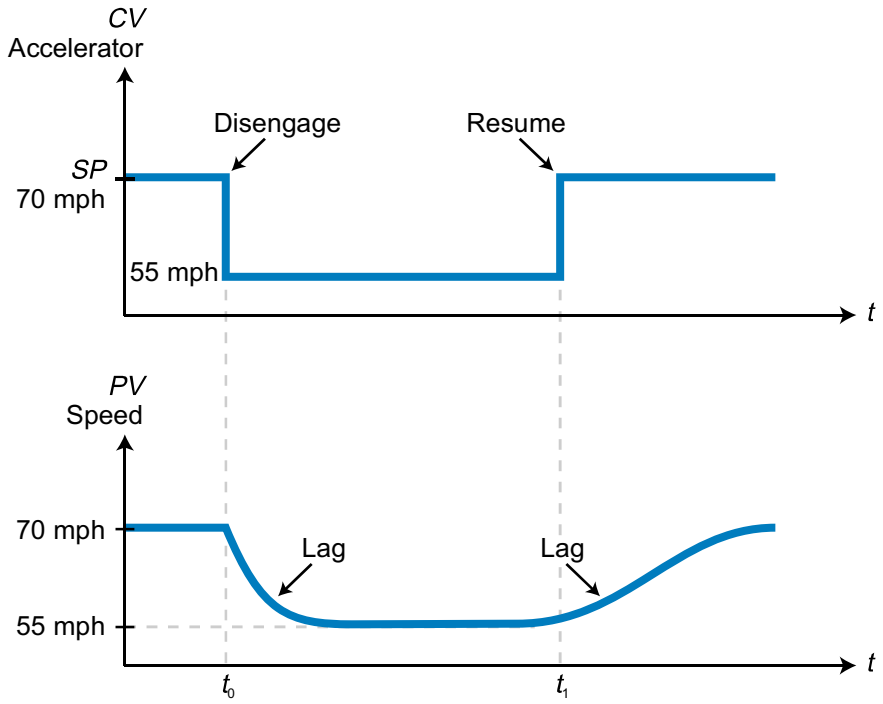


Figure 14-29. Lag time in a car's cruise control mechanism.

A process generally has one of two types of lag, which are the result of its transfer function:

- first-order lag
- second-order lag

First-order lag time is the lag a process variable exhibits in response to a rapid change in the control variable. *Second-order lag time* is the oscillating response of a process variable as it settles to its steady-state value after a step change in the input. Figure 14-30 illustrates both first- and second-order lags. Note how the two processes respond differently to the same change in input because of the different types of lags. A first-order lag is also called a **first-order response**, while a second-order lag is called a **second-order response**. Both types of responses are called *system transient responses*.

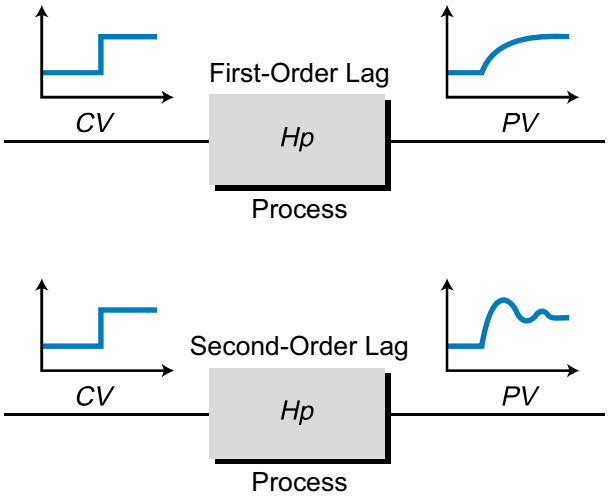


Figure 14-30. First- and second-order lags.

14-4 LAPLACE TRANSFORM BASICS

As mentioned previously, the response of a process is tied to the transfer function of the process itself. Each section of a control system has a transfer function that can be described mathematically. This includes one transfer function for the controller, one for the process, and one for the total system in either an open-loop or closed-loop configuration. As shown in Figure 14-31, the controller's transfer function (*Hc*) is the ratio of its output (*CV*) to its input, which is the error between the set point and the process variable. The process's transfer function, *Hp*, is also the ratio of its output (*PV*) to its input, the control variable.

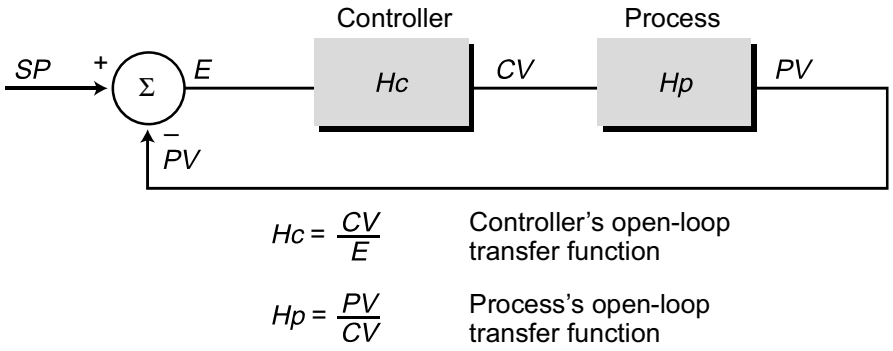


Figure 14-31. Transfer functions in a control system.

Mathematically, transfer functions are expressed through **Laplace transforms**. Laplace transforms are mathematical functions that are used to solve complex differential equations by converting them into easy-to-manage algebraic equations. It is beyond the scope of this book to explain the

mathematical derivation of Laplace transforms. However, we will discuss how they are used in process control functions, to aid in the understanding of first- and second-order systems and their transient responses.

TRANSFER FUNCTIONS

As explained earlier, a system can have either a first-order response or a second-order response. These responses (transfer functions) are expressed as complex differential equations, with first-order system responses characterized by first-order differential equations and second-order systems characterized by second-order differential equations. A first-order differential equation is a mathematical statement that expresses the rate of change of a function with respect to its independent variable. A second-order differential equation expresses the rate of change of a first-order term with respect to its independent variable (i.e., the rate of change of the rate of change). The notations used to express first- and second-order differential equations are as follows:

$$\text{First-order equation} \quad y = \frac{dx}{dt} + x$$

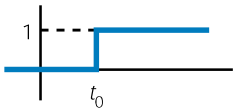
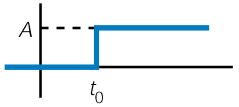
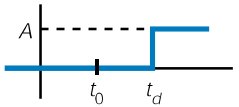
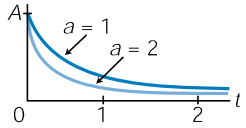
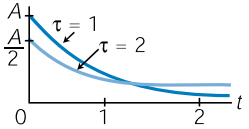
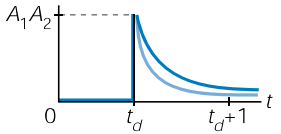
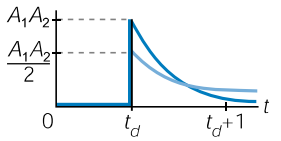
$$\text{Second-order equation} \quad y = \frac{d^2x}{dt^2} + \frac{dx}{dt} + x$$

Most control processes found in industrial applications can be described as either first order or second order. For more complex processes with third-order responses, a second-order equation can be used to approximate the process response.

Laplace transforms use known substitutions for complex differential equations to change them into more easily solvable algebraic equations. To accomplish this, Laplace transforms convert differential equations from the time (t) domain—the process response as a function of time—to the frequency (s) domain—the process response as a function of frequency. Table 14-2 shows some of the most common Laplace transforms found in process control applications. This table also includes inverse Laplace transforms, which are used to convert Laplace equations back into time domain responses. Table 14-2 also includes the time domain process responses to a step input.

DERIVATIVE LAPLACE TRANSFORMS

Laplace transforms replace the derivative terms in both first-order and second-order differential equations with their respective frequency domain s terms. Table 14-3 shows the Laplace transforms for both first- and second-

DESCRIPTION	LAPLACE TRANSFORM	TIME FUNCTION	PROCESS RESPONSE
Any Function f	$F(s)$	$f(t)$	
Unit Step Input	$\frac{1}{s}$	$u(t)$	 $u(t) = 0 \quad t < t_0$ $u(t) = 1 \quad t \geq t_0$
Step Input	$\frac{A}{s}$	$A(t)$	 $A(t) = 0 \quad t < t_0$ $A(t) = A \quad t \geq t_0$
Delay (Dead Time) to Step Input	$Ae^{-t_d s}$	$A(t - t_d)$	 $A(t - t_d) = 0 \quad t < t_d$ $A(t - t_d) = A \quad t \geq t_d$
First-Order Equations			
First-Order Response	$\frac{A}{s + a}$	Ae^{-at}	
First-Order Response with Lag	$\frac{A}{\tau s + 1}$	$\frac{A}{\tau} e^{\frac{-t}{\tau}}$	
First-Order Response plus Dead Time	$\frac{A_1 A_2}{s + a} e^{-t_d s}$	$A_1 A_2 e^{-at}$ for $t \geq t_d$	
First-Order Response with Lag plus Dead Time	$\frac{A_1 A_2}{\tau s + 1} e^{-t_d s}$	$A_1 A_2 e^{\frac{-t}{\tau}}$ for $t \geq t_d$	

	$\tau = 1$	$\tau = 2$	For $A=1$
$t = 0$	1.000	1.000	
$t = 1$	0.368	0.135	
$t = 2$	0.135	0.018	

	$\tau = 1$	$\tau = 2$	For $A=1$
$t = 0$	1.000	0.500	
$t = 1$	0.368	0.303	
$t = 2$	0.135	0.184	

	$a = 1$	$a = 2$	For $A_1A_2=1$
$t = t_d$	1.000	1.000	
$t = t_d + 1$	0.368	0.135	
$t = t_d + 2$	0.135	0.018	

	$\tau = 1$	$\tau = 2$	For $A_1A_2=1$
$t = t_d$	1.000	0.500	
$t = t_d + 1$	0.368	0.135	
$t = t_d + 2$	0.135	0.018	

Table 14-2. Laplace transforms.

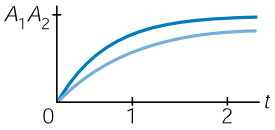
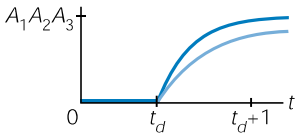
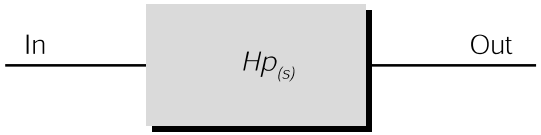
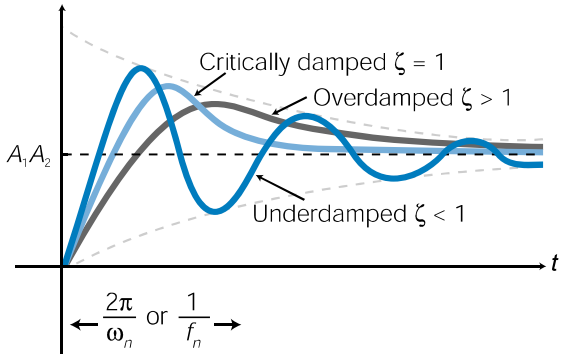
DESCRIPTION	LAPLACE TRANSFORM	TIME FUNCTION	PROCESS RESPONSE
First-Order Response to Step Input (A_1/s) with Lag	$\frac{A_1A_2}{s(\tau s+1)}$	$A_1A_2(1-e^{\frac{-t}{\tau}})$	 For $A_1A_2=1$: $\tau=1$: $t=0$ 0.000, $t=1$ 0.632, $t=2$ 0.865; $\tau=2$: $t=0$ 0.000, $t=1$ 0.393, $t=2$ 0.632
First-Order Response to Step Input (A_1/s) with Lag plus Dead Time	$\frac{A_1A_2A_3}{s(\tau s+1)}e^{-t_d s}$	$A_1A_2A_3(1-e^{\frac{-t}{\tau}})$ for $t \geq t_d$	 For $A_1A_2A_3=1$: $\tau=1$: $t=t_d$ 1.000, $t=t_d+1$ 0.632, $t=t_d+2$ 0.865; $\tau=2$: $t=t_d$ 1.000, $t=t_d+1$ 0.393, $t=t_d+2$ 0.632
Second-Order Equations			
Second-Order Transfer Function ($Hp_{(s)}$) for $\zeta < 1$ (Underdamped)	$\frac{A\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$	$\frac{A\omega_n e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_n\sqrt{1-\zeta^2}t\right)$	 $Hp_{(s)}=\frac{\text{Out}}{\text{In}}$
Second-Order Transfer Function ($Hp_{(s)}$) for $\zeta = 1$ (Critically Damped)	$\frac{A}{(\tau s+1)^2}$	$\frac{At}{\tau^2}e^{\frac{-t}{\tau}}$	
Second-Order Transfer Function ($Hp_{(s)}$) for $\zeta > 1$ (Overdamped)	$\frac{A}{(\tau_1 s+1)(\tau_2 s+1)}$	$\frac{A}{\tau_1-\tau_2}(e^{\frac{-t}{\tau_1}}-e^{\frac{-t}{\tau_2}})$	
Second-Order Step Response (A_1/s) for $\zeta < 1$ (Underdamped)	$\frac{A_1A_2\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$	$A_1A_2\left[1+\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_n\sqrt{1-\zeta^2}t-\psi\right)\right]$ where $\psi=\tan^{-1}\frac{\sqrt{1-\zeta^2}}{-\zeta}$ $(0<\psi<\pi)$	
Second-Order Step Response (A_1/s) for $\zeta = 1$ (Critically Damped)	$\frac{A_1A_2}{s(\tau s+1)^2}$	$A_1A_2\left(1-\frac{\tau-t}{\tau}e^{\frac{-t}{\tau}}\right)$	
Second-Order Step Response (A_1/s) for $\zeta > 1$ (Overdamped)	$\frac{A_1A_2}{s(\tau_1 s+1)(\tau_2 s+1)}$	$A_1A_2\left(1+\frac{\tau_1 e^{\frac{-t}{\tau_2}}-\tau_2 e^{\frac{-t}{\tau_2}}}{\tau_2-\tau_1}\right)$	

Table 14-2 continued.

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	Time (*t*) Domain	Laplace (*s*) Domain
Function	$x_{(t)}$	$X_{(s)}$
First Derivative	$\frac{dx}{dt}$	$sX_{(s)} - x_{(t=0)}$
Second Derivative	$\frac{d^2x}{dt^2}$	$s^2X_{(s)} - s\frac{dx_{(t=0)}}{dt} - x_{(t=0)}$
Integral	$\int_0^t Adt$	$\frac{A}{s}$

Table 14-3. Derivative Laplace transforms.

order derivative terms. In the frequency domain, a first-order derivative term becomes an *s* term times the function in the frequency domain minus a constant, which is the value of the function at $t = 0$ in the time domain. A second-order derivative becomes an s^2 Laplace term times the Laplace function minus s times the value of the time domain first derivative at $t = 0$ minus the value of the function at $t = 0$ in the time domain. Therefore, a simple first-order differential equation of the form:

$$y_{(t)} = \frac{dx}{dt} + x_{(t)}$$

becomes the following equation in Laplace form:

$$Y_{(s)} = sX_{(s)} - x_{(t=0)} + X_{(s)}$$

Assuming that the value of the function $x_{(t=0)}$ is zero, the equation becomes:

$$\begin{aligned} Y_{(s)} &= sX_{(s)} + X_{(s)} \\ &= X_{(s)}(s+1) \end{aligned}$$

If $X_{(s)}$ represents the Laplace output of the process and $Y_{(s)}$ represents the input, as shown in Figure 14-32, the equation for the transfer function of the process (output divided by input) in Laplace form becomes:

$$\frac{X_{(s)}}{Y_{(s)}} = \frac{1}{s+1}$$

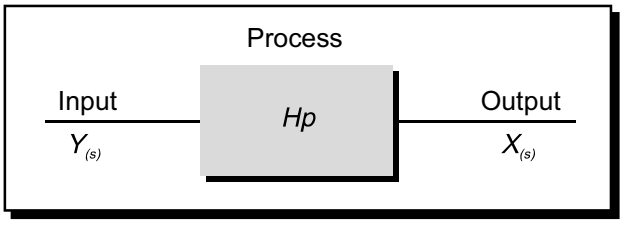


Figure 14-32. Process inputs and outputs in Laplace form.

This indicates that the Laplace output $X_{(s)}$ is equal to the Laplace input times the transfer function in Laplace form:

$$X_{(s)} = Y_{(s)}\left(\frac{1}{s+1}\right)$$

So, by working in the frequency domain instead of the time domain, the solving of the differential equation is reduced to an algebraic manipulation. In the Laplace domain, the exponent of the *s* term indicates the order of the transfer function—in this case, a first-order transfer function.

A second-order differential equation of the form:

$$y_{(t)} = \frac{d^2x}{dt^2} + \frac{dx}{dt} + x_{(t)}$$

becomes the following equation in Laplace form:

$$Y_{(s)} = \left[s^2X_{(s)} - s\frac{dx_{(t=0)}}{dt} - x_{(t=0)}\right] + \left[sX_{(s)} - x_{(t=0)}\right] + X_{(s)}$$

Assuming that the values of the initial parameters are zero, this equation becomes:

$$\begin{aligned} Y_{(s)} &= s^2X_{(s)} + sX_{(s)} + X_{(s)} \\ &= X_{(s)}\left(s^2 + s + 1\right) \end{aligned}$$

Since $Y_{(s)}$ represents the input and $X_{(s)}$ represents the output, the transfer function for a second-order response in Laplace form is:

$$\frac{X_{(s)}}{Y_{(s)}} = \frac{1}{\left(s^2 + s + 1\right)}$$

Note that the exponent of the *s* term indicates a second-order transfer function. As a result of the different mathematical formats of first-order and second-order transfer functions, a second-order system will have an oscillating response, while a first-order system will have a smooth response toward its final steady-state value. We will discuss these response curves in more detail later.

The substitutions used in Laplace transforms are the result of computations performed on electrical circuit networks (e.g., resistors, capacitors, and inductors) that create transfer functions. Mechanical and hydraulic systems also use Laplace transforms to represent system transfer functions, primarily because of their similarity in mathematical representation to electrical systems. The following example illustrates the derivation of a transfer function for a resistor/capacitor (R/C) network.

EXAMPLE 14-3

Figure 14-33 represents an R/C (resistor/capacitor) electrical network. Find **(a)** the differential equation that represents this network, **(b)** the network transfer function in Laplace, and **(c)** the equivalent closed-loop block diagram. Refer to Appendix G for information about the characteristics of electrical circuit elements.

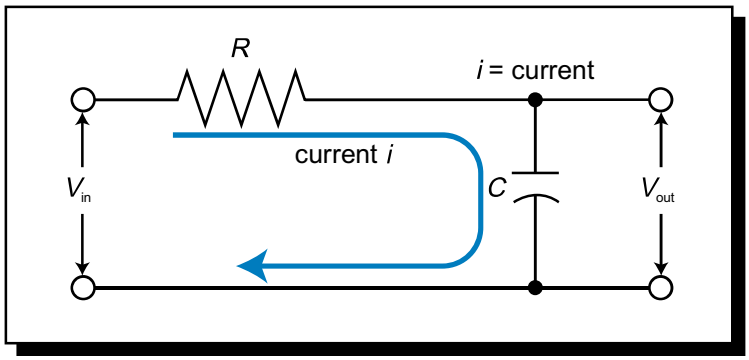


Figure 14-33. R/C electrical network diagram.

SOLUTION

The term V_C represents the voltage across the capacitor (C). This term is equivalent to V_{out} and is represented by:

$$V_{out} = V_C = \frac{1}{C}\int_0^t i\,dt$$

Kirchhoff's voltage law states that the voltage at the output of an electrical circuit is equal to the input voltage minus the voltage across the resistor (V_R), which is equal to the current (i) times the resistance (R). Therefore:

$$\begin{aligned} V_{out} &= V_{in} - V_R \\ &= V_{in} - iR \end{aligned}$$

Solving for the current yields:

$$\begin{aligned} V_{out} &= V_{in} - iR \\ iR &= V_{in} - V_{out} \\ i &= \left(V_{in} - V_{out}\right)\left(\frac{1}{R}\right) \end{aligned}$$

Replacing the current term in the capacitor's output voltage equation with this value of i produces:

$$
\begin{aligned}
V_{out} &= \frac{1}{C}\int_0^t i\,dt \\
&= \frac{1}{C}\int_0^t \left(V_{in} - V_{out}\right)\frac{1}{R}\,dt \\
&= \frac{1}{C}\int_0^t \frac{V_{in}}{R}\,dt - \frac{1}{C}\int_0^t \frac{V_{out}}{R}\,dt \\
&= \frac{1}{RC}\int_0^t V_{in}\,dt - \frac{1}{RC}\int_0^t V_{out}\,dt \\
&= \frac{1}{RC}\left[\int_0^t V_{in}\,dt - \int_0^t V_{out}\,dt\right]
\end{aligned}
$$

(a) To obtain the differential equation of this network, we must take the derivative of both sides of the output voltage equation to eliminate the integral terms:

$$
\begin{aligned}
\frac{dV_{out}}{dt} &= \frac{1}{RC}\left(V_{in} - V_{out}\right) \\
V_{in} &= RC\frac{dV_{out}}{dt} + V_{out}
\end{aligned}
$$

(b) The Laplace form of this first-order differential equation is:

$$V_{in(s)} = RCsV_{out(s)} - V_{out(t=0)} + V_{out(s)}$$

Since there is no initial value at $t = 0$, this equation becomes:

$$
\begin{aligned}
V_{in(s)} &= RCsV_{out(s)} + V_{out(s)} \\
&= V_{out(s)}\left(RCs + 1\right)
\end{aligned}
$$

The transfer function is:

$$\frac{V_{out(s)}}{V_{in(s)}} = \frac{1}{sRC + 1}$$

The equation for a first-order system with lag is represented by:

$$Hp_{(s)} = \frac{1}{\tau s + 1}$$

Therefore, this electrical network is a first-order lag system where the time constant τ is equal to RC (the resistance times the capacitance).

(c) Figure 14-34 shows the block diagram of this closed-loop system.

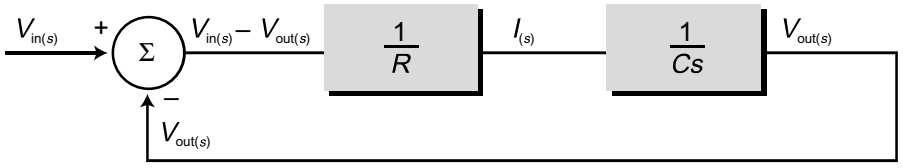


Figure 14-34. Block diagram of the R/C network.

Referring to the hot-water heater example used in Section 14-3, let's assume that the differential equation that describes the process, otherwise known as the heating system's *enthalpy balance equation*, is given by:

$$\text{Steam flow} = A\frac{dT}{dt} + BT_{(t)}$$

where A and B are constants and T is the temperature. Since the steam flow is directly related to the control variable and the temperature is the process variable, we can rewrite this equation as:

$$CV_{(t)} = A\frac{dPV}{dt} + BPV_{(t)}$$

Taking the Laplace transform of this equation, we get:

$$CV_{(s)} = AsPV_{(s)} - APV_{(t=0)} + BPV_{(s)}$$

Assuming that the value of PV at $t = 0$ is zero, this equation becomes:

$$\begin{aligned} CV_{(s)} &= AsPV_{(s)} + BPV_{(s)} \\ &= PV_{(s)}(As + B) \end{aligned}$$

The process's transfer function, which is equal to the process's output divided by its input, is:

$$\frac{PV_{(s)}}{CV_{(s)}} = \frac{1}{As + B}$$

To obtain a standard first-order lag equation (i.e., a fraction whose denominator is $\tau s + 1$), we can divide both the numerator and the denominator by B:

$$\begin{aligned} \frac{PV_{(s)}}{CV_{(s)}} &= \frac{\left(\frac{1}{B}\right)}{\left(\frac{As+B}{B}\right)} \\ &= \frac{\left(\frac{1}{B}\right)}{\left(\frac{A}{B}s + 1\right)} \end{aligned}$$

Previously, we calculated experimentally that the gain for this process is 0.8°C/% and that it took 15 minutes to achieve the final steady-state value. Furthermore, as we will discuss in Section 14-6, it is a given that in a first-order system with lag the output variable will be at 99.33% of the input variable when $t = 5\tau$. The observed value of the response at 15 minutes is 100%, which is close to the 99.33% at 5τ. Therefore, we can approximate the value of τ as:

$$5\tau \approx 15 \text{ min}$$

$$\tau \approx \frac{15 \text{ min}}{5} \approx 3 \text{ min}$$

Thus, we can represent the process's transfer function as:

$$\frac{PV_{(s)}}{CV_{(s)}} = \frac{0.8}{3s+1}$$

INTEGRAL LAPLACE TRANSFORMS

Aside from a simple constant gain, an integral transfer function represents the simplest of all process responses (see Figure 14-35). It is represented by the equation:

$$\frac{\text{Output}}{\text{Input}} = \frac{PV}{CV} = \int_0^t A\,dt$$

$$= A\int_0^t dt$$

$$PV = A\int_0^t CV\,dt$$

where A is a constant gain. In Laplace, the term ($\frac{A}{s}$) replaces the time domain integral term. Therefore, the Laplace transform of this equation is:

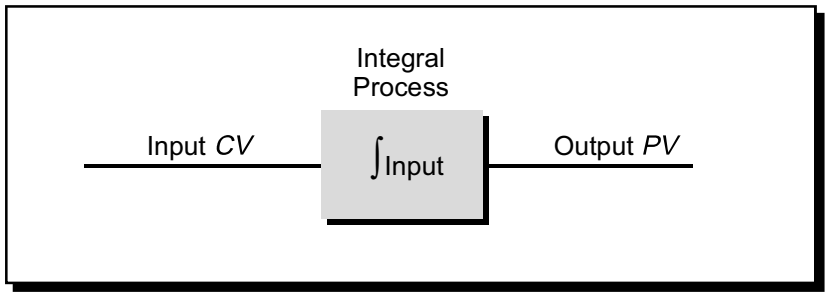


Figure 14-35. Integral transfer function.

$$PV_{(s)} = \left(\frac{A}{s}\right)CV_{(s)}$$

An integral process integrates the input with the process over time. This implies that the rate of change of the process output varies according to the input, as shown in the following equation:

$$PV = A\int_0^t CVdt$$
$$dPV = ACVdt$$
$$\frac{dPV}{dt} = ACV$$

Therefore, in an integral process, a change in the control variable will produce a rate of change in the process variable over time.

In process systems, changes in the control variable are usually simulated by producing a step change with amplitude *B*. In Laplace form, a step change to an integral process is represented as:

$$PV = CV\left(\frac{A}{s}\right)$$
$$= \left(\frac{B}{s}\right)\left(\frac{A}{s}\right)$$
$$= \frac{AB}{s^2}$$

where:

$A =$ the constant integral gain
$B =$ the amplitude of the step change

The inverse Laplace (time domain) response of this equation is:

$$PV_{(t)} = ABt$$

This implies that the rate of increase of the process variable is *AB* per second (or per minute, depending on time units). Figure 14-36 illustrates this integral response curve, which is a ramp-type function.

As we discussed briefly in Chapter 8, the following integro-differential equation describes the output (i.e., control variable) of a typical proportional-integral-derivative (PID) controller (see Figure 14-37):

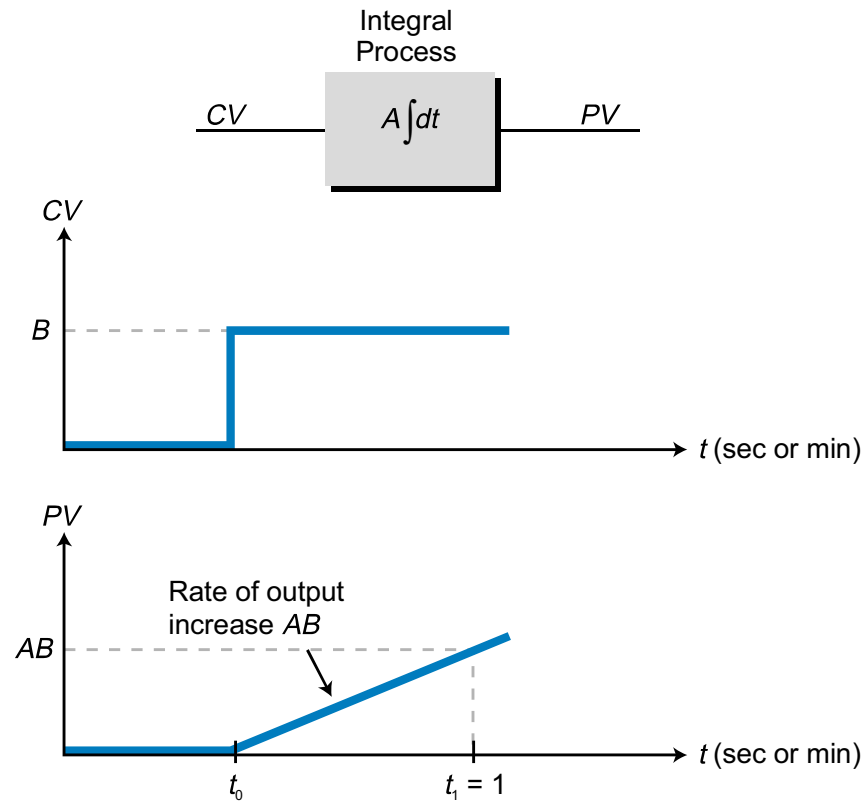


Figure 14-36. Integral response to a step change.

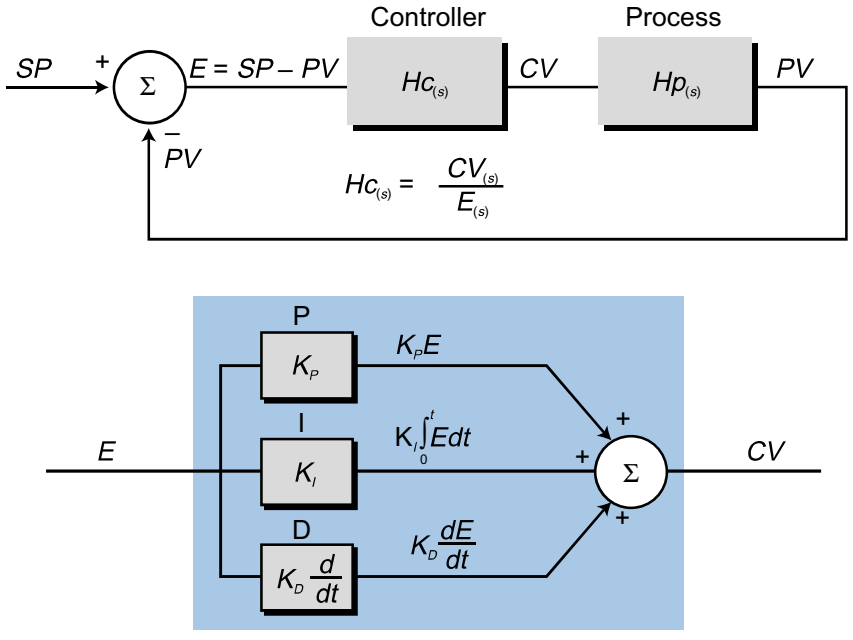


Figure 14-37. PID process control loop.

$$CV_{(t)} = K_P E + K_I \int_0^t E dt + K_D \frac{dE}{dt} + CV_{(t=0)}$$

where:

$CV_{(t)}$ = the control variable output
K_P = the proportional gain
E = the error ($SP - PV$)
K_I = the integral gain
K_D = the derivative gain

To find the Laplace transform of this PID equation, we must make the appropriate substitutions, assuming that the initial parameters are zero, which yields:

$$CV_{(s)} = K_P E_{(s)} + \frac{K_I E_{(s)}}{s} + K_D E_{(s)} s$$

Therefore, the transfer function of a PID controller (Hc) in Laplace is:

$$Hc_{(s)} = \frac{\text{Out}}{\text{In}} = \frac{CV_{(s)}}{E_{(s)}}$$

$$CV_{(s)} = K_P E_{(s)} + \frac{K_I E_{(s)}}{s} + K_D E_{(s)} s$$

$$\frac{CV_{(s)}}{E_{(s)}} = K_P + \frac{K_I}{s} + K_D s$$

14-5 DEAD TIME RESPONSES IN LAPLACE FORM

Until now, we have only discussed Laplace transforms of ideal processes. In reality, however, no process is ideal. Most processes contain either dead time, lag time, or both. Therefore, these factors must be accounted for when analyzing a process's transfer function and performing its Laplace transform.

Dead time involves a shift, or displacement, of the time variable t, meaning that the process input occurs at time t but the output does not occur until time t_d. The Laplace transform of a dead time factor is $e^{-t_d s}$, where t_d is the delay of the output response and e is a constant equal to 2.718.

Figure 14-38 illustrates a transfer function of a simple system with dead time (and no lag), which receives a step (OFF-to-ON) input with amplitude A. Note that the value of the process gain is 1, because an input of magnitude

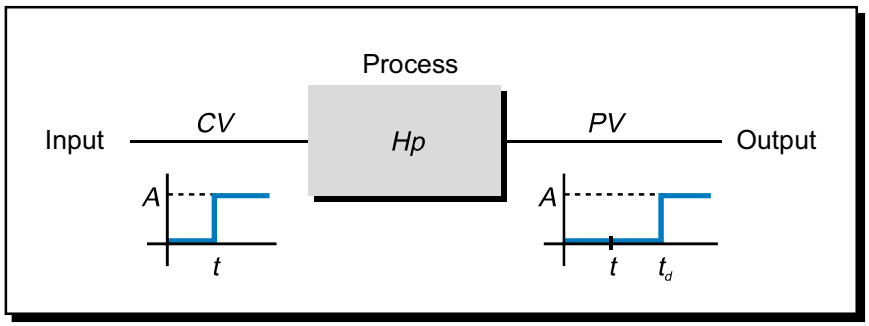


Figure 14-38. Transfer function of a system with dead time.

A yields an output of magnitude *A*. The output, however, has a dead time equal to t_d. Therefore, the transfer function *Hp* is equal to the process gain times the dead time. If the process gain equals 1, then the Laplace transform of this transfer function is:

$$Hp_{(s)} = e^{-t_d s}$$

The input (*CV*) to the process in Figure 14-38 is a step input of amplitude *A*. In the previous section, we explained that the Laplace transform of a step input is:

$$CV = \frac{A}{s}$$

Knowing these two equations, along with the fact that *Hp* is equal to the output divided by the input, we can find the value, in Laplace, of the process variable:

$$Hp = \frac{PV}{CV}$$
$$PV = (CV)(Hp)$$
$$= \left(\frac{A}{s}\right)e^{-t_d s}$$

14-6 LAG RESPONSES IN LAPLACE FORM

Now that we have explained the basics of Laplace transforms, let's examine the various responses exhibited by processes. Most processes exhibit either a first-order or second-order lag response with one or two time lags, respectively (see Figure 14-39). Note that this figure shows open-loop processes whose inputs are the control variable from the controller.

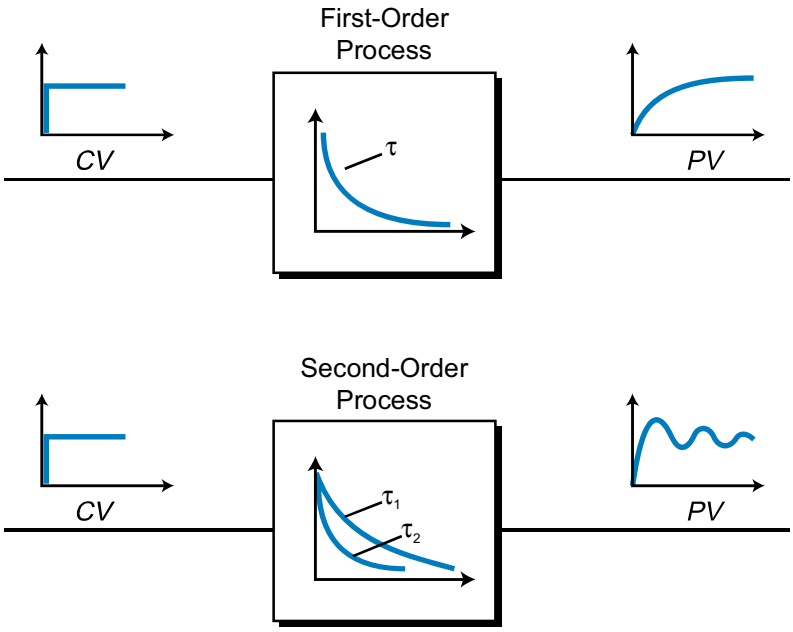


Figure 14-39. First- and second-order lags.

FIRST-ORDER LAG RESPONSES

First-order lag is one of the most common types of process responses. In this type of system, the process variable response lags behind a rapid change (step change) in the control variable. In the time domain, a first-order response to a step input is represented by the equation:

$$V_{out} = V_{in}\left(1 - e^{\frac{-t}{\tau}}\right)$$

where:

V_{in} = the step input to the process
V_{out} = the output of the process
t = time
τ = the time constant

As shown in Figure 14-40, this response is an exponential function, meaning that the value of V_{out} increases rapidly with time to equal V_{in}. The time constant τ describes how quickly the output catches up with the input value—the smaller the value of τ, the faster V_{out} equals V_{in} and vice versa. So, first-order responses with smaller time constants have shorter lag times.

For a first-order system with a step input and lag, the value of the output, given that $t = \tau$, is 63.2% of the input is:

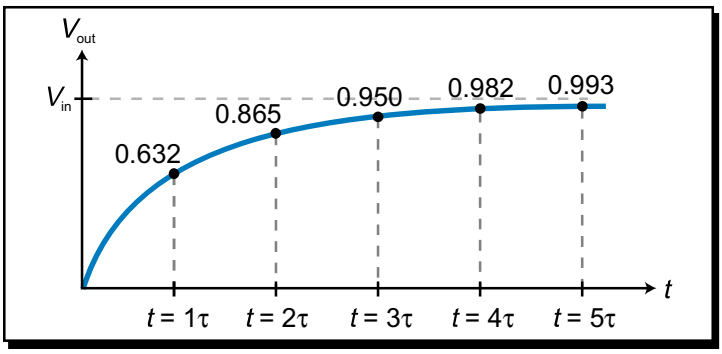


Figure 14-40. First-order lag process response.

$$
\begin{aligned}
V_{out} &= V_{in}\left(1 - e^{\frac{-t}{\tau}}\right) \\
&= V_{in}\left(1 - e^{\frac{-\tau}{\tau}}\right) \\
&= V_{in}\left(1 - e^{-1}\right) \\
&= V_{in}(1 - 0.368) \\
&= 0.632 V_{in}
\end{aligned}
$$

As the value of t increases, the value of V_{out} becomes closer to 100% of V_{in}. As shown in Figure 14-40, at the time $t = 4\tau$, the value of V_{out} will be over 98% of that of V_{in}, and when the time variable reaches 5τ, V_{out} will be over 99% of V_{in}.

In the Laplace domain, a first-order system transfer function with lag is represented by the equation (see Figure 14-41):

$$Hp_{(s)} = \frac{A_1}{\tau s + 1}$$

where:

A_1 = the process amplitude gain
τ = the system's lag time

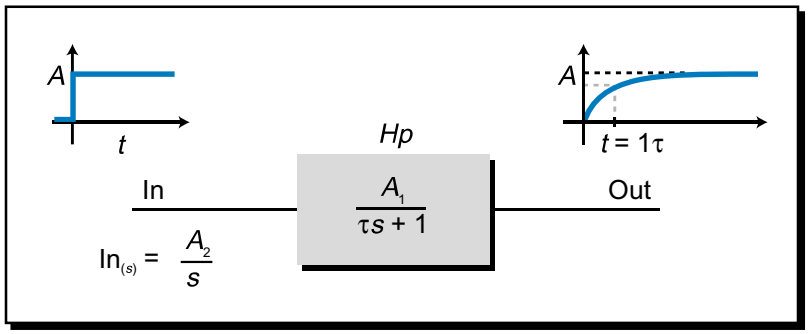


Figure 14-41. First-order transfer function with lag in the Laplace domain.

Using Table 14-2, the inverse Laplace transform (represented by $\mathcal{L}^{-1}$) of this transfer function, which turns the Laplace equation into a time-based transfer function, is:

$$^{-1}[Hp_{(s)}] = Hp_{(t)} = \left(\frac{A_1}{\tau}\right)e^{\frac{-t}{\tau}}$$

This response, as shown in Figure 14-42, has a decaying form (i.e., it decreases over time) due to the term e. This same system given a step input with amplitude A_2 would have the equation:

$$\begin{aligned}\text{Out}_{(s)} &= \left(\text{In}_{(s)}\right)\left(Hp_{(s)}\right)\\ &= \left(\frac{A_2}{s}\right)\left(\frac{A_1}{\tau s+1}\right)\\ &= \frac{A_1A_2}{s(\tau s+1)}\end{aligned}$$

Again using Table 14-2, the inverse Laplace response, or real-time response of the system ($\text{Out}_{(t)}$), is:

$$\text{Out}_{(t)} = A_1A_2\left(1-e^{\frac{-t}{\tau}}\right)$$

where the value of the output will be $0.63A_1A_2$ (63% of A_1A_2) when $t = \tau$. If the input is a unit step (that is, the amplitude A_2 equals 1) and the gain of $Hp_{(s)}$ equals 1 ($A_1 = 1$), then the output will be:

$$\begin{aligned}\text{Out}_{(t)} &= A_1A_2\left(1-e^{\frac{-t}{\tau}}\right)\\ &= (1)(1)\left(1-e^{\frac{-t}{\tau}}\right)\\ &= \left(1-e^{\frac{-t}{\tau}}\right)\end{aligned}$$

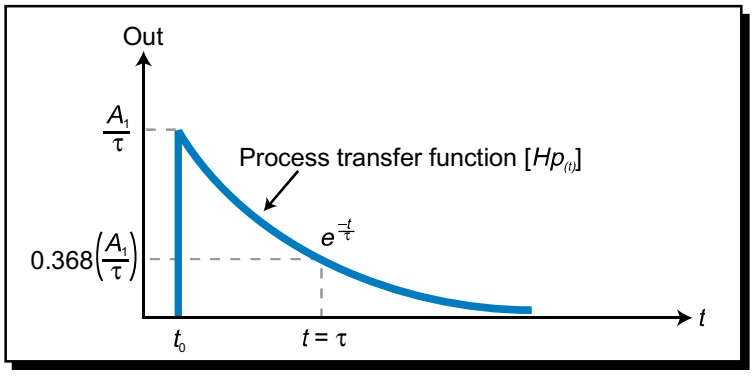


Figure 14-42. Process transfer function in the time domain.

Figure 14-43 graphs the response of the process variable $\text{Out}_{(t)}$ for $A_1 = A_2 = 1$. This curve, representing a first-order response to a step input plus lag, is a function of the system's transfer function, which is the step value (1) minus the system's curve term ($e^{\frac{-t}{\tau}}$).

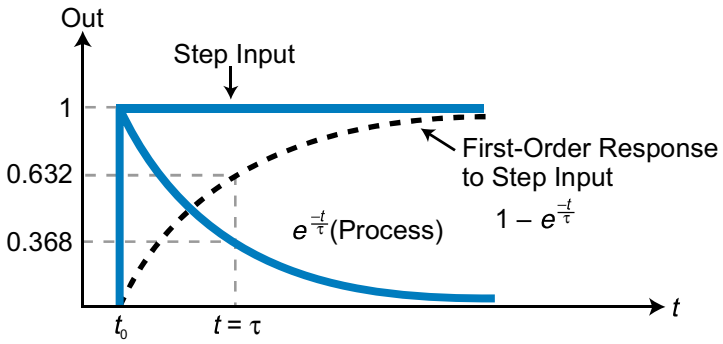


Figure 14-43. Process variable's lag response.

Adding a simple dead time term ($e^{-t_d s}$) to a first-order step response with lag generates the Laplace transfer function:

$$\text{Out}_{(s)} = \left(\frac{A_1 A_2}{s(\tau s + 1)}\right) e^{-t_d s}$$

where t_d is the dead time. Figure 14-44 shows the graph of this function in the time domain. The value of the output is:

$$\text{Out}_{(t)} = A_1 A_2 \left(1 - e^{\frac{-t}{\tau}}\right) \qquad \text{for } t \geq t_d$$

$$\text{Out}_{(t)} = 0 \qquad \text{for } t < t_d$$

Note that the first output response equation is valid for t values greater than t_d, the dead time. $\text{Out}_{(t)}$ will be zero for time values before the dead time t_d.

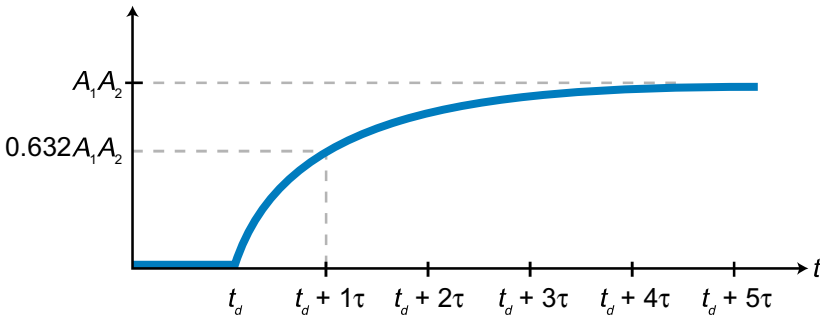


Figure 14-44. First-order step response with dead time and lag.

EXAMPLE 14-4

A process system has a first-order response with a time constant of 10.8 minutes. **(a)** Calculate how long it will take for the value of the output V_{out} to be at 90% of the input V_{in}. **(b)** Calculate the value of V_{out} at 90% of V_{in} given a 5 minute dead time.

SOLUTION

(a) A first-order system with lag has a response of:

$$V_{out} = V_{in}\left(1 - e^{\frac{-t}{\tau}}\right)$$

The value of τ is 10.8 seconds and the required ratio of output over input is 90%, or 0.90; therefore:

$$V_{out} = V_{in}\left(1 - e^{\frac{-t}{\tau}}\right)$$

$$\frac{V_{out}}{V_{in}} = 1 - e^{\frac{-t}{\tau}}$$

$$0.90 = 1 - e^{\frac{-t}{10.8}}$$

$$e^{\frac{-t}{10.8}} = 1 - 0.90$$

$$= 0.10$$

Solving for t by taking the natural logarithm (ln) of both sides of the equation yields:

$$e^{\frac{-t}{10.8}} = 0.10$$

$$\frac{-t}{10.8} = \ln 0.10$$

$$-t = (10.8)(\ln 0.10)$$

$$= (10.8)(-2.303)$$

$$= 24.87 \text{ minutes}$$

So, in 24.87 minutes, the value of the output will be at 90% of the value of the input.

(b) The dead time will simply add to the time required to achieve the 90% value. Therefore, with a lag of 5 minutes, the system will reach a value of 90% final output in 29.87 minutes (24.87 min + 5 min).

SECOND-ORDER LAG RESPONSES

A second-order lag response exhibits oscillations that occur while the output signal is settling into its final steady-state value. This type of response is caused by a step change in the input or a disturbance in the process.

A second-order transfer function with lag is characterized by a second-order differential equation that is represented in Laplace form as:

$$Hp_{(s)} = \frac{\text{Out}}{\text{In}} = \frac{A\omega_n^2}{\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right)}$$

where:

A = the gain

ω_n = the resonant, or natural, frequency of oscillation in radians/second

ζ = the damping coefficient

Figure 14-45 illustrates this second-order, oscillating response to a step input. The frequency term ω_n is the factor that determines how quickly the response oscillates above and below the desired outcome. The damping coefficient ζ is the factor that suppresses the oscillation over time, so that the response finally levels off at the desired outcome value. The complete numerator term $A\omega_n^2$ represents the system gain (K_{sys}), which specifies the total amplitude of the response signal given its frequency.

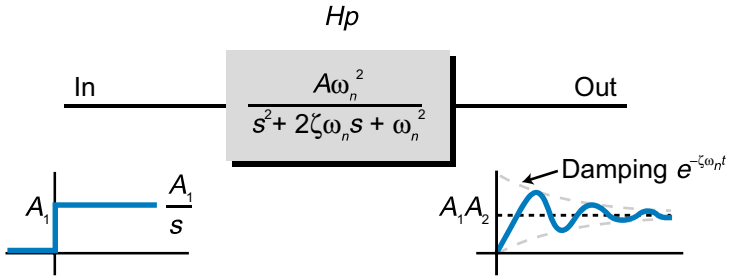


Figure 14-45. Second-order response to a step input.

The amplitude of the oscillation of a second-order response dies off exponentially due to the damping of the factor $e^{-\zeta\omega_n t}$, which is part of the inverse Laplace transform representation (time domain) of the system. If the damping coefficient (ζ) is equal to 0, then the term $e^{-\zeta\omega_n t}$ will be 1 and the response will oscillate indefinitely in a sinusoidal manner at a frequency of ω_n, instead of leveling out. Thus, the damping coefficient determines the shape of the response (see Figure 14-46).

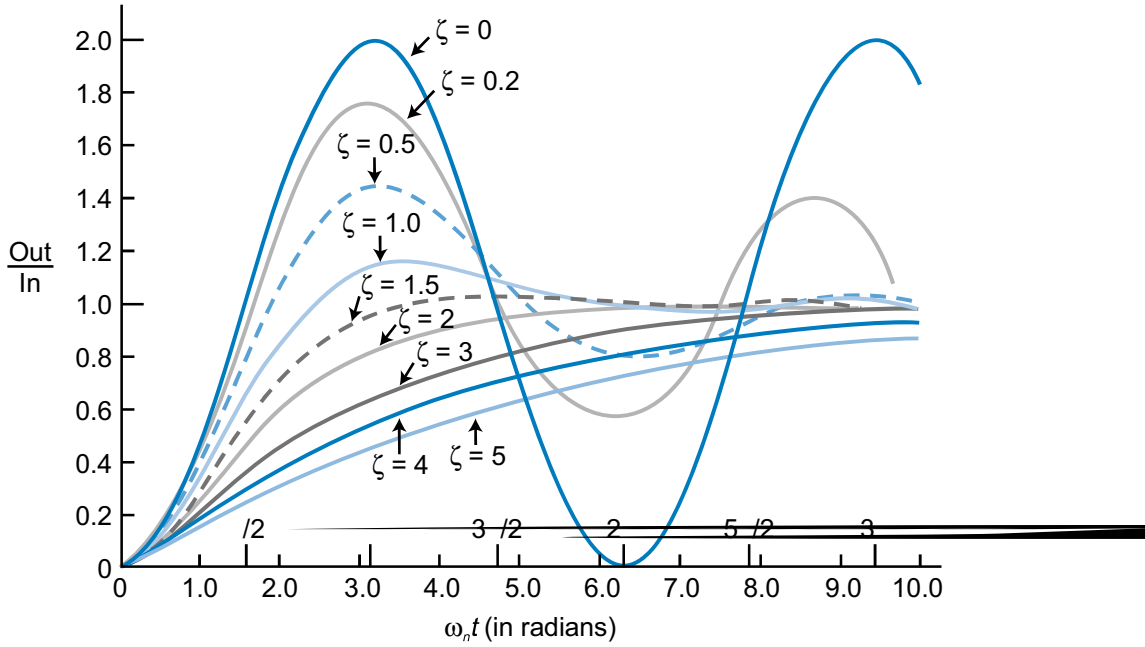


Figure 14-46. Damping coefficient effect on the oscillation of a second-order response.

Unlike a first-order system, a second-order system has two lag times (τ_1 and τ_2), which are related to the frequency of oscillation (ω_n). These two lag times combine to create a system second-order time constant τ_{sys}, which is equal to:

$$\tau_{sys} = \frac{1}{\omega_n}$$

As used in Laplace and time domain second-order response equations, the term ω_n represents frequency. This frequency is expressed in radians per second. However, this frequency can also be expressed in degrees. A second-order response is a sinusoidal response, meaning that it fluctuates above and below the final outcome (set point) value once every 2π periods (see Figure 14-47). Therefore, the response period is characterized by the equation $\frac{2\pi}{\omega_n}$ (see Figure 14-48). In degrees, this same period is expressed as $\frac{1}{f_n}$, where f_n is the frequency in hertz. Therefore:

$$\frac{2\pi}{\omega_n} = \frac{1}{f_n}$$

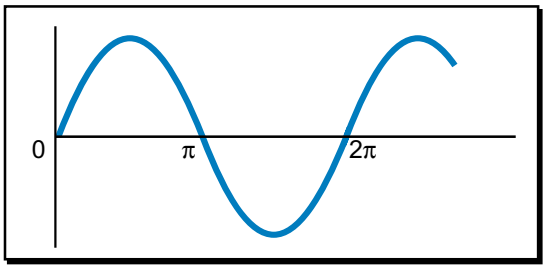


Figure 14-47. Sinusoidal response of a second-order system around the set point.

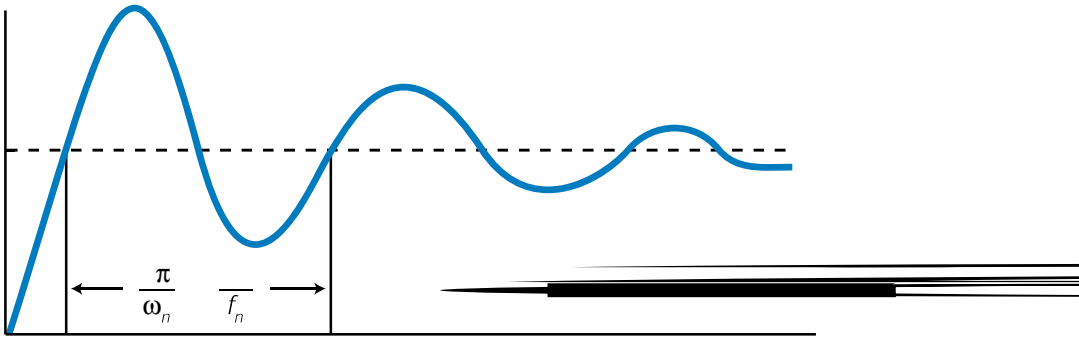


Figure 14-48. Response period of a sinusoidal curve.

Solving for ω_n yields:

$$\omega_n = 2\pi f_n$$

So, the radian/sec frequency term ω_n is equivalent to the degree frequency term $2\pi f_n$.

14-7 TYPES OF SECOND-ORDER RESPONSES

A second-order system can exhibit one of three types of responses:

- overdamped ($\zeta > 1$)
- critically damped ($\zeta = 1$)
- underdamped ($\zeta < 1$)

These responses differ in how they reach the final steady-state value, or set point, over time due to the value of their damping coefficients (see Figure 14-49). An underdamped response oscillates around the set point because its time domain transfer function contains the damping term $e^{-\zeta\omega_n t}$. Critically damped and overdamped responses do not contain this term, so they overshoot the set point and then settle back to it.

OVERDAMPED RESPONSES

An **overdamped response** is a second-order response with lag whose damping coefficient (ζ) is greater than 1. By algebraically manipulating the transfer function of a second-order system with lag (see Section 14-5), the transfer function of an overdamped response can be expressed as:

$$Hp_{(s)} = \left(\frac{A_1}{\tau_1 s + 1}\right)\left(\frac{A_2}{\tau_2 s + 1}\right)$$

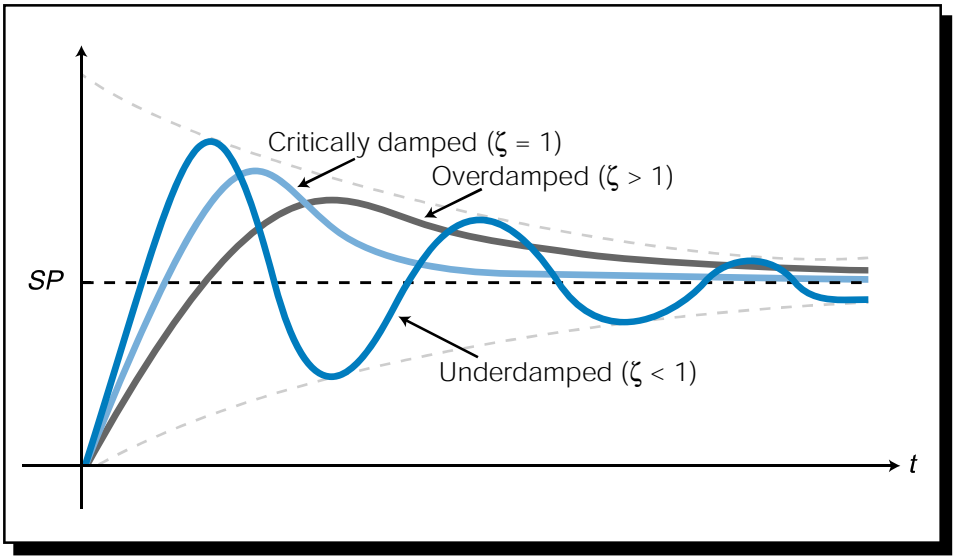


Figure 14-49. Overdamped, critically damped, and underdamped responses.

In this equation, which is a function of two first-order systems (i.e., two time lags), the terms A_1 and A_2 represent the gains. By substituting the term K_{OD} for the total overdamped system gain A_1A_2, this function can be simplified to:

$$Hp_{(s)} = \frac{A_1A_2}{(\tau_1 s + 1)(\tau_2 s + 1)}$$
$$= \frac{K_{OD}}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

EXAMPLE 14-5

For an overdamped system (ζ > 1), solve for **(a)** K_{sys}, **(b)** ω_n, **(c)** τ_{sys}, and **(d)** ζ using the transfer functions for a second-order response and an overdamped response.

$$Hp_{(s)} = \frac{A\omega_n^2}{\left(S^2 + 2\zeta\omega_n S + \omega_n^2\right)} \quad \text{(Second-order transfer function)}$$

$$Hp_{(s)} = \left(\frac{A_1}{\tau_1 S + 1}\right)\left(\frac{A_2}{\tau_2 S + 1}\right) \quad \text{(Overdamped response)}$$

SOLUTION

Multiplying the terms in the overdamped transfer function yields the equation:

$$Hp_{(s)} = \frac{A_1 A_2}{(\tau_1 s + 1)(\tau_2 s + 1)}$$
$$= \frac{A_1 A_2}{\left[\tau_1 \tau_2 s^2 + (\tau_1 + \tau_2)s + 1\right]}$$

Dividing by the term $\tau_1\tau_2$ generates the following equation, which has a denominator in the form of a second-order lag transfer function:

$$Hp_{(s)} = \frac{\left(\frac{A_1 A_2}{\tau_1 \tau_2}\right)}{\left(s^2 + \frac{(\tau_1+\tau_2)}{\tau_1\tau_2} s + \frac{1}{\tau_1\tau_2}\right)}$$

Therefore, this equation is equal to the second-order lag equation:

$$Hp_{(s)} = \frac{A\omega_n^2}{\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right)}$$

because the denominator of the polynomial can be separated into two real factors, where τ_1 and τ_2 are the two time constants. Thus, the relationship of the terms in these two equations is:

$$Hp_{(s)} = \frac{\overbrace{\left(\frac{A_1 A_2}{\tau_1 \tau_2}\right)}^{K_{sys}}}{\underbrace{\left(s^2 + \frac{(\tau_1+\tau_2)}{\tau_1\tau_2} s + \frac{1}{\tau_1\tau_2}\right)}_{2\zeta\omega_n \qquad \omega_n^2}}$$

(a) Knowing that the term K_{OD} is equal to the overdamped system gain A_1A_2, the term K_{sys} for an overdamped system is:

$$K_{sys} = A\omega_n^2 = \frac{A_1 A_2}{\tau_1 \tau_2} = \frac{K_{OD}}{\tau_1 \tau_2}$$

(b) For an overdamped system, ω_n^2 is equal to:

$$\omega_n^2 = \frac{1}{\tau_1 \tau_2}$$

Solving for ω_n generates:

$$\omega_n^2 = \frac{1}{\tau_1 \tau_2}$$
$$\omega_n = \sqrt{\frac{1}{\tau_1 \tau_2}}$$
$$= \frac{1}{\sqrt{\tau_1 \tau_2}}$$

(c) Earlier, we explained that τ_{sys} is equal to 1 over the frequency ω_n. Using the information from part (b), τ_{sys} for an overdamped system is:

$$\begin{aligned}\tau_{sys} &= \frac{1}{\omega_n} \\ &= \frac{1}{\left(\frac{1}{\sqrt{\tau_1\tau_2}}\right)} \\ &= \sqrt{\tau_1\tau_2}\end{aligned}$$

(d) The damping coefficient terms for a second-order system with lag and an overdamped system relate as follows:

$$2\zeta\omega_n = \frac{\left(\tau_1 + \tau_2\right)}{\tau_1\tau_2}$$

Solving for ζ yields:

$$\begin{aligned}2\zeta\omega_n &= \frac{\left(\tau_1 + \tau_2\right)}{\tau_1\tau_2} \\ \zeta &= \frac{\left(\tau_1 + \tau_2\right)}{\left(\tau_1\tau_2\right)\left(2\omega_n\right)} \\ &= \frac{\left(\tau_1 + \tau_2\right)}{\left(\tau_1\tau_2\right)(2)\left(\frac{1}{\sqrt{\tau_1\tau_2}}\right)} \\ &= \frac{\left(\tau_1 + \tau_2\right)}{\left(\frac{2\tau_1\tau_2}{\sqrt{\tau_1\tau_2}}\right)} \\ &= \frac{\left(\tau_1 + \tau_2\right)}{\left(2\sqrt{\tau_1\tau_2}\right)}\end{aligned}$$

An overdamped second-order transfer function in real time (the time domain) is described by the inverse Laplace transform (see Table 14-2):

$$H_{(t)} = \frac{K_{OD}}{\tau_1 - \tau_2}\left(e^{\frac{-t}{\tau_1}} - e^{\frac{-t}{\tau_2}}\right)$$

This indicates two exponential decaying responses—one at a rate of τ_1 and the other at the rate of τ_2. Figure 14-50 illustrates the form of these two exponential responses, along with the response of $H_{(t)}$, which is a function of a combination of these two responses. Note that, as indicated in the time domain transfer function term ($e^{\frac{-t}{\tau_1}} - e^{\frac{-t}{\tau_2}}$), the curve of $H_{(t)}$ is equal to the curve of the τ_1 response minus the curve of the τ_2 response.

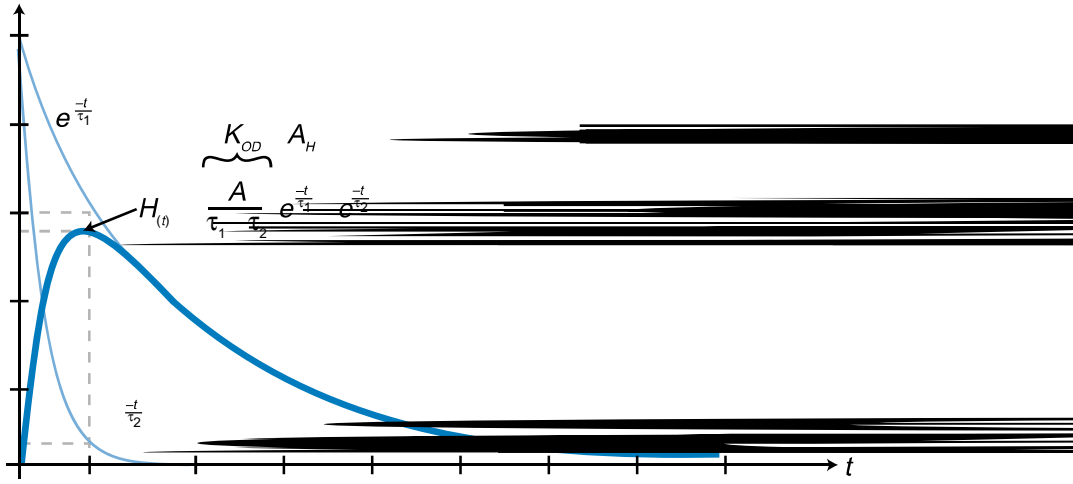


Figure 14-50. Real-time transfer function of an overdamped second-order process.

Figure 14-51 illustrates a second-order system response to a step input with amplitude *B*. As shown in this figure, the output of the time domain transfer function in response to a step input is similar in form to the second-order response curve shown in Figure 14-50. The overdamped response may also follow the shape of a first-order system response curve if one of the time

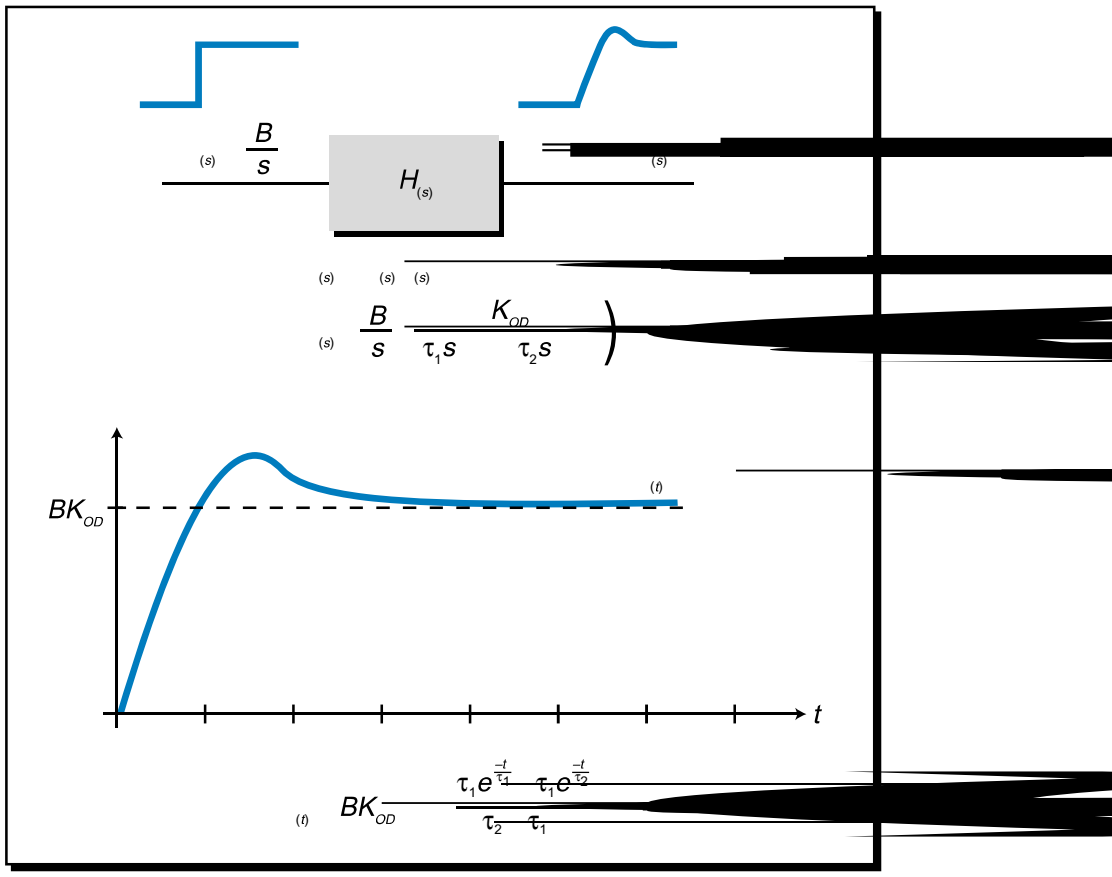


Figure 14-51. Second-order response to a step input.

constants is significantly longer than the other (i.e., $\tau_1 >> \tau_2$). Figure 14-52 illustrates a case like this, where one of the exponential components (τ_2) dies out much more rapidly than the other (τ_1). Thus, the response to the unit step is heavily damped, causing a sluggish response similar to a first-order one. The system response is heavily damped because the value of ζ in this system becomes large (see the value of ζ in Example 14-4). Two first-order systems with different time lags, which are connected in series (or *cascaded*), will produce this type of overdamped second-order response (see Figure 14-53). In a cascaded system, the output of one part of the system depends on the input to another part of the system.

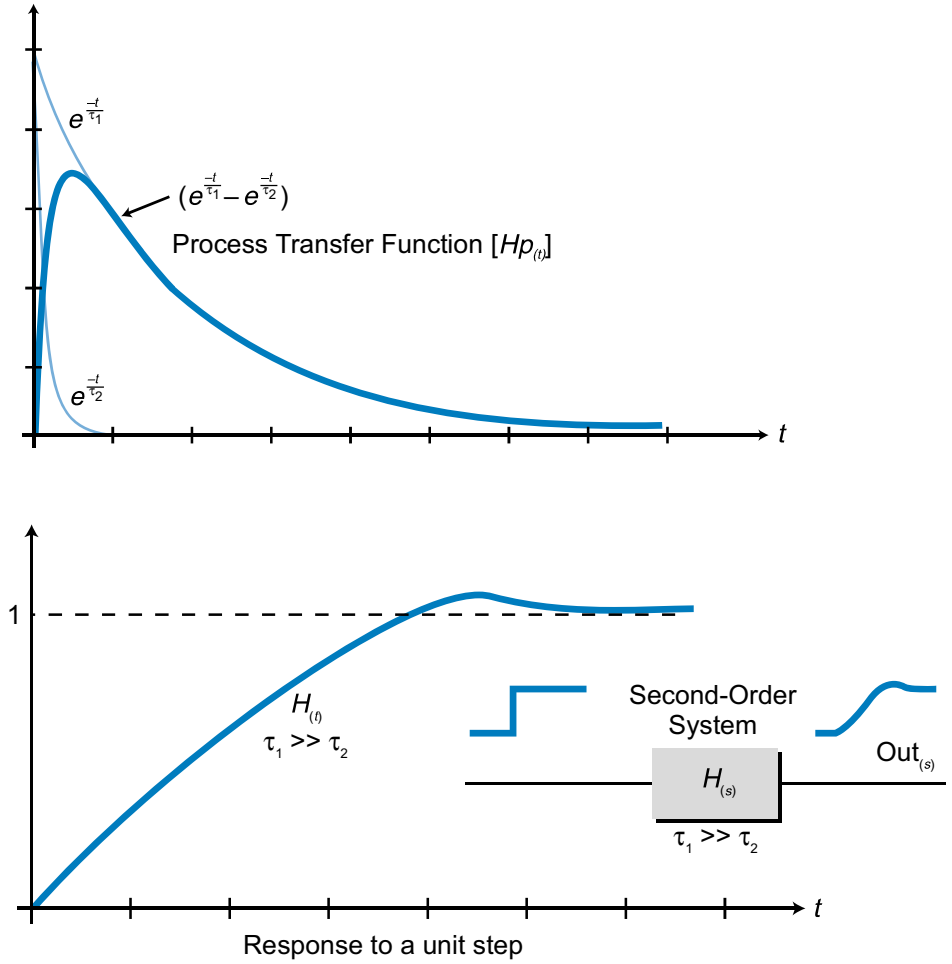


Figure 14-52. A heavily damped response.

Critically damped responses, which are second-order responses where $\zeta = 1$, are the result of second-order transfer functions where $\tau_1 = \tau_2$. Therefore, the transfer function for this type of response is ($\tau = \tau_1 = \tau_2$):

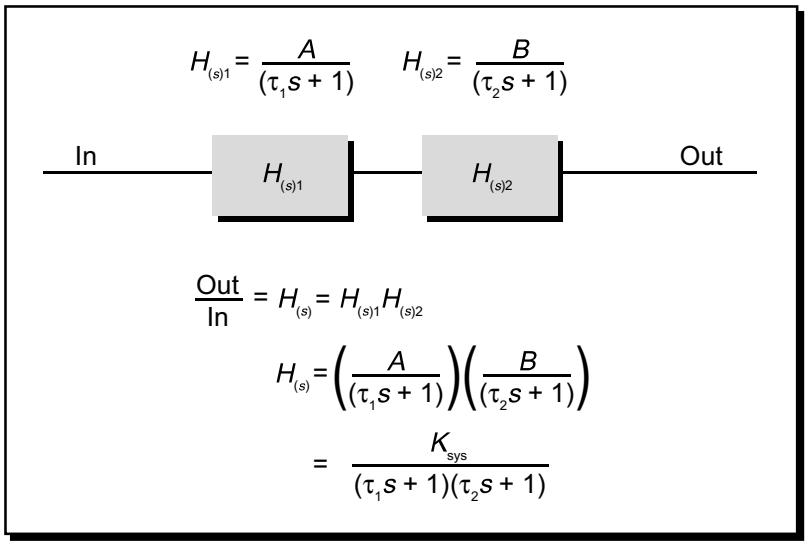


Figure 14-53. Two first-order systems with different lag times cascaded to form an overdamped second-order system.

$$H_{(s)} = \frac{A}{(\tau_1 s + 1)(\tau_2 s + 1)}$$
$$= \frac{A}{(\tau s + 1)^2}$$

This computation comes from substituting $\zeta = 1$ in the second-order lag transfer function, making the denominator a second-order polynomial of the form $(s^2 + 2\omega_n s + \omega_n^2)$:

$$H_{(s)} = \frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$
$$= \frac{A\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2}$$
$$= \frac{A\omega_n^2}{(s + \omega_n)^2}$$

Therefore, substituting K_{sys} for the system gain (nominator) and τ_{sys} for the system lag time produces:

$$H_{(s)} = \frac{K_{sys}}{(\tau_{sys} s + 1)^2}$$

EXAMPLE 14-6

Show **(a)** how to derive the second-order critically damped transfer function from the second-order lag transfer function and **(b)** how to conclude that ζ is equal to 1.

$$H_{(s)} = \frac{A\omega_n^2}{\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right)} \quad \text{(Second-order transfer function)}$$

$$H_{(s)} = \frac{K_{sys}}{(\tau s + 1)^2} \quad \text{(Critically damped transfer function)}$$

SOLUTION

(a) Given that ζ = 1 for a critically damped response, dividing the numerator and denominator of the second-order function by ω_n^2 yields:

$$\begin{aligned} H_{(s)} &= \frac{A\omega_n^2}{s^2 + 2s\omega_n + \omega_n^2} \\ &= \frac{\left(\frac{A\omega_n^2}{\omega_n^2}\right)}{\left(\frac{s^2}{\omega_n^2} + \frac{2s\omega_n}{\omega_n^2} + \frac{\omega_n^2}{\omega_n^2}\right)} \\ &= \frac{A}{\left(\frac{1}{\omega_n^2}s + \frac{2}{\omega_n}s + 1\right)} \end{aligned}$$

Replacing the term $\frac{1}{\omega_n}$ with $\frac{1}{\tau_{sys}}$ generates the equation:

$$\begin{aligned} H_{(s)} &= \frac{A}{\frac{1}{\left(\frac{1}{\tau_{sys}^2}\right)}s + \frac{2}{\left(\frac{1}{\tau_{sys}}\right)}s + 1} \\ &= \frac{A}{\tau_{sys}^2 s + 2\tau_{sys}s + 1} \end{aligned}$$

Substituting K_{sys} for A, this equation forms the critically damped second-order response:

$$\begin{aligned} H_{(s)} &= \frac{K_{sys}}{\tau_{sys}^2 s + 2\tau_{sys}s + 1} \\ &= \frac{K_{sys}}{(\tau_{sys}s + 1)^2} \end{aligned}$$

(b) From Example 14-4, we know that:

$$\tau_{sys} = \sqrt{\tau_1 \tau_2}$$

$$\zeta = \frac{\tau_1 + \tau_2}{2\sqrt{\tau_1 \tau_2}}$$

Because $\tau_1 = \tau_2$ in a critically damped system, τ_{sys} becomes:

$$\tau_{sys} = \sqrt{\tau_1^2} = \tau_1$$

or

$$\tau_{sys} = \sqrt{\tau_2^2} = \tau_2$$

Since both τ_1 and τ_2 are equal, τ_{sys} equals τ. Using the τ_1 and τ_2 equality in the damping coefficient equation proves that $\zeta = 1$:

$$\zeta = \frac{2\tau}{2\sqrt{\tau^2}}$$

$$= \frac{2\tau}{2\tau}$$

$$= 1$$

A critically damped system where $\zeta = 1$ indicates that the response of the system will be rapid as compared to a more sluggish overdamped response where $\zeta > 1$. Cascading two first-order systems with the same (or nearly the same) lag time ($\tau_1 = \tau_2$) will produce a critically damped second-order system response (see Figure 14-54).

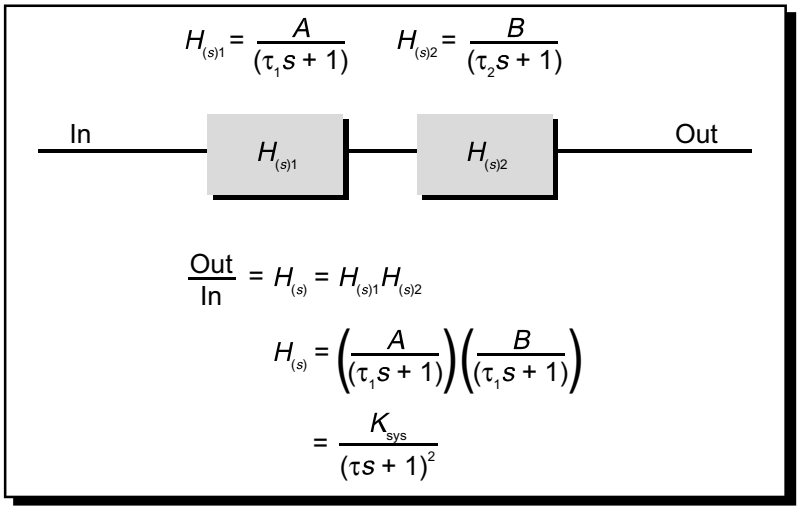


Figure 14-54. Two first-order systems with the same lag time cascaded to form a critically damped second-order system.

The inverse Laplace transform of a second-order critically damped system's response to a unit step, given that the gain K_{sys} equals A_1A_2, is (from Table 14-2):

$$\mathcal{L}^{-1}\left[\frac{K_{sys}}{s(\tau s+1)^2}\right] = K_{sys}\left[1-\left(\frac{\tau-t}{\tau}\right)e^{\frac{-t}{\tau}}\right]$$

A critically damped system achieves a steady-state value quicker than the other two types of second-order systems. However, the amplitude of the overshoot of a critically damped response is larger than that of an overdamped system.

UNDERDAMPED RESPONSES

Second-order **underdamped responses** exhibit an over and undershoot signal (oscillating response) at a natural resonant frequency of ω_n in radians/second. This oscillation is the result of a damping factor (ζ) that is less than 1. This means that instead of being able to factor the denominator of the second-order lag transfer function into a polynomial (i.e., $s^2 + 2\zeta\omega_n s + \omega_n^2$), the denominator becomes a complex-root quadratic equation. The inverse Laplace transform of this equation produces an exponential, decreasing sinusoidal response to a unit step input ($\frac{1}{s}$) represented by (from Table 14-2):

$$\mathcal{L}^{-1}\left[\left(\frac{1}{s}\right)\left(\frac{A\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}\right)\right] = A\left[1+\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_n\sqrt{1-\zeta^2}\,t-\psi\right)\right]$$

For small values of ζ, this response exhibits a behavior to a unit step approximate to:

$$\underset{\text{unit step}}{\text{Out}_{(t)}} = A\left[1+e^{-\zeta\omega_n t}\sin(\omega_n t)\right]$$

This makes the transfer function response approximate to:

$$H_{(t)} \approx e^{-\zeta\omega_n t}\sin(\omega_n t) \approx \sin(\omega_n t)\Big|_{\zeta=0}$$

which is the form of a sine curve. The response of this equation, shown in Figure 14-55, illustrates the damping factor ζ of the sinusoidal response. The closer the damping factor is to 1 (critical damping), the lower the frequency of oscillation and the sooner it will level off (see Figure 14-56a). Remember that if $\zeta = 0$, the response will oscillate forever as a sinusoidal response at a

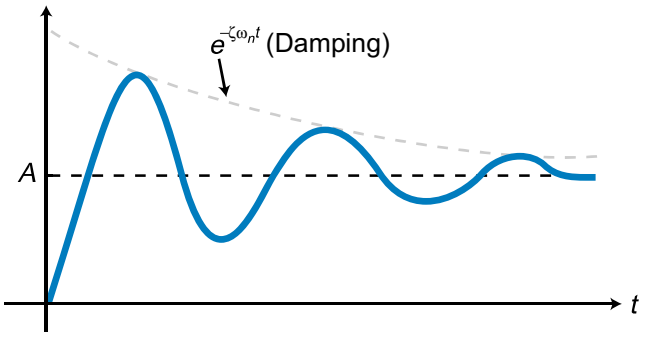


Figure 14-55. Underdamped response.

frequency ω_n; therefore, the closer the value of ζ gets to zero (see Figure 14-56b), the higher the frequency and the longer the oscillations will last (τ becomes longer).

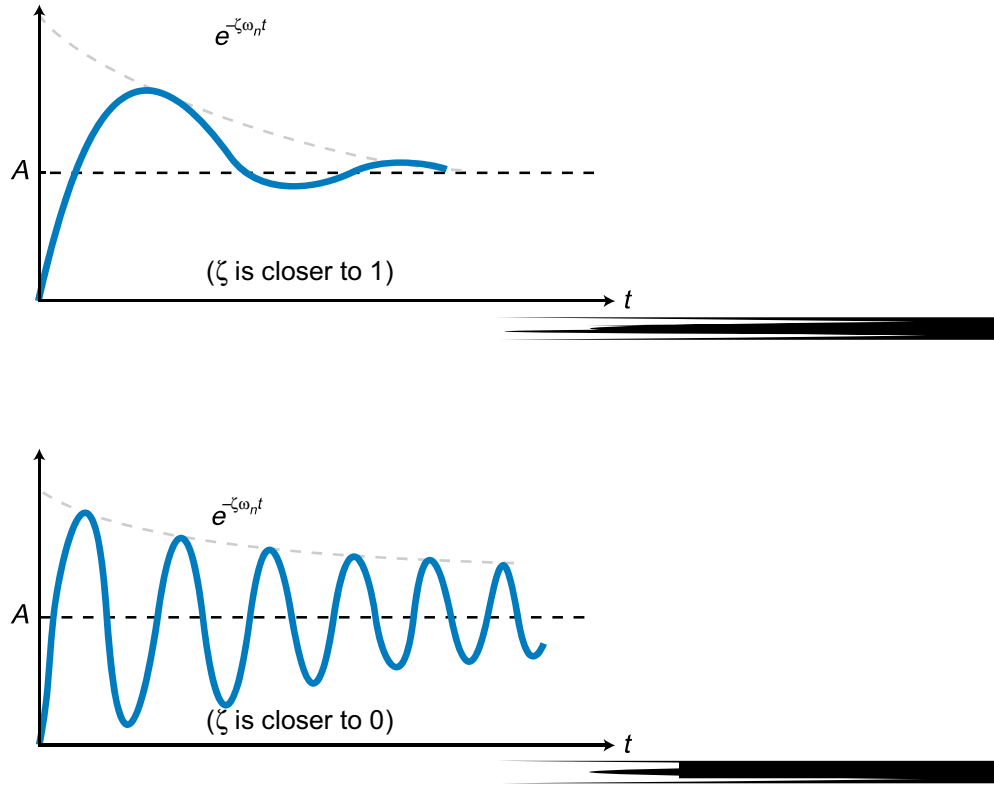


Figure 14-56. Frequency of oscillation is **(a)** lower when ζ is closer to 1 and **(b)** higher when ζ is closer to 0.

The exponential sinusoidal response of an underdamped second-order system will settle to 5% of its steady-state value within 3τ (three time constants), to 2% within 4τ, and to 0.5% within 5τ. The second-order lag response (τ_{sys}) for an underdamped system is defined as:

$$\tau_{sys} = \frac{1}{\zeta\omega_n}$$

which indicates that the lag time constant depends on the value of ζ. Figure 14-57 illustrates some typical parameters used to describe underdamped second-order systems.

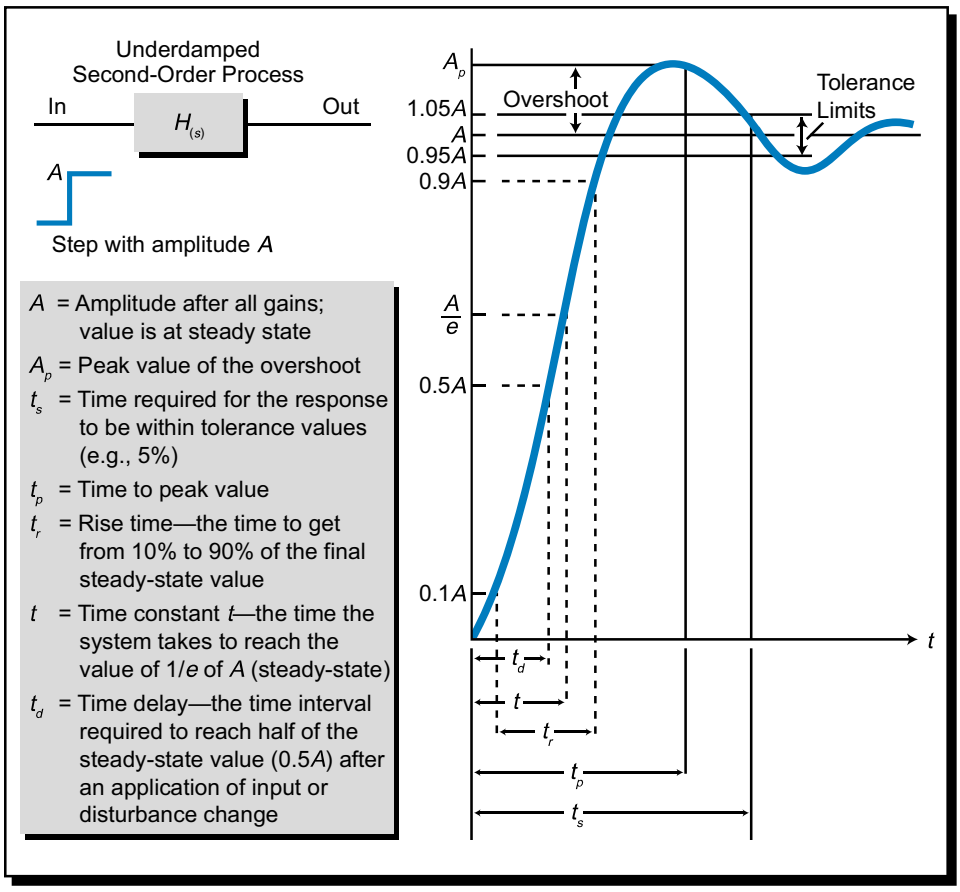


Figure 14-57. Parameters of an underdamped second-order system.

EXAMPLE 14-7

Compare the relationship between the first-order time response term $e^{\frac{-t}{\tau}}$ and the second-order, sinusoidal, exponential decay term $e^{-\zeta\omega_n t}$, which is used in the underdamped transfer function:

$$e^{-\zeta\omega_n t}\sin(\omega_n t)$$

SOLUTION

The time constant τ_{sys} for an underdamped system is equal to:

$$\tau_{sys} = \frac{1}{\zeta\omega_n}$$

or

$$\zeta\omega_n = \frac{1}{\tau_{sys}}$$

The response of a first-order exponential system is $e^{\frac{-t}{\tau}}$, where τ is the system's time constant. Therefore, the decaying term $e^{-\zeta\omega_n t}$ in the equation:

$$e^{-\zeta\omega_n t}\sin(\omega_n t)$$

is equal to:

$$e^{\frac{-t}{\tau_{sys}}}\sin(\omega_n t)$$

This indicates that, in an underdamped second-order system, the value of τ_{sys} (lag time) becomes larger as the value of ζ becomes smaller ($\tau_{sys} = \frac{1}{\zeta\omega_n}$) This is similar to the behavior of a first-order system with a long lag, because the time to reach the steady-state value will be long. In a second-order underdamped system, the oscillation continues for a longer time since the term $e^{\frac{-t}{\tau_{sys}}}$ provides less damping. If $\zeta = 1$, the value of τ_{sys} becomes $\frac{1}{\omega_n}$, which is the lag time of a critically damped system.

14-8 SUMMARY

The objective of a process control system is to maintain the process variable (process output) at a desired target value, referred to as the set point. The system provides this control by implementing a feedback loop, meaning that it reads the process variable and compares it to the set point value. The controller then uses the difference between these values, as computed by $E = SP - PV$, to determine how much corrective action it must take. The error, which the controller calculates as a percentage of the full range of the process variable, can be caused by changes in the set point or by disturbances to the process.

Open-loop systems are systems in which the process variable is not fed back into the control system for reference. Closed-loop systems, on the other hand, do receive process variable feedback. Most process control systems are closed-loop systems that receive negative feedback. In a negative feedback system, the controller determines the error by subtracting the process variable from the set point.

Process dynamics refers to changes in the process that occur due to disturbances or changes in the set point. Process gain changes are a result of gains in the process variable value created by changes in the control variable output. The dynamics of a process also includes dead time and lag time. Dead time is the delay that occurs between the moment a change is made in the control variable and the moment the process variable begins to react to the control variable change. Lag time is the delay associated with the time required by the process control loop to bring the process variable to the set point by adjusting the final control element. The lag time is a finite time required by the control system to physically adjust the final control element (e.g., a steam valve).

A transient is the process variable response to a change in set point or to the creation of a disturbance (e.g., a load change). The transient response depends not only on the dynamics of the process, but also on the characteristics of the process itself. These characteristics are the result of the transfer functions of the controller and the process. A transfer function is the mathematical representation of a system's response, where the response is computed by dividing the output by the input. Transfer functions are expressed in the frequency domain using Laplace transforms, to allow easy algebraic manipulations of the equations. The inverse Laplace transform of a transfer function converts a frequency-based Laplace response into a time-based response.

Each element in a control system loop has a transfer function associated with it—the controller has one and the process has one. The combined controller/process system also has a transfer function. Transfer functions are categorized as either first-order or second-order responses. First-order systems have one lag time associated with the process, while second-order systems have two lag times. Laplace transforms are used to mathematically represent both first- and second-order process transfer functions, as well as controller transfer functions and the combination of both process and controller functions in a closed-loop configuration. Although it is difficult to obtain the actual transfer function of a process, a knowledge of the type of transfer function expected from a process response is extremely useful, especially when tuning the controller.

First-order systems have one lag time, resulting in an exponential, decaying response. When the system receives a step input change, its open-loop output will have the following time domain response, which smoothly follows the input:

$$V_{\text{out}} = V_{\text{in}}\left(1 - e^{\frac{-t}{\tau}}\right)$$

The time constant τ specifies the time the output takes to achieve 63.2% of the final steady-state value. The time constant τ is sometimes referred to as the 63% response time. After 5τ periods have elapsed, the value of the output

response will be at 99.33% of its final value. In Laplace form, the transfer function of a first-order system has the form:

$$H_{(s)} = \frac{\text{Out}}{\text{In}} = \frac{A_1}{\tau s + 1}$$

Second-order systems have two lag times and are described by the transfer equation:

$$H_{(s)} = \frac{\text{Out}}{\text{In}} = \frac{A\omega_n^2}{\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right)}$$

Second-order systems can have three types of responses, depending on their damping coefficient: overdamped ($\zeta > 1$), critically damped ($\zeta = 1$), and underdamped ($\zeta < 1$). Each of these types of responses have different inverse Laplace transforms, translating into different time domain responses. Overdamped responses ($\zeta > 1$) have two different time constants (τ_1 and τ_2), and their response over time in reaching the set point is sluggish. Critically damped systems ($\zeta = 1$) have two lag times, or time constants, that are equal ($\tau_1 = \tau_2$). These systems reach the set point much faster than overdamped systems. Underdamped responses ($\zeta < 1$) produce a faster response than either overdamped or critically damped responses, resulting in an overshoot and undershoot of the final value that dies off exponentially as the steady-state value is approached. An underdamped response has two time constants that are imaginary, or mathematically speaking, that have complex roots produced by their quadratic equation.

Although some processes have more complicated responses (third- and fourth-order responses), these processes' transfer functions can be approximated by second-order system transfer functions. Most manufacturing processes can be classified as either first-order or second-order systems. In the next chapter, we will discuss how to use PID control to adjust the inputs to these complex systems to obtain a desired output.



KEY TERMS

control element
control loop
control variable
critically damped response
dead time
error
error deadband
first-order response
lag time
Laplace transforms

overdamped response
process control
process gain
process variable
second-order response
set point
steady state
step response
step test
transfer function
transient response
underdamped response

CHAPTER
FIFTEEN

PROCESS CONTROLLERS AND LOOP TUNING

Confusion worse confounded.

—John Milton



CHAPTER HIGHLIGHTS

In the previous chapter, we explained some important topics, such as process variable responses and transfer functions, that are elemental to the understanding of process control. In this chapter, we will continue our discussion of process control by explaining how a controller regulates a process. We will discuss the different types of controllers available, their advantages and disadvantages, and the effects that they have on the processes being controlled. We will also examine several tuning methods that are used to stabilize the process and determine the controller's tuning constants. After you finish this chapter, you will be ready to integrate PID control into a PLC application.

15-1 INTRODUCTION

The behavior of a closed-loop control system depends not only on the characteristics of the process and its transfer function, but also on the type of controller and the design decisions that occur during the selection of tuning parameters. As we explained previously, in a process control system, the process receives control information from the controller in the form of the control variable, which acts on the control element or process actuator (e.g., valve). The normal value of the control variable is usually at 50% of its range, so that it can either increase or decrease to accommodate for changes in the process variable.

The effect that a controller has on the process is the result of its *action*, or operational mode. Like the process itself, a controller also has a transfer function, which can be represented mathematically by Laplace transforms. The interaction between the controller and the process comprises the true essence of closed-loop process control.

In process control, the controller is responsible for the stability of the control system. Figure 15-1 illustrates three types of stability responses:

- stable
- conditionally stable
- unstable

Stable responses have an asymptotic characteristic, meaning that, as time increases, the response of the system approaches some finite value (see Figure 15-1a). Conversely, *conditionally stable responses* have a sinusoidal-type wave shape (see Figure 15-1b). This sinusoidal response has a low amplitude and may be acceptable in noncritical control loops, but not in the control of critical processes. *Unstable responses*, as the name implies, are system responses that are not acceptable because they create an unstable, or "runaway," condition (see Figure 15-1c). The sinusoidal amplitude of an unstable response increases as time increases.

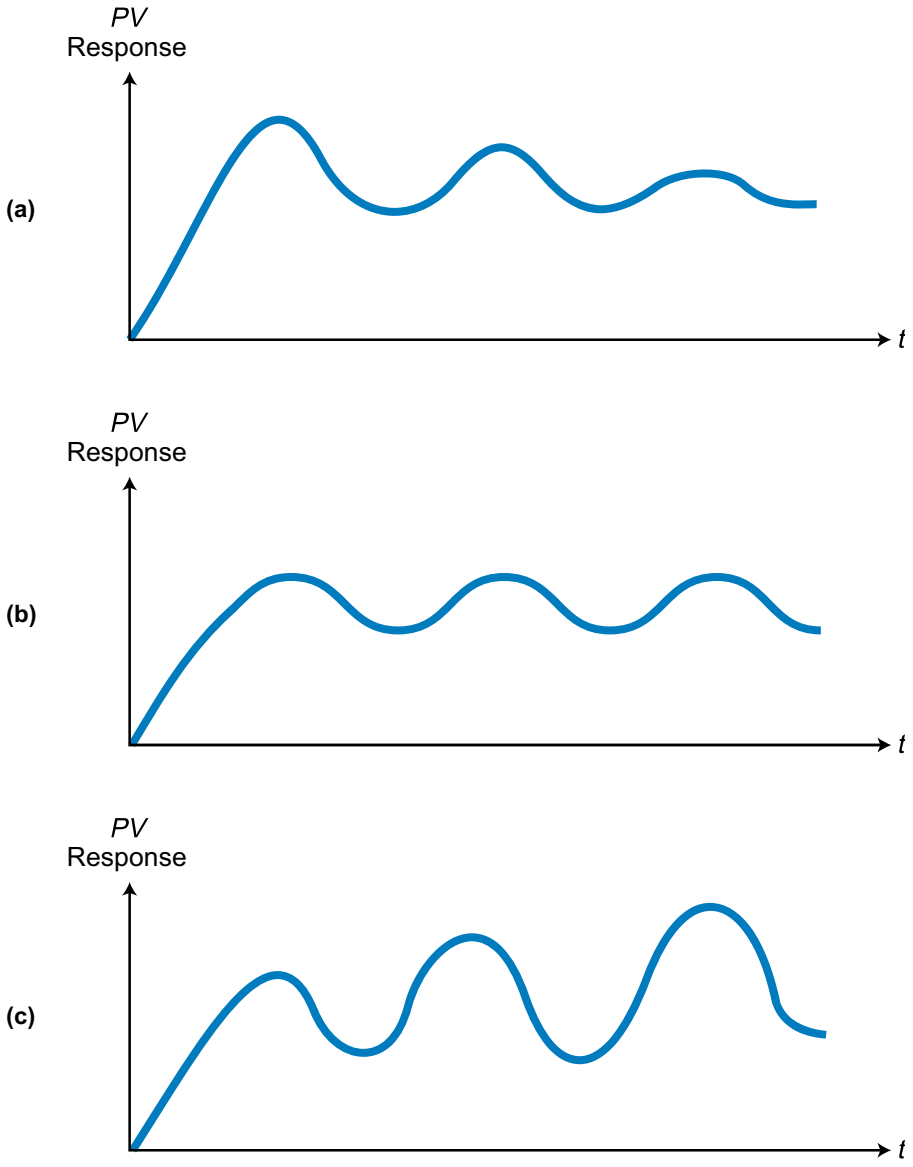


Figure 15-1. (a) Stable, **(b)** conditionally stable, and **(c)** unstable responses.

15-2 CONTROLLER ACTIONS

In the previous chapter, we explained how a process's transfer function indicates the process's behavioral response to an input change. Now, we will explain the controller's transfer function, *Hc*, along with the types of controller modes used to control the process. Figure 15-2 illustrates an open-loop configuration of the controller and process transfer functions by themselves, in which the controller's output (the control variable *CV*) is the input to the process. The transfer function for this open-loop configuration is:

$$\frac{\text{Output}}{\text{Input}} = \frac{PV_{(s)}}{\text{Input}_{(s)}} = \left(Hc_{(s)}\right)\left(Hp_{(s)}\right)$$

where $Hc_{(s)}$ and $Hp_{(s)}$ are the controller and process transfer functions in Laplace form.

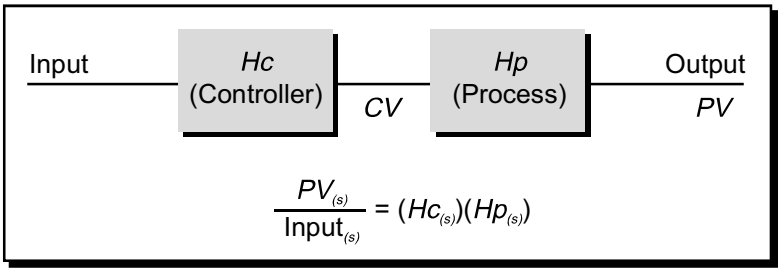


Figure 15-2. Open-loop system configuration.

As in an open-loop system, the controller in a closed-loop system also regulates the process variable value. However, a closed-loop controller provides either a *direct action* or a *reverse action* to the process it is controlling (see Figure 15-3). The difference between these two types of actions is the effect that the control variable (*CV*) from the controller (*Hc*) has on the process variable (*PV*) of the process (*Hp*). The type of process behavior required by an application determines the type of controller action needed in the system. For instance, a heating system and a cooling system behave differently, so their controller actions must behave differently as well.

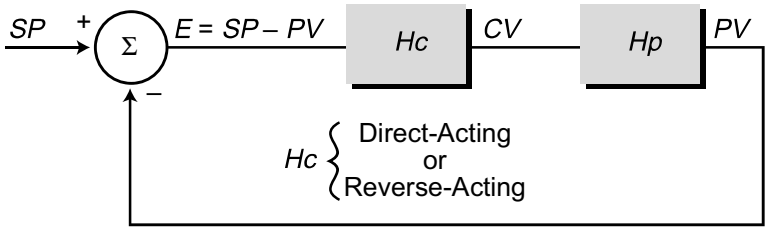


Figure 15-3. Closed-loop system configuration.

Note that, in a closed-loop control system, the key variable input to the controller is the error signal. After interpreting the error signal, the controller sends commands to the process via the control variable to bring the error to zero. In this chapter, we will refer to the error signal as the input to the controller and to the control variable as the controller output.

DIRECT-ACTING CONTROLLERS

A **direct-acting controller** is a closed-loop controller whose control variable output increases in response to an increase in the process variable. This is the type of action exhibited by a typical air-cooling system. As the temperature (*PV*) increases (i.e., it becomes warmer), the controller increases the value of its output (i.e., it increases the output of the air-conditioning compressor) to bring the process variable back to the set point.

Figure 15-4 illustrates another example of a direct-acting controller in which two materials are mixed in an exothermic (heat-producing) batch. Cold water flowing through the tank jacket cools the batch. A temperature sensor measures the temperature process variable, which has a cool set point.

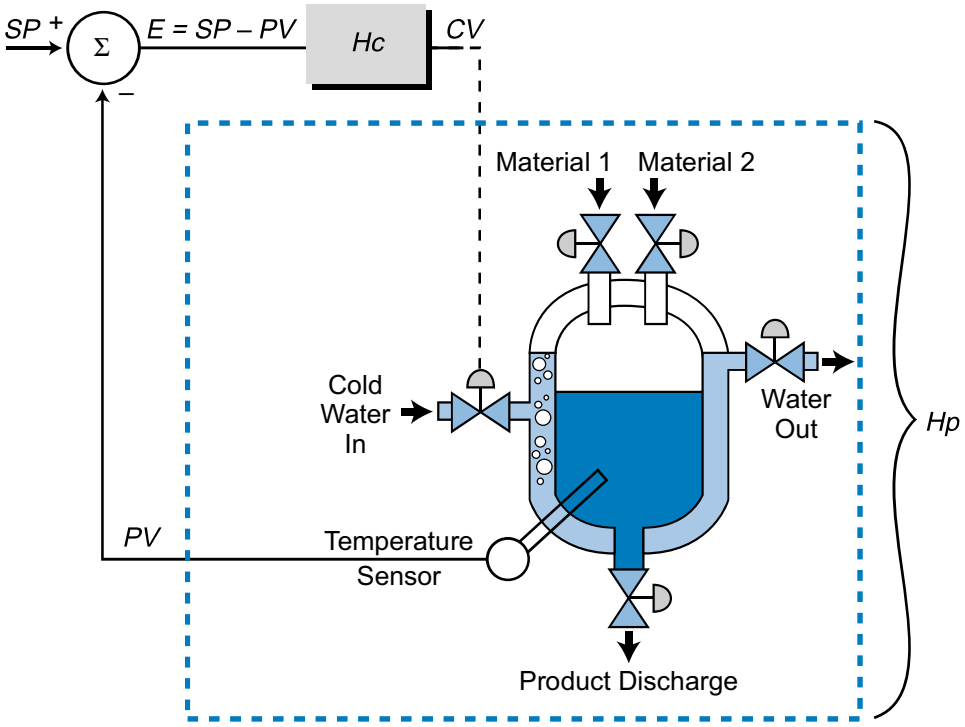


Figure 15-4. Direct-acting controller controlling the temperature in a batch-cooling process.

Figure 15-5 shows the reaction of the process in Figure 15-4. If the control variable that controls the cold water valve is open 100% (full open), the temperature of the batch will be 100°F; if the cold water valve is at 0% (closed), the temperature of the batch will be at 200°F. The desired set point of this process is 150°F, which corresponds to a 50% controller output. Therefore, in this process, if the cold water control valve opening increases, the system temperature decreases and vice versa.

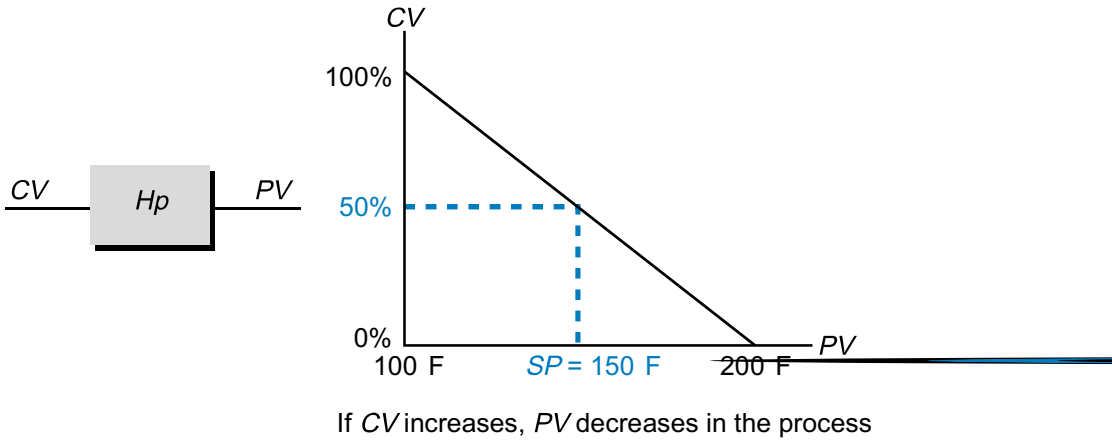


Figure 15-5. Process variable's reaction to a direct-acting controller.

Figure 15-6 shows the reaction of the controller to the process variable. If the controller senses that the temperature is too hot, it opens the cold water valve more to cool off the batch. Conversely, if the temperature is too cold, the controller decreases the opening of the control valve to warm up the temperature. Therefore, the controller in this system is a direct-acting controller because, as the process variable (temperature) increases, the controller increases its control variable output (opens the valve for more cold water) to bring *PV* closer to the set point, thus bringing the error to zero. In terms of error, the equation $E = SP - PV$ indicates that a direct-acting controller will increase its output as the error value in the system becomes more negative (as *PV* increases, *E* becomes more negative) and will decrease it as the error becomes more positive (as *PV* decreases, *E* becomes more positive).

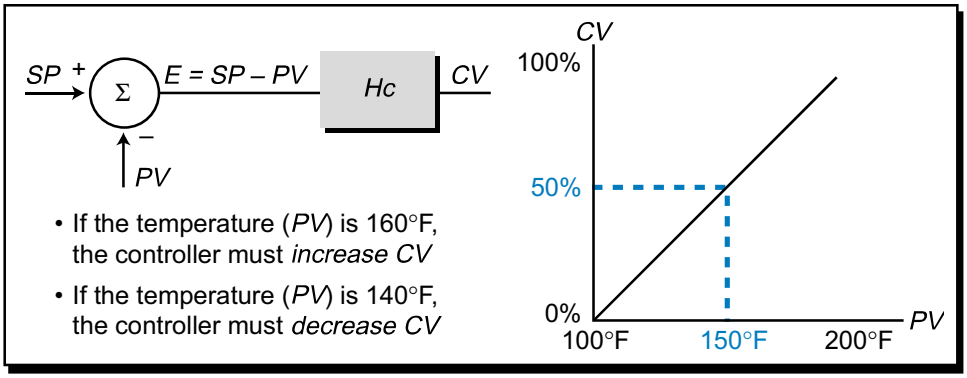


Figure 15-6. Relationship between *CV* and *PV* in a direct-acting controller.

In process control applications, a direct-acting controller is sometimes said to respond to a positive increase in error with an increase in the control variable (increase in controller output). The term "positive error," however, can be deceiving because it refers to the error change in the process variable, not to the change in the actual system error value. For example, referring to Figure 15-6, if the temperature (*PV*) increases from the set point of 150°F to 160°F, the direct-acting controller will increase the control variable because the process variable has increased by +10°F. This change in the process variable is a positive error because the actual *PV* value has changed in a positive direction. The system error (*E*), on the other hand, will become more negative due to this same change. When *PV* equaled the set point, the system error was 0 (150°F – 150°F). When the process variable increased to 160°F, however, the system error became –10°F (150°F – 160°F). Regardless of the terminology used, a direct-acting controller senses the direction of change in both the process variable and error and responds appropriately.

REVERSE-ACTING CONTROLLERS

A **reverse-acting controller** behaves oppositely of a direct-acting controller—if the controller detects an increase in the process variable, it will respond by decreasing the control variable. This behavior is typical of a

heating system. As the temperature becomes warmer (*PV* increases), the controller decreases the amount of heat the furnace produces to maintain the temperature at the set point.

Figure 15-7 illustrates a heating process in which a steam control valve allows heat to enter into the tank jacket of the batch system. The graph illustrated in Figure 15-8 shows that, in this heating system, if the steam control valve (*CV*) is at 100%, the temperature (*PV*) of the batch will be 200°F. Conversely, if the steam valve is at 0%, or completely closed, the batch temperature will drop to 100°F. To maintain the set point at 150°F, the controller must maintain the control variable output at 50% of its range. Figure 15-9 shows the relationship between the control variable and the process variable for the controller in this system. Because it is a reverse-acting controller, if the process variable (temperature) increases, the controller decreases its output to bring the error closer to zero.

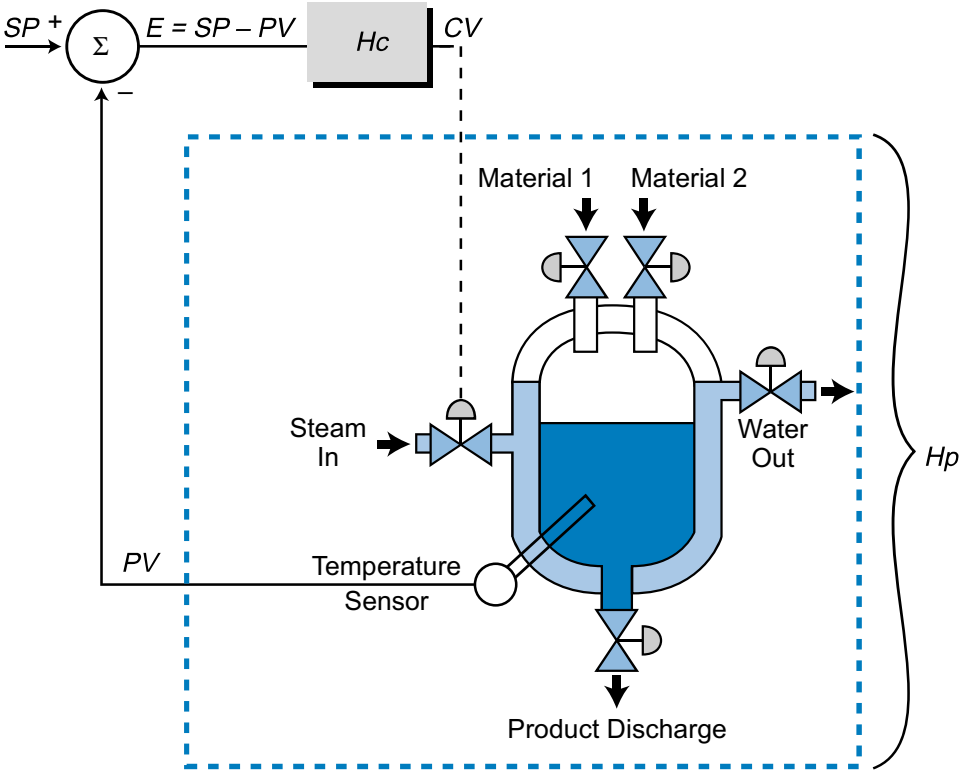


Figure 15-7. Reverse-acting controller controlling the temperature in a batch-heating process.

The selection of a direct- or reverse-acting controller depends on the behavior of the process itself. If the process reacts in a direct manner (*PV* increases as *CV* increases), the system's controller must provide reverse action, as in the case of a heating system, to control the process. If the process reacts in an inverse manner (*PV* increases as *CV* decreases), the controller must use direct action to control the process. Some single-loop controllers have a toggle

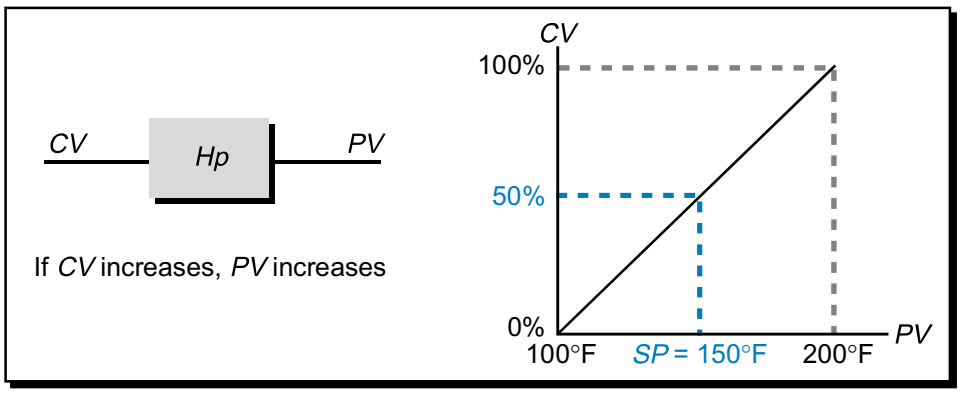


Figure 15-8. Process variable's reaction in a reverse-acting controller.

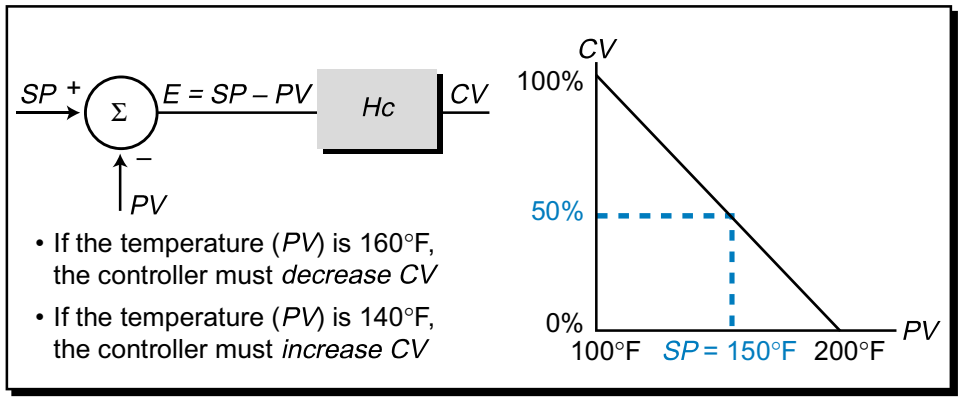


Figure 15-9. Relationship between *CV* and *PV* in a reverse-acting controller.

switch that can be used to select the desired action of the controller (direct or reverse). The control switch on a home thermostat is an example of this type of switch. During the winter, when the switch is set to heat, the system operates in a reverse-acting mode. During the summer, when the switch is set to cool, the system operates in a direct-acting mode. The closed-loop system remains the same, except for the behavior of the controller. The process behavior, which changes from winter to summer, necessitates the switch from reverse to direct action.

15-3 DISCRETE-MODE CONTROLLERS

A controller can have one of two modes that describes its output signal:

- discrete (ON/OFF) mode
- continuous (analog) mode

In discrete mode, the controller produces a discontinuous ON/OFF signal as its output (the control variable), which serves as the input to the process (see Figure 15-10a). In continuous mode, the controller emits an analog output

signal (see Figure 15-10b). In this section, we will discuss discrete-mode controllers. In the next section, we will explore continuous-mode controllers.

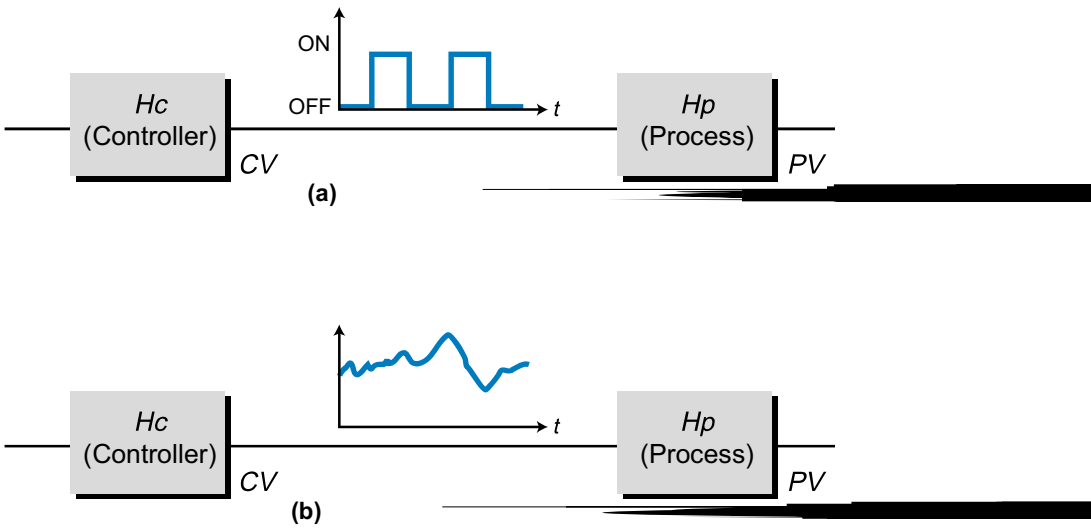


Figure 15-10. (a) Discrete- and **(b)** continuous-mode controllers.

Due to the nature of their signal, **discrete-mode controllers** produce a conditionally stable response. This means that the system error fluctuates between a predetermined deadband, creating a low-amplitude sinusoidal response. These controllers are used in systems where this type of response is acceptable. A noncritical heating system that uses an ON/OFF signal to control the heater is an example of a discrete-mode controller. A home air-conditioning and heating system is another example of a discrete-mode system, because the process variable cycles between two values on either side of the set point when the air conditioner or heater is turned ON or OFF. The two most common types of discrete-mode controllers are:

- two-position controllers
- three-position controllers

TWO-POSITION DISCRETE CONTROLLERS

A **two-position controller**, also called an *ON/OFF controller*, is the most basic type of process controller. As the name implies, it provides an ON/OFF signal to the process's control element (see Figure 15-11). A typical example of an ON/OFF controller is a home heating system. The heater turns ON when the temperature is below the set point and turns OFF when the temperature reaches an acceptable level. Ideally, if the set point temperature is 70°F, the heater will turn ON when the temperature is less than 70°F and turn OFF when it is greater than 70°F, as the heater tries to keep the error ($SP - PV$) at

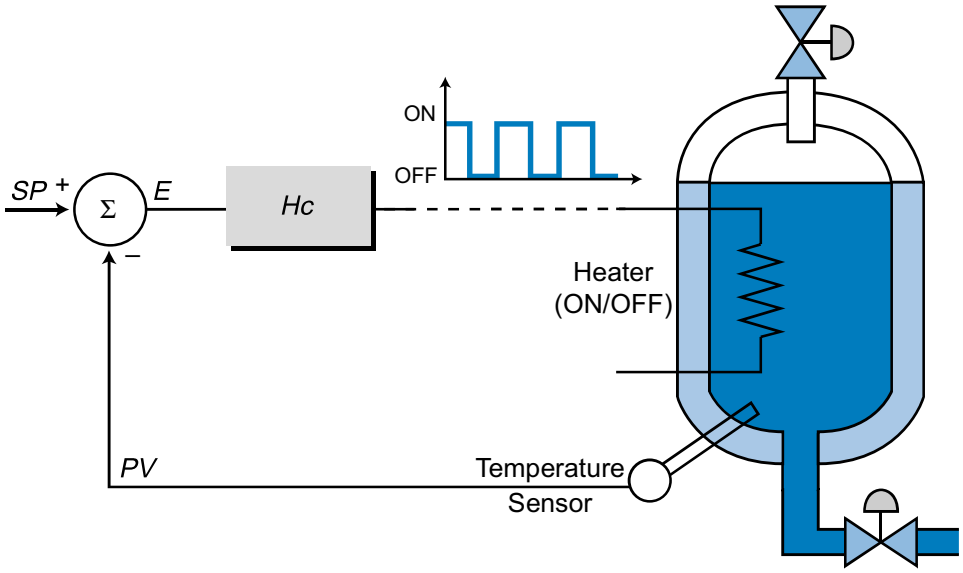


Figure 15-11. Two-position discrete controller controlling a heater.

zero. However, most heating systems have an error deadband, meaning that the heater will turn OFF at a value just above the target temperature and turn ON at a value just below it. So, if the heater in our example has a deadband of 68°F to 72°F, the heater will turn OFF when the temperature reaches 72°F and turn ON when it falls to 68°F. This deadband range avoids the constant ON/OFF action associated with trying to keep the process variable at one exact set point.

The example heating system has a reverse-acting controller, because if the controller senses that the process variable (temperature) decreases, it will increase its output to 100% (ON), as shown in Figure 15-12. As a result of the error deadband, the heater will turn ON when the temperature drops to 68°F and turn OFF when it reaches 72°F. Furthermore, note that the controller is sometimes ON and sometimes OFF within the error deadband. This depends

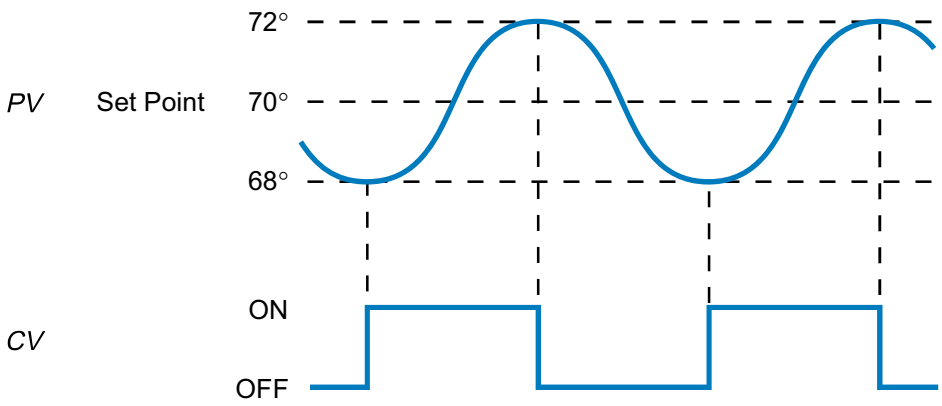


Figure 15-12. Behavior of the process and control variables in the heating system.

on the direction of the process variable. If the value of the process variable is decreasing within the deadband, then the controller is OFF, since it senses that the temperature is still at an acceptable level (reverse-acting). When the process variable reaches the lower limit of the deadband, then the controller will turn ON, causing the direction of the process variable to change (i.e., to increase). The controller will remain ON until *PV* reaches the upper limit of the deadband. At that time, the controller will turn OFF, *PV* will begin to decrease, and the cycle will repeat.

The action of an ON/OFF controller can be described by:

$$CV = 100\%\ (\text{ON}) \quad \text{IF error} > -\Delta E$$
$$CV = 0\%\ (\text{OFF}) \quad \text{IF error} < +\Delta E$$

where $\pm\Delta E$ represents the error deadband. Graphically, this controller output can be represented as shown in Figure 15-13, where the deadband of $\pm\Delta E$ is equal to ±2°F. If the process variable "comes" from the positive side (i.e., the error declines to a value less than $-\Delta E$, or 68°F), the controller output will turn ON at point 1 and remain ON (point 2) until the error reaches $+\Delta E$ (point 3).

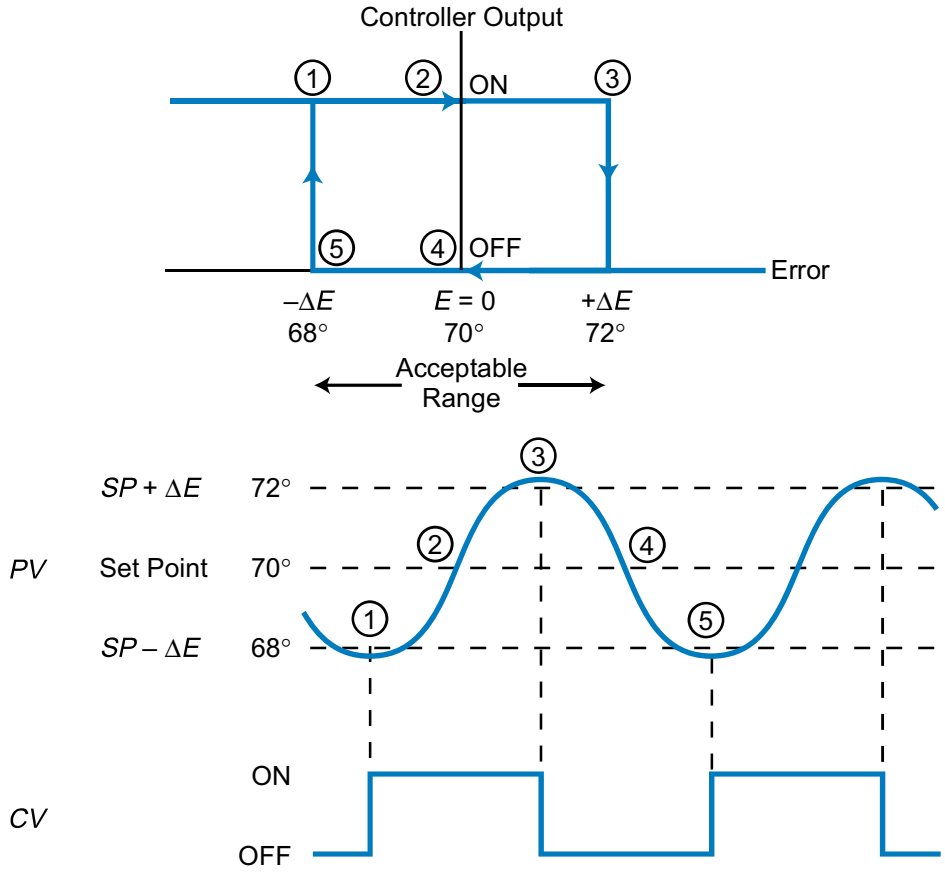


Figure 15-13. Controller output of the heating process.

At that time, the controller will turn OFF and remain OFF (through point 4) until the error drops to $-\Delta E$ (point 5), causing the cycle to repeat. This deadband curve is said to have *hysteresis*, meaning that the reaction of the system depends on its previous actions. It also produces an oscillating response, which is acceptable in this case. Also note that the curve of the ON/OFF controller signal will tend to overshoot the $SP + \Delta E$ value and undershoot the $SP - \Delta E$ value of the heater system due to finite warm up and cool off times (lag times).

ON/OFF controllers are appropriate for applications where large-scale, sudden changes are uncommon and the process reaction rate is slow. If the error deadband of the controller is reduced, then the amount of error in the system will decrease; however, the frequency of the ON/OFF and process variable cycles will increase. Conversely, if the deadband is increased, the oscillation frequency will decrease, but the error will be maximized. Thus, a trade-off exists between the desired error deadband and the frequency of the ON/OFF activation of the control element. The control element (e.g., valve, compressor, etc.) and other system components may be seriously damaged if they are turned ON and OFF too rapidly. Therefore, the system must be configured to compromise between the error allowance and the frequency of oscillation.

EXAMPLE 15-1

A two-position discrete-mode controller controls a cooling system, maintaining the system at a set point of 70°F. The controller has a deadband of ±3°F to allow for deviations from the set point.

(a) Plot the relationship between the controller's ON/OFF output, the process variable response, and the error curve, disregarding any overshoot or undershoot conditions. **(b)** Determine whether this is a direct- or reverse-acting controller.

SOLUTION

(a) Figure 15-14a illustrates the response of the process variable (temperature) to the controller's ON/OFF output. Figure 15-14b shows the hysteresis curve of the controller output versus the error.

(b) This controller is a direct-acting one, because as the process variable increases (passes $+\Delta E$ of SP), the controller will increase the control variable from 0% (OFF) to 100% (ON).

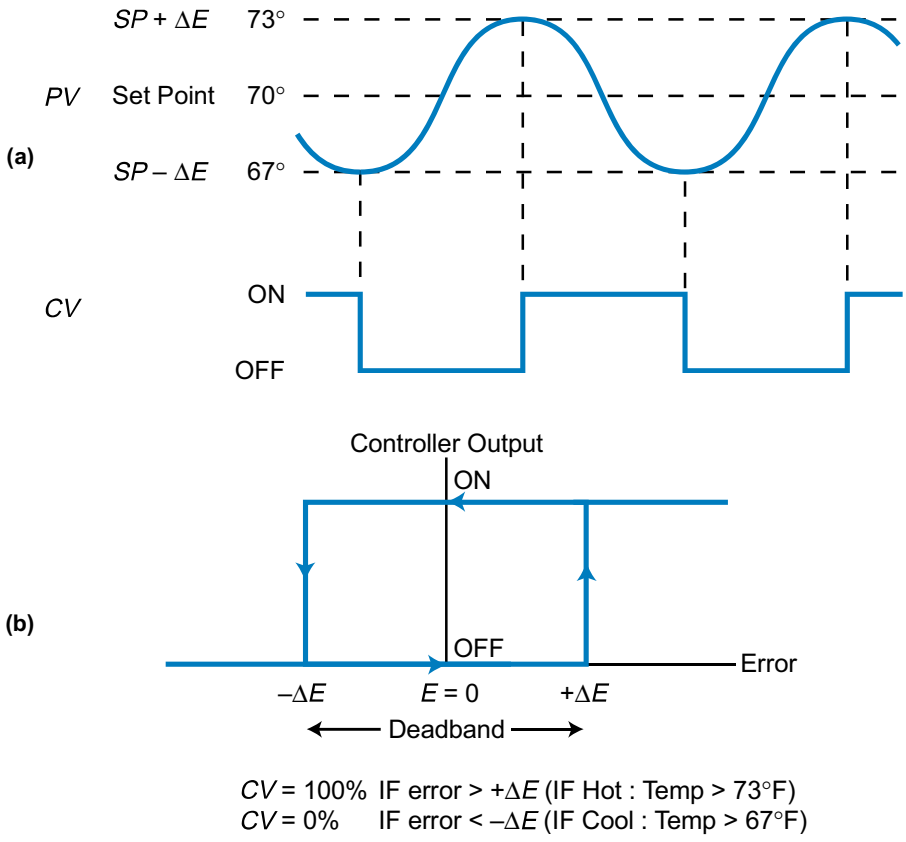


Figure 15-14. Example 15-1 **(a)** process variable response and **(b)** hysteresis curve.

EXAMPLE 15-2

Figure 15-15 shows a mixer tank that is heated by an ON/OFF heating control system. The set point temperature is 200°F with a deadband deviation of ±5% from the set point. When the heater is not on, the

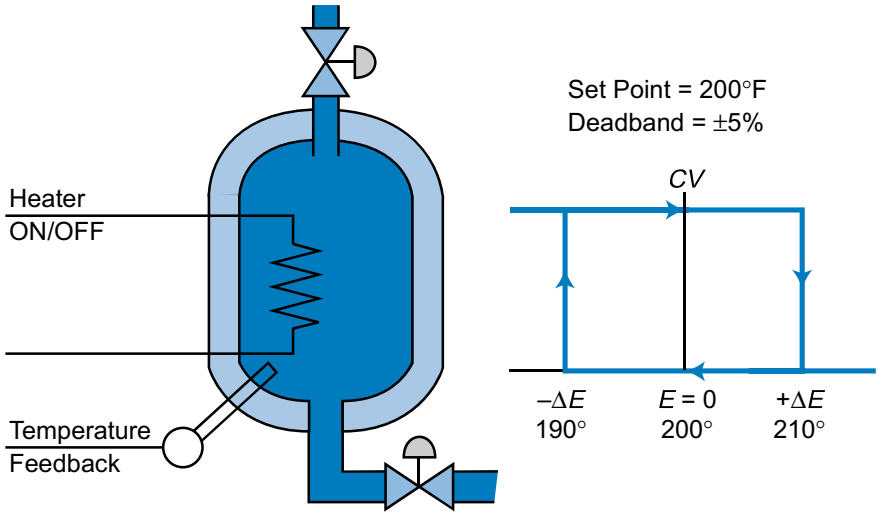


Figure 15-15. Mixer tank heated by an ON/OFF control system.

system linearly loses (cools) 4°F per minute; when the heater is applied, the system gains 8°F per minute. The system starting point is at the set point temperature with the heater in the OFF mode.

(a) Plot the oscillation response (cycle period) of the system and controller, and **(b)** calculate the response in part (a) taking into consideration a heater lag time of 30 seconds (0.5 min).

SOLUTION

(a) Figure 15-16 illustrates the response of the process variable over time, along with the controller's output status. The upper value of the deadband (ΔE = +5%) is 210°F, while the lower value (ΔE = –5%) is 190°F. This curve starts at 200°F (*SP*) and declines at a rate of 4°F/min until the temperature equals 190°F ($SP - \Delta E$). At 190°F, the controller turns ON and starts heating the system at a rate of 8°F/min until the temperature reaches 210°F ($SP + \Delta E$), at which point, the controller turns off the heater. The process variable starts to cool off again at the rate of 4°F/min until the temperature reaches $SP - \Delta E$, where the cycle is repeated.

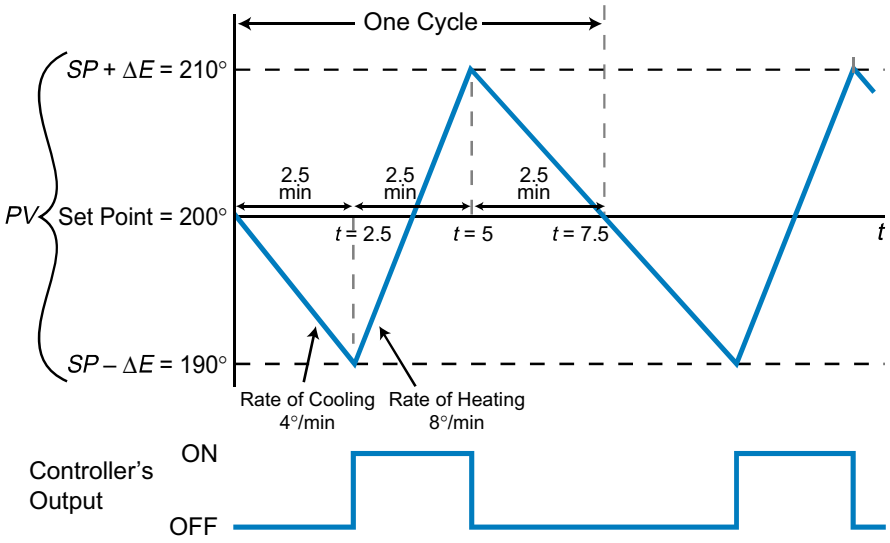


Figure 15-16. Process variable response for Example 15-2.

This system's response curve can be represented by two equations, one for when the controller is OFF and another for when the controller is ON:

$$\text{Temp1}_{(t)} = -4(t - t_1) + \text{Temp}_{(t_1)} \quad \text{(OFF mode)}$$

$$\text{Temp2}_{(t)} = 8(t - t_2) + \text{Temp}_{(t_2)} \quad \text{(ON mode)}$$

where:

$\text{Temp1}_{(t)}$ = the value of the *PV* curve (temperature) when the controller is OFF

$\text{Temp2}_{(t)}$ = the value of the *PV* curve when the controller is ON

$\text{Temp}_{(t_1)}$ = the value of *PV* at time t_1

$\text{Temp}_{(t_2)}$ = the value of *PV* at time t_2

t = time

Note that the first curve has a slope of –4°F/min and that the second curve has a slope of +8°F/min. So, the time required to reach $SP - \Delta E$ (190°) at $t_1 = 0$ and $\text{Temp}_{(t_1)} = 200°$ is:

$$\begin{aligned} \text{Temp1}_{(t)} &= -4(t - t_1) + \text{Temp}_{(t_1)} \\ 190 &= -4(t - 0) + 200 \\ 190 &= -4t + 200 \\ 4t &= 10 \\ t &= \frac{10}{4} = 2.5 \text{ min} \end{aligned}$$

At 2.5 minutes, the controller will turn ON. So, knowing that $\text{Temp2}_{(t)}$ is equal to 190°F at $t = 2.5$, we need to find the time at t_2. $\text{Temp}_{(t_2)}$ is equal to 210°F, because that is the time when the controller will turn OFF again; therefore:

$$\begin{aligned} \text{Temp2}_{(t)} &= 8(t - t_2) + \text{Temp}_{(t_2)} \\ 190 &= 8(2.5 - t_2) + 210 \\ 190 &= 20 - 8t_2 + 210 \\ 8t_2 &= 40 \\ t_2 &= \frac{40}{8} = 5 \text{ min} \end{aligned}$$

So, the temperature value will be 210°F when $t = 5$ minutes. Thus, the time from the moment the controller turns ON the heater at $SP - \Delta E$ (190°F) to the moment the controller turns OFF the heater at $SP + \Delta E$ (210°F) is 2.5 minutes. This is the time at 210°F minus the time it took to get to 190°F (5 min – 2.5 min = 2.5 min).

To complete the calculation of the oscillation cycle period, we must find the amount of time required for the temperature to cool off to the set point value again. This is equal to half of the time it takes for the

temperature to go from 210° to 190°. This value is the same as the time calculated for the curve $Temp1_{(t)}$, which is 2.5 minutes. Therefore, the frequency of oscillation will be 7.5 minutes.

Another way to calculate the time for each curve is to determine the difference between the temperatures and divide this difference by the rate required to get from one temperature to another. The time required for the first half of the curve, the OFF mode, to decline from the set point (200°F) to the lower limit of the deadband (190°F) is:

$$200°F \rightarrow 190°F \text{ @ rate} - 4°F/\text{min}$$
$$-10°F \text{ change @ rate} - 4°F/\text{min}$$
$$t = \frac{-10°F}{-4°F/\text{min}} = 2.5 \text{ min}$$

The same calculation for the OFF-to-ON state of the controller is:

$$190°F \rightarrow 210°F \text{ @ rate } 8°F/\text{min}$$
$$20°F \text{ change @ rate } 8°F/\text{min}$$
$$t = \frac{20°F}{8°F/\text{min}} = 2.5 \text{ min}$$

Finally, the time required for the next part of the curve is:

$$210°F \rightarrow 190°F \text{ @ rate} - 4°F/\text{min}$$
$$-20°F \text{ change @ rate} - 4°F/\text{min}$$
$$t = \frac{-20°F}{-4°F/\text{min}} = 5 \text{ min}$$

However, to compute the oscillation period, the system only requires half of this last time calculation, 2.5 minutes, to complete the cycle (i.e., return to the set point). Thus, the total time for the oscillation response is 7.5 min (2.5 min + 2.5 min + 2.5 min).

(b) A lag of 30 seconds, or 0.5 minutes, will cause the ON/OFF response to undershoot and overshoot the deadband, slightly affecting the frequency of oscillation (see Figure 15-17). This 0.5-minute lag may be due to the cooling off and heating up times associated with the heating element. The oscillation frequency for the first part of the curve can be calculated as follows:

$$200°F \rightarrow 190°F \text{ @ rate} - 4°F/\text{min}$$
$$t = \frac{-10°F}{-4°F/\text{min}} = 2.5 \text{ min}$$

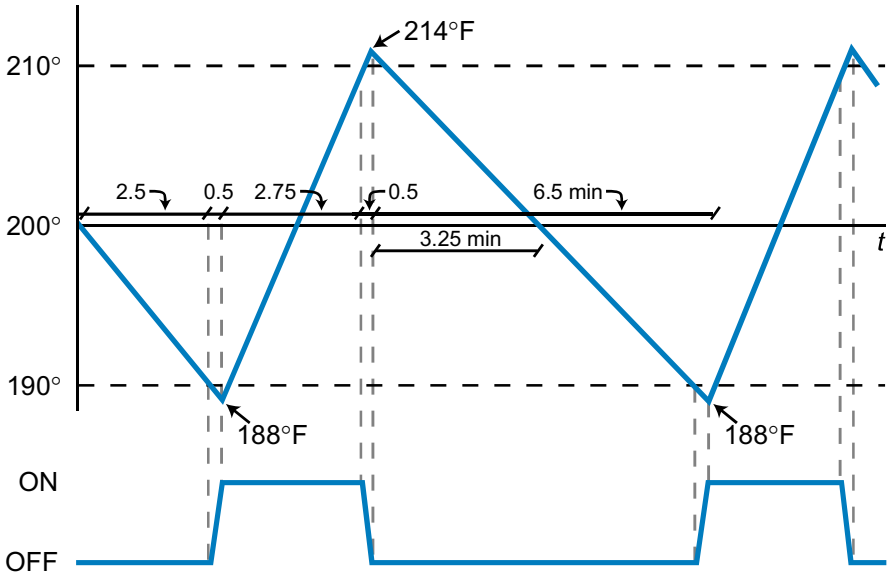


Figure 15-17. Undershoot and overshoot of the error deadband due to lag.

However, once the temperature reaches 190°F and the heater turns ON, another 0.5 minutes will elapse while the heating element heats up. Meanwhile, the temperature will continue to drop. So, during that 0.5-minute lapse, the temperature will drop another 2°F; making the final low-limit temperature 188°F:

$$t = \frac{\Delta \text{ temperature}}{\text{rate}}$$

$$0.5 \text{ min} = \frac{\Delta \text{ temperature}}{-4°\text{F/min}}$$

$$\Delta \text{ temperature} = (0.5)(-4) = -2°\text{F}$$

$$190°\text{F} - 2°\text{F} = 188°\text{F}$$

This lag will cause an undershoot of the deadband. Once the controller is ON, it will heat the tank at a rate of 8°F/min, reaching the 210°F upper temperature level in 2.75 minutes:

$$188°\text{F} \rightarrow 210°\text{F} \text{ @ rate } 8°\text{F/min}$$

$$t = \frac{22°\text{F}}{8°\text{F/min}} = 2.75 \text{ min}$$

The 0.5-minute lag will cause an overshoot of 4°F:

$$0.5 \text{ min} = \frac{\Delta \text{ temperature}}{8°\text{F/min}}$$

$$\Delta \text{ temperature} = (0.5)(8)$$

$$= 4°\text{F}$$

This will make the upper temperature 214°F (210°F + 4°F). The final period of oscillation, which is the last half of the curve, is the cooling off period between the upper temperature limit (214°F) and the lower limit (188°F):

$$214°F \rightarrow 188°F \text{ @ rate } 4°F/\text{min}$$

$$t = \frac{26°F}{4°F/\text{min}} = 6.5 \text{ min}$$

The half point of this curve, where *PV* equals the set point, will occur at 3.25 minutes. Thus, the total period of this system with lag, as shown in Figure 15-17, is:

$$\text{Period} = (2.5 \text{ min} + 0.5 \text{ min}) + (2.75 \text{ min} + 0.5 \text{ min}) + 3.25 \text{ min}$$
$$= 9.5 \text{ min}$$

The addition of a 0.5-minute lag to this system will increase the frequency from 7.5 minutes to 9.5 minutes.

THREE-POSITION DISCRETE CONTROLLERS

A **three-position controller** provides three output levels, instead of just two output levels like a two-position controller. Basically, this controller has an additional ON setting at 50% of the full ON range. The use of a three-position controller tends to reduce the cycling behavior of the process variable because it provides an intermediate output level, rather than just the two level settings of an ON/OFF controller. The controller stops at the intermediate 50% setting when the set point is achieved. In fact, a controller's output is usually designed so that its half output, or 50%, coincides with the level required by the process to maintain the process variable at the set point, minimizing the error in the system. Figure 15-18 illustrates a three-position, direct-acting controller's output (*CV*) according to the error present in the system. This output can be represented mathematically as:

$$CV = 100\% \quad \text{IF error} > +\Delta E$$
$$CV = 50\% \quad \text{IF} -\Delta E < \text{error} < +\Delta E$$
$$CV = 0\% \quad \text{IF error} < -\Delta E$$

Because three-position field devices are not as widely available as two-position ON/OFF devices, analog field devices are often used to implement three-position control. A PLC may also implement a three-position output using a contact output interface (see Figure 15-19). The incoming sides of

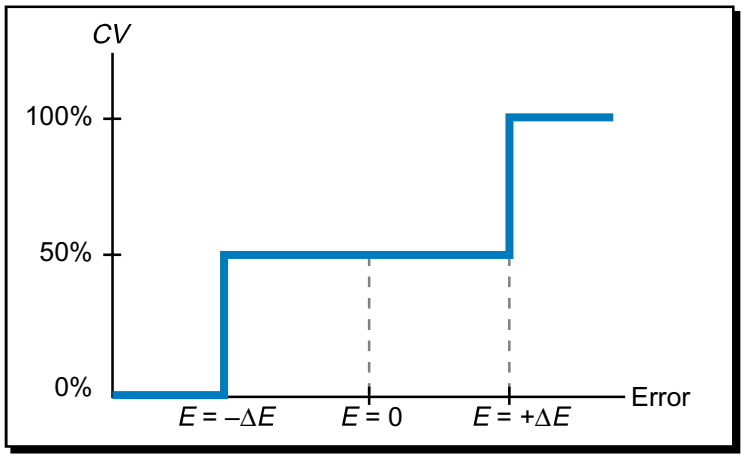


Figure 15-18. Three-position, direct-acting controller's output in response to error.

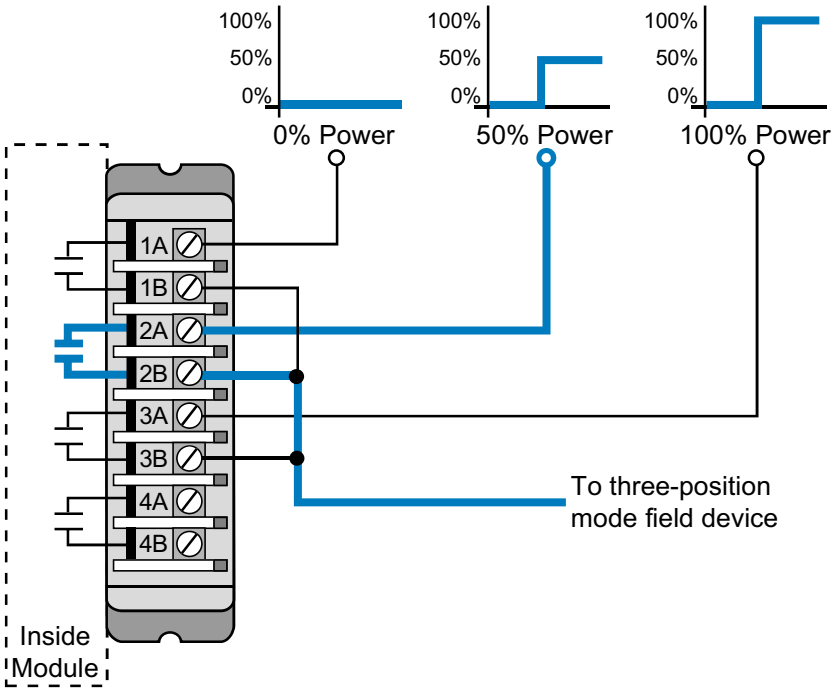


Figure 15-19. Contact output interface implementing three-position control.

three of the contact module's terminals are connected to 0%, 50%, and 100% power signals, respectively. The other sides of the contacts join together and connect to the field device (e.g., valve). In this configuration, a PLC with a contact output module can be interfaced with analog signals set at 0%, 50%, and 100% of the full range of the field control device to implement three-position control. Discrete-mode controllers with more than three positions (e.g., five positions, multipositions, etc.) may also be implemented this way in some control applications. Since output field devices with more than three positions are not readily available, analog output devices are often used in

conjunction with multiple output contact cards to obtain multiposition control. Figure 15-20 illustrates an example of this type of direct-acting control system with five possible output settings.

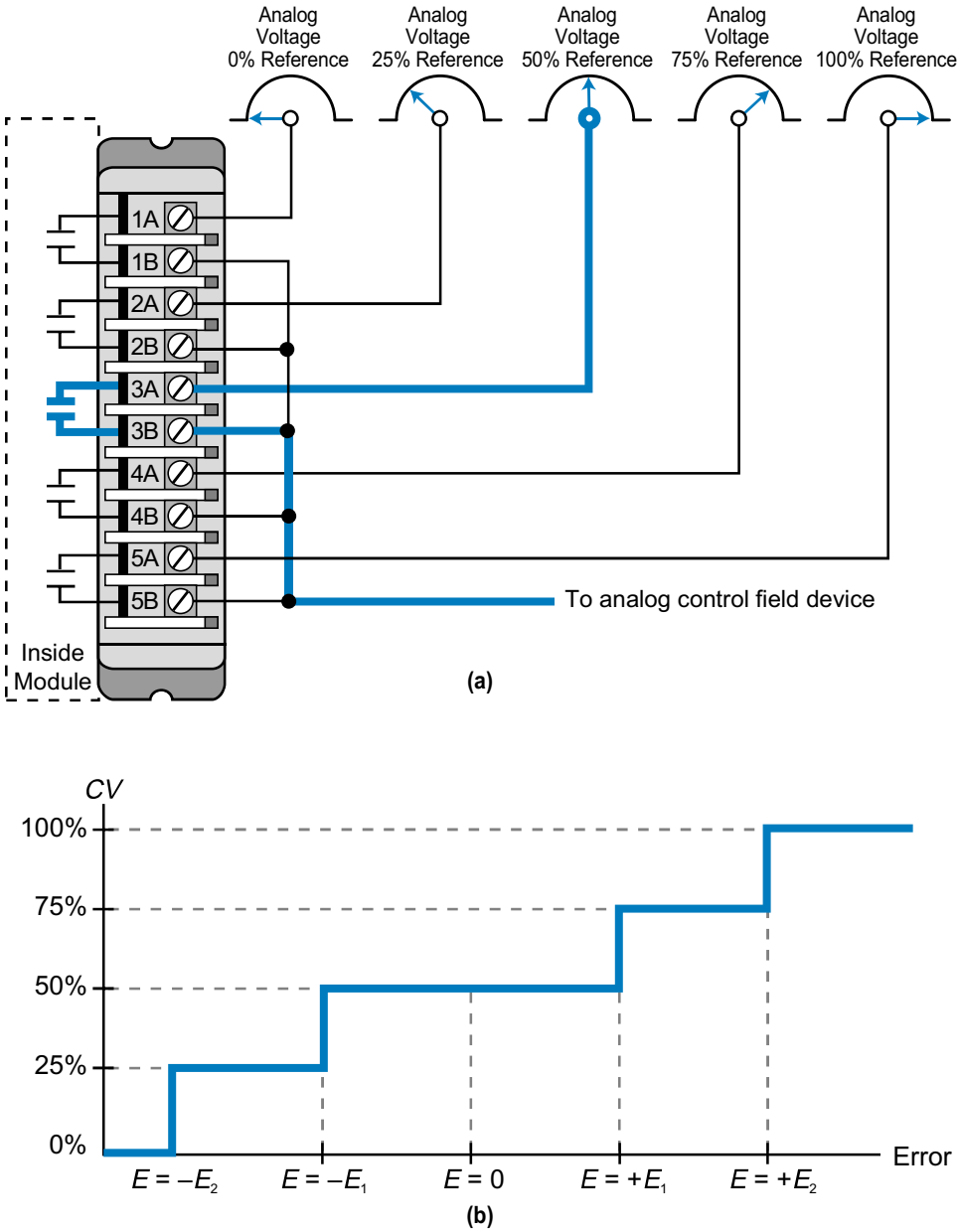
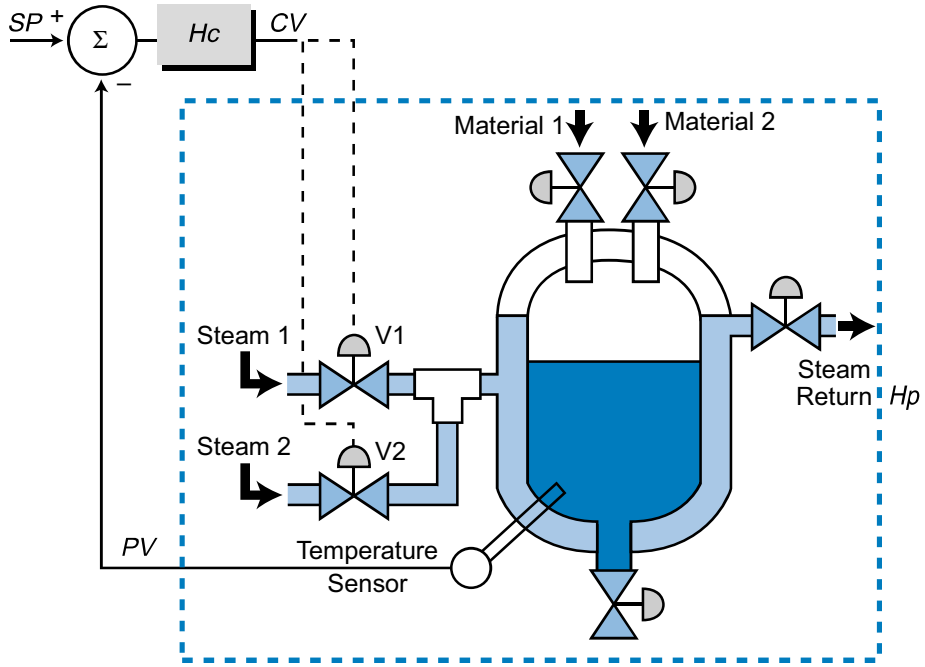


Figure 15-20. (a) A discrete mode, multiposition controller and **(b)** its output.

EXAMPLE 15-3

Graphically illustrate the reaction of a three-position controller output to the steam heating system shown in Figure 15-21. Include the effect of the controller's lag.



V2	V1	Steam
OFF	OFF	0%
OFF	ON	50%
ON	ON	100%

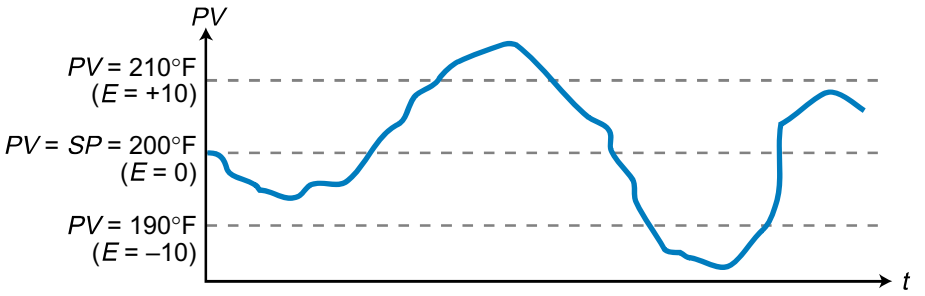


Figure 15-21. Steam heating system controlled by a three-position controller.

SOLUTION

Figure 15-22 shows the controller output (*CV*) for this three-position discrete-mode controller. Note that the response plot (see Figure 15-22a) shows that an overshoot is present, indicating a lag in steam actuation. The same lag also creates an undershoot condition when both V1 and V2 are ON. As shown, the system overshoot and undershoot pass the deadband during the lag periods.

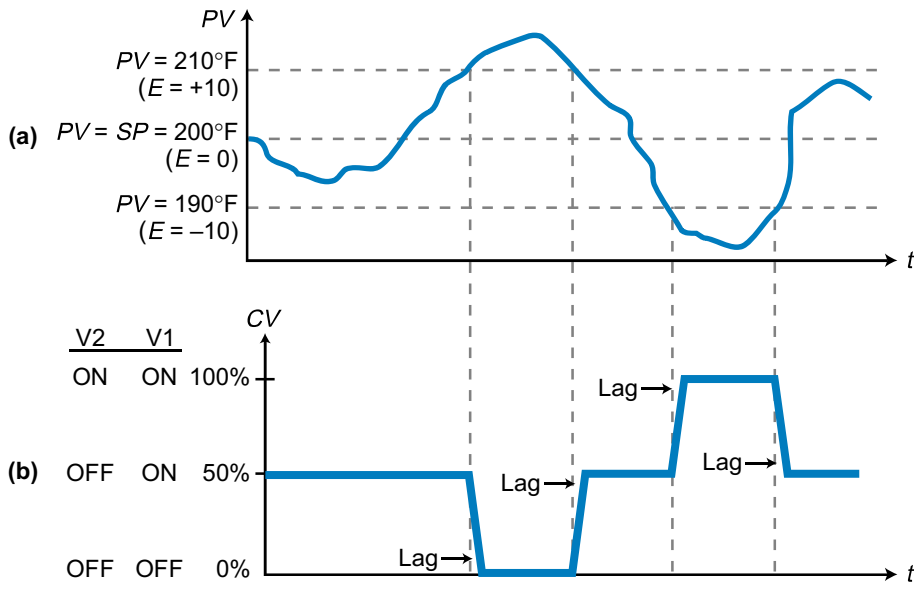


Figure 15-22. (a) Response plot and **(b)** controller output of the heating system in Figure 15-21.

15-4 CONTINUOUS-MODE CONTROLLERS

Most process control applications employ **continuous-mode controllers**, instead of discrete-mode controllers, to avoid the oscillatory system response caused by ON/OFF control. A continuous-mode controller sends an analog signal to the process control field device (see Figure 15-23) to regulate the process variable, bringing the error signal to zero in a closed-loop system. A continuous-mode controller behaves like a multiposition controller with an infinite number of positions. In a PLC-based system, the controller may be an intelligent I/O interface or software routine instructions that use standard I/O analog modules.

Continuous-mode controllers use three different modes to control the process:

- proportional control mode
- integral control mode
- derivative control mode

These three modes are also referred to as controller *actions*, each one reacting differently to the error present in the system in a direct- or reverse-acting fashion. The *proportional mode* provides a control variable adjustment that is proportional to the error deviation. The *integral mode* (or *reset mode*) provides a change in the control variable based on the time history of the error. The *derivative mode* (or *rate mode*) provides a change in the control variable based on the rate of change of the error signal. The combination of all three

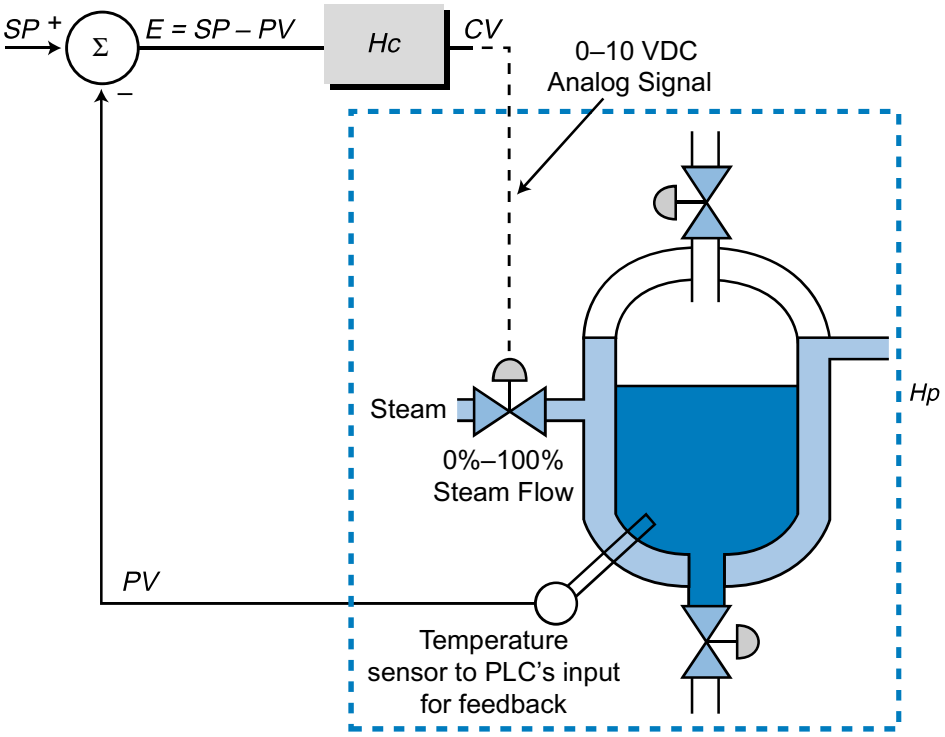


Figure 15-23. Block diagram of a continuous-mode controller.

modes in one controller forms the industry standard known as PID control. Table 15-1 shows the different possible combinations of continuous controller modes. Note that the derivative action is not used as a stand-alone mode



Table 15-1. Continuous controller modes.

in applications. This is due to the derivative action's response, which produces a high output but only for a short period of time. This has little effect on the process and, consequently, does not provide any process control.

15-5 PROPORTIONAL CONTROLLERS (P MODE)

A **proportional controller** adjusts the control variable output in a manner proportional to the error. As shown in Figure 15-24, the controller (*Hc*) receives feedback information from the process (*Hp*) in the form of the process variable, which is then compared to the set point. The error created, either positive or negative, tells the controller what percentage of output (*CV*) to provide to bring the error to zero. Figure 15-25 illustrates a typical proportional controller transfer function for a direct-acting controller (e.g., a cooling system). As the error becomes more negative (*PV* > *SP*), the controller will increase the control variable in proportion to the error. This will cause the process variable (from *Hp*) to decrease, thus pushing the error to zero. If the error becomes more positive, the opposite occurs.

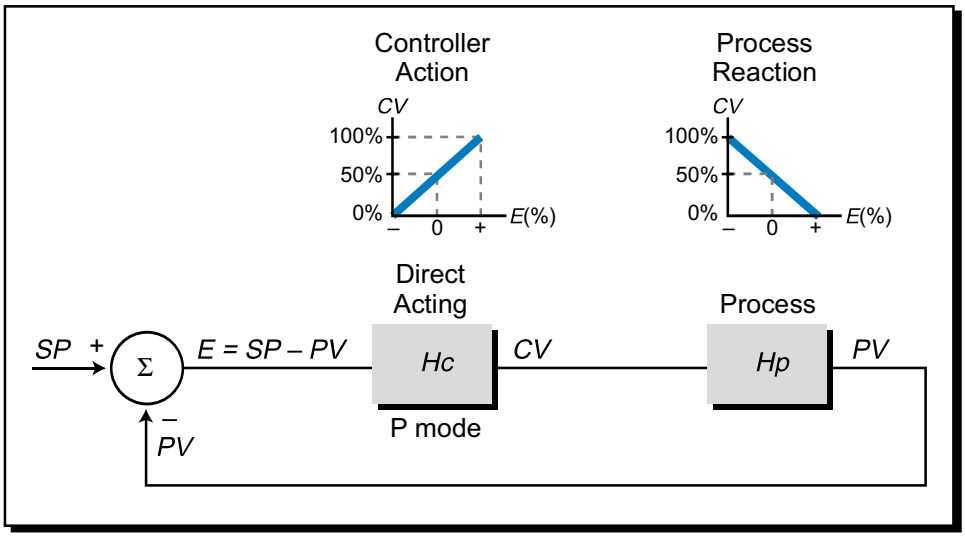


Figure 15-24. Proportional closed-loop control.

The control variable output ($CV_{(t)}$) of a proportional controller, starting from the set point value, is expressed by:

$$CV_{(t)} = K_P E + CV_{(E=0)}$$

where:

K_P = the proportional gain of the controller

E = the current error

$CV_{(E=0)}$ = the controller output when the error equals 0

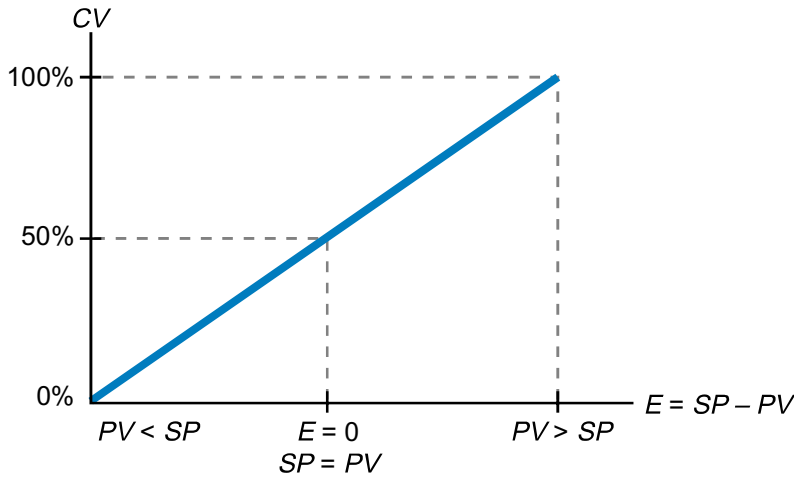


Figure 15-25. Direct-acting proportional controller transfer function.

Using this equation, a proportional controller can adjust the value of the control variable according to time and error by replacing the $CV_{(E=0)}$ term with the previous value of CV:

$$CV_{\text{new}} = K_P E + CV_{\text{old}}$$

So, if a controller with a control variable output value of 50% senses that the proportional error in the system ($K_P E$) is 20%, its new output will be 70%:

$$\begin{aligned} CV_{\text{new}} &= 20\% + 50\% \\ &= 70\% \end{aligned}$$

This value indicates a linear correspondence between the control variable and the error, as was depicted in the graph in Figure 15-25. This graphic representation is called the *proportional band*, and it shows the error values associated with the full range of the controller output. The slope of this graph, the proportional gain K_P, is computed by dividing the percentage change in output by the percentage change in error:

$$K_P = \frac{\%\text{ change in } CV}{\%\text{ change in } E}$$

The proportional gain, therefore, is expressed in units of %/%. For example, a gain of 1 indicates that a 1% change in error will cause a 1% change in controller output. Note that the direction of the slope of the proportional gain (the positive or negative response of the control variable to a change in the error) depends on whether the controller is direct acting or reverse acting.

The proportional gain relationship between the error and the control variable depends on the width of the band upon which the controller is acting. For example, the temperature control system in Figure 15-26 has a temperature

response that spans from 60°F to 180°F, equaling a range of 120°F (180°F – 60°F). However, if the controller only needs to exert control from 90°F to 150°F with the set point at 120°F, it will only be controlling a range of 60°F (150°F – 90°F) over the total range of 120°F. Therefore, the proportional band of the controller is 60°F over the 120°F range. Accordingly, the proportional band (*PB*) of control as a percentage of the full process variable range is represented as:

$$PB = \frac{PV_{max} - PV_{min}}{PV_{(max\ range)} - PV_{(min\ range)}}$$

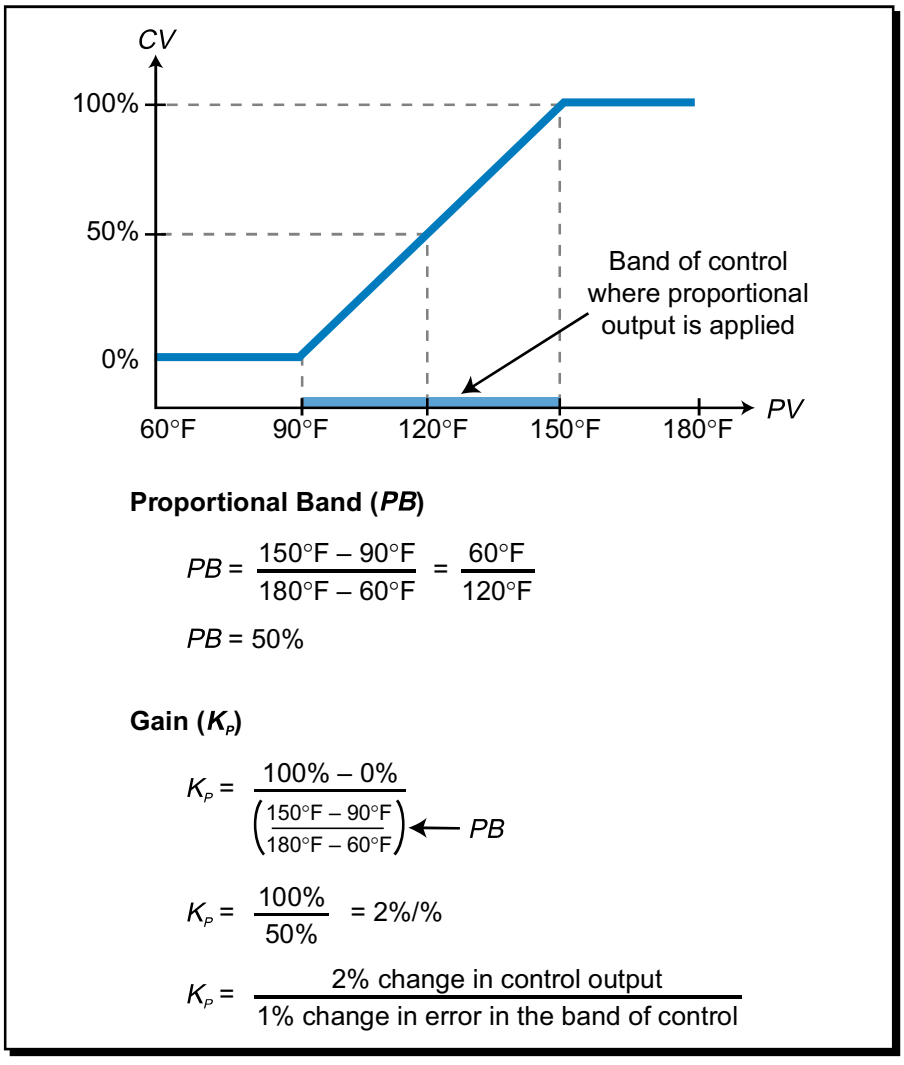


Figure 15-26. Proportional band and gain calculations.

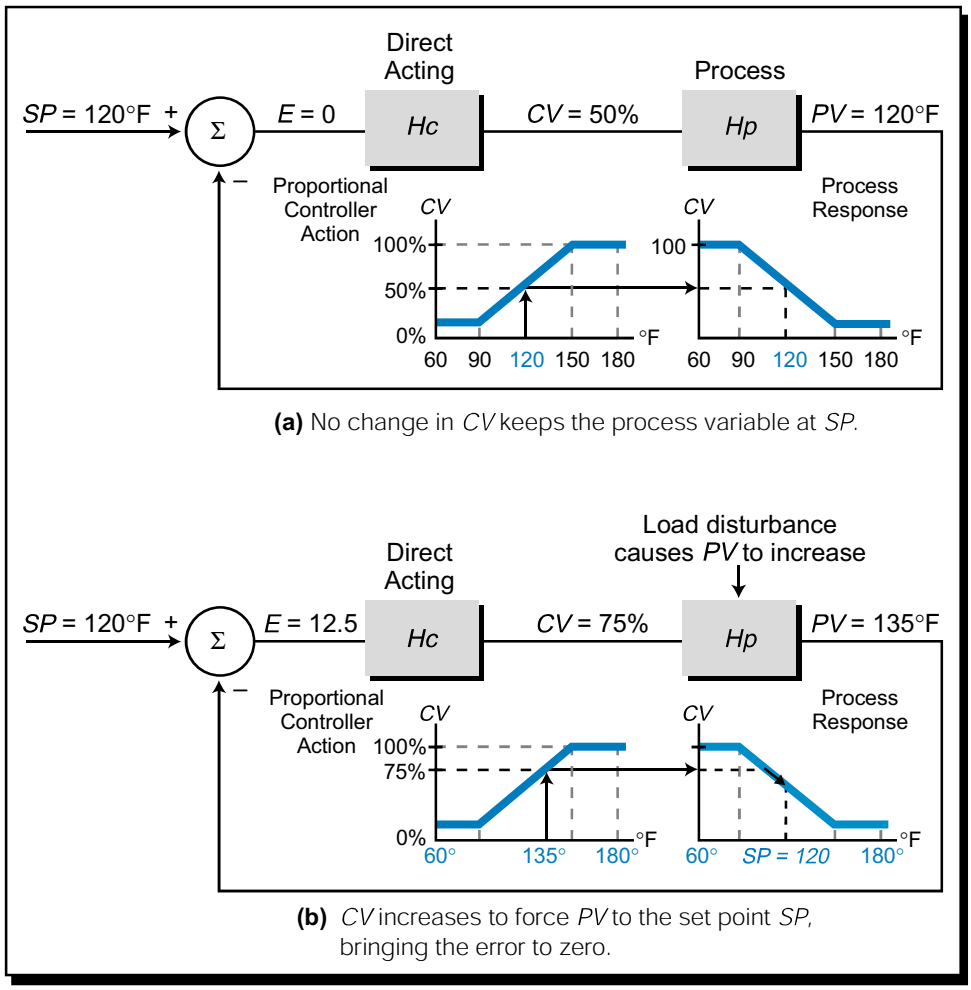
As shown in the calculations in Figure 15-26, the proportional band of the temperature control system is 50%. The proportional gain of the system is defined by how much the control variable output changes for each percent of

error within the control band. The error percentage range is equal to the proportional band percentage, because both express how much the process variable can deviate from the set point. The gain, according to Figure 15-26, will be 2, meaning that the controller's output will change 2% for every 1% change in error. This controller is a direct-acting controller, since the control variable will change in the same direction (+ or –) as the percentage of error. Note that the gain and proportional band are inversely related, meaning that:

$$K_P = \frac{1}{PB}$$

$$PB = \frac{1}{K_P}$$

If the process variable in the previous example is at the set point of 120%, then the controller must only maintain *CV* at 50% to keep the error at 0 (see Figure 15-27a). However, if the process variable increases to 135°F, the error incurred over the total temperature span will be:



(a) No change in *CV* keeps the process variable at *SP*.

(b) *CV* increases to force *PV* to the set point *SP*, bringing the error to zero.

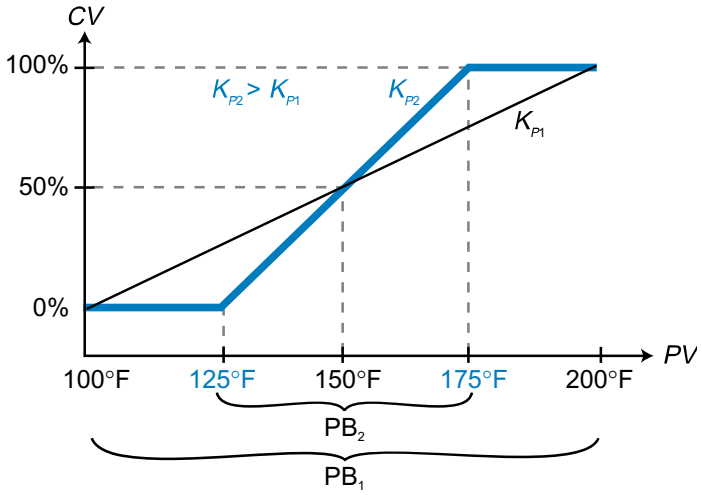
Figure 15-27. Process with **(a)** no change and **(b)** change in *CV*.

$$E = \frac{SP - PV}{PV_{min} - PV_{max}}$$
$$= \frac{120°F - 135°F}{60°F - 180°F}$$
$$= \frac{-15°F}{-120°F}$$
$$= 12.5\%$$

This error (12.5% above the set point over the *PV* range) in the controller equation, as well as in the graphic in Figure 15-27b, indicates that the new output of the controller will be 75%:

$$CV_{new} = K_P E + CV_{old}$$
$$= (2\%)(12.5\%) + 50\%$$
$$= 25\% + 50\%$$
$$= 75\%$$

The gain of a controller indicates how sensitive the controller is to error. The proportional band also indicates this sensitivity, since the gain and the band are related. Figure 15-28 illustrates two controllers with gains of $K_{P1} = 1$ and



$$PB_1 = \frac{200° - 100°}{200° - 100°} = 1 = 100\%$$

$$PB_2 = \frac{175° - 125°}{200° - 100°} = 0.5 = 50\%$$

$$K_{P1} = \frac{CV_{max} - CV_{min}}{PB_1} = \frac{100\% - 0\%}{100\%} = 1\ \%/\%$$

$$K_{P2} = \frac{CV_{max} - CV_{min}}{PB_2} = \frac{100\% - 0\%}{50\%} = 2\ \%/\%$$

Figure 15-28. Two controllers with gains of 1 and 2.

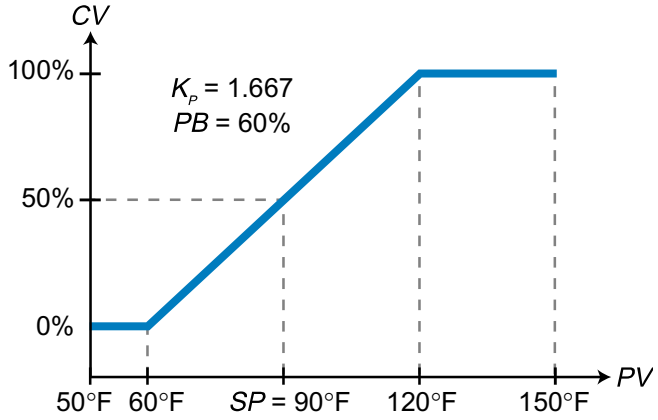
$K_{P2} = 2$ that have proportional bands of 100% and 50%, respectively. The system with a gain of 1 will change the controller output 1% for every 1% of error, while the system with a gain of 2 will have twice the sensitivity, changing *CV* 2% for every 1% error.

EXAMPLE 15-4

Graph the transfer function for a proportional controller with a proportional band of 60% over a process variable range of 50°F to 150°F. The proportional band is centered around a set point of 90°F at a 50% controller output.

SOLUTION

Figure 15-29 illustrates the controller's transfer function. Note that, in this system, the set point (90°F) is not at the center of the total process variable range.



$$PB = \frac{120°F - 60°F}{150°F - 50°F} = \frac{60°F}{100°F} = 60\%$$

$$K_P = \frac{1}{PB} = \frac{1}{60\%} = 1.667$$

Figure 15-29. Example controller's transfer function.

The proportional gain for a reverse-acting controller (see Figure 15-30) is calculated using the same equations as used for a direct-acting one; however, the sign of the gain will be negative due to the slope of the curve. The proportional band for the reverse-acting controller in Figure 15-30 is:

$$PB = \frac{200°F - 100°F}{200°F - 100°F}$$
$$= 100\%$$

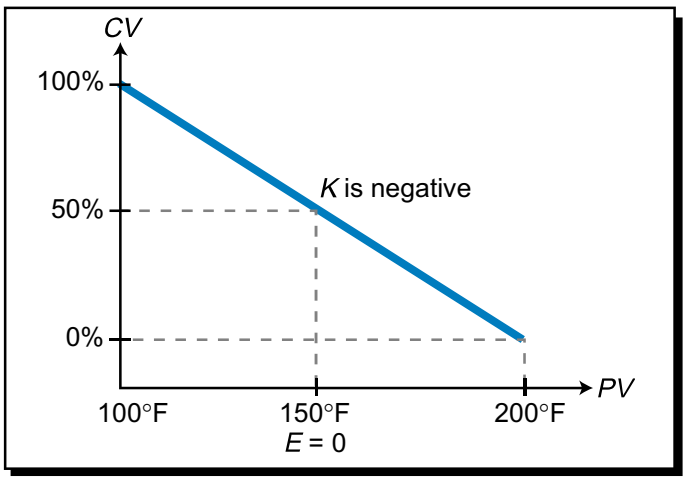


Figure 15-30. Proportional gain for a reverse-acting controller.

The gain is:

$$\begin{aligned} K_P &= \frac{0\% - 100\%}{PB} \\ &= \frac{-100\%}{100\%} \\ &= -1\%/\% \end{aligned}$$

If the process variable temperature is 160°F, the percentage error over the full variable range will be:

$$\begin{aligned} E &= \frac{SP - PV}{PV_{\min} - PV_{\max}} \\ &= \frac{150°\text{F} - 160°\text{F}}{100°\text{F} - 200°\text{F}} \\ &= \frac{-10°\text{F}}{-100°\text{F}} \\ &= 10\% \end{aligned}$$

This indicates that the value of *PV* is 10% more than the *SP* value. Assuming that the previous *CV* output was at the set point (50%), the controller's new output will be:

$$\begin{aligned} CV_{\text{new}} &= K_P E + CV_{\text{old}} \\ &= (-1)(10\%) + 50\% \\ &= -10\% + 50\% \\ &= 40\% \end{aligned}$$

Thus, the controller will reduce the value of its output to 40% of its range. This will affect the process by reducing the process variable to the set point value.

CLOSED-LOOP PROPORTIONAL CONTROL

Figure 15-31a illustrates a typical open-loop process control system where the process variable and transfer function in Laplace form are represented by the equation:

$$PV_{(s)} = \left(SP_{(s)}\right)\left(Hc_{(s)}\right)\left(Hp_{(s)}\right)$$

$$\frac{PV_{(s)}}{SP_{(s)}} = Hc_{(s)}Hp_{(s)}$$

Figure 15-31b shows this same system in a closed-loop configuration. For the closed-loop system, the process variable is defined by:

$$PV_{(s)} = \left(E_{(s)}\right)\left(Hc_{(s)}\right)\left(Hp_{(s)}\right)$$

Replacing $E_{(s)}$ with $SP_{(s)} - PV_{(s)}$ yields:

$$PV_{(s)} = \left(SP_{(s)} - PV_{(s)}\right)Hc_{(s)}Hp_{(s)}$$

$$PV_{(s)} = \left(SP_{(s)}Hc_{(s)}Hp_{(s)}\right) - \left(PV_{(s)}Hc_{(s)}Hp_{(s)}\right)$$

$$PV_{(s)}\left(1 + Hc_{(s)}Hp_{(s)}\right) = SP_{(s)}Hc_{(s)}Hp_{(s)}$$

Solving for $PV_{(s)}$ over $SP_{(s)}$ yields the closed-loop transfer function of this process control system:

$$\frac{PV_{(s)}}{SP_{(s)}} = \frac{Hc_{(s)}Hp_{(s)}}{Hc_{(s)}Hp_{(s)} + 1}$$

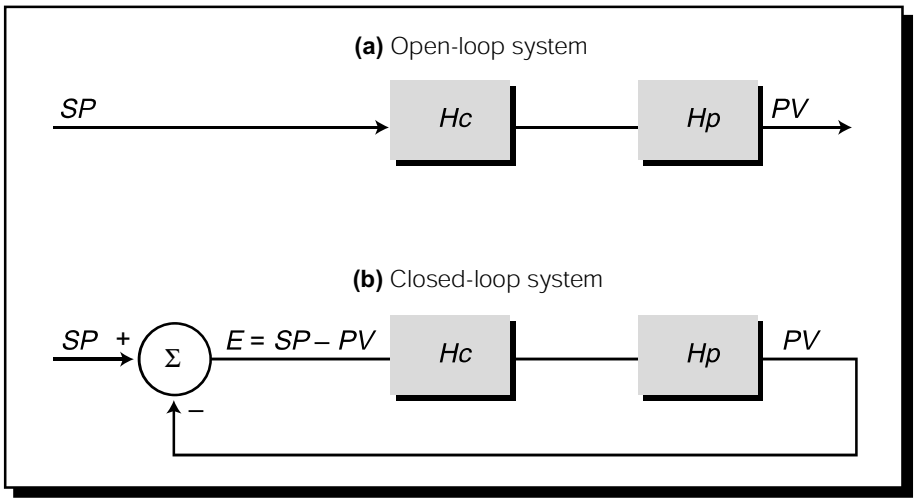


Figure 15-31. (a) Open-loop and **(b)** closed-loop process control systems.

The term $Hc_{(s)}$ represents the controller's transfer function, while the term $Hp_{(s)}$ represents the process's transfer function. The process's transfer function may take the form of a first-order response or a second-order response (overdamped, underdamped, or critically damped). As we will discuss later, the ideal controller transfer function for a system with a second-order process plus lag and dead time is one with proportional, integral, and derivative (PID) components.

A closed-loop system's response to a proportional controller creates an error that cannot be eliminated. This error is referred to as *offset*. Figure 15-32 shows the graph of a proportional controller's transfer function, in which a 50% *CV* output keeps the process variable at the set point. If a load disturbance occurs, the error will increase and the controller will change the output variable to try to bring the error back to zero. However, if the load disturbance requires a permanent output change in the controller (CV_{new}), the one-to-one relationship between the controller and error will prohibit a zero error value, because the original function curve is changed. For instance, if the process variable in Figure 15-33a increases due to a load disturbance, the control variable (assuming a direct-acting controller) will increase proportionally to try to bring the error back to zero. If the load disturbance causes a permanent change in the required controller output, the new output level (CV_{new}) will cause the process variable to return to the set point value. However, as *PV* begins to approach the set point (see Figure 15-33b), the controller will reduce its output because the error is diminishing. The reduced *CV* output will cause the error to increase, because the process requires a control variable output at the level of CV_{new} to maintain the process variable at the set point. Thus, error will always be present in the system.

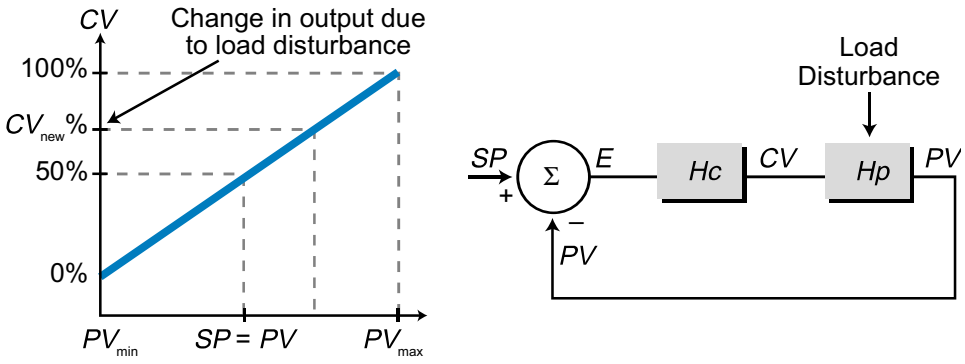


Figure 15-32. An offset caused by a load disturbance in a closed-loop system.

In process applications, the need for a permanent change in *CV* is typical, thus proportional controllers always produce a small amount of error. This error limits the use of proportional controllers to applications that include a

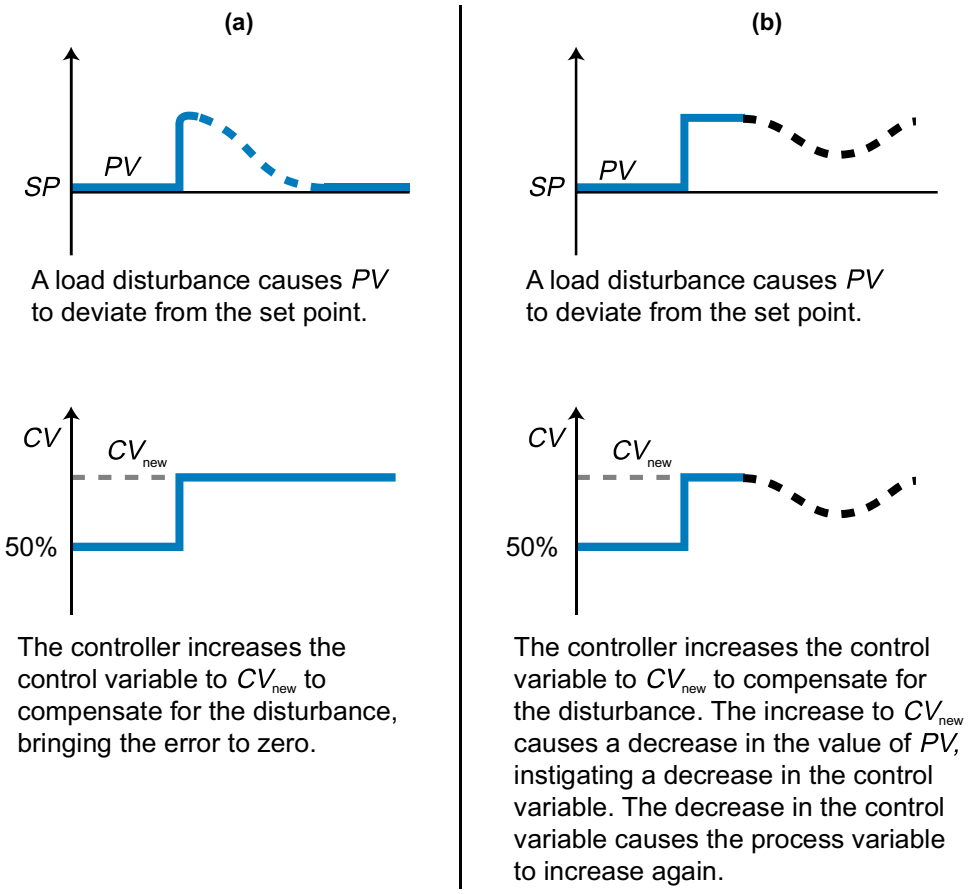


Figure 15-33. (a) Desired response of a proportional controller to a load disturbance and **(b)** its actual response.

manual reset, allowing the operator to change the *bias*, or operating point, of the controller. This changes the level of controller output associated with $E = 0$ from CV to CV_{new}. Another way to minimize the system error is to increase the gain, K_P, thus reducing the proportional band. This method is precarious, however, because too much gain will make the system oscillate like an ON/OFF controller with a small deadband. The reason for this is that if a small error occurs around the set point, the controller will have a large output swing (CV) to correct for the process variable (PV). This will push the error in the opposite direction. When the error goes the opposite direction of the set point, the controller will quickly respond with another large output change, thus forcing the process variable to return to its original direction. Hence, the system will behave like an ON/OFF system if the proportional gain is too large.

The following example illustrates the effect of error in a proportional controller controlling a first-order system. Note that the step change in set point is permanent, simulating a permanent disturbance or change.

EXAMPLE 15-5

The closed-loop system shown in Figure 15-34 has a first-order process (*Hp*) with a gain of 5 and a time constant of τ = 30 seconds. The proportional controller (*Hc*) has a proportional gain of 8. **(a)** Find the closed-loop transfer function, **(b)** calculate the response to a unit step $\frac{1}{s}$, and **(c)** plot the response, indicating the system time constant (τ_{sys}) and the steady-state value.

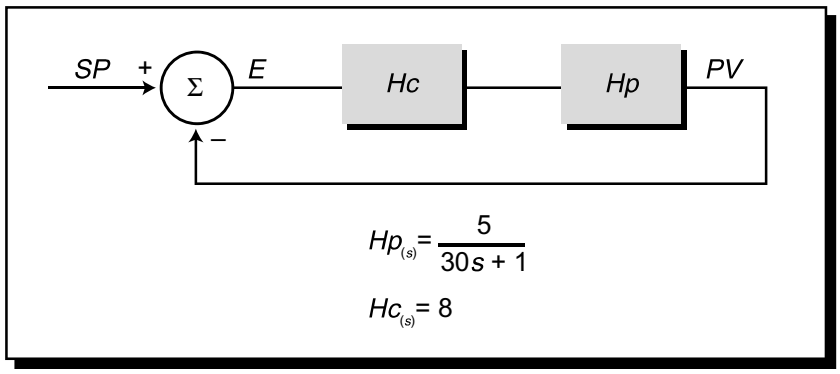


Figure 15-34. Closed-loop system.

SOLUTION

(a) The transfer function of a closed-loop system is expressed as:

$$\frac{PV_{(s)}}{SP_{(s)}} = \frac{Hc_{(s)}Hp_{(s)}}{Hc_{(s)}Hp_{(s)} + 1}$$

As discussed in Chapter 14, the Laplace transfer function of a first-order process with lag is:

$$Hp_{(s)} = \frac{A}{\tau s + 1}$$

So the process's transfer function is:

$$Hp_{(s)} = \frac{5}{30s + 1}$$

A proportional controller's Laplace transfer function is simply the value of its gain, so:

$$Hc_{(s)} = 8$$

Therefore, the closed-loop transfer function of the entire system is:

$$\frac{PV_{(s)}}{SP_{(s)}} = \frac{(8)\left(\frac{5}{30s+1}\right)}{\left[(8)\left(\frac{5}{30s+1}\right)\right]+1}$$

$$= \frac{\left(\frac{40}{30s+1}\right)}{\left(\frac{40}{30s+1}+1\right)}$$

$$= \frac{\left(\frac{40}{30s+1}\right)}{\left(\frac{40+30s+1}{30s+1}\right)}$$

$$= \frac{40}{40+30s+1}$$

$$= \frac{40}{30s+41}$$

This transfer function indicates that this is a first-order system. To express it in the form of a first-order system, we must divide the numerator and denominator by 41 to obtain:

$$\frac{PV_{(s)}}{SP_{(s)}} = \frac{\left(\frac{40}{41}\right)}{\left(\frac{30s}{41}+\frac{41}{41}\right)}$$

$$= \frac{0.976}{0.732s+1}$$

(b) The response to a unit step change in the set point is given by:

$$SP_{(s)} = \frac{1}{s} \quad \text{(unit step)}$$

Therefore, using the previous equation, the process variable response will be:

$$PV_{(s)} = SP_{(s)}\left(\frac{0.976}{0.732s+1}\right)$$

Thus:

$$PV_{(s)} = \left(\frac{1}{s}\right)\left(\frac{0.976}{0.732s+1}\right)$$

$$PV_{(s)} = \frac{0.976}{s(0.732s+1)}$$

According to Table 14-1, this response is in the form of the inverse Laplace transform of a first-order response to a step input with lag:

$$\mathcal{L}^{-1}\left[\frac{A}{s(\tau s+1)}\right] = A\left(1-e^{\frac{-t}{\tau_{sec}}}\right)$$

Hence, in the time domain, the process variable response will be equal to:

$$PV_{(t)} = 0.976\left(1-e^{\frac{-t}{0.732\,sec}}\right)$$

(c) Figure 15-35 illustrates the time response of the closed-loop system to a unit step change in the set point. Note that the gain of the system is 0.976, meaning that the process variable in the system will not reach the value of the unit step input. Instead, the system will respond only 0.976 to the unit step change of 1. The process variable steady-state value (PV_{ss}), which is the final value of *PV*, can be computed using the final value theorem:

$$\begin{aligned} PV_{SS} &= \lim_{t\to\infty}\left[0.976\left(1-e^{\frac{-t}{0.732}}\right)\right] \\ &= \lim_{t\to\infty}[0.976(1-0)] \\ &= 0.976 \end{aligned}$$

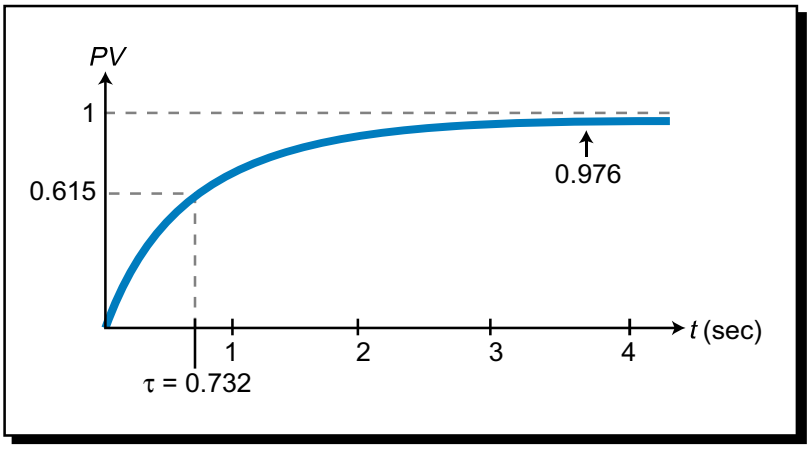


Figure 15-35. System response to step change.

Therefore, the system will always have a residual error of 2.4% (1.0 – 0.976 = 0.024). The system time constant, τ_{sys} = 0.732, indicates that the system will take 0.732 seconds to reach 0.615, 63% of the steady-state value.

The open-loop response of the process to a step change would have a steady-state value of 5 (see Figure 15-36a), where 1τ would occur at 30 seconds. The open-loop response of the controller and process to a unit step would have a gain of 40 with this same τ constant (30 sec).

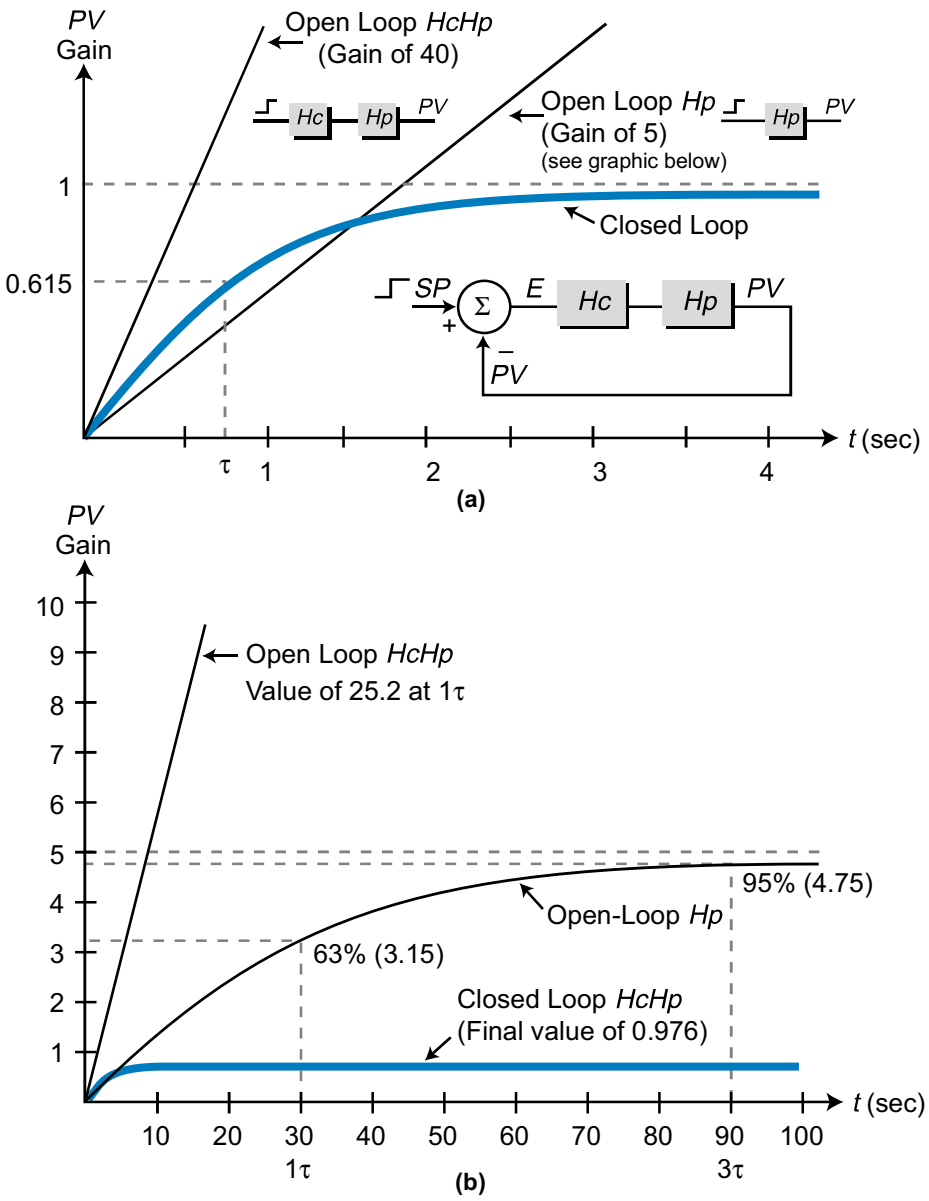


Figure 15-36. Open- and closed-loop responses shown in **(a)** detail and **(b)** 10× detail.

The closed-loop response of the controller and process, however, would have a much smaller gain (in this case, 0.976) due to the negative feedback in the control loop.

The value of the gain in a proportional controller influences the response of a second-order closed-loop system as shown in Figure 15-37. The higher the gain, the faster the process responds; however, cycling and overshoot occur. Lowering the gain makes the response much slower and the value at steady state smaller. For example, the value of $K_P = 1$ in Figure 15-37 will cause a slow, closed-loop response to a unit step change in the set point with a final steady-state value of 0.5.

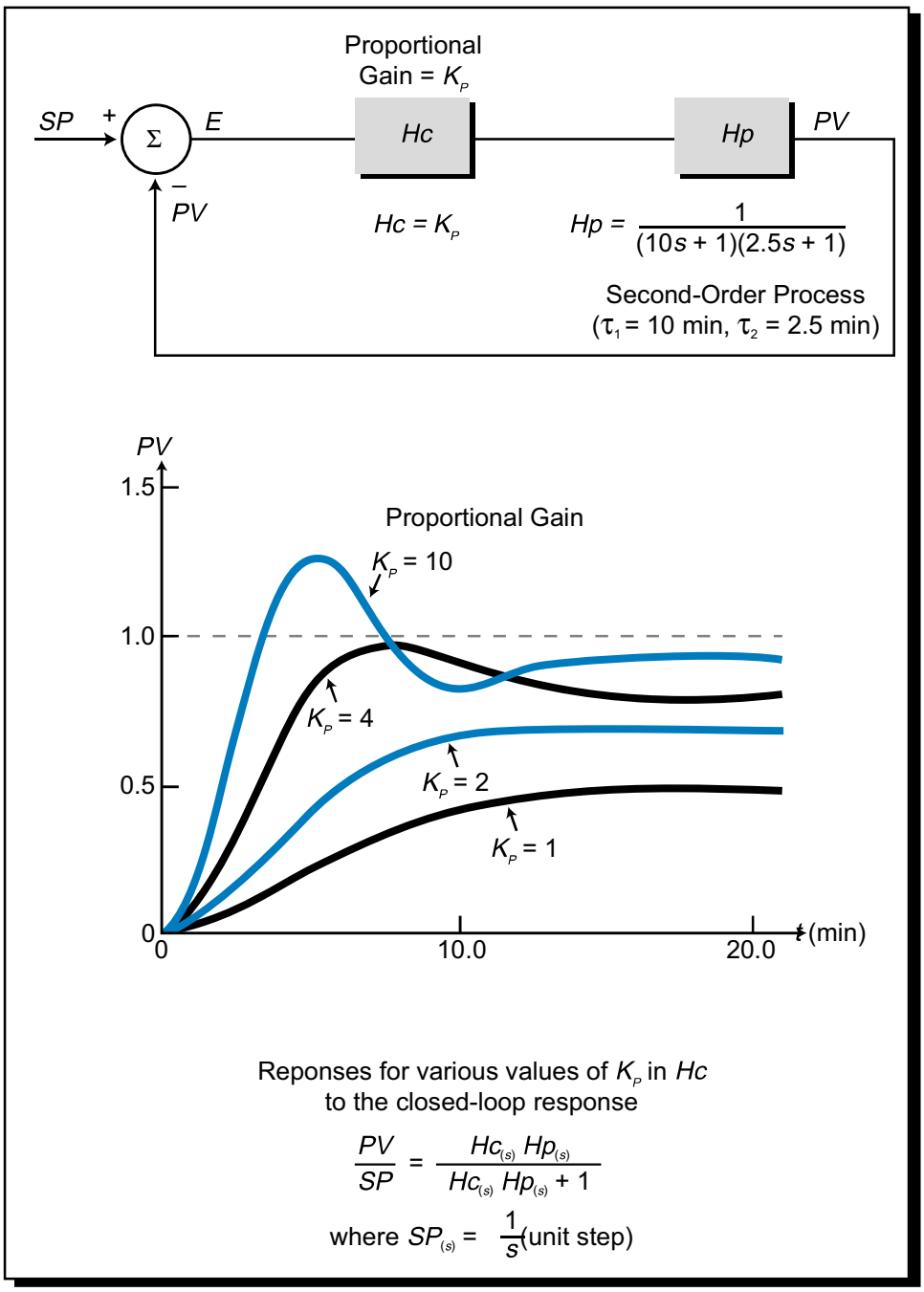


Figure 15-37. Responses of a second-order closed-loop system to different values of proportional gain.

15-6 INTEGRAL CONTROLLERS (I MODE)

An **integral controller** provides an output whose rate of change is proportional to the error deviation. This means that the larger the error, the faster the controller's output changes and vice versa. An integral controller will stop adjusting its output once the error becomes zero. When used in conjunction

with a proportional controller, an integral controller will bring the system's residual error to zero. An integral controller's output (*CV*) is represented by:

$$\frac{dCV}{dt} = K_I E$$

where:

$\frac{dCV}{dt}$ = the rate of change in controller output in % over seconds

K_I = the integral gain in % of the controller output per second per % error

E = the error in %

This differential equation indicates that the controller's output $CV_{(t)}$ can be obtained by taking the integral over time of both sides of the equation so that:

$$\frac{dCV}{dt} = K_I E$$
$$dCV = K_I E dt$$
$$\int_0^t dCV = \int_0^t K_I E dt$$
$$CV_{(t)} - CV_{(t=0)} = K_I \int_0^t E dt$$
$$CV_{(t)} = K_I \int_0^t E dt + CV_{(t=0)}$$

where the term $CV_{(t=0)}$ is the value of the output at $t = 0$. When an integral controller is used in a closed-loop system, it calculates $CV_{(t)}$ for every change in error. So, if the value of the error changes after the controller has calculated a previous value $CV_{(t)}$, then it will use this previous value of $CV_{(t)}$ as the $CV_{(t=0)}$ value and calculate a new $CV_{(t)}$ output based on the new error.

The integral gain K_I (see Figure 15-38) indicates the sensitivity of the output's rate of change to the percentage of error that occurs over time. A large value of K_I means that a small error will produce a large rate of change in the controller output. Conversely, a small value of K_I means that a small error produces a small rate of change in the controller output. In Figure 15-38, the rate of change of K_{I1} is greater than that of K_{I2}.

Figure 15-39 illustrates the reaction of an integral controller's output to a change in the process variable due to a load disturbance. Note that, at the moment the error occurs, the controller starts the integration of the error

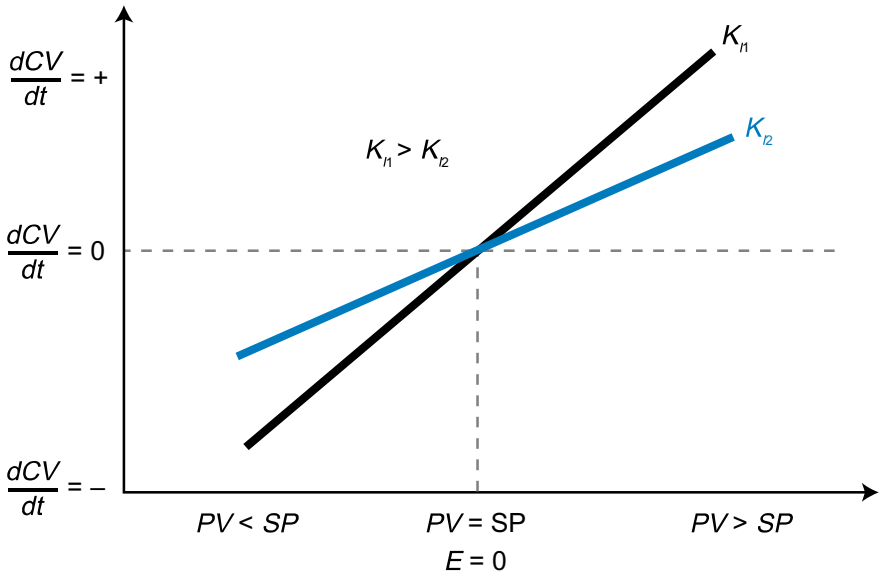


Figure 15-38. Integral gain.

value, meaning that the control variable begins to increase as a function of the magnitude of the error. The error in Figure 15-39 is constant, creating a ramp integration of the controller's output. That is, the amount of error remains constant over time, so the control variable increases at a steady rate.

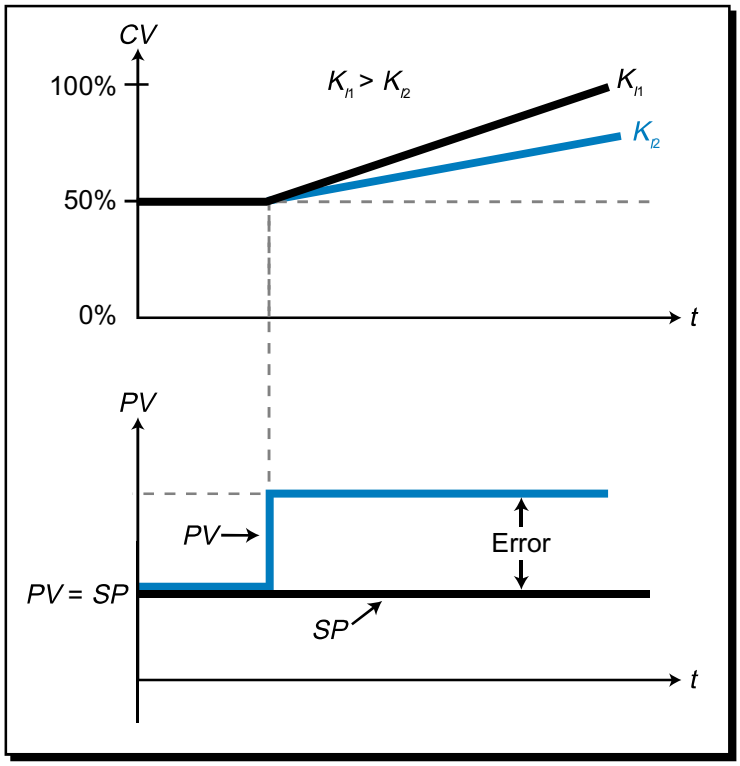


Figure 15-39. Integral controller's response to a step change in the process variable.

To further illustrate the effect of the measured error on the control variable output, let's examine Figure 15-40, which shows the graph of a direct-acting integral controller's output response to a change in error. If the error makes a large jump (1), the controller will respond with a steep increase in output. As the error begins to decrease (2), the rate of increase of the output variable will also decrease (less ramping). When the error becomes zero (3), then the controller will keep its output at its previous level. As the error increases again, but in the opposite direction (4), the output will begin to decrease. As the error decreases, but still remains negative (5), the control variable will continue to decrease but at a less rapid rate. Furthermore, if the error increases positively (6), then the output will increase again. Finally, as the error goes to zero and remains there (7), the controller will level out the control variable and make no more changes to its output level. Thus, an integral controller can adjust its output level to bring the error to zero. An integral controller does not exhibit the limitations of the linear relationship of a proportional controller; thus, it is able to keep a zero error at an output value other than 50% of the controller output.

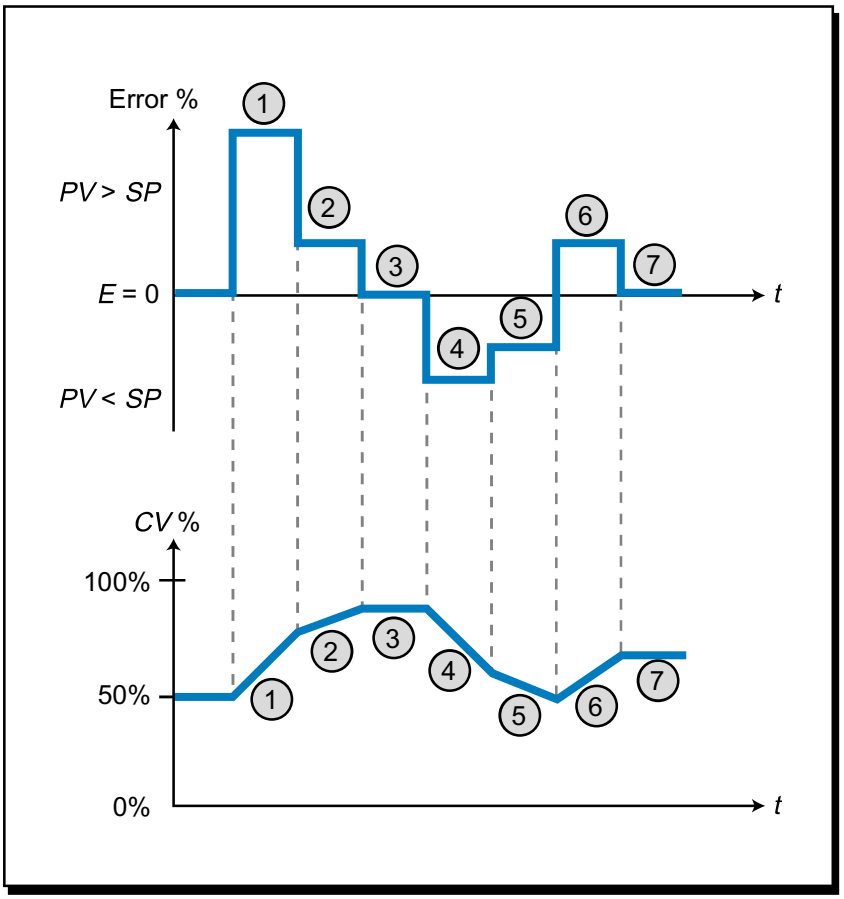


Figure 15-40. Output response to changes in error.

The gain of an integral controller (K_I) is defined by the equation:

$$K_I = \frac{\left(\frac{dCV}{dt}\right)}{E}$$
$$= \frac{\text{change in \% of } CV \text{ per second}}{\text{change in \% of error}}$$

A value of $K_I = 0.2$ indicates that the controller will change 0.2% per second for every 1% of error present in the system. So, if the 1% error in the system lasts for 2 seconds and then goes to zero, the controller will increase its output 0.4%.

As discussed earlier, different values of K_I have different curves, or slopes, associated with them. Figure 15-41a shows the curves for two integral gains, K_{I1} and K_{I2}. For both gain values, the controller will make no change to the output (CV) if the error equals 0. However, if the error increases to $PV_{\max}$, the controller will change its output at a rate of 25% per second if the integral gain equals K_{I1}, while it will only change its output at a rate of 15% per second if the integral gain equals K_{I2}. Likewise, if the error drops to $PV_{\min}$, the controller will change CV at a rate of –25% per second for the K_{I1} value and –15% per second for the K_{I2} value. Both K_{I1} and K_{I2} can be thought of as belonging to a "family" of curves that expresses the value of the control variable over time for given integral gain and error values (see Figure 15-41b). For example, if K_IE equals 1.25 (K_I = 0.5 and E = 2.5%), then in 8 seconds the value of CV will change by 10%. The family of curves illustrates the speed of the control variable change for different error values.

The value of the integral gain K_{I1} in Figure 15-41 can be computed as:

$$K_I = \frac{\%\text{ change in } \left(\frac{dCV}{dt}\right)}{\%\text{ error over full range}}$$
$$= \frac{\left(\frac{dCV_{\max}}{dt} - \frac{dCV_{\min}}{dt}\right)}{\left(\frac{PV_{\max} - PV_{\min}}{PV_{(\text{max range})} - PV_{(\text{min range})}}\right)}$$
$$= \frac{25\,\%/\text{sec} - (-25\,\%/\text{sec})}{\left(\frac{200°\text{F} - 100°\text{F}}{200°\text{F} - 100°\text{F}}\right)}$$
$$= \frac{50\,\%/\text{sec}}{100\%}$$
$$= 0.5\ \text{sec}^{-1}$$

The value K_I = 0.5 sec^{-1} indicates that the controller will gain 0.5% in output per second for each percentage of error present. If the error is 50%

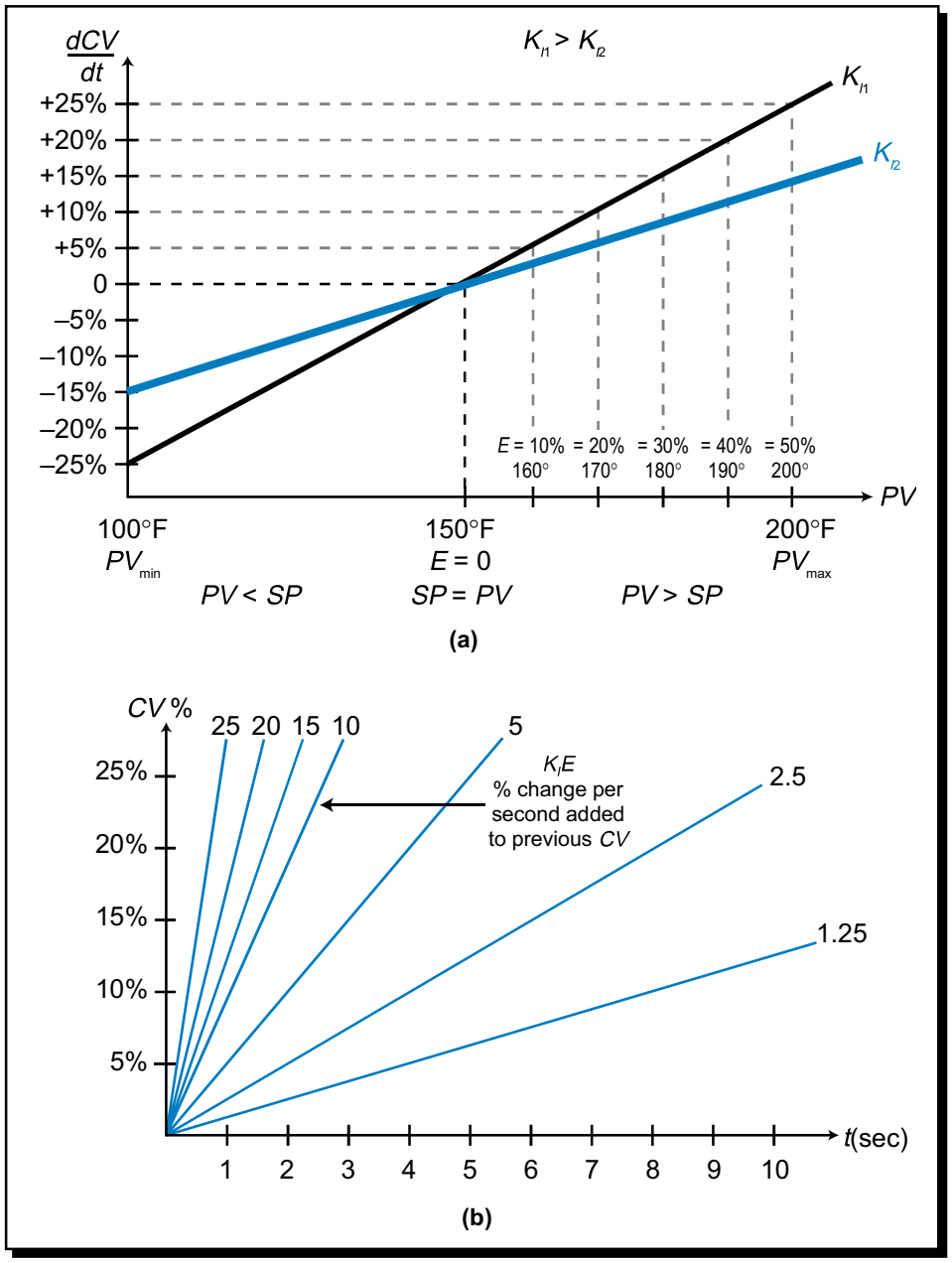


Figure 15-41. (a) Integral gain curves and **(b)** the family of curves for an integral controller.

(i.e., $PV = 200°F$), then after one second the controller's output will be 75% (see Figure 15-42):

$$\begin{aligned} CV_{(t=1)} &= K_I \int_0^t E dt + CV_{(t=0)} \\ &= K_I E t \Big|_{t=0}^{t=1} + CV_{(t=0)} \\ &= K_I E(t - t_0) + CV_{(t=0)} \\ &= \left(0.5\ \text{sec}^{-1}\right)(50\%)(1-0) + 50\% \\ &= 75\% \end{aligned}$$

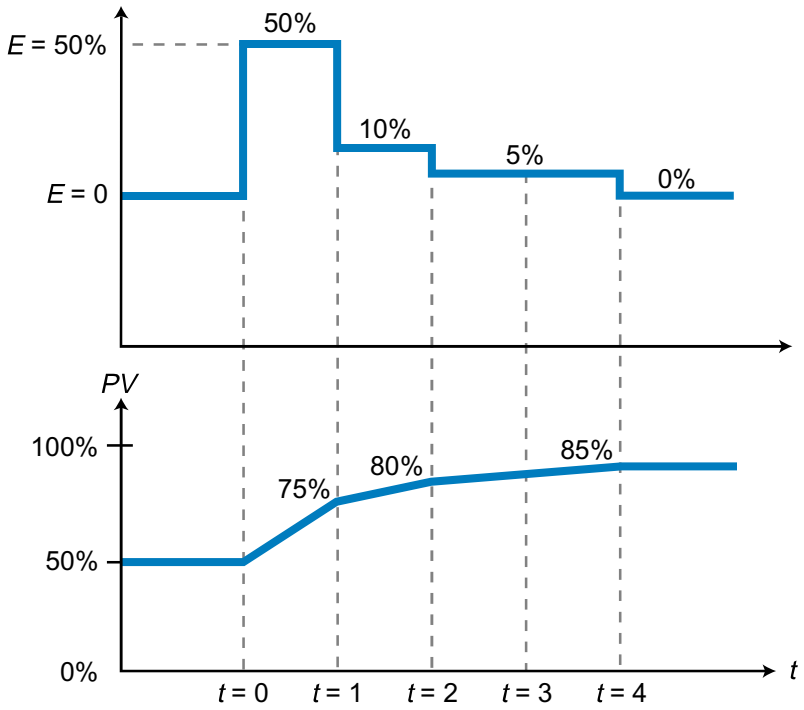


Figure 15-42. Integral controller output response to a change in error.

If the error as shown in Figure 15-42 drops to 10%, the output will be:

$$\begin{aligned} CV_{(t=2)} &= K_I \int_0^t E dt + CV_{(t=1)} \\ &= K_I E t\Big|_{t=1}^{t=2} + CV_{(t=1)} \\ &= \left(0.5\ \text{sec}^{-1}\right)(10\%)(2-1) + 75\% \\ &= 5\% + 75\% \\ &= 80\% \end{aligned}$$

Therefore, the new control variable output from $t = 1$ to $t = 2$ will be 80%. If the controller error drops to 5% for the next two seconds (from $t = 2$ to $t = 4$), the controller output will increase steadily from 80% (at $t = 2$) to 85% (at $t = 4$). After $t = 4$, the error is 0%, therefore, the controller output stays at 85%. Note that the family of curves shown in Figure 15-41b is the product of $K_I E$ for a particular value of error. If the error stays constant for t seconds, then the change in the value of CV over that time period will follow the $K_I E$ curve for that error value.

The inverse of the gain term K_I is referred to as the *integral time* (T_I), or *reset time*, in seconds. The integral time is the time it takes for the control variable (CV) to change 1% for a 1% change in error. It is expressed as:

$$T_I = \frac{1}{K_I}$$

The T_I variable is used by some manufacturers to allow the user to indirectly enter the integral gain into the controller. If the integral time must be specified in minutes, as is required by some manufacturers, a simple conversion can change T_I from seconds to minutes:

$$T_I = \frac{1}{K_I} \text{ (in seconds) or } \frac{1}{(K_I)\left(\frac{60 \text{ sec}}{\text{min}}\right)} \text{ (in minutes)}$$

So, for the previous example, the reset time is equal to:

$$\begin{aligned} T_I &= \frac{1}{K_I} \\ &= \frac{1}{0.5 \text{ sec}^{-1}} \\ &= 2 \text{ seconds} \end{aligned}$$

The integral controller mode is also referred to as *reset action*, because it automatically resets the error to zero over time.

EXAMPLE 15-6

Illustrate the transfer function of an integral controller with a gain of $K_I = 0.2$ sec^{-1} over a process variable range of 100°F to 200°F. Plot the response of the controller's output for an error due to a permanent load disturbance of +10% above the set point of 150°F over the full range. Two seconds after the controller increases its output, the error will drop by 5%. After 3 more seconds, the error will become zero. Find the value of *CV* after 5 seconds.

SOLUTION

Figure 15-43 shows the integral controller's transfer function. When the error is +10% above the set point, the process variable will be at 160°F, which will cause the controller's output to increase at a rate of 2% per second:

$$\begin{aligned} \frac{dCV}{dt} &= K_I E \\ &= (0.2)(10) \\ &= 2\% \text{ per second} \end{aligned}$$

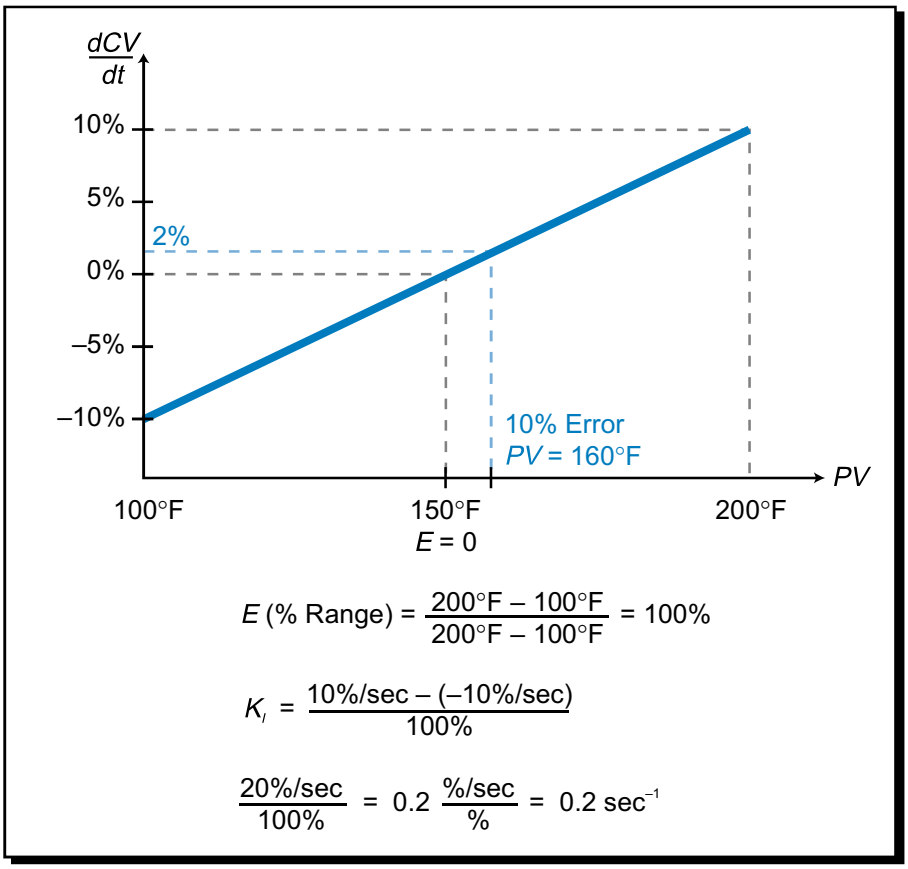


Figure 15-43. Integral controller's transfer function.

As Figure 15-44 illustrates, after 2 seconds of integral action, the controller output will be 54%:

$$\begin{aligned} CV_{(t=2)} &= K_I \int_{t=0}^{t=2} E\,dt + CV_{(t=0)} \\ &= (0.2)Et\Big|_{t=0}^{t=2} + 50\% \\ &= [0.2][10(2-0)] + 50\% \\ &= 54\% \end{aligned}$$

After the 2 seconds have elapsed, the error will drop to 5% and the controller will integrate at a rate of:

$$\begin{aligned} \frac{dCV}{dt} &= K_I E \\ &= (0.2)(5\%) \\ &= 1\% \text{ per second} \end{aligned}$$

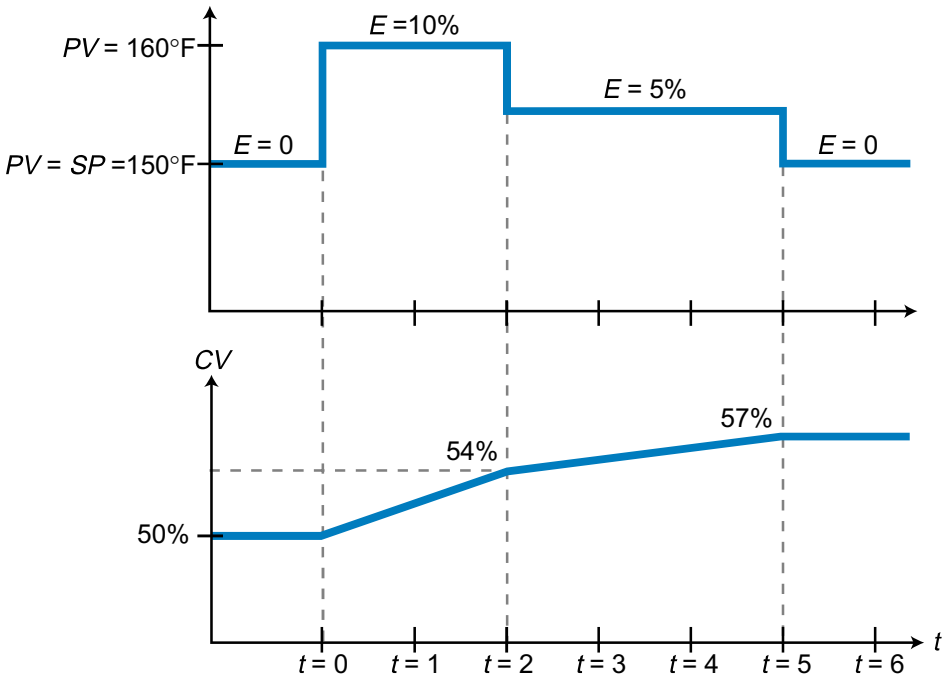


Figure 15-44. Controller output.

Therefore, at the end of the next 3 seconds, the controller output will be 57%:

$$\begin{aligned} CV_{(t=5)} &= K_I \int_{t=2}^{t=5} E dt + CV_{(t=2)} \\ &= (0.2) Et \Big|_{t=2}^{t=5} + 54\% \\ &= [0.2][5(5-2)] + 54\% \\ &= 57\% \end{aligned}$$

At this point, the error will drop to zero, so the controller will stop changing the *CV*, maintaining its output at a new zero error value of 57%.

15-7 PROPORTIONAL-INTEGRAL CONTROLLERS (PI MODE)

Although an integral controller does not have the residual error at steady state that a proportional controller has, its response action to a step change in input (step in error) is often too slow to be used in real-life applications. This slow speed, as compared with the immediate response of a proportional controller, is due to the ramping effect of the integral action as the controller increases its output. Therefore, proportional action is normally added to an

integral controller (see Figure 15-45) to form a **proportional-integral (PI) controller**. This type of controller has a fast response time (proportional action), plus it eliminates all residual error (integral action). As illustrated in Figure 15-46, a PI controller can have one of two configurations:

- parallel
- series

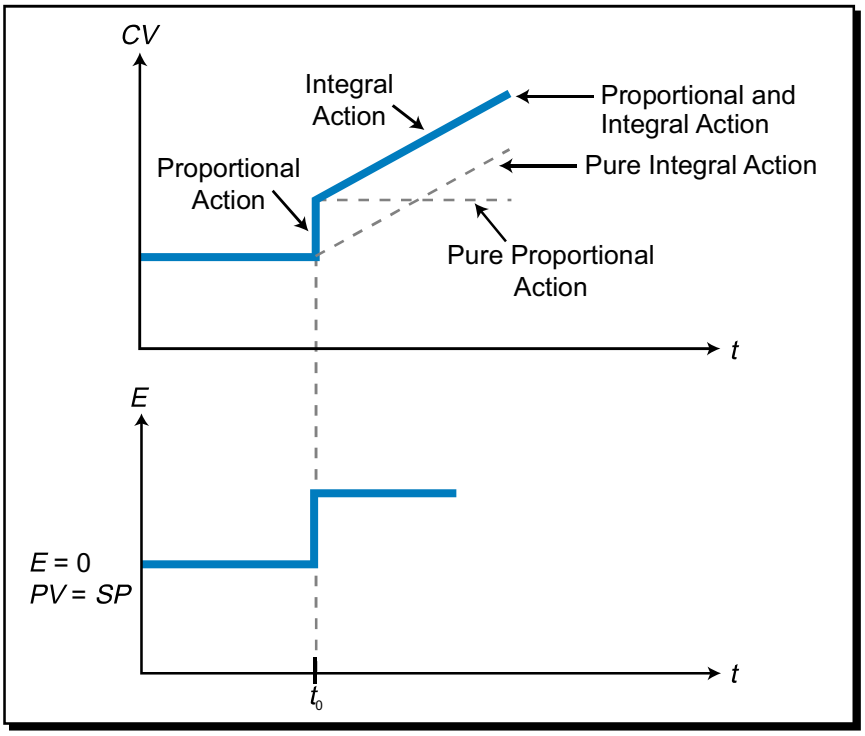


Figure 15-45. Proportional-integral action.

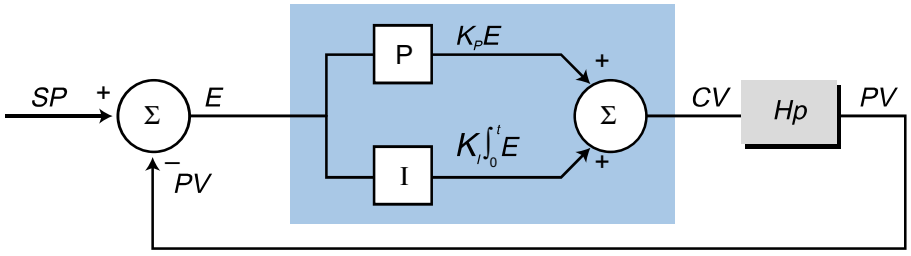
In a *parallel PI controller*, the proportional and integral actions occur independently of each other, so the controller's output (*CV*) is equal to the proportional action plus the integral action:

$$CV_{\text{new}} = K_P E + K_I \int_0^t E dt + CV_{\text{old}}$$

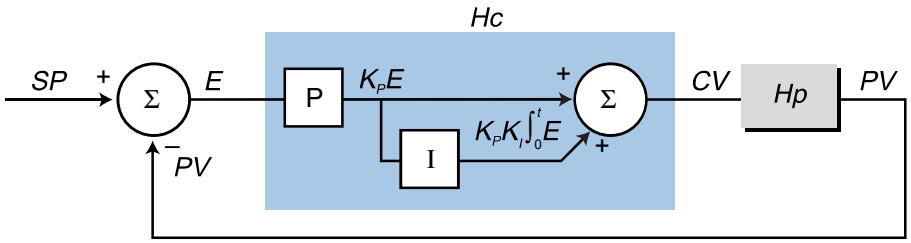
In a *series PI controller*, on the other hand, the integral action occurs after the proportional action. Therefore, the input to the integral action is not the system error *E*, but rather the result of the proportional action $K_P E$. Accordingly, a series PI controller's output is defined by:

$$CV_{\text{new}} = K_P E + K_I \int_0^t K_P E dt + CV_{\text{old}}$$

$$= K_P E + K_P K_I \int_0^t E dt + CV_{\text{old}}$$



(a) Parallel PI controller



(b) Series PI controller

Figure 15-46. (a) Parallel and **(b)** series PI controllers.

Both of these types of PI controllers eliminate error offset and have a faster response time than an integral-only controller. However, series PI controllers multiply the integral gain times the proportional gain, producing an effect called *repeating*. In repeating, the effect of the proportional gain (K_PE) is repeated during every integral time period T_I, causing the integral action of the controller to equal that of the proportional action. This means that a series PI controller responds faster to a change in error than a parallel controller when their proportional gains are greater than one ($K_P > 1$) and their integral times are the same.

The term *repeats* is used when referring to how many times the proportional amount is repeated in one minute. If the value of T_I is less than 1 minute, then the integral gain is repeated more than one time per minute. This can be seen in the equation:

$$\begin{aligned} CV_{\text{new}} &= K_PE + K_PK_I\int_0^t Edt + CV_{\text{old}} \\ &= K_PE + K_PK_IEt\Big|_{t=0}^{t=1} + CV_{\text{old}} \\ &= K_PE + K_PK_IEt + CV_{\text{old}} \\ &= K_PE + K_PEK_It + CV_{\text{old}} \\ &= K_PE + K_PE\left(\frac{1}{T_I}\right)t + CV_{\text{old}} \end{aligned}$$

When $t = T_I$, the term $K_P E$ is repeated once by the integral action, in this case in the period from $t = 0$ to $t = 1$ minute. Figure 15-47a illustrates the integral gain of a PI controller with a repeat of 1 (i.e., $T_I = 1$). After 1 minute, the term $K_P E$ is repeated. Figure 15-47b illustrates a PI controller with an integral time of $T_I = 0.333$, indicating that the term $K_P E$ will be repeated three times in one minute.

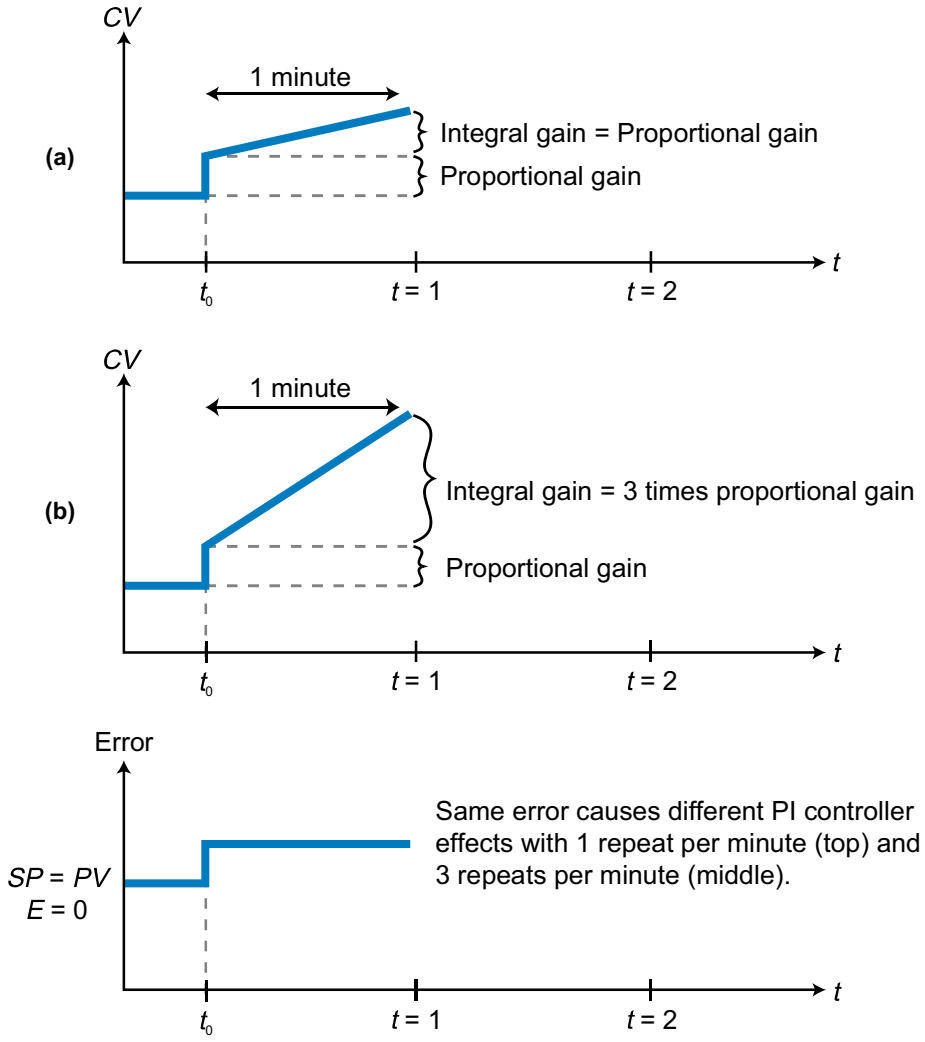


Figure 15-47. PI controller with an integral time of **(a)** 1 and **(b)** 1/3.

EXAMPLE 15-7

(a) Graph the value of the control variable after 1 minute for a series PI controller given that the proportional gain is 2 and the integral gain is 0.01 sec^{-1}. The process variable changes from the set point of 150°F to 155°F over a process variable range of 100°F to 200°F. At the set point, the controller has an output of 50%. **(b)** How long will it take for the integral gain to equal the proportional gain?

SOLUTION

(a) The error created in the system over the *PV* range is:

$$E(\%) = \frac{155°F - 150°F}{200°F - 100°F}$$
$$= 5\%$$

The given values of the proportional and integral actions are:

$$K_P = 2\%/\%$$
$$K_I = 0.01\,\text{sec}^{-1}$$

Thus, the value K_I in minutes is:

$$K_I = \left(\frac{0.01}{\text{sec}}\right)\left(\frac{60\,\text{sec}}{1\,\text{min}}\right)$$
$$= \frac{0.6}{\text{min}}$$
$$= 0.6\,\text{min}^{-1}$$

The control variable for this series PI controller is defined as:

$$CV_{new} = K_P E + K_P K_I \int_0^t E dt + CV_{(t=0)}$$
$$= \left[\left(2\tfrac{\%}{\%}\right)(5\%)\right] + \left[\left(2\tfrac{\%}{\%}\right)\left(0.6\,\text{min}^{-1}\right)\left(5\%\Big|_{t=0}^{t=1}\right)\right] + 50\%$$
$$= 10\% + \left[(10\%)\left(0.6\,\text{min}^{-1}\right)(1\,\text{min} - 0\,\text{min})\right] + 50\%$$
$$= 10\% + 6\% + 50\%$$
$$= 66\%$$

Figure 15-48 illustrates this control variable response.

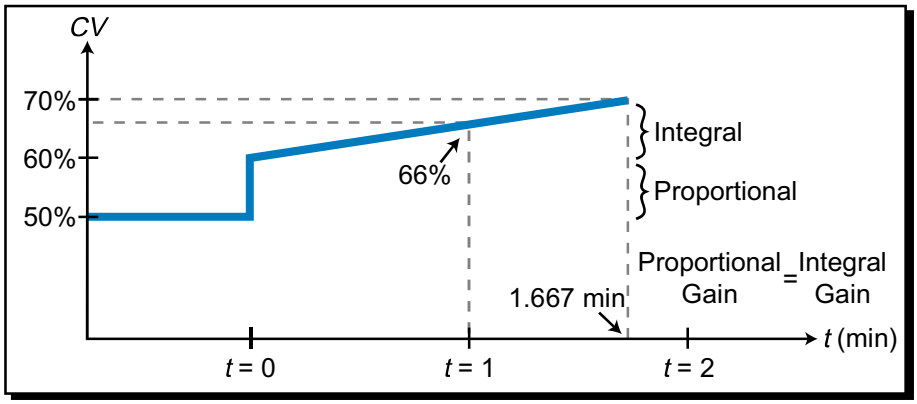
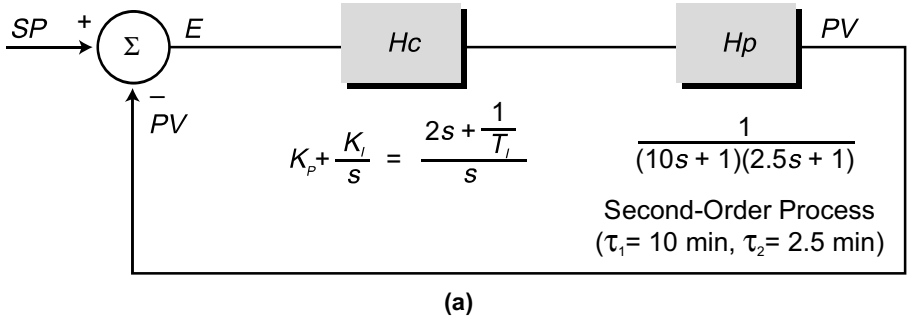


Figure 15-48. Control variable response.

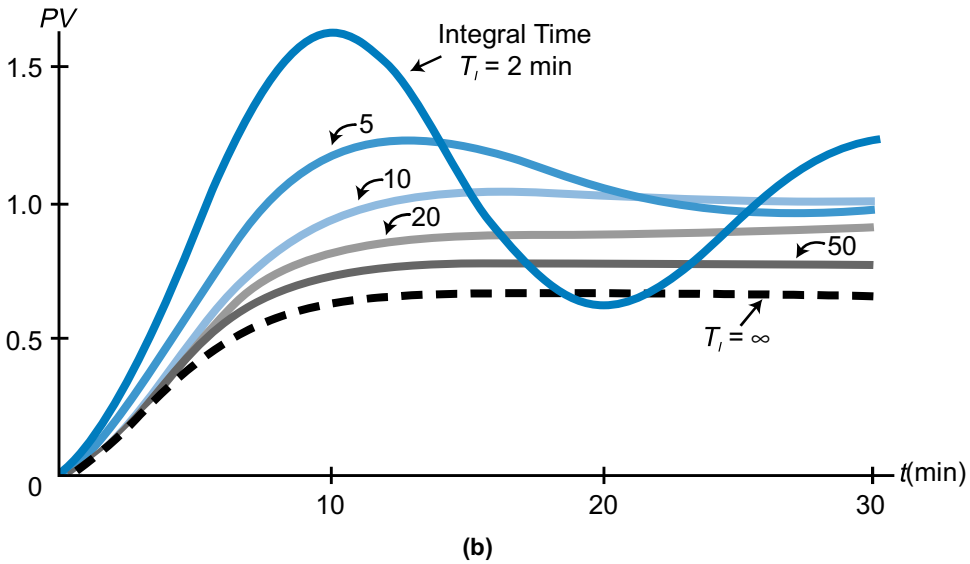
(b) The integral gain will equal the proportional gain in 1.667 minutes:

$$K_P E = K_P K_I E t$$
$$(2\%)(5\%) = (2\%)\left(0.6\ \text{min}^{-1}\right)(5\%)(t)$$
$$t = \frac{1}{\left(0.6\ \text{min}^{-1}\right)}$$
$$= 1.667\ \text{min}$$

The integral, or reset, time of a PI controller influences the ultimate closed-loop response of the system (see Figure 15-49). As the reset time decreases, the response speed increases, creating an overshoot. The overshoot in the response will cause the proportional action to initiate a negative increase (reduction of output), producing an oscillating response.



(a)



(b)

Figure 15-49. (a) A series PI controller (*Hc*) controlling a second-order process and **(b)** the normalized response of the process variable to a change in set point for various values of T_I. The proportional gain K_P is equal to 2%/% for all values of T_I.

The signs of the proportional gain (K_PE) and the combined proportional-integral gain ($K_PK_I\int Edt$) terms are important when determining the integral gain curves for reverse-acting and direct-acting series PI controllers. In the direct-acting mode (see Figure 15-50a), the signs of K_P and K_I are both positive. Therefore, a negative error will make both the K_P and K_I terms negative, resulting in proper controller action. Similarly, a positive error will make both terms positive, again resulting in proper direct action control.

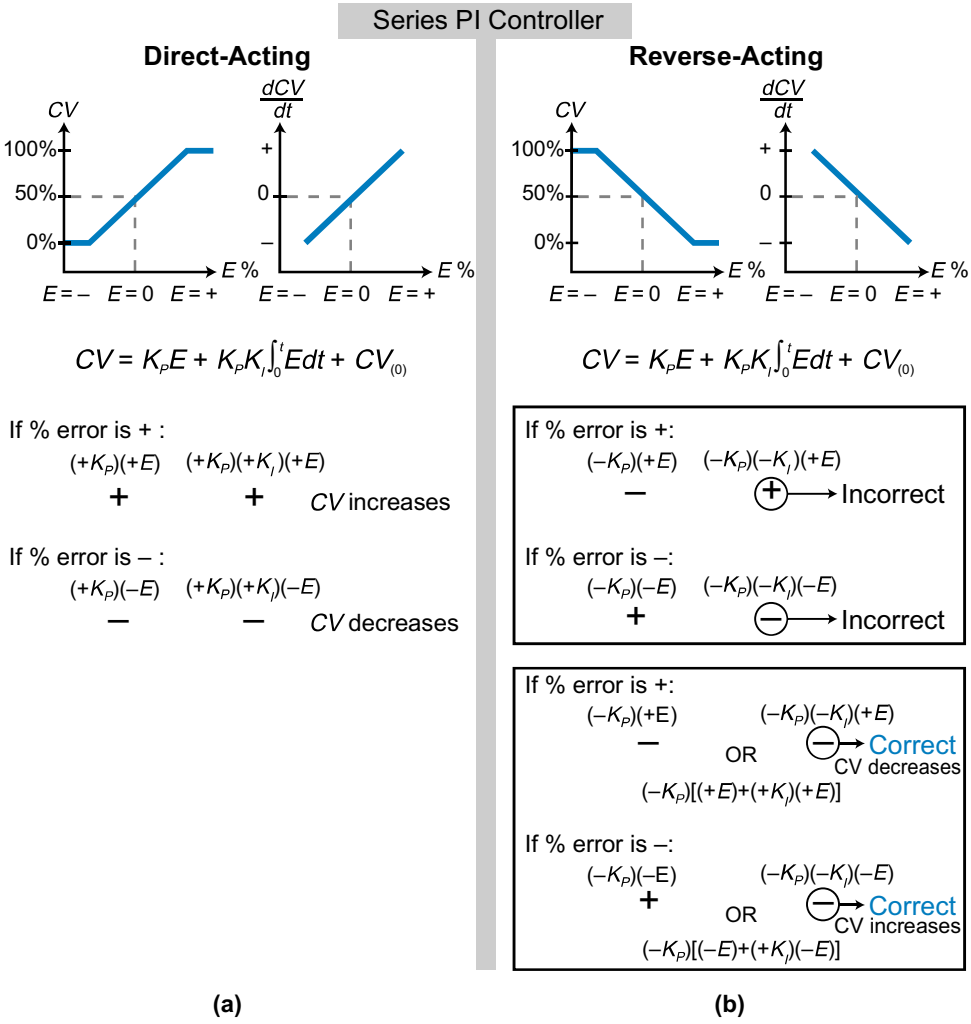


Figure 15-50. Gain curves for **(a)** direct- and **(b)** reverse-acting series PI controllers.

In a reverse-acting series PI controller, both the proportional gain and the combined proportional-integral gain (K_PK_I) must be negative ($-K_PK_I$) for the controller to correctly implement a reverse action (see Figure 15-50b). This means that the integral gain must be positive—a negative integral gain would result in a positive combined gain term. Since the proportional gain must be negative, the output of a reverse-acting series PI controller can be expressed as:

$$CV_{\text{new}} = -K_P\left(E + K_I\int_0^t Edt\right) + CV_{(t=0)}$$

The negative sign in the proportional gain term ensures that the controller will operate as reverse-acting. In a PLC system, the user enters the values for K_P and K_I; therefore, some manufacturers of series PI controllers allow the user to select a reverse-acting controller by specifying the proportional gain as a negative value. In this type of system, the controller takes care of all other computational signs, to ensure proper controller action and a proper control variable response. Otherwise, when the error is positive, one term (proportional) reduces the value of CV, while the other (integral) adds to it and vice versa if the error is negative.

The following example illustrates how a PI controller ultimately brings the error in a closed-loop system to zero at steady state. This example is an extension of Example 15-5, which used only proportional control and, as a result, had an offset error.

EXAMPLE 15-8

The closed-loop system in Example 15-5 has a first-order process with a gain of 5 and a time constant of τ = 30 seconds. The controller has a proportional gain of K_P = 8. If the controller also has an integral action with a gain of K_I = 0.1 sec^{-1}, forming a PI parallel controller, find **(a)** the closed-loop transfer function of the system and **(b)** the steady-state value of the response to a unit step change in set point.

SOLUTION

(a) The process's transfer function is defined by:

$$Hp_{(s)} = \frac{5}{30s+1}$$

The controller's transfer function is expressed as:

$$Hc_{(t)} = K_P E + K_I\int_0^t Edt$$

$$Hc_{(s)} = K_P + \frac{K_I}{s}$$

$$= \frac{K_P s + K_I}{s}$$

$$= \frac{8s + 0.1}{s}$$

Therefore, the closed-loop transfer function is:

$$\frac{PV_{(s)}}{SP_{(s)}} = \frac{Hc_{(s)}Hp_{(s)}}{Hc_{(s)}Hp_{(s)} + 1}$$

$$= \frac{\left[\left(\frac{8s+0.1}{s}\right)\left(\frac{5}{30s+1}\right)\right]}{\left[\left(\frac{8s+0.1}{s}\right)\left(\frac{5}{30s+1}\right) + 1\right]}$$

$$= \frac{\left(\frac{40s+0.5}{30s^2+s}\right)}{\left(\frac{40s+0.5+30s^2+s}{30s^2+s}\right)}$$

$$= \frac{40s + 0.5}{30s^2 + 41s + 0.5}$$

(b) The response of the process variable to a step change in set point is represented by:

$$PV_{(s)} = SP_{(s)}\left(\frac{40s + 0.5}{30s^2 + 41s + 0.5}\right)$$

$$= \left(\frac{1}{s}\right)\left(\frac{40s + 0.5}{30s^2 + 41s + 0.5}\right)$$

The final value of the process variable at steady state can be computed by taking the inverse Laplace transform of $PV_{(s)}$ to obtain $PV_{(t)}$ and then evaluating the response value as t approaches infinity ($t \rightarrow \infty$). However, obtaining the inverse Laplace transform of this response can be very cumbersome. So, as an alternative, we can use the final value theorem and apply it to the equation in the Laplace, or frequency, domain:

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF_{(s)}$$

The steady-state value of the process variable in response to a unit step input change can be found by multiplying the Laplace equation times s and evaluating it as s approaches zero. Therefore:

$$\lim_{s\to 0} sPV_{(s)} = (s)\left(\frac{1}{s}\right)\left(\frac{40s + 0.5}{30s^2 + 41s + 0.5}\right)$$

$$= \frac{40(0) + 0.5}{30(0)^2 + 41(0) + 0.5}$$

$$= \frac{0.5}{0.5}$$

$$= 1$$

Thus, the error will be zero at steady state:

$$E = SP - PV$$
$$= 1 - 1$$
$$= 0$$

A PI controller may create a situation in which it *saturates* the control variable output. Saturation occurs when the control variable output remains pegged at its maximum value (100%). The control variable will remain saturated even if the error starts to come down (see Figure 15-51). The integral action will not change direction until the percentage of error becomes negative ($PV > SP$). This situation is called **integral windup**, or *reset windup,* and it can be damaging to the process. It occurs when a large error is present in a system with a slow response (large time constant). In this situation, the controller will keep increasing the control variable value because the error remains constant due to the lag's effect on the integral corrective action. Eventually, the control variable will saturate at 100%. In other words, the controller's corrective action continues to occur when the process takes too long to respond. The start-up of a batch process is a typical example of a situation in which a reset windup can occur. As we'll discuss later, this condition can be prevented.

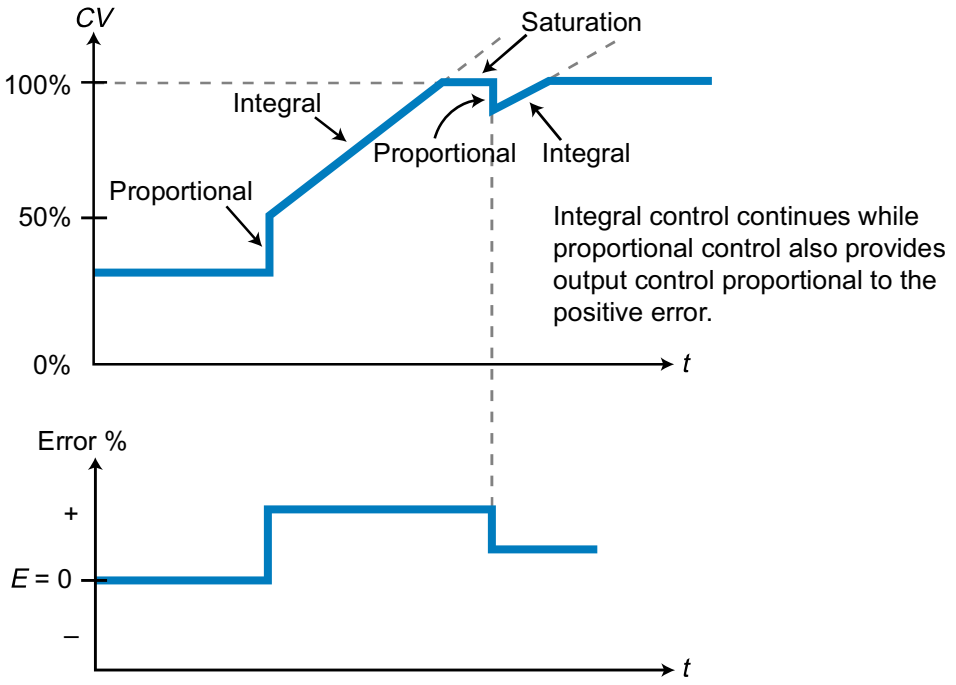


Figure 15-51. Saturation of the control variable output.

15-8 DERIVATIVE CONTROLLERS (D MODE)

STANDARD DERIVATIVE CONTROLLERS

The output of a **derivative controller** is proportional to the rate of change of the error in the system, which is expressed as $\frac{dE}{dt}$ (see Figure 15-52). This derivative action, also referred to as *rate mode*, is expressed mathematically as:

$$CV_{\text{new}} = K_D \frac{dE}{dt} + CV_{\text{old}}$$

where:

CV_{new} = the control variable
CV_{old} = the previous value of CV
K_D = the derivative gain constant in %(sec/%)
$\frac{dE}{dt}$ = the rate of change of error over the duration of change in %/sec

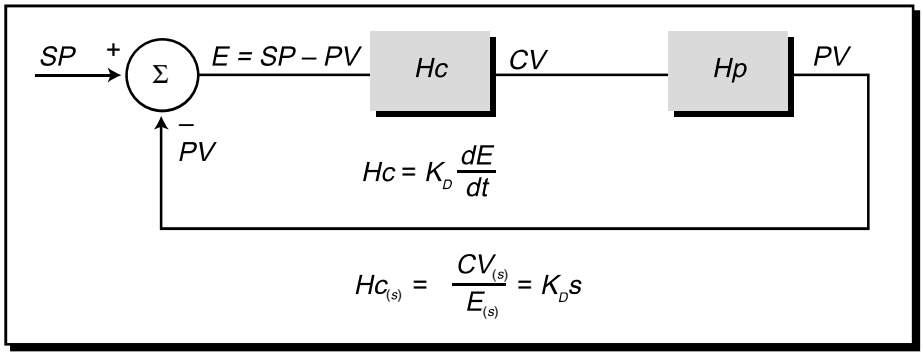


Figure 15-52. Derivative controller action.

The derivative gain constant (K_D) is also referred to as the *rate time*. It can be expressed in seconds or minutes as:

$$K_D = T_D \text{ seconds} \quad \text{(rate time)}$$

or

$$K_D = \frac{T_D}{60} \text{ minutes} \quad \text{(if } T_D \text{ is given in seconds)}$$

In Laplace form, the derivative controller transfer function takes the form:

$$Hc_{(s)} = \frac{CV_{(s)}}{E_{(s)}}$$
$$= K_D s = T_D s$$

Figure 15-53 illustrates the derivative gain transfer function in a direct-acting system by indicating the corresponding controller outputs for different rates of change ($\frac{dE}{dt}$) in error. Like in the integral mode, the rates of error change form several family curves (see Figure 15-53b). For example, if the error increases at a rate of 1.0%/sec, the controller will apply a derivative action that makes its output jump from 50% to 70% (see Figure 15-53a). If the rate of increase slows down to 0.5%/sec, the controller will decrease its output to 60%. When the rate of change of error equals zero, the controller will decrease its output to 50% again (see Figure 15-54). Note that the derivative action is based on the rate at which the error changes, not the actual value of the error.

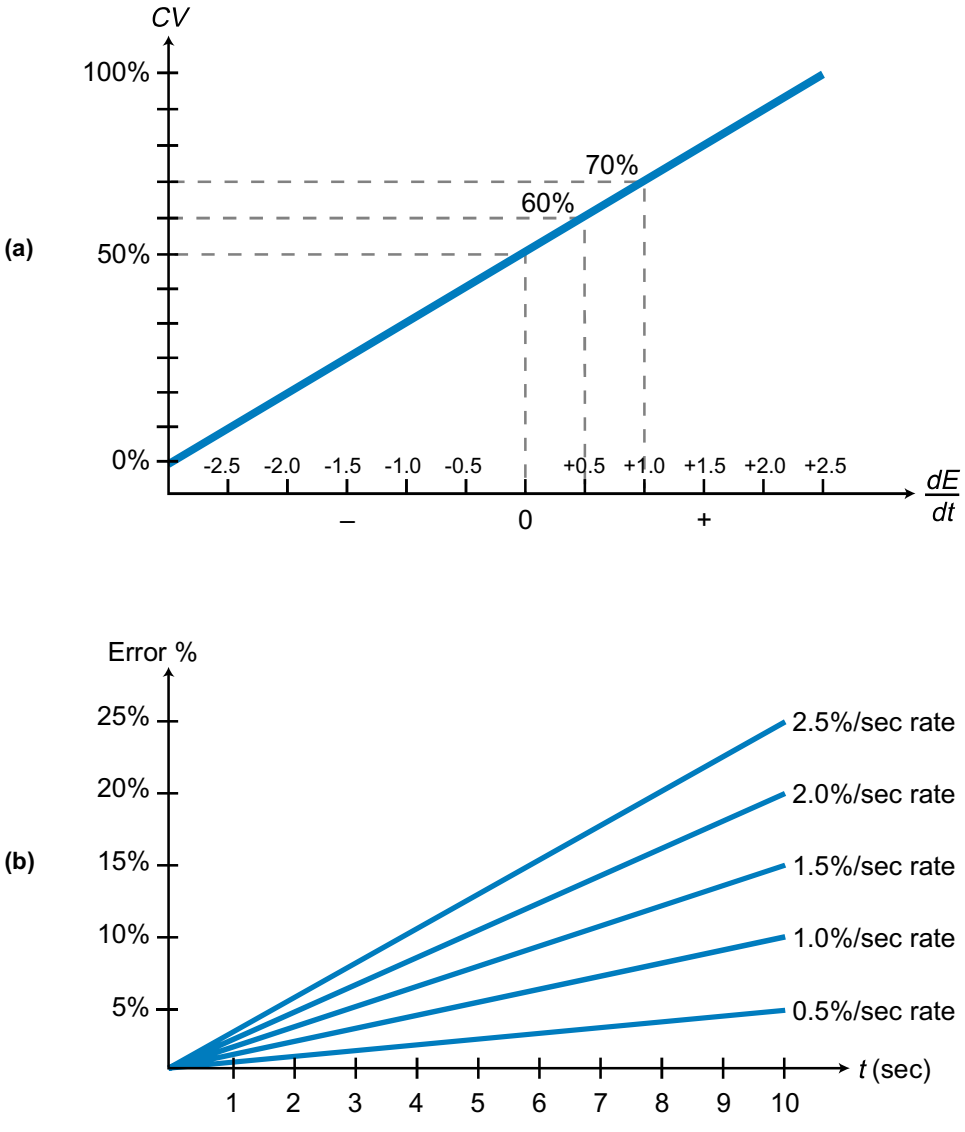


Figure 15-53. (a) Derivative controller transfer function and **(b)** its family of curves.

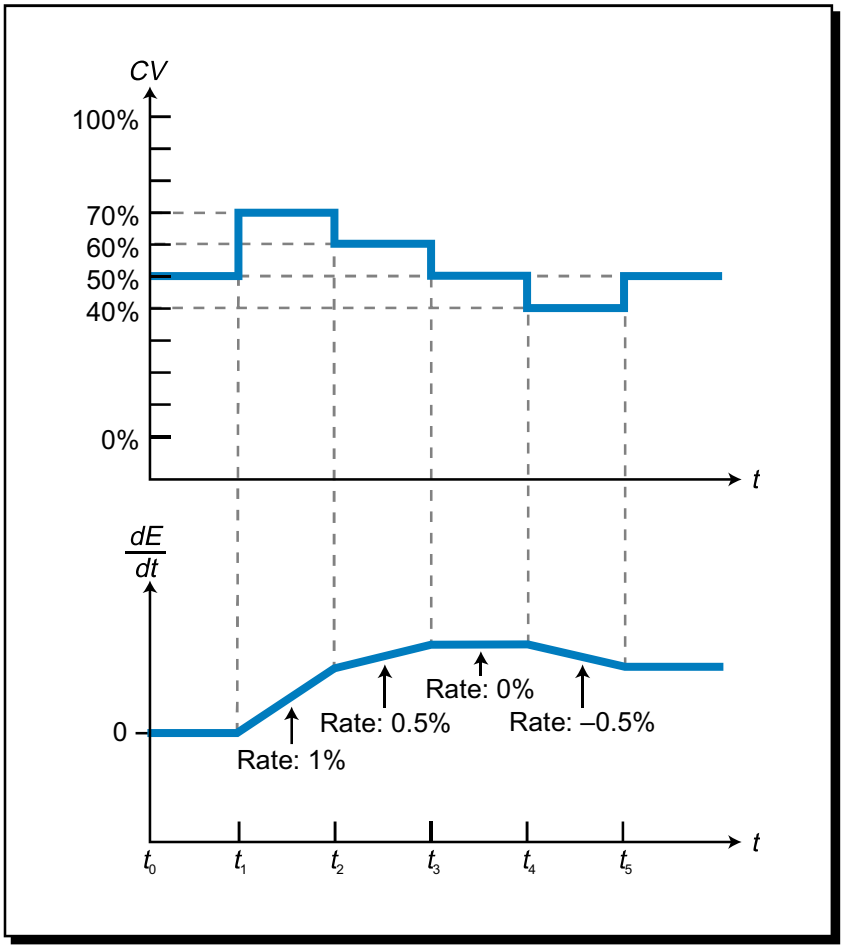


Figure 15-54. Derivative controller response for the transfer function shown in Figure 15-53a.

Derivative action is not used by itself in a controller; rather, it is used in combination with proportional and proportional-integral actions. There are several reasons for this. First, the derivative action response to a step change (see Figure 15-55a) creates an infinite change in error over time ($\frac{dE}{dt} = \infty$), causing the output of the controller to have 100% saturation for an instant (point 1 in Figure 15-55b). If the error remains at its stepped up value, the controller will sense no change and will return the control variable to 50% (between points 1 and 2). At point 2, when the error drops in a step fashion (see Figure 15-55a), the control variable will again have an infinite change over time, thus causing a 0% output (point 2 in Figure 15-55b).

The second reason why derivative action is not used alone is that it only produces a change in output if there is a change in the rate of error (points 3, 5, 6, and 7 in Figure 15-55). If a large error remains constant, the controller will maintain the control variable at 50% of its range (point 8), thus the error will not be corrected.

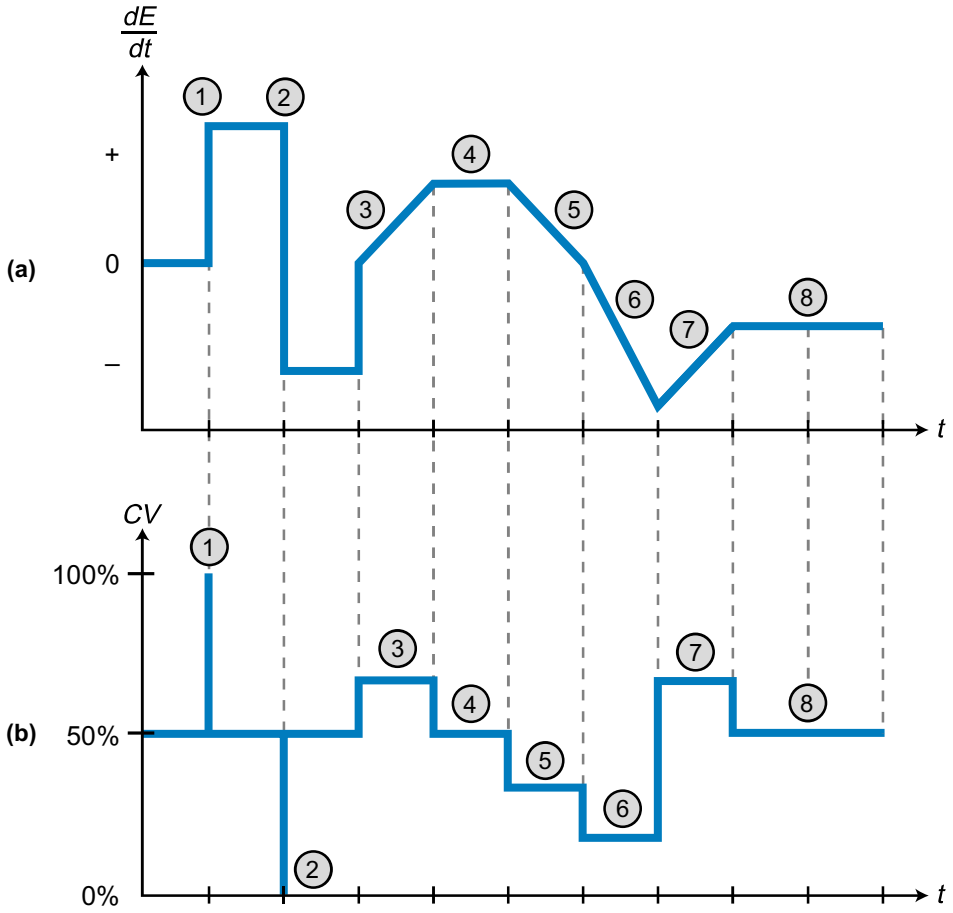


Figure 15-55. (a) Step changes and **(b)** their corresponding derivative responses.

MODIFIED DERIVATIVE CONTROLLERS

Derivative action may also be expressed in terms of the change in the process variable rate over time:

$$CV_{\text{new}} = -K_D \frac{dPV}{dt} + CV_{\text{old}}$$

This type of derivative action, used by some PLCs, avoids the saturation of the control variable in response to a step change in the set point. In this type of controller, the control variable tracks the process variable, which is very unlikely to change in a step fashion.

Note that the sign of the K_D term for a modified derivative controller is negative. It is derived from:

$$E = SP - PV$$

$$CV_{(t)} = K_D \frac{dE}{dt} + CV_{(t=0)}$$

$$= K_D\left[\left(\frac{dSP}{dt}\right) - \left(\frac{dPV}{dt}\right)\right] + CV_{(t=0)}$$

$$= \left[K_D\left(\frac{dSP}{dt}\right) - K_D\left(\frac{dPV}{dt}\right)\right] + CV_{(t=0)}$$

Since the set point is a constant value, the term $K_D(\frac{dSP}{dt})$ will equal 0. Therefore, the control variable is equal to:

$$CV_{(t)} = -K_D \frac{dPV}{dt} + CV_{(t=0)}$$

As shown in Figure 15-56, a modified derivative controller cannot be used by itself because the error signal is not fed back to the controller for error correction. Therefore, this type of derivative controller must be used in combination with either a proportional or proportional-integral controller.

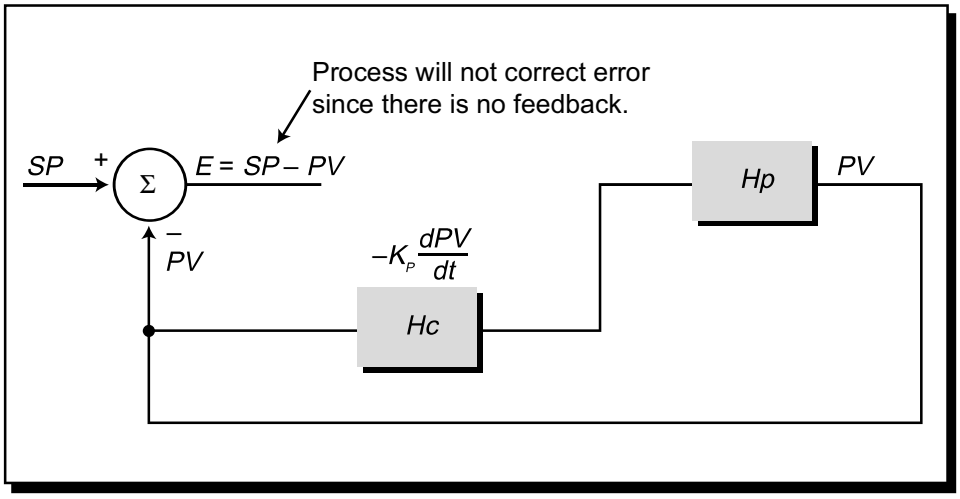


Figure 15-56. Modified derivative controller.

15-9 PROPORTIONAL-DERIVATIVE CONTROLLERS (PD MODE)

Proportional-derivative controllers are composite controllers that combine the actions of proportional and derivative controllers. A PD controller's output equation is represented as (see Figure 15-57):

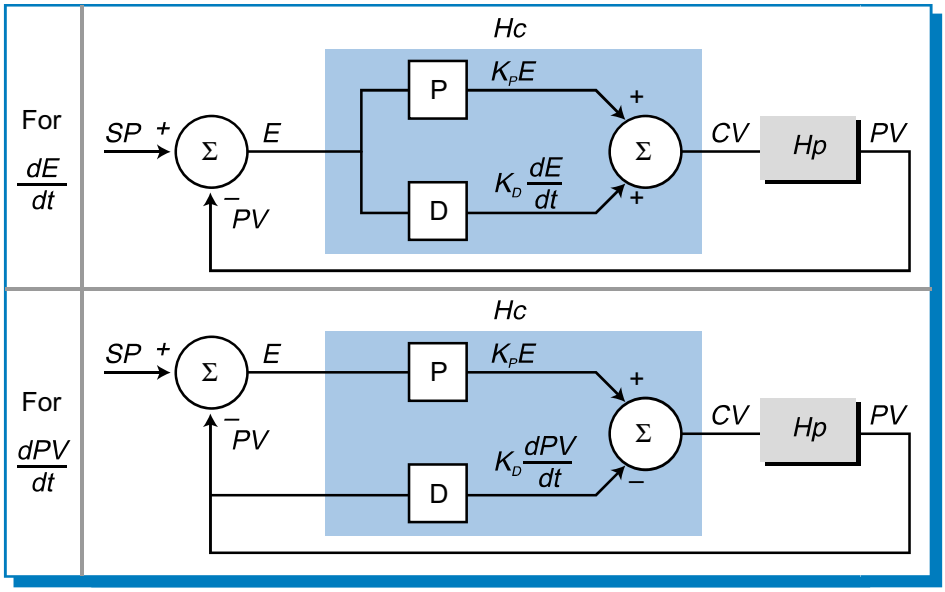
$$\text{For } \frac{dE}{dt}: \quad CV_{(t)} = K_P E + K_D \frac{dE}{dt} + CV_{(t=0)} \qquad \text{(parallel)}$$

$$CV_{(t)} = K_P E + K_P K_D \frac{dE}{dt} + CV_{(t=0)} \qquad \text{(series)}$$

$$\text{For } \frac{dPV}{dt}: \quad CV_{(t)} = K_P E - K_D \frac{dPV}{dt} + CV_{(t=0)} \qquad \text{(parallel)}$$

$$CV_{(t)} = K_P E - K_P K_D \frac{dPV}{dt} + CV_{(t=0)} \qquad \text{(series)}$$

(a) Parallel PD controller



(b) Series PD controller

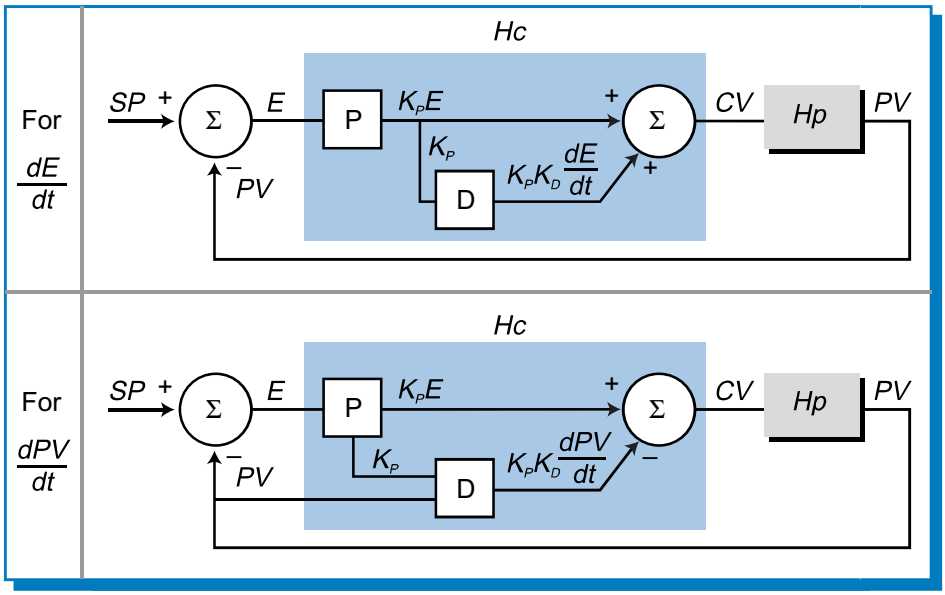


Figure 15-57. (a) Parallel and **(b)** series PD controllers.

These equations are formed by adding the equations for the proportional and derivative actions. Sometimes, the term K_D is replaced with the term T_D, since both have their units in time (seconds or minutes). The term K_D (or T_D) in a series PD controller (see Figure 15-58) indicates the time it takes for the proportional action to equal the derivative action, in other words, for the controller to *repeat* the derivative action.

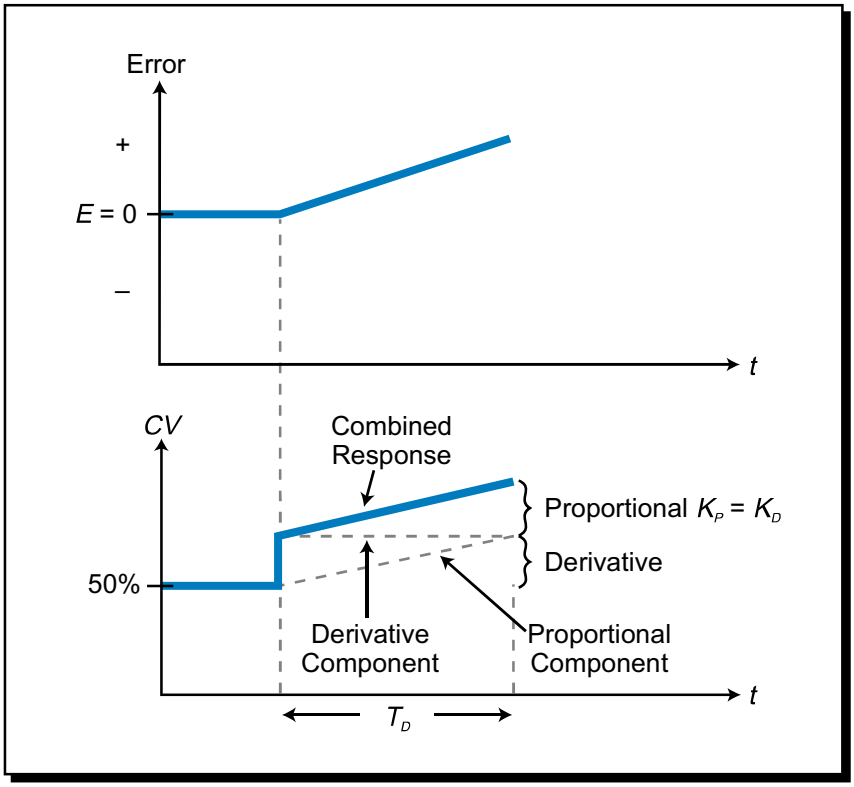


Figure 15-58. Proportional-derivative controller's response to an error.

The derivative component of a PD controller provides a faster response than just the proportional action alone, since it provides an immediate response to an error change that behaves in ramp form (see Figure 15-59). The proportional response to a ramping error is slower than the anticipatory response of a derivative action. The proportional action increases the output as it reads the error level. Since the proportional action only senses the amount of error and not its rate of change, it does not anticipate the top error value until that point is reached. A derivative action, on the other hand, anticipates the error value because it evaluates the rate at which the error is changing and, correspondingly, provides an extra amount of controller output. Therefore, when the error changes in ramp form instead of step form, the derivative gain compensates for the proportional control's delay in action. Although the derivative gain offsets the integral delay in a PD controller, it does not eliminate the offset error at steady state, which is shown in Figure 15-59 at t_5.

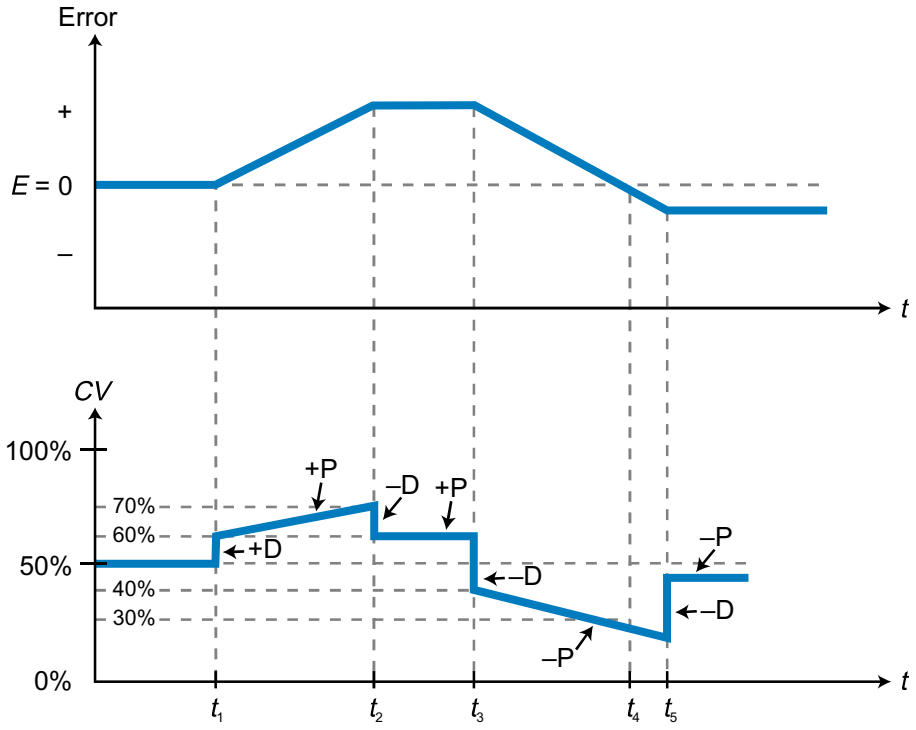


Figure 15-59. PD responses to step changes in error.

The derivative action in a PD controller adds stability to a closed-loop system by reducing the amount of overshoot and undershoot in the system's response. The derivative component acts as a "brake" in the system, slowing the proportional response as the process variable approaches its set point. The speed of response, however, also slows down. To observe this braking effect, let's examine the reaction of the closed-loop system in Figure 15-60 to a unit step. This is a second-order system with a proportional gain of K_P = 8 and no derivative gain (switch open). The addition of derivative action to this system will help to stabilize the overshoot and undershoot of the response to a change in error.

If the set point in Figure 15-60 changes, the proportional controller will try to bring the error to zero by making *PV* equal *SP*. The error at the start (t_0) is 1, and as *PV* approaches *SP*, this error becomes smaller. In this proportional action, the controller output is positive (direct acting), which makes the *PV* value become more positive. The slope of *PV* is also positive, as seen at point A. This positive value of $\frac{dPV}{dt}$ can be approximated as shown in Figure 15-60. If derivative action were present in this system (switch closed), then the value of $\frac{dPV}{dt}$ would be negative:

$$PV_{(t)} = K_P E - K_P K_D \frac{dPV}{dt} + PV_{(t=0)}$$

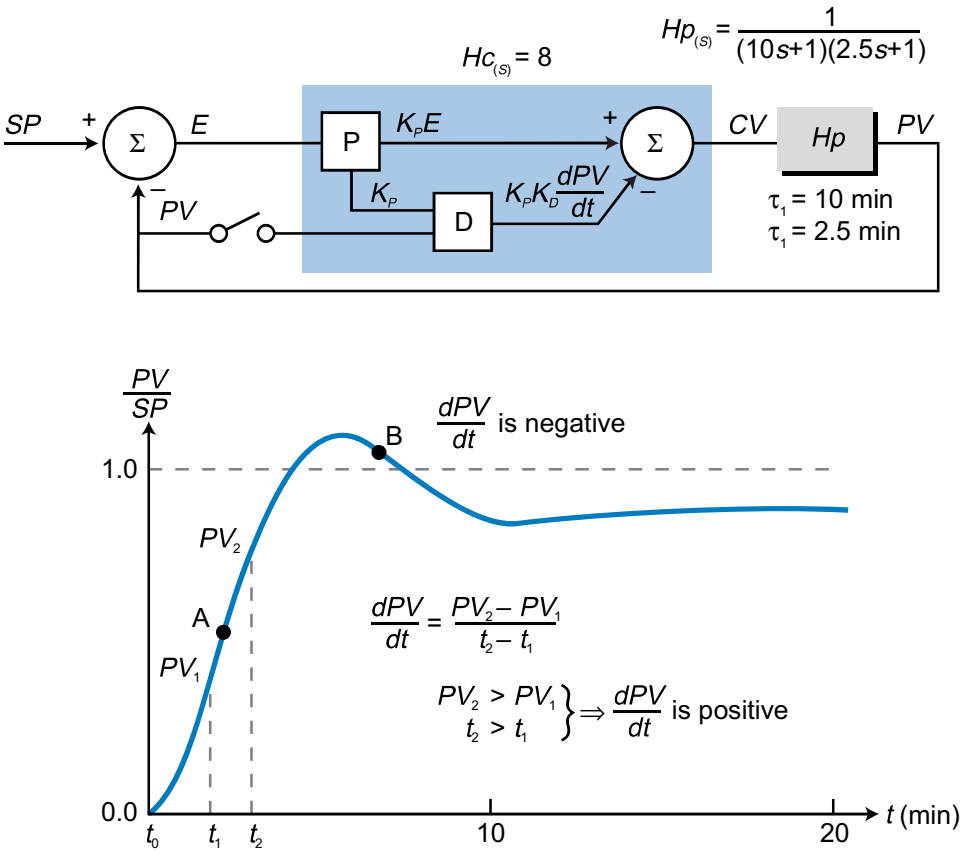


Figure 15-60. Reaction of a closed-loop system to a unit step.

thus having the opposite sign of the proportional gain. Therefore, with derivative action, the output of $PV_{(t)}$ (at point A) would be less than a pure proportional controller without derivative action. In fact, a positive $\frac{dPV}{dt}$ term (slope) would make the derivative term in the PD system equation negative. This indicates that the derivative action of the PD controller will brake the response of the proportional action, therefore reducing the amount of overshoot. The same holds true when the slope is negative, which occurs when the response of the pure proportional action starts to decrease (point B). When the proportional response becomes negative, the derivative term becomes positive, thus braking the undershoot:

$$CV_{(t)} = K_P(-E) - K_PK_D\left(-\frac{dPV}{dt}\right) + CV_{(t@b)}$$

$$= -K_PE + K_PK_D\frac{dPV}{dt} + CV_{(t@b)}$$

So, by adding the derivative action to this closed-loop system (switch closed in block diagram), it is possible to reduce the overshoot and undershoot through the braking effect of the derivative action. Figure 15-61 illustrates

this closed-loop system for several values of derivative gain K_D (or T_D, derivative time). As the gain of the derivative action increases, the overshoot and undershoot decrease. However, the system response also slows down.

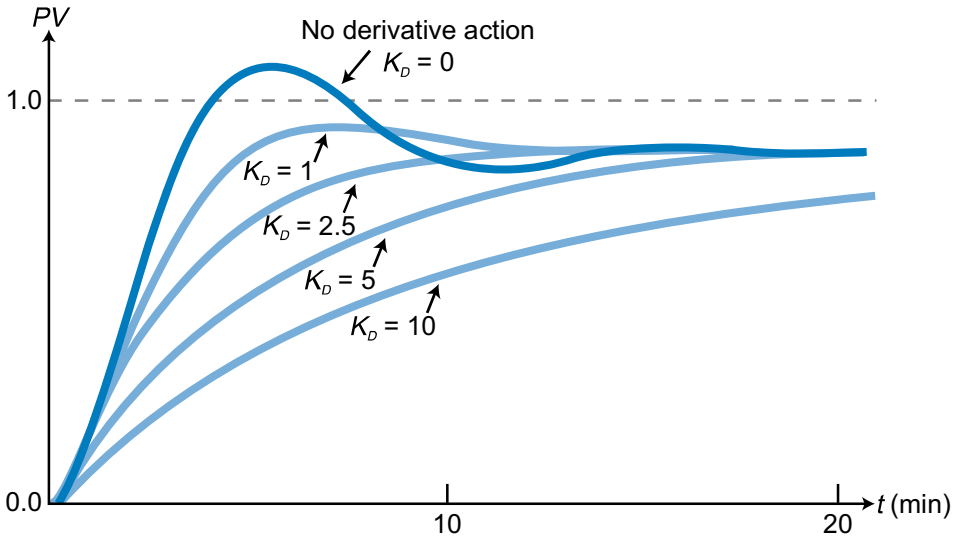


Figure 15-61. Closed-loop process response to a proportional-derivative controller for several values of K_D.

If a proportional-derivative controller has too much derivative gain, the system response will start to look like the graph in Figure 15-62. This indicates that the derivative action is no longer effective in restoring the desired stability margin.

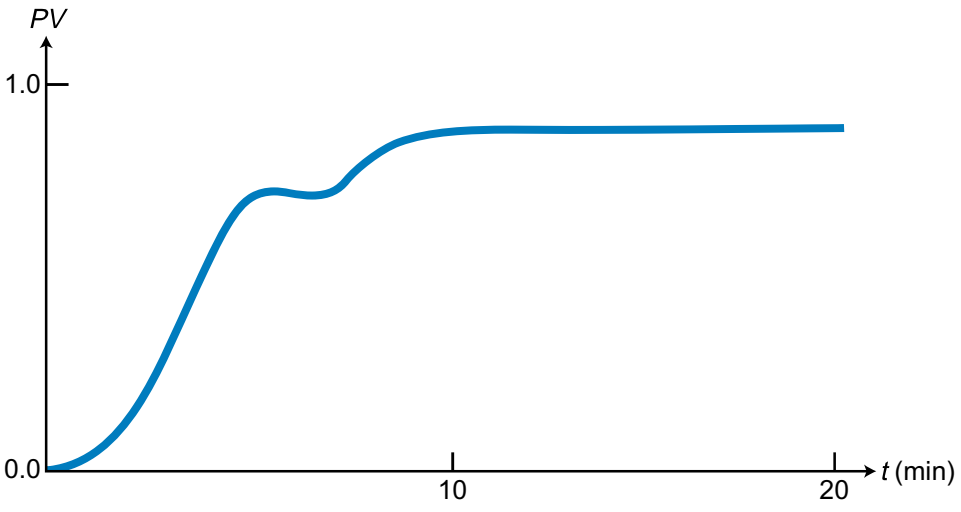


Figure 15-62. Process reponse of a proportional-derivative controller with too much derivative gain.

EXAMPLE 15-9

The closed-loop system described in Examples 15-5 and 15-8 employed proportional and proportional-integral controllers, respectively, to control a first-order system with a gain of 5 and a time constant of τ = 30 seconds. Given that the system utilizes a proportional-derivative controller with a proportional gain of $K_P = 8$ and a derivative gain of 2 minutes (120 seconds), find **(a)** the closed-loop transfer function of the system and **(b)** the steady-state value of the response to a step input ($\frac{1}{s}$).

SOLUTION

(a) The transfer function of the process is:

$$Hp_{(s)} = \frac{5}{30s+1}$$

The controller's transfer function is:

$$\begin{aligned} Hc_{(s)} &= K_P + K_D s \\ &= 8 + 120s \end{aligned}$$

Therefore, the closed-loop system transfer function is:

$$\begin{aligned} \frac{PV_{(s)}}{SP_{(s)}} &= \frac{Hc_{(s)}Hp_{(s)}}{Hc_{(s)}Hp_{(s)}+1} \\ &= \frac{\left[\left(8+120s\right)\left(\frac{5}{30s+1}\right)\right]}{\left[\left(8+120s\right)\left(\frac{5}{30s+1}\right)+1\right]} \\ &= \frac{\left(\frac{40+600s}{30s+1}\right)}{\left(\frac{40+600s}{30s+1}\right)+1} \\ &= \frac{\left(\frac{40+600s}{30s+1}\right)}{\left(\frac{40+600s+30s+1}{30s+1}\right)} \\ &= \frac{\left(\frac{40+600s}{30s+1}\right)}{\left(\frac{41+630s}{30s+1}\right)} \\ &= \frac{40+600s}{41+630s} \end{aligned}$$

(b) The system response to a step input change is represented by:

$$PV_{(s)} = SP_{(s)}\left(\frac{40+600s}{41+630s}\right)$$
$$= \left(\frac{1}{s}\right)\left(\frac{40+600s}{41+630s}\right)$$

Applying the final value theorem to this Laplace function yields the following value:

$$\lim_{s\to 0} sPV_{(s)} = \lim_{s\to 0} s\left(\frac{1}{s}\right)\left(\frac{40+600s}{41+630s}\right)$$
$$= \frac{40+600(0)}{41+630(0)}$$
$$= \frac{40}{41}$$
$$= 0.976$$

So, the final value of the process variable at steady state will be 0.976, producing an offset error of 2.4%.

15-10 PROPORTIONAL-INTEGRAL-DERIVATIVE CONTROLLERS (PID MODE)

A **proportional-integral-derivative (PID) controller** combines the actions of all three controller modes. A PID controller, also called a *three-mode controller*, can be used to control almost any process that involves lags and dead times. A PID controller can be arranged in either a series or parallel configuration using either a standard or modified derivative action ($\frac{dE}{dt}$ and $\frac{dPV}{dt}$, respectively). The basic expression for the process variable output for a standard parallel PID controller (see Figure 15-63) is:

$$PV_{(t)} = K_P E + K_I \int_0^t E dt + K_D \frac{dE}{dt} + PV_{(t=0)}$$

In Laplace form, this controller's transfer function is represented as:

$$Hc_{(s)} = \frac{CV_{(s)}}{E_{(s)}}$$
$$= K_P + \frac{K_I}{s} + K_D s$$

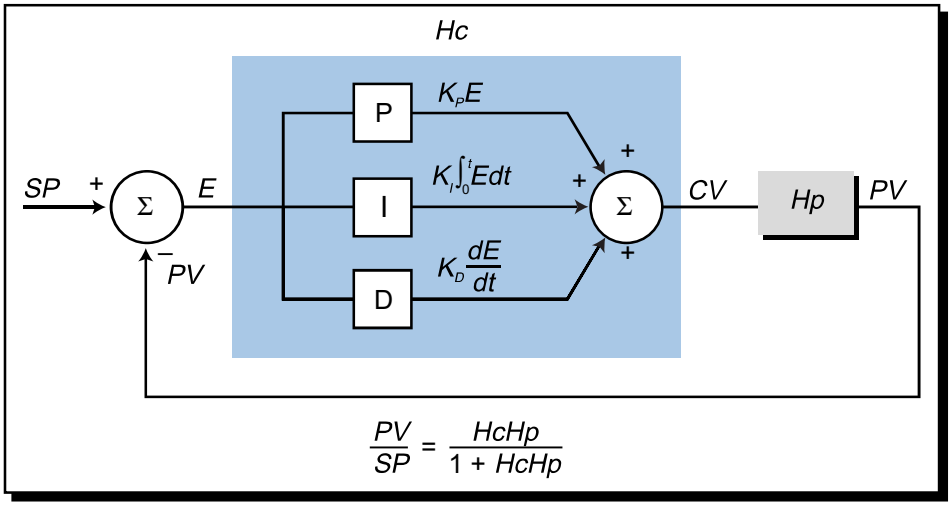


Figure 15-63. Standard parallel PID controller.

Figure 15-64 illustrates the serial and parallel system configurations for a PID controller, along with their respective closed-loop equations.

PID control eliminates the offset of the proportional action through its integral action and suppresses oscillation with its derivative action. When properly tuned (see Section 15-12), a PID controller will smoothly regulate the response of a complex system or process.

ORIGINS OF PID CONTROL

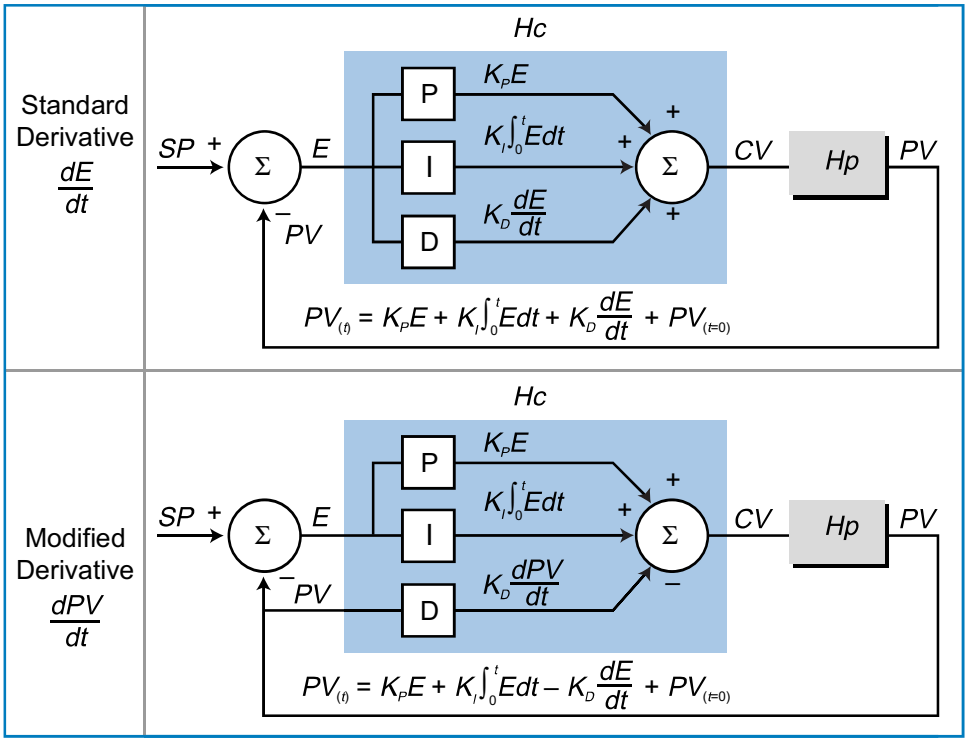
In this section, we will explain why the PID controller is the perfect controller for a typical process. To illustrate the relationship between a PID controller and a process, we will examine a typical second-order process system. For computational purposes, a second-order system can be thought of as including a first-order system, in order to determine what type of controller will make the process in an open-loop system have a transfer function equal to one. We will discuss this in more detail shortly.

It is very difficult to determine the exact transfer function ($Hp_{(s)}$) of a real-life process (i.e., a manufacturing process). However, it can be approximated by a second-order system with two lag times and a dead time delay. In Laplace form, this transfer function is defined as:

$$\frac{PV_{(s)}}{CV_{(s)}} = \frac{Ae^{-t_d s}}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

The $e^{-t_d s}$ term, the dead time delay, can be omitted from this equation, since we know that this term only indicates that there is a shift in time in the response. For practical purposes, the dead time will cause the response to behave in the same manner, only displaced in time by the delay. So, for

(a) Parallel PID controller



(b) Serial PID controller

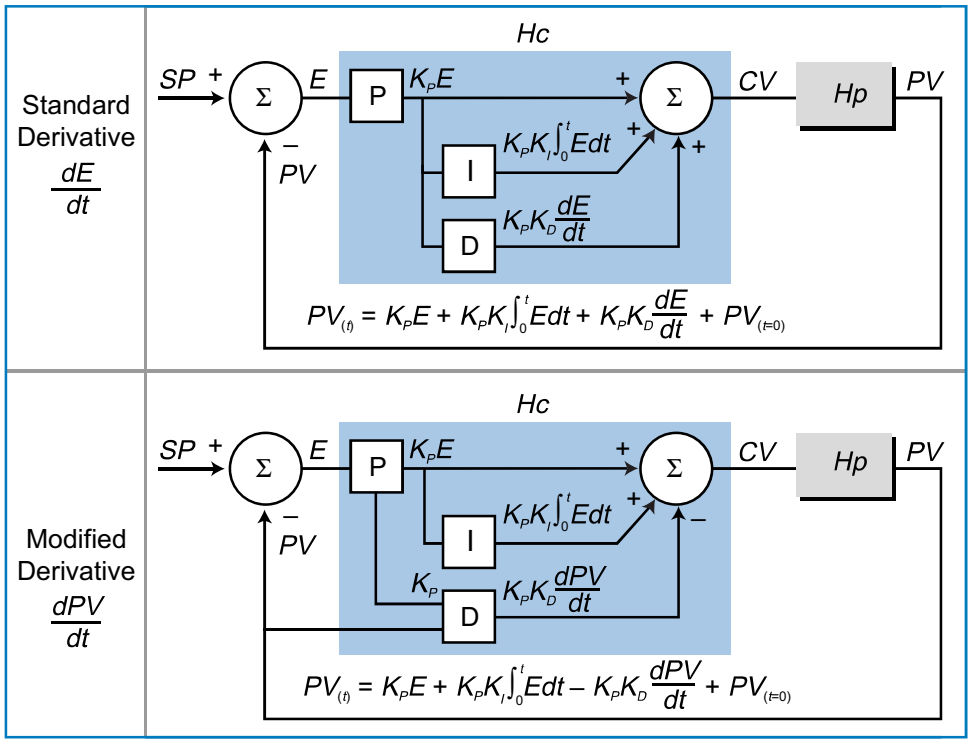


Figure 15-64. (a) Parallel and **(b)** serial PID controllers.

purposes of obtaining an equation for the best controller to govern this system, the delay can be omitted during the initial controller calculations. However, we should remember that there is a dead time response in the system.

The *A* term in the second-order system transfer function indicates the process gain and the τ_1 and τ_2 terms are the two lag times (see Figure 15-65). This second-order system is said to be inclusive of a first-order system, meaning that if one of the lag times is zero, the second-order equation will represent a first-order system. The ideal transfer function of a perfect process control system ($\frac{PV}{SP}$) should equal one, indicating that the output of the process (*PV*) immediately follows any changes in the set point without requiring negative feedback to correct the error since there is no error. In other words, if there is a step change in the set point from 0 to 1, the process will respond immediately with a change from 0 to 1. The controller-process relationship in a perfect system is such that they complement each other perfectly. Therefore, the transfer function of the process variable over the set point will be one. Accordingly, the equation for a perfect open-loop system is (see Figure 15-66):

$$\frac{PV_{(s)}}{SP_{(s)}} = Hc_{(s)}Hp_{(s)} = 1$$

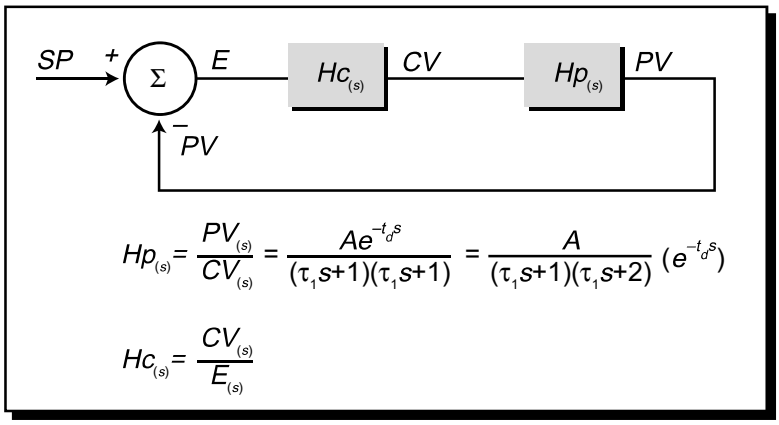


Figure 15-65. Second-order system.

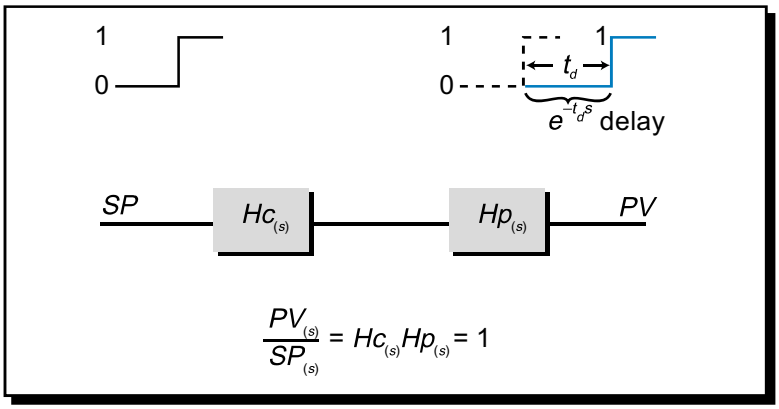


Figure 15-66. Perfect open-loop system.

So, for a perfect system, the controller's transfer function should be the inverse of the processor's transfer function. Therefore, the controller's transfer function in a perfect system using a typical process approximation is:

$$\begin{aligned} Hc_{(s)} &= \frac{1}{Hp_{(s)}} \\ &= \frac{1}{\left(\frac{A}{(\tau_1 s+1)(\tau_2 s+1)}\right)} \\ &= \left(\frac{1}{A}\right)(\tau_1 s+1)(\tau_2 s+1) \\ &= \left(\frac{1}{A}\right)(\tau_1\tau_2 s^2+\tau_1 s+\tau_2 s+1) \\ &= \left(\frac{1}{A}\right)\left[\tau_1\tau_2 s^2+(\tau_1+\tau_2)s+1\right] \end{aligned}$$

The term $\frac{1}{A}$ is a constant; therefore, it can be renamed as A_1:

$$Hc_{(s)} = A_1\left[\tau_1\tau_2 s^2+(\tau_1+\tau_2)s+1\right]$$

Dividing each term in the bracket by s yields:

$$\begin{aligned} Hc_{(s)} &= A_1\left[\frac{\tau_1\tau_2 s^2}{s}+\frac{(\tau_1+\tau_2)s}{s}+\frac{1}{s}\right] \\ &= A_1\left[\tau_1\tau_2 s+(\tau_1+\tau_2)+\frac{1}{s}\right] \end{aligned}$$

Multiplying the A_1 term and rearranging the equation produces:

$$Hc_{(s)} = A_1(\tau_1+\tau_2)+\frac{A_1}{s}+A_1\tau_1\tau_2 s$$

P I D

The terms in this equation indicate that the controller has a proportional gain, an integral action ($\frac{1}{s}$), and a derivative component (s). Therefore, the perfect system controller exhibits proportional, integral, and derivative actions. Note that the constant gain term in this equation does not imply that the gains should be the same for each of the PID actions. Rather, it indicates that these gains must be present and specified. Because a PID-type controller is the natural derivation from a perfect system, PID is considered a universal

type of control for manufacturing processes. In fact, of all the PID configurations shown in Figure 15-64, perhaps the most commonly used in PLCs is the serial, modified derivative, PID configuration.

DIGITAL IMPLEMENTATION OF PID IN A PLC

A programmable controller system implements the PID control action using a discrete, or digital, algorithm to update the control variable (*CV*). For example, a modified serial PID controller may use the following digital algorithm, where the current control variable output (CV_n) is represented as:

$$CV_n = CV_{(n-1)} + K_P\left(E_n - E_{(n-1)}\right) + K_P K_I T_s E_n - \left(\frac{K_P K_D}{T_s}\right)\left(PV_n - 2PV_{(n-1)} + PV_{(n-2)}\right)$$

where:

CV_n = the controller output at the *n*th update
$CV_{(n-1)}$ = the controller output at the *n*th minus one update
K_P = the proportional gain (in seconds, where appropriate)
K_I = the integral gain (in seconds, where appropriate)
K_D = the derivative gain (in seconds, where appropriate)
E_n = the error at the *n*th update
$E_{(n-1)}$ = the error at the *n*th minus one update
T_s = the loop sample time in seconds
PV_n = the process variable at the *n*th update
$PV_{(n-1)}$ = the process variable at the *n*th minus one update
$PV_{(n-2)}$ = the process variable at the *n*th minus two update

The loop sample time (T_s) is the frequency of how often the PLC reads and executes the integration and derivative terms in the algorithm equation. In PLCs, this time can be selected from a range of 0.1 seconds to several hundred seconds (e.g., 600 seconds, or 10 minutes) Figure 15-67 illustrates several sampling rates. A small value of T_s (fast update time) is desirable in a process application where the process variable responds rapidly to control variable changes. However, because large values of T_s are necessary to evoke a stable derivative action, the trade-off between a low and high T_s value must be balanced carefully to ensure a correct system response. Otherwise, the derivative action can produce a bumpy action.

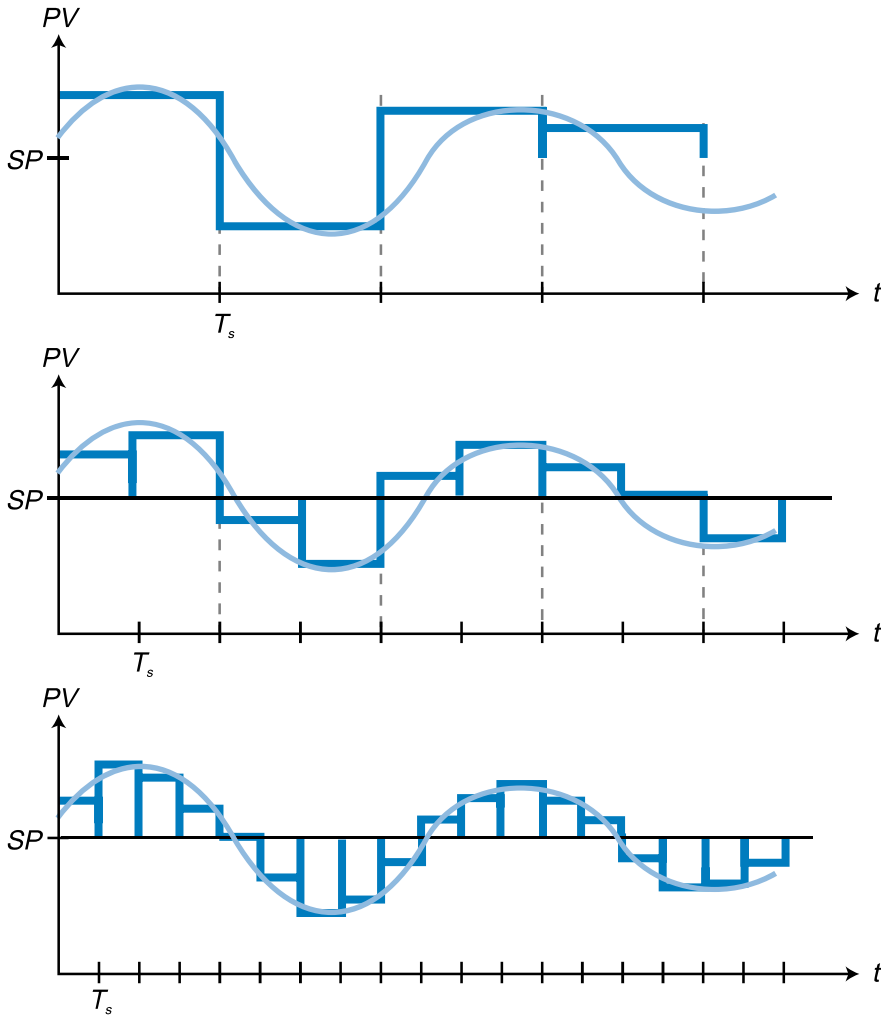


Figure 15-67. Loop sample rates.

The digital PID algorithm implemented in PLC systems calculates the error by approximating the area between the process variable and the set point (see Figure 15-68). This area calculation provides an approximate value of error.

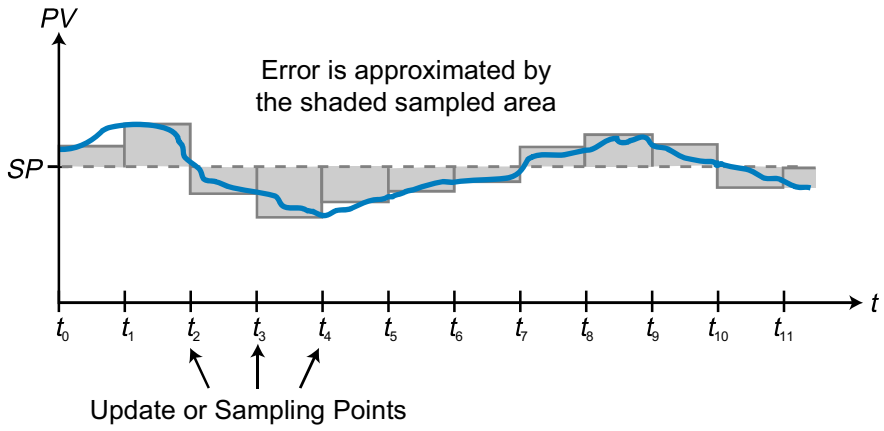


Figure 15-68. Error approximation using loop sample times.

INTEGRAL (RESET) WINDUP

As discussed in Section 15-7, integral (or reset) windup is a problematic condition that occurs in PI and PID controllers, resulting in the saturation of the controller's output ($CV = CV_{max}$). Integral windup is typical in startup situations during batch processes. To avoid integral windup, some PLC manufacturers offer PID interfaces that prevent integral action when the controller's output reaches 100%. These interfaces accomplish this by forcing the error input to the integrator section of the PID controller to zero. The block diagram in Figure 15-69 illustrates this method of integral windup prevention.

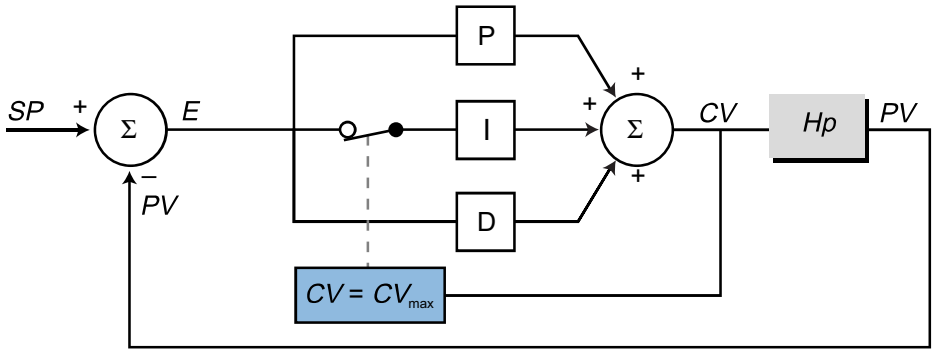


Figure 15-69. PID controller with integral windup prevention.

PID BUMPLESS AUTO/MANUAL TRANSFER

Most PLC applications that implement PID control employ automatic/manual control stations that allow the operator to switch between manual and PLC process control. To prevent a step change or "bump" during this switch, the control station must ensure that both controllers, the manual controller and the PLC (automatic), send the same output (CV) to the process. Otherwise, the process may receive a change in the control variable, which could produce a transient response in the system.

Figure 15-70 illustrates a PLC system that uses a PID controller interface with a manual control station that allows for bumpless transfer. Basically, the automatic (PLC) and manual controllers must follow each others outputs when they are operating. Figure 15-71 illustrates this configuration in block diagram form for a modified serial PID controller. When the system is in manual mode, the PID controller tracks the manual controller's output, so that when the transfer from manual to automatic occurs, both controller outputs are the same. A similar operation takes place during an automatic-to-manual transfer.

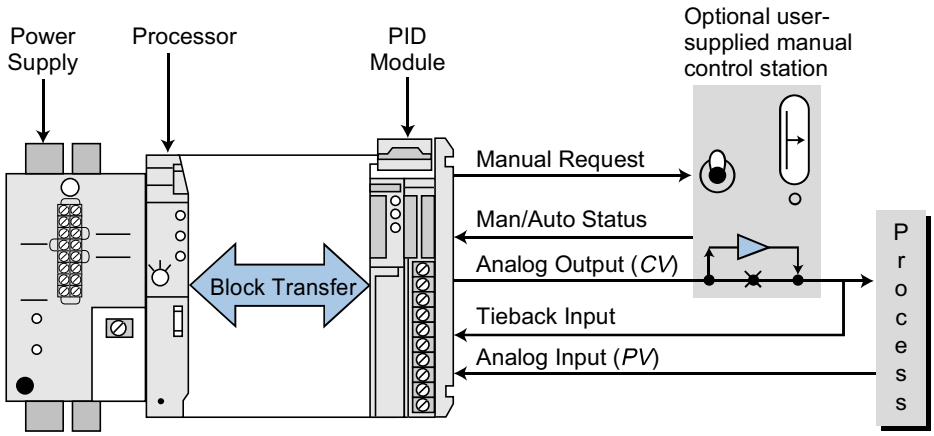


Figure 15-70. PID interface with a manual control station for bumpless transfer.

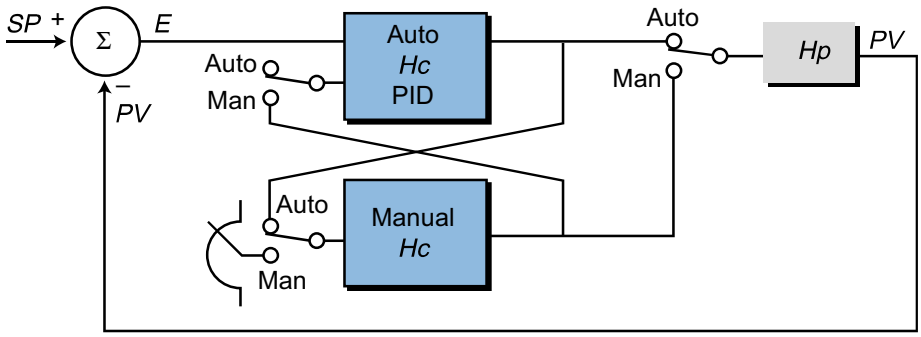


Figure 15-71. Auto/manual control station block diagram.

During a manual-to-automatic transfer in a PLC system, the PID interface processor may also set the set point equal to the process variable. This forces the system error (*SP* – *PV*) to zero, ensuring that a bump does not occur during the transfer. The control variable output of the PLC controller, which tracks the manual *CV* output, is left unchanged during the transfer. After the transfer, the PID processor returns the set point to its original value.

15-11 ADVANCED CONTROL SYSTEMS

While conventional PID control provides universal control for most processes, other techniques can increase the performance of a process control system. One of the most commonly practiced techniques used to increase process control performance is cascade control.

CASCADE CONTROL

Cascade control uses two controllers configured so that the output of one feedback loop becomes the set point for the other one. Figure 15-72a illustrates a temperature control batch system that utilizes a single PID controller, while Figure 15-72b shows the same system with cascade

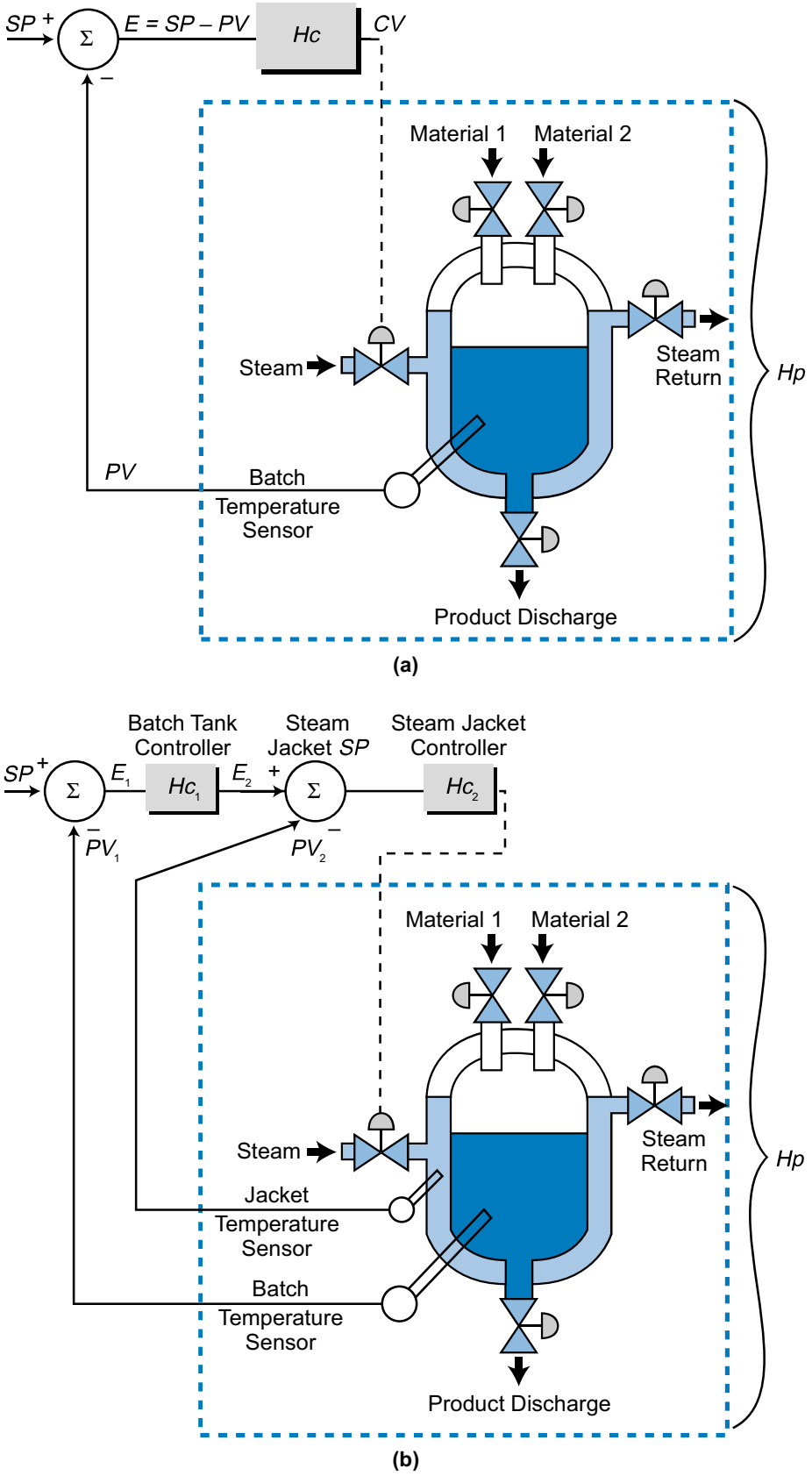


Figure 15-72. Temperature control systems with **(a)** single and **(b)** cascaded PID controllers.

control. In the cascade configuration, the batch tank controller provides the set point for the steam jacket temperature controller, which in turn, actuates the steam valve. The batch tank loop is called the *primary loop*, since the main process variable (the batch temperature) is the primary control concern (see Figure 15-73). The steam jacket temperature loop is called the *secondary loop,* or *inner loop,* since the jacket temperature is of secondary interest in the control system.

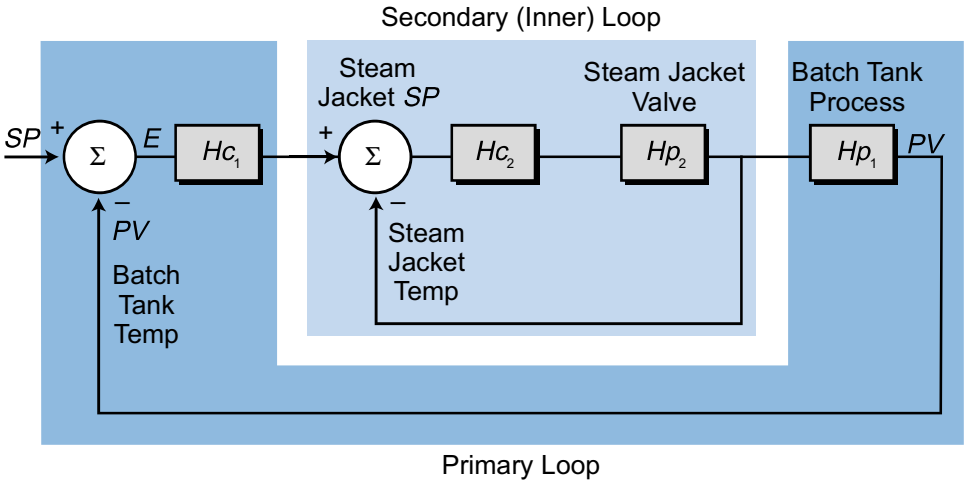


Figure 15-73. Primary and secondary loops of the temperature control system.

The greatest advantage of cascade control systems, and in fact the main reason for their use, is that they respond quicker than single-controller systems to disturbances that affect the primary loop. In cascade control, the secondary loop response to a disturbance generally occurs first, before the primary loop starts to respond. In the batch system shown in Figure 15-72b, a change in steam temperature will affect the tank's jacket temperature before it affects the main batch temperature due to the lag and dead times associated with the batch process. The steam jacket secondary control loop will respond first to this disturbance and try to correct it, thus minimizing the effect of the disturbance on the main batch system. This fast response of the secondary loop enhances the performance of cascaded control systems as compared to single-loop control systems.

Most programmable controller systems allow cascade control directly to the PID intelligent interface or analog input modules (if PID is implemented in the main PLC processor). Therefore, the user must only identify the input to the secondary loop (see Figure 15-74). The secondary loop cascade input is also referred to as the *remote input* in conventional, single-loop controllers.

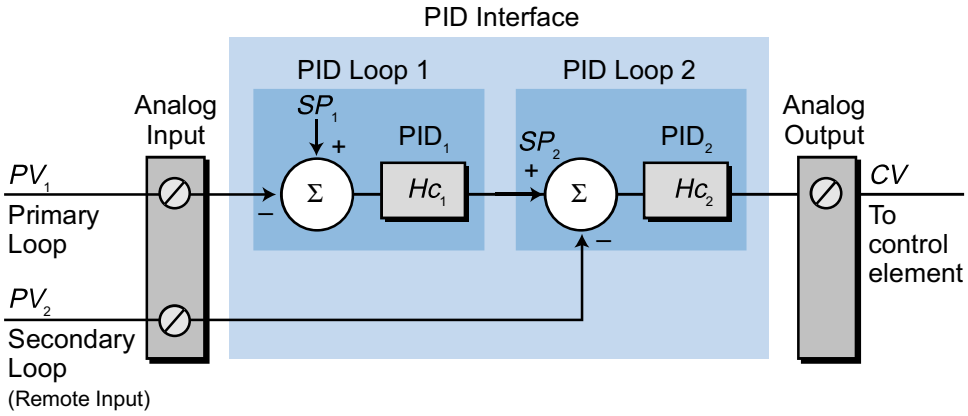


Figure 15-74. Cascade control directly to a PID interface.

BUMPLESS CASCADE CONTROL

PLC systems also provide bumpless transfer in cascade control configurations. Most often, the transfer sequence from manual to automatic is initiated in the secondary loop (see Figure 15-75). Once the secondary loop is placed in automatic mode, the secondary loop set point is set to the value of the secondary process variable PV_2. Then, the primary loop is placed in automatic mode and the primary loop set point is set to the value of the primary process variable PV. The primary loop's output is left unchanged. This is the initial operation that avoids a bump. From there, the primary loop's set point returns to the desired value, and the primary's output adjusts the set point of the secondary loop controller. In general, the primary loop cannot be activated unless the secondary loop is already active or in the AUTO mode.

The tuning of cascade controllers, which we'll explain in the next section, must be performed in a sequential, logical fashion. In most systems, the user must tune the secondary loop first, with the primary loop in manual mode. After the secondary loop is tuned, the tuning of the primary loop can begin.

15-12 CONTROLLER LOOP TUNING

For a process control system to work correctly, its control loop(s) must be tuned. **Loop tuning** involves selecting the constants [K_P, K_I (or T_I), and K_D (or T_D)] that will be used with the proportional, integral, and derivative actions of a controller. With these constants at the proper levels, the controller can effectively and efficiently regulate the process variable to the set point.

A process often experiences disturbances caused by changes in the set point or the process load (see Figure 15-76). These disturbances cause an error in the system, thereby changing the controller output, which in turn, impacts

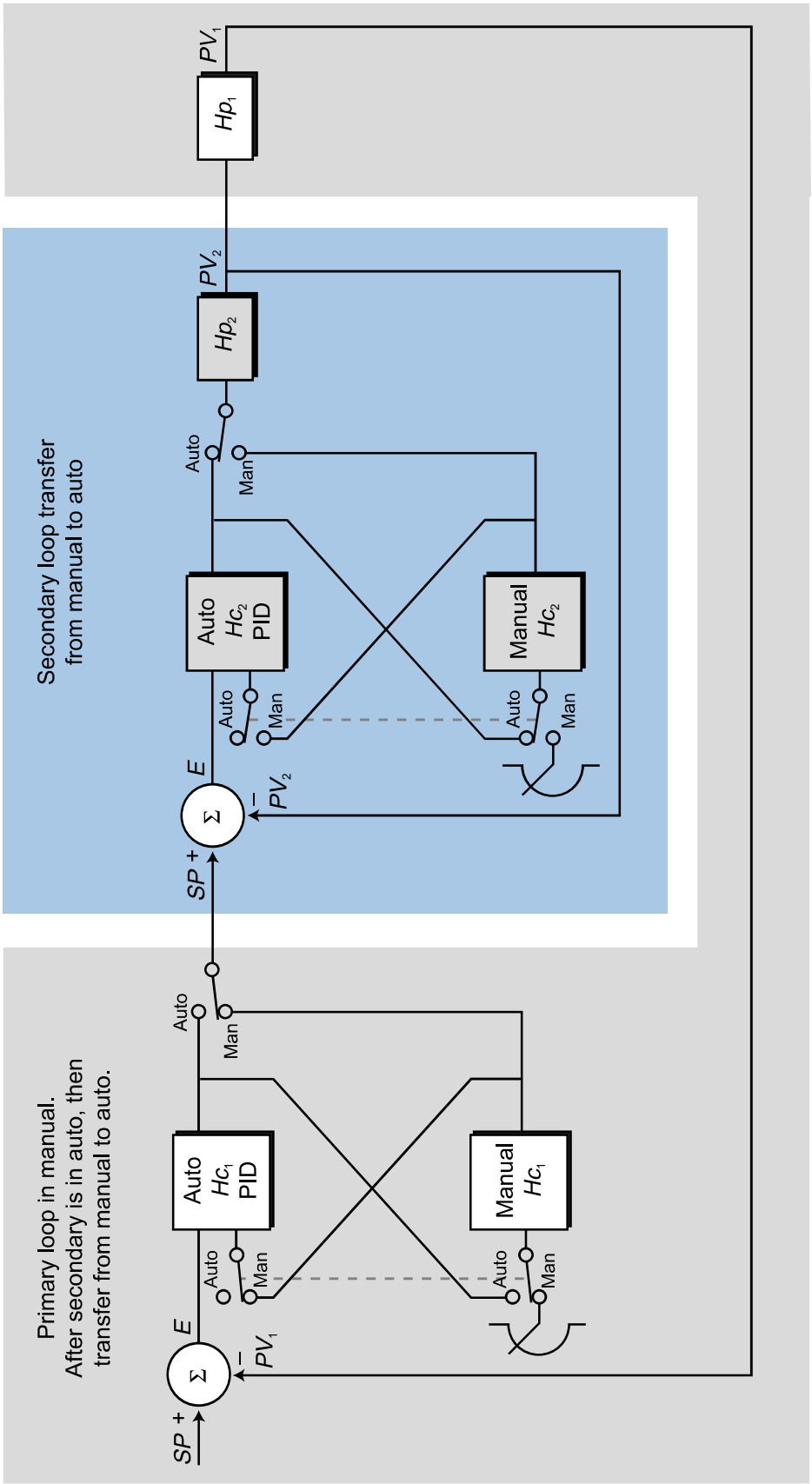


Figure 15-75. Bumpless transfer in cascaded PID controllers.

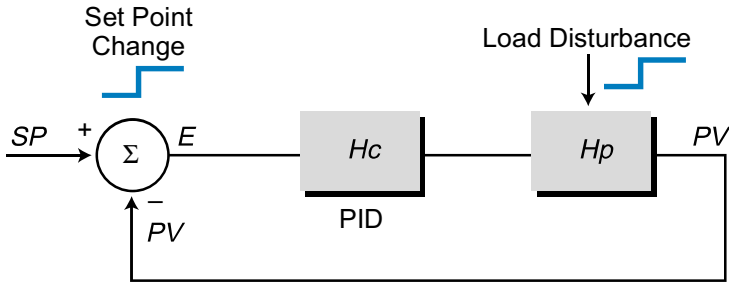


Figure 15-76. Process disturbances.

the process variable. The process variable response to both system disturbances and the controller action must be stable; that is, it must not oscillate or grow in value without limit. This process variable response can typically be categorized as either overdamped, critically damped, or underdamped (see Figure 15-77a). However, another type of process variable response is a **quarter-amplitude response**, also called a *quarter-delay ratio response* (see Figure 15-77b). This response, which is the desired response after closed-loop tuning, reduces the *PV* overshoot by one-quarter each cycle.

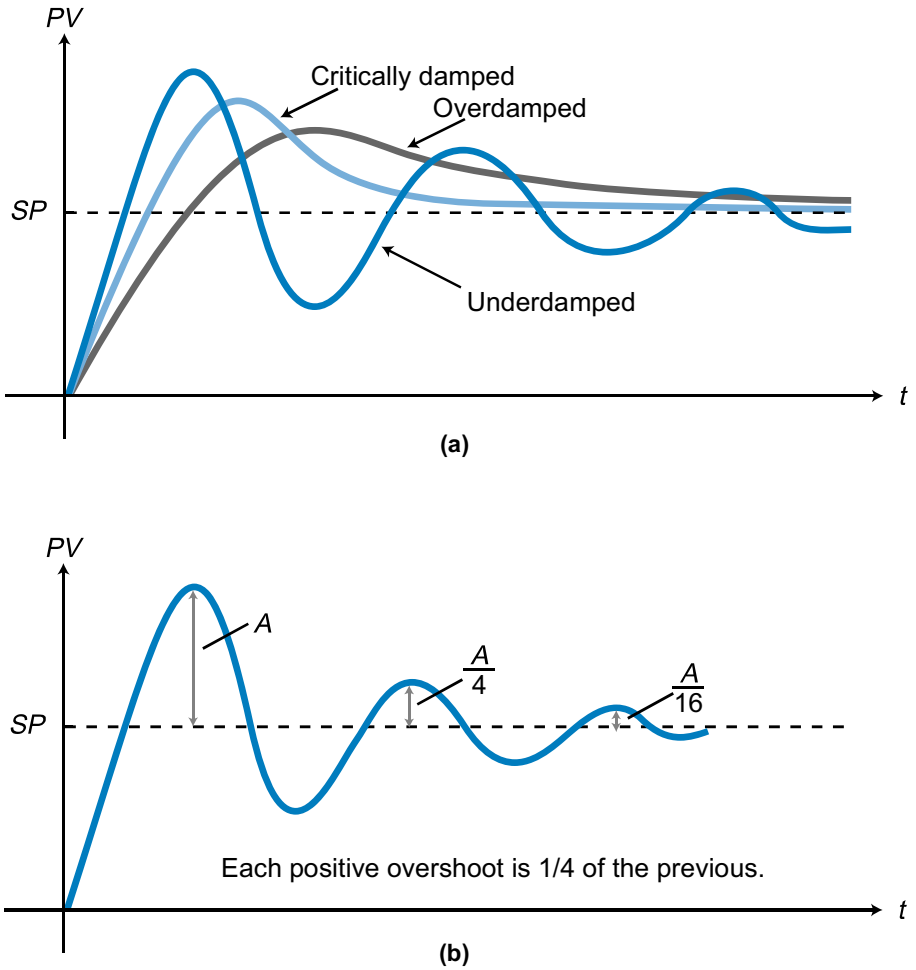


Figure 15-77. Process variable reponses: **(a)** overdamped, critically damped, underdamped, and **(b)** quarter-amplitude.

Table 15-2 shows the characteristics of each type of process variable response. The tuning parameters of the system will have a decisive impact on the type of process response exhibited by the system.

Response Type	Quality of Response
Overdamped	This response approaches the set point smoothly without oscillation. The maximum deviation from the set point is less than that of a critically damped response.
Critically Damped	This response approaches the set point smoothly, but faster than an overdamped response. This results in a larger maximum deviation from the set point. This is a relatively good response for applications in which oscillation is not acceptable.
Underdamped	This response cycles, producing several deviations over and under the set point before settling to a steady-state value. It gives a fast response to a disturbance change.
Quarter Amplitude	This response is the desired tuned system response. The closed-loop system reduces the magnitude of each overshoot by one-quarter of the previous one.

Table 15-2. Process variable response characteristics.

There are several mathematical methods for determining the tuning constants of a PID controller and for analyzing the stability of a system. One method is a *Bode plot analysis*, which analyzes amplitude (gain) and frequency response (phase shifts) to tune the system. This method is useful when the process transfer function is known. However, in continuous manufacturing processes, this is rarely the case. Therefore, we will present three other practical methods for determining the tuning constants that will produce a quality stable response. These methods are:

- the Ziegler-Nichols open-loop tuning method
- the integral of time and absolute error (ITAE) open-loop tuning method
- the Ziegler-Nichols closed-loop tuning method

The open-loop methods test the process response while the controller is in manual mode without any feedback connections (i.e., *PV* is not being fed back to the controller). Batch control processes are typical applications for open-loop tuning methods. The closed-loop technique tests the process response when the controller is in automatic mode. This tends to produce a better result, since the controller and the process are operating normally. Servo and positioning control processes are typical applications for closed-loop tuning methods. These processes cannot be tuned without feedback; therefore, they cannot use open-loop tuning techniques.

ZIEGLER–NICHOLS OPEN-LOOP TUNING METHOD

John Ziegler and Nathaniel Nichols developed the **Ziegler-Nichols open-loop tuning method** in 1942, and it remains a popular technique for tuning controllers that use proportional, integral, and derivative actions. The Ziegler-Nichols open-loop method is also referred to as a *process reaction method*, because it tests the open-loop reaction of the process to a change in the control variable output (see Figure 15-78). This basic test requires that the response of the system be recorded, preferably by a chart recorder or plotter. Once certain process response values are found, they can be plugged into the Ziegler-Nichols equation with specific multiplier constants for the gains of a controller with either P, PI, or PID actions.

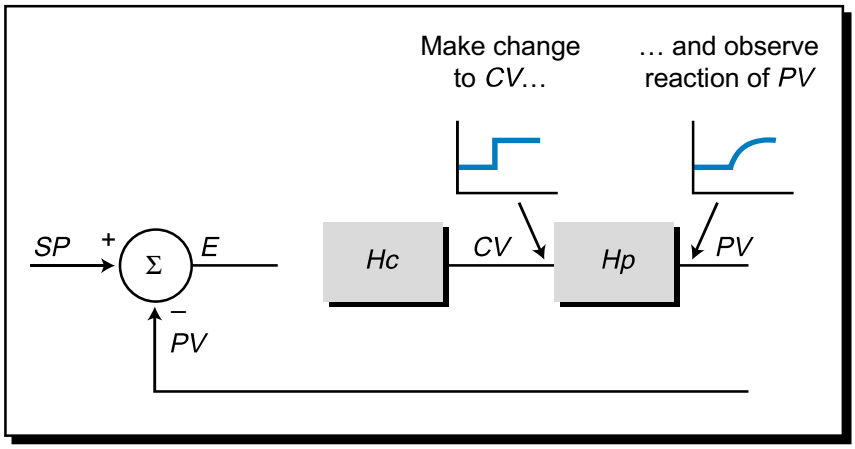


Figure 15-78. Ziegler-Nichols open-loop tuning method.

To use the Ziegler-Nichols open-loop tuning method, you must perform the following steps, which we will illustrate using the system in Figure 15-79:

1. **Bring *PV* to 50%.** With the controller in manual mode (see Figure 15-79), vary the controller's output (*CV*) so that the process variable is at 50% of its range. Turn on the chart recorder and let the system stabilize. For the system in Figure 15-79, let's assume that the control variable must be increased from 50% to 55% to increase *PV* from 40% to 50% of its range.

2. **Step change the *CV* output by 10%.** Manually step the controller's output (*CV*) by 10%. Record on the chart the time value when the step occurs. Observe the process variable response. In Figure 15-79, *CV* steps from 55% to 65% at t_1. The final value of *PV* in response to this change is 165°F, or 65% of the full range, at steady state.

   Figure 15-80 illustrates the process variable response. This response provides important information about the lag time and the rate of change of *PV*.

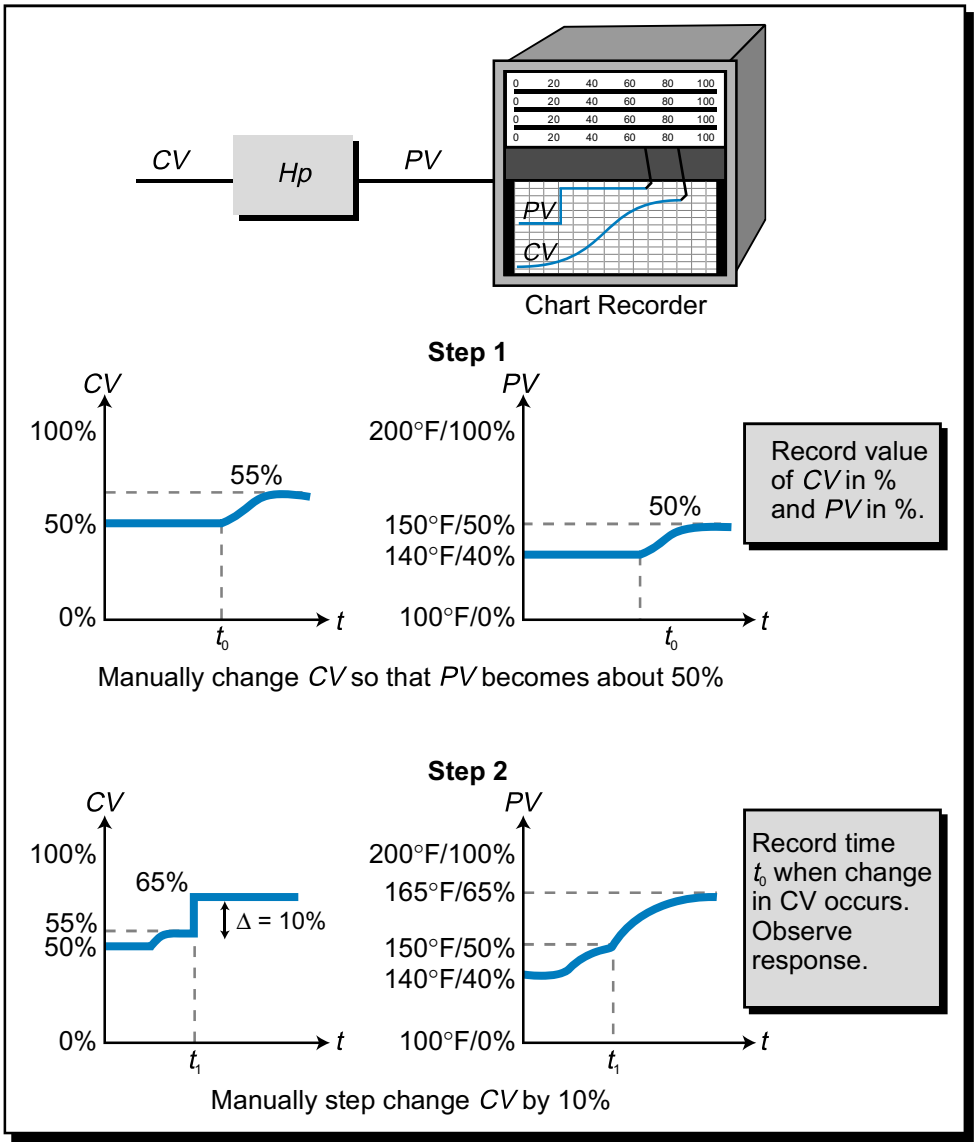


Figure 15-79. Steps 1 and 2 of the Ziegler-Nichols open-loop tuning method.

3. **Find the reaction rate.** Extend the line for the process variable response before the step change (see point A in Figure 15-80). Draw a tangent (point B) to the *PV* response to the *CV* step change at the steepest rise point on the graph to determine the *reaction rate* (*N*) of the process variable. The reaction rate is equal to the change in the process variable over the change in time. It is found by making a right triangle from the tangent line (point C) and finding the tangent of angle θ (the value of the opposite side of the triangle divided by the value of the adjacent side).

4. **Calculate the lag time.** To determine the lag time L_t, find the point at which the tangent line intersects the extension of the original

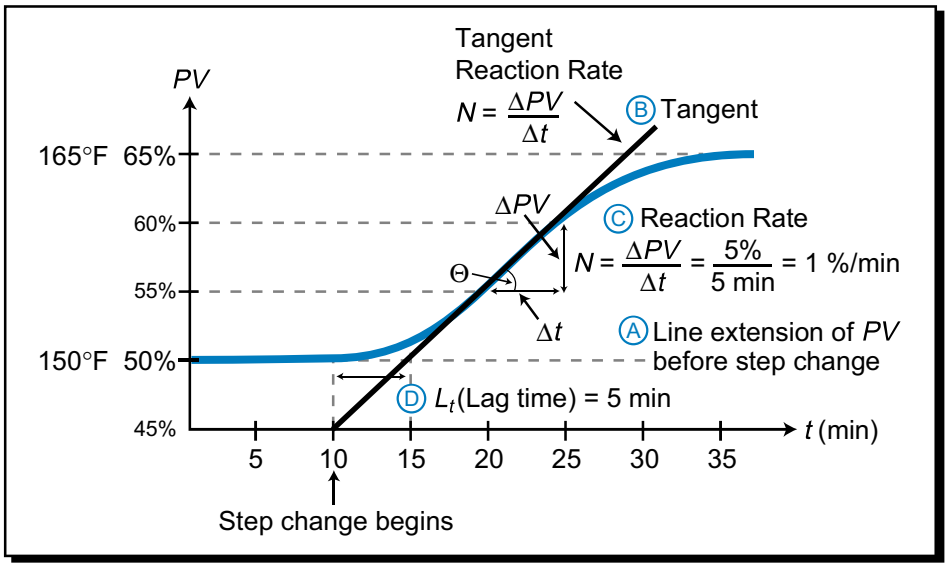


Figure 15-80. Process variable response to step change.

process variable response line (point D). Subtract the time at which the step change occurred from the time at which the tangent and response extension lines crossed.

5. **Determine the loop tuning constants.** Plug in the reaction rate and lag time values to the Ziegler-Nichols open-loop tuning equations for the appropriate type of controller—P, PI, or PID—to calculate the controller constants. Table 15-3 shows these tuning equations. For our example, the constant values for a P, PI, and PID controller would be:

$$\text{P mode}: \quad K_P = \frac{\Delta CV}{(L_t)(N)} = \frac{10\%}{(5\text{ min})(1\frac{\%}{\text{min}})} = 2\frac{\%}{\%}$$

$$\text{PI mode}: \quad K_P = \frac{(0.9)\Delta CV}{(L_t)(N)} = \frac{(0.9)(10\%)}{(5\text{ min})(1\frac{\%}{\text{min}})} = 1.8\frac{\%}{\%}$$

$$T_I = (3.33)(L_t) = (3.33)(5\text{ min}) = 16.65\text{ min}$$

$$\text{PID mode}: K_P = \frac{(1.2)\Delta CV}{(L_t)(N)} = \frac{(1.2)(10\%)}{(5\text{ min})(1\frac{\%}{\text{min}})} = 2.4\frac{\%}{\%}$$

$$T_I = (2)(L_t) = (2)(5\text{ min}) = 10\text{ min}$$

$$T_D = (0.5)(L_t) = (0.5)(5\text{ min}) = 2.5\text{ min}$$

The objective of these tuning constants is to produce a quarter-amplitude response in the process variable.

Type of Controller	Loop Tuning Constant	Tuning Equation
Proportional (P)	K_P	$K_P = \frac{\Delta CV}{L_t \ N}$
Proportional-Integral (PI)	K_P	$K_P = \frac{\Delta CV}{L_t \ N}$
	T_I	$T_I = \ L_t$
Proportional-Integral-Derivative (PID)	K_P	$K_P = \frac{\Delta CV}{L_t \ N}$
	T_I	$T_I = \ L_t$
	T_D	$T_D = \ L_t$

Table 15-3. Ziegler-Nichols open-loop tuning equations.

There are two problems with the Ziegler-Nichols open-loop tuning method. The first problem is that the ratio of the derivative time T_D to the integral time T_I in the equations is designed for a quarter-amplitude response:

$$\frac{T_D}{T_I} = \frac{0.5L_t}{2L_t} = \frac{1}{4} \quad \Rightarrow \quad T_I = 4T_D$$

This does not allow for small changes in T_I and/or T_D. For instance, a system with a PID controller and a process that has a large error at stabilization and a tendency to overshoot and undershoot the set point requires an increase in integral action and an increase in derivative action at the same time. In a PID controller's output equation, the derivative time and the integral time are inversely related:

$$CV_{(t)} = K_P E + \frac{K_P}{T_I}\int_0^t Edt - K_P T_D \frac{dPV}{dt} + CV_{(t=0)}$$

For the derivative action to increase, the derivative gain must increase; yet, for the integral action to increase, the integral gain must decrease. However, in the open-loop, quarter-amplitude tuning equations, the relationship between T_I and T_D is $T_I = 4T_D$, meaning that an increase in T_I causes an increase in T_D. Thus, this relationship causes an imbalance in the controller's PID output equation.

The second problem occurs in systems that have processes in which the lag time L_t equals the dead time D_T. In this situation, the derivative time T_D is equal to:

$$T_D = 0.5L_t$$

or

$$T_D = 0.5D_T \quad (\text{when } D_T = L_t)$$

This means that, as the dead time gets larger, the derivative time T_D also increases. In a process with a long dead time, the opposite is required. As the dead time increases, the derivative action should decrease to compensate for it. This is due to the fact that the dead time is a time delay, which changes the derivative action's effect on the overshoot from the desired negative feedback braking effect into an aggravating effect similar to an undesired positive feedback loop. This is similar to driving a car on an icy surface—the car can veer out of control due to the driver's delay in steering correction because of the slippery road.

EXAMPLE 15-10

The Ziegler-Nichols open-loop tuning method was used to obtain the process response shown in Figure 15-81. Find the tuning parameters for a serial PID controller given that the control variable change that caused this response was 11%.

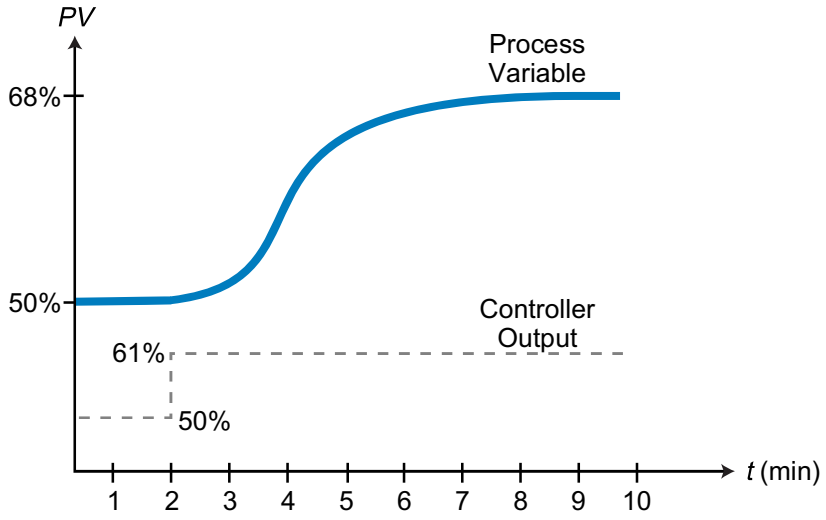


Figure 15-81. Process response obtained by the Ziegler-Nichols open-loop tuning method.

SOLUTION

Figure 15-82 shows the tangent used to determine the tuning values. The lag time L_t is estimated at 1.15 minutes (3.15 min – 2 min). The value of the reaction time N is calculated by finding the tangent of the

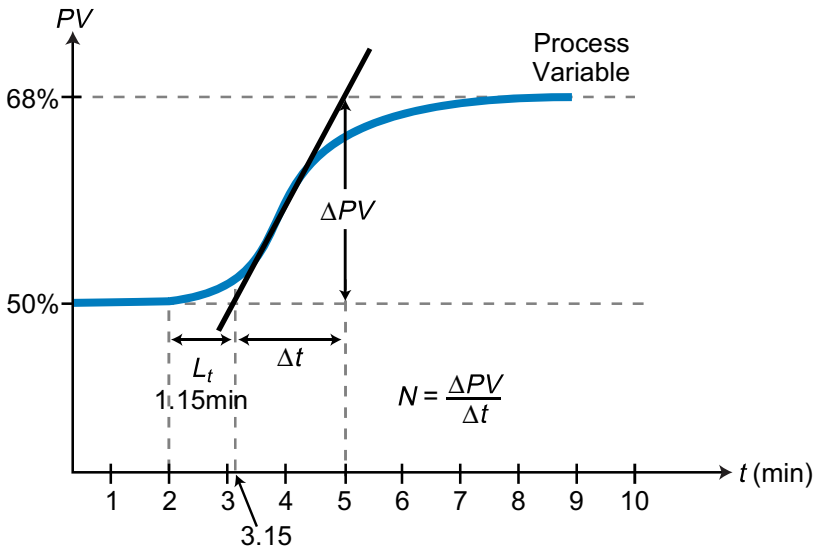


Figure 15-82. Tangent used to determine tuning values.

angle formed by the intersection of the tangent line with the *PV* line extension:

$$N = \frac{\Delta PV}{\Delta t}$$

$$= \frac{-}{-}$$

$$=$$

$$=$$

The PID tuning constants, using the Ziegler-Nichols open-loop equations from Table 15-3, are:

$$K_P = \frac{(1.2)\Delta CV}{(L_t)(N)} = \frac{(1.2)(11\%)}{(1.15\ \text{min})(9.73\ \frac{\%}{\text{min}})} = 1.18\ \frac{\%}{\%}$$

$$T_I = (2)(L_t) = (2)(1.15\ \text{min}) = 2.3\ \text{min}$$

$$T_D = (0.5)(L_t) = (0.5)(1.15\ \text{min}) = 0.575\ \text{min}$$

ITAE OPEN-LOOP TUNING METHOD

The **integral of time and absolute error (ITAE) open-loop tuning method** produces less response oscillation than the Ziegler-Nichols open-loop method and also minimizes the problems associated with it. This method can be used to calculate the tuning constants for processes, such as pH control, that

cannot tolerate as much oscillation as produced by the quarter-amplitude response. The ITAE method, which is based on the minimization of the integral of time and the absolute error of the response, is represented by:

$$\text{ITAE} = \int_0^\infty E_{(t)} t dt$$

where $E_{(t)}$ is the absolute error as a function of time.

The minimization of the overshoot error of a quarter-amplitude response, such as the one achieved with the open-loop Ziegler-Nichols method, occurs during the first overshoot in an ITAE-tuned controller, bringing the system response close to the behavior of a critically damped response. The controller's ITAE tuning settings result in a response that minimizes the first and second amplitude overshoots and virtually eliminates the third (see Figure 15-83). In fact, the ratio of the damping of the second overshoot to the first overshoot is less than 1/8 (second overshoot divided by first overshoot); therefore, the response approximates critically damped behavior. Consequently, the damping in the ITAE method is much better than that in the Ziegler-Nichols open-loop method.

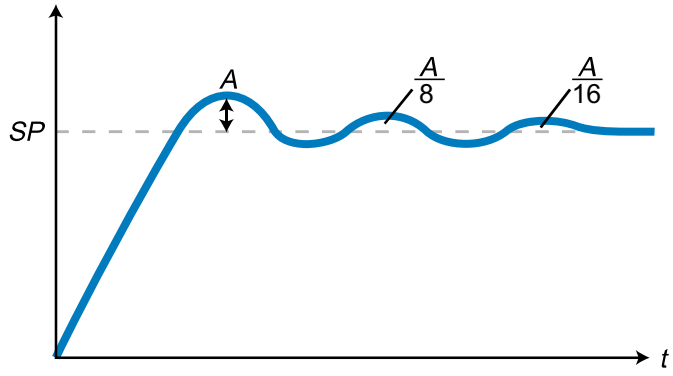


Figure 15-83. ITAE-tuned controller with minimized overshoot.

The procedure for obtaining the controller's tuning constants using the ITAE method is the same as that for the Ziegler-Nichols open-loop method, except that the data interpretation of the graphic response is more detailed. As an example, let's examine the response obtained previously with the Ziegler-Nichols open-loop method and apply the ITAE techniques to obtain the new equation values.

Figure 15-84 illustrates the same response as before with new values for the dead time (D_T) and the process lag (τ, previously called L_t). The tangent to the response is drawn at the steepest rise (like in the Ziegler-Nichols open-loop method). This tangent line determines D_T and τ. Note that τ is calculated from the intersection of the tangent and the *PV* value extension to the time

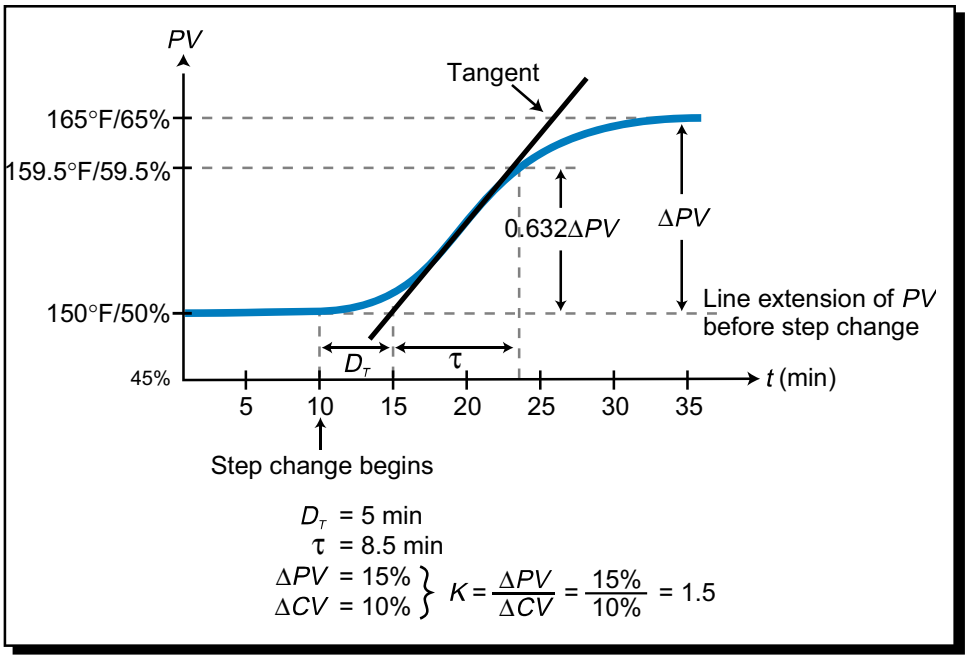


Figure 15-84. Process variable response to step change.

where the actual response, not the tangent, has a value of 63.2% of the final steady-state value. Whereas the final value of the response at steady-state was not used in the Ziegler-Nichols open-loop method, it is used in the ITAE method to determine the percentage gain in the process variable's response. This gain in the process value (ΔPV) is used to determine the gain value K, which will be used in the tuning equations. The gain K is equal to ΔPV divided by ΔCV, where the term ΔCV is the manual change (in percentage of range) executed by the controller's output (process input) over the controlling element (e.g., steam valve). Table 15-4 shows the tuning equations for the ITAE open-loop tuning method.

For the example shown in Figure 15-84, the values of the tuning constants will be:

$$\text{Process gain}: \quad K = \frac{\Delta PV}{\Delta CV} = \frac{15\%}{10\%} = 1.5$$

$$\text{P mode}: \quad K_P = \frac{0.490}{K}\left(\frac{D_T}{\tau}\right)^{-1.084} = \frac{0.490}{1.5}\left(\frac{5}{8.5}\right)^{-1.084} = 0.581$$

$$\text{PI mode}: \quad K_P = \frac{0.586}{K}\left(\frac{D_T}{\tau}\right)^{-0.916} = \frac{0.586}{1.5}\left(\frac{5}{8.5}\right)^{-0.916} = 0.635$$

$$T_I = \frac{\tau}{1.03 - 0.165\left(\frac{D_T}{\tau}\right)} = \frac{8.5}{1.03 - 0.165\left(\frac{5}{8.5}\right)} = 9.111 \text{ min}$$

Type of Controller	Loop Tuning Constant	Tuning Equation
Proportional (P)	K_P	$K_P = \frac{0.490}{K}\left(\frac{D_T}{\tau}\right)^{-1.084}$
Proportional-Integral (PI)	K_P	$K_P = \frac{0.586}{K}\left(\frac{D_T}{\tau}\right)^{-0.916}$
	T_I	$T_I = \frac{\tau}{1.03 - 0.165\left(\frac{D_T}{\tau}\right)}$
Proportional-Integral-Derivative (PID)	K_P	$K_P = \frac{0.965}{K}\left(\frac{D_T}{\tau}\right)^{-0.855}$
	T_I	$T_I = \frac{\tau}{0.796 - 0.147\left(\frac{D_T}{\tau}\right)}$
	T_D	$T_D = 0.308\tau\left(\frac{D_T}{\tau}\right)^{0.929}$
$K = \frac{\Delta PV}{\Delta CV}$		

Table 15-4. ITAE open-loop tuning equations.

$$\text{PID mode}: \quad K_P = \frac{0.965}{K}\left(\frac{D_T}{\tau}\right)^{-0.855} = \frac{0.965}{1.5}\left(\frac{5}{8.5}\right)^{-0.855} = 1.013$$

$$T_I = \frac{\tau}{0.796 - 0.147\left(\frac{D_T}{\tau}\right)} = \frac{8.5}{0.796 - 0.147\left(\frac{5}{8.5}\right)} = 11.980 \text{ min}$$

$$T_D = 0.308\tau\left(\frac{D_T}{\tau}\right)^{0.929} = (0.308)(8.5)\left(\frac{5}{8.5}\right)^{0.929} = 1.599 \text{ min}$$

In the ITAE loop tuning method, the controller settings ensure a damping ratio of less than 1/8 for the P and PI modes. The PID mode, however, still presents a problem in systems with large dead times, although this problem is not as severe as it is in the Ziegler-Nichols open-loop method. This problem stems from the fact that the exponent of the derivative action (0.929) term T_D is close to the value of 1, which makes an approximate value of T_D be 0.308 times the value of the dead time:

$$T_D \approx 0.308\tau\left(\frac{D_T}{\tau}\right)^{\approx 1}$$

$$\approx 0.308\tau\left(\frac{D_T}{\tau}\right)$$

$$\approx 0.308 D_T$$

As discussed earlier, a derivative term proportional to the dead time can cause an aggravating response in the system that affects the overshoot. This problem in the ITAE method, however, is less pronounced than it is in the Ziegler-Nichols open-loop method where $T_D = 0.5D_T$ when $L_t = D_T$. Note also that the ITAE method does not contain a fixed ratio constant of T_D/T_I, which is the case in the Ziegler-Nichols open-loop method (T_D/T_I = 1/4). Therefore, the ITAE method cannot cause a potential imbalance in the PID controller equation.

Derivative control action should not be used in processes with large dead times. As a de facto rule of thumb, a large dead time is one in which the ITAE open-loop test produces a value of D_T that is greater than τ. If this is the case, then T_D should be set to zero, implementing control without derivative action.

ZIEGLER-NICHOLS CLOSED-LOOP TUNING METHOD

The **Ziegler-Nichols closed-loop tuning method** is used to obtain the controller constants [K_P, K_I (or T_I), and K_D (or T_D)] in a system with feedback. This technique allows for the tuning of processes, such as servo positioning systems, that cannot run in an open-loop environment.

The main objective of the Ziegler-Nichols closed-loop method is to find the value of the proportional-only gain that causes the control loop to oscillate indefinitely at a constant amplitude (see Figure 15-85). This gain, which causes steady-state oscillations, is called the *ultimate proportional gain* (K_{PU}). Another important value associated with this proportional-only control tuning method is the *ultimate period* (T_U). The ultimate period is the time required to complete one full oscillation once the response begins to oscillate at a constant amplitude. These two parameters, K_{PU} and T_U, are used to find the loop-tuning constants of the controller (P, PI, or PID). To find the values of these parameters and to calculate the tuning constants, you must do the following:

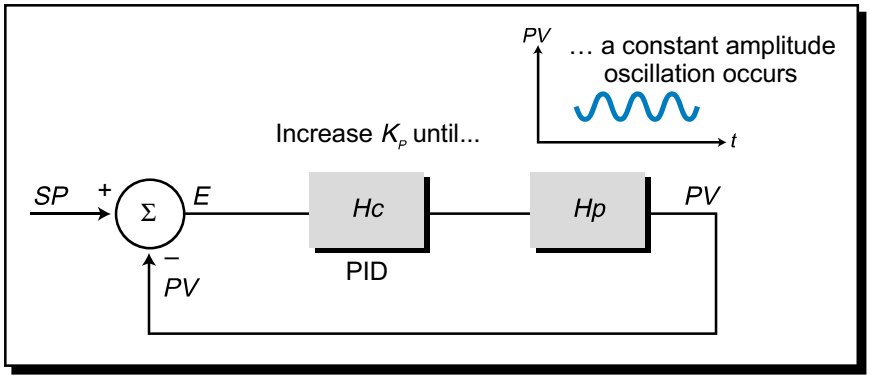


Figure 15-85. Ziegler-Nichols closed-loop tuning method.

1. **Implement proportional-only control.** Remove all integral and derivative actions. In most controllers, the removal of integral time (T_I) is done by setting T_I equal to 999 (or its largest number) or by setting K_I equal to 0. To remove the derivative action, set K_D (or T_D) to 0. Place the controller in automatic mode with the control variable and process variable at 50%.

2. **Create a disturbance in the system.** Create a small disturbance in the loop by slightly changing the set point (see point A in Figure 15-86). Start increasing the proportional gain, or lowering the percentage proportional band (%*PB*), until the process variable begins to oscillate (point B). Continue to increase and decrease the gain until the oscillations have a constant amplitude (point C). Record this response and determine the ultimate proportional gain and ultimate period.

   In the example system in Figure 15-86, the set point of 150°F is slightly changed to 155°F while the gain is increased to $K_D = 3$ (point A). Once the oscillation starts, the set point is returned to 150°F. The oscillation begins to decay at t_2, so the gain is increased again to $K_P = 4$ (point B), However, the response starts to grow in amplitude at t_3, so the gain is reduced to $K_P = 3.5$. At this point, the response exhibits a constant amplitude oscillation (point C). Therefore, the ultimate gain (K_{PU}) is 3.5 and the ultimate period (T_U) is 10 minutes.

3. **Calculate the constants.** Plug the K_{PU} and T_U values into the Ziegler-Nichols closed-loop tuning equations to determine the settings for the controller to be used. Table 15-5 provides the tuning equations for this closed-loop method.

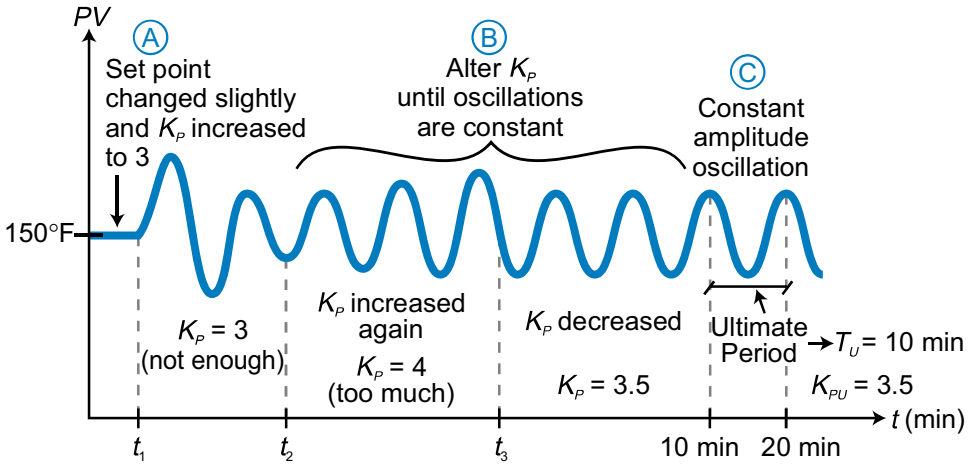


Figure 15-86. System tuned using the Ziegler-Nichols closed-loop tuning method.

Type of Controller	Loop Tuning Constant	Tuning Equation
Proportional (P)	K_P	$K_P = (0.5)(K_{PU})$
Proportional-Integral (PI)	K_P	$K_P = (0.45)(K_{PU})$
	T_I	$T_I = \frac{T_U}{1.2}$
Proportional-Integral-Derivative (PID)	K_P	$K_P = (0.6)(K_{PU})$
	T_I	$T_I = \frac{T_U}{2}$
	T_D	$T_D = \frac{T_U}{8}$
Note: $\%PB = \frac{1}{K_P};\ K_I = \frac{1}{T_I};\ K_D = T_D$		

Table 15-5. Ziegler-Nichols closed-loop tuning equations.

For the example system in Figure 15-86, the tuning constants for each controller mode will be:

$$\text{P mode}: \quad K_P = (0.5)(K_{PU}) = (0.5)(3.5) = 1.75$$

$$\text{PI mode}: \quad K_P = (0.45)(K_{PU}) = (0.45)(3.5) = 1.575$$

$$T_I = \frac{T_U}{1.2} = \frac{10 \text{ min}}{1.2} = 8.33 \text{ min}$$

$$\text{PID mode}: \quad K_P = (0.6)(K_{PU}) = (0.6)(3.5) = 2.10$$

$$T_I = \frac{T_U}{2} = \frac{10 \text{ min}}{2} = 5 \text{ min}$$

$$T_D = \frac{T_U}{8} = \frac{10 \text{ min}}{8} = 1.25 \text{ min}$$

The magnitude of the constant oscillation amplitude is not important in the Ziegler-Nichols closed-loop tuning equations; however, all the elements in the loop must be within operating range. For example, the control variable must not vary from fully open to fully closed to create the oscillation.

The Ziegler-Nichols closed-loop method provides a quarter-amplitude response. This response is acceptable for P and PI modes; however, in PID mode, it presents the same equation imbalance as experienced in the Ziegler-Nichols open-loop technique. Again, this is due to the fixed ratio of the derivative time to the reset time.

Another problem with this closed-loop technique is that the majority of process control loops in manufacturing operations cannot tolerate oscillations for long periods of time, especially if many trials are necessary. The time required to obtain a steady-state oscillating response is typically 30–60 minutes, but it can take up to several hundred minutes. The Ziegler-Nichols closed-loop method can be slightly altered to avoid this time problem.

ALTERED ZIEGLER-NICHOLS CLOSED-LOOP TUNING METHOD

The altered version of the Ziegler-Nichols closed-loop method provides an approximation of the desired quarter-amplitude response. The test procedure in this method reduces the time required to observe the process variable's steady-state response to a change in controller output. Instead of changing the value of *CV* (through the inverse of the proportional gain) until *PV* exhibits a constant amplitude oscillation, this altered method compares the responses of several trials until an approximate quarter-amplitude response is obtained. The procedures and steps used in this altered method are the same as in the standard Ziegler-Nichols closed-loop method, where several trials are performed by changing the proportional gain until the response approximates a quarter-amplitude response.

Once the process variable response approximates a quarter-amplitude response, the chart recorder records the period of decaying oscillation (see Figure 15-87), as well as the gain of the proportional action. The gain and decaying oscillation period for this quarter-amplitude test are called $K_{1/4}$ (quarter-amplitude gain) and $T_{1/4}$ (quarter-amplitude period), instead of K_{PU} and T_U. These $K_{1/4}$ and $T_{1/4}$ parameters are then plugged into the standard Ziegler-Nichols closed-loop tuning equations (shown previously in Table 15-5) in place of K_{PU} and T_U, respectively. The values of K_{PU} and T_U will be:

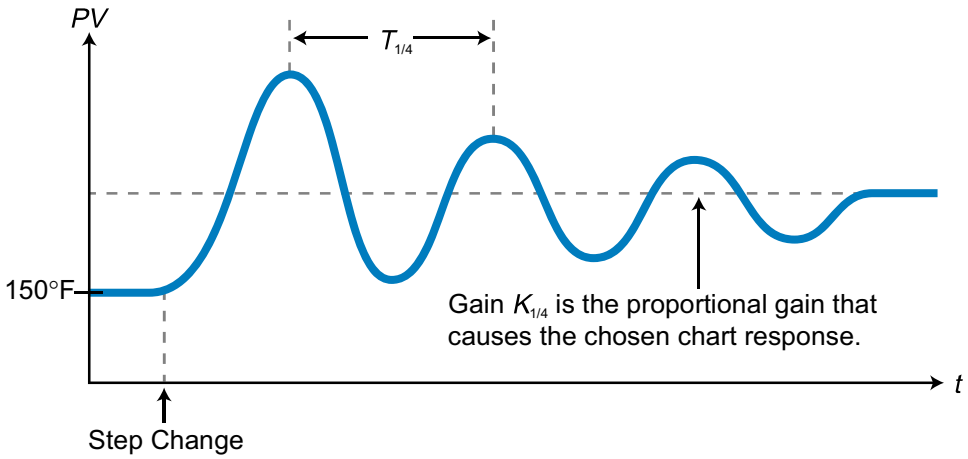


Figure 15-87. Quarter-amplitude test.

$$K_{PU} = 2K_{1/4}$$
$$T_U = T_{1/4}$$

This is due to the fact that the proportional gain in the standard Ziegler-Nichols method is:

$$\begin{aligned} K_P &= 0.5K_{PU} \\ &= 0.5\left(2K_{1/4}\right) \\ &= K_{1/4} \end{aligned}$$

The values that will be obtained for T_I in the integral action of the PI mode and for T_I and T_D in the integral and derivative actions of the PID mode will be slightly larger than those values calculated through the unaltered Ziegler-Nichols closed-loop method, since the $T_{1/4}$ reading will be slightly larger than the T_U reading. Although this method avoids the problem of long test periods, it does present another problem—the determination of the period of decaying oscillation. Reading a decaying oscillation period is more difficult than reading a period of constant amplitude oscillation, because it is harder to ascertain where the decaying sinusoidal curve's middle section is.

SOFTWARE TUNING METHODS

Another method for tuning PID controllers is the use of software tuning systems. These software packages run on personal computers using Unix, Windows, or another platform (see Figure 15-88). They connect to the controller or PLC either directly or via a DDE (dynamic data exchange) interface. These software systems reduce the tuning time and, at the same time, optimize control loop performance.

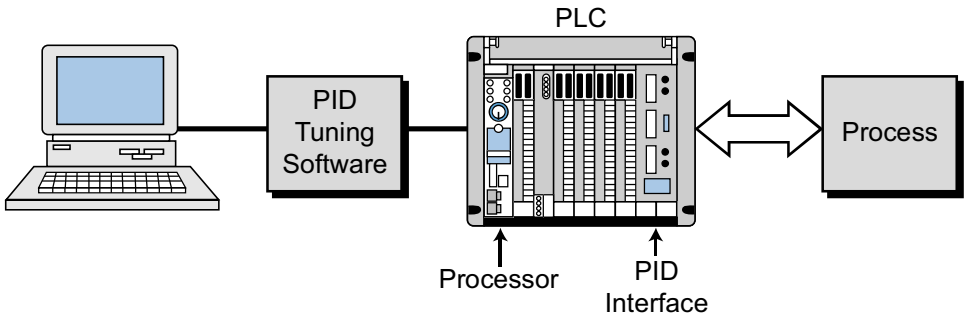


Figure 15-88. Software loop tuning.

Software tuning programs provide numerous viewing selections, windows, and on-line help screens that show process characteristics including simulations, modeling, plots, and frequency responses (see Figure 15-89). Additionally, they provide important information about the process itself that can be extremely difficult, if not impossible, to obtain manually. For example,

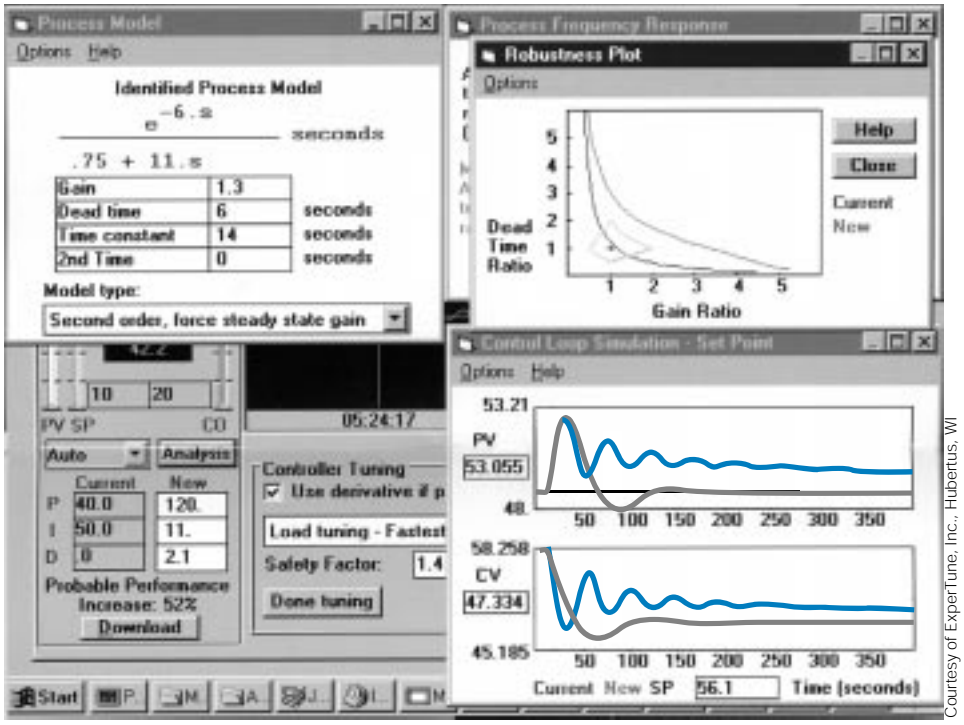


Figure 15-89. Software tuning program screens showing process characteristics.

ExperTune®, by ExperTune, Inc., identifies the transfer function of the process during the tuning test (see Figure 15-90), thus providing information such as process gain, dead time, and lag time constants. This software optimizes the PID tuning parameters automatically for even the most complicated of processes. It also performs tuning for cascade PID loops. Additionally, it provides a "robustness" plot, which shows how a change in the process dead time and/or gain will affect the closed-loop system's stability given the current loop tuning constants. In fact, the robustness plot shows all the gain and dead time values that will allow for stable closed-loop operation. For example, it shows how a change in gain will affect the stability of the closed-loop system as it is tuned. Thus, the robustness plot provides a quick look at the trade-offs between tuning and stability.

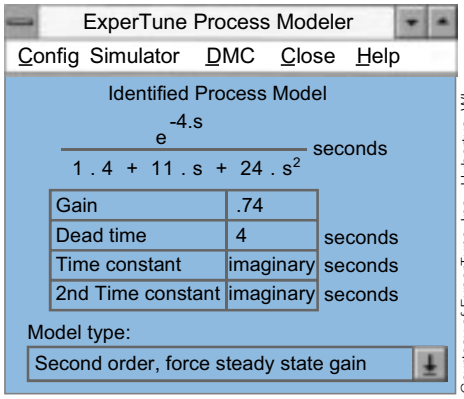


Figure 15-90. Process transfer function obtained through software loop tuning.

Software tuning systems also allow "what if" analysis, meaning that they suggest new controller constants given hypothetical process values. They can also provide PID tuning parameters from ASCII data files about the plant process. These features reduce the amount of time required to tune the system because they can produce a database of tuning constants for a variety of process scenarios. These software tuning systems greatly benefit users wanting to simplify their process tuning efforts while reducing the amount of time required for manual tuning.

15-13 SUMMARY

A controller in a process control system receives data about the set point value and the actual process variable value and then compares these values to generate an error value. The controller uses this error value according to a control algorithm to manipulate the control variable. The control variable directs a final control element (e.g., a valve) to bring the process variable to the desired set point, eliminating the system error. A controller is direct acting if its output increases in response to an increase in the process variable; it is reverse acting if its output decreases in response to an increase in the process variable.

There are two types of controller modes: discrete and continuous. Some of the most commonly used discrete-mode controllers are the two-position, or ON/OFF, mode and the three-position mode. Two-position controllers turn the output ON (100% open) or OFF (0% open) once the process variable crosses an error deadband around the set point. Three-position controllers are an extension of two-position ones in the sense that they have one more output level. This type of discrete controller provides 0%, 50%, and 100% controller output levels.

Continuous-mode controllers include proportional controllers, integral controllers, and derivative controllers. These controllers can also be combined to provide proportional-integral (PI), proportional-derivative (PD), and proportional-integral-derivative (PID) controllers.

In proportional control, the corrective output action is proportional to the size of the error deviation ($E = SP - PV$). This type of control provides a fast response and is relatively simple to implement. Proportional control, however, always leaves some offset error between the desired and actual values of the process variable. If the proportional band in a proportional controller is set too wide, the offset error will be larger than if the proportional band is narrow. However, too narrow of a proportional band will create oscillation and, thus, system instability.

Integral control provides corrective action as a function of the integral of the error (i.e., the sum of the error over time). It provides its highest gain, or corrective action, at low frequencies (i.e., in a slowly changing process).

Integral action tends to ignore high-frequency changes, such as noise or rapid transients, in the process. Although this control mode eliminates the inherent offset error present in a proportional controller, it adversely affects stability. If the integration time is reduced, the response during the slow period of the process will become faster, inducing cycling.

Derivative control provides corrective action as a function of either the rate of change of the error or the process variable. A derivative controller provides its highest gain, or corrective action, at high frequencies. Hence, it provides an anticipatory response to the process variable change. The derivative mode cannot be used alone and does not eliminate residual error.

A proportional-integral, or PI, controller combines the fast response of proportional action with the offset error elimination of integral action. A PI controller with an integral time that is too long will exhibit a response that will take a long time to return to the set point. Conversely, a shorter integral time will cause the process variable to cross the set point faster, resulting in damped oscillations. An integral time that is too short, however, will produce continuous oscillations.

A proportional-derivative, or PD, controller provides better response stability than a PI controller, but it does not provide offset error elimination. A PD controller is useful in applications where the process has a long lag time delay in its recovery from a disturbance. The derivative action in this controller provides a lead function, which cancels some of the process lag and allows the proportional band to become narrower. This improves response and stability. A PD controller does not eliminate offset error, but a narrower proportional band can reduce the amount of residual error in the system. A derivative time constant (K_D or T_D) that is too long will cause the process variable to change too rapidly and overshoot the set point with damped oscillation. Conversely, if the derivative constant is too short, the process variable will take too long to reach the set point.

A proportional-integral-derivative, PID, controller combines the increased stability of a PD controller with the eliminated offset feature of a PI controller. A PID controller can be used to control almost any type of process, including those with long lag times. The gains for each of the control actions in a PID controller can be derived experimentally utilizing several tuning methods.

An integral, or reset, windup situation occurs when an integral action saturates a controller's output at 100%. Integral windup usually happens during the start-up of a process. This condition occurs when the error in a slow-responding process system is large. The proportional action tries to bring the process variable closer to the set point, but the slow speed of the process response and the presence of error induces the integral action to continue, keeping the controller at 100% output. This condition can be prevented by disabling the integral action once the controller's output reaches 100%.

Bumpless transfer refers to a controller's ability to switch from manual to automatic control and vice versa without a step change in the input to the process. In a bumpless transfer, the manual controller station tracks the automatic controller's output and vice versa to keep the control variable output constant.

Cascade control refers to an advanced control technique where the output of one controller is the set point input to another controller. Cascade control provides more precise control than noncascaded control, since the secondary (or inner) loop will react quickly to a disturbance before it starts to affect the primary (or outer) loop.

Controller loop tuning is the process of manipulating the parameters (gains) in a PID controller so that the response of the process system is satisfactory. A satisfactory response is one that exhibits the desired speed of response, yet meets the required accuracy and stability criteria. Control processes are generally tuned under operating conditions, as opposed to start-up conditions, so that the process variable is stable at an operating point. Since the transfer function of a process is rarely known, experimental measurements and tests can be made to obtain parameters that will help determine the desired controller gains for PID control. These experimental measurements are known as the "modeling" of the system. Some of the most popular experimental tuning methods are the Ziegler-Nichols open-loop method, the integral of time and absolute error (ITAE) open-loop method, and the Ziegler-Nichols closed-loop method.

During the modeling of a process, a known disturbance is created and the resulting response is observed and recorded. The disturbance should be one that actually occurs during process operation (e.g., a change in load, flow rate, or speed of the system). However, the creation of this type of disturbance is impractical to implement in a real-life situation; therefore, a change in set point is most often used as the disturbance to the process. The values obtained from this disturbance are then plugged into the tuning method's equations to obtain the values for the proportional, integral, and derivative gain terms. These values are then used as the starting parameters for the controller, which will provide process control by minimizing the error in the system.



KEY TERMS

cascade control
continuous-mode controller
derivative controller
direct-acting controller
discrete-mode controller
integral controller
integral of time and absolute error open-loop tuning method (ITAE)
integral windup
loop tuning

proportional controller
proportional-derivative controller
proportional-integral controller
proportional-integral-derivative controller
quarter-amplitude response
reverse-acting controller
three-position controller
two-position controller
Ziegler-Nichols closed-loop tuning method
Ziegler-Nichols open-loop tuning method